SURFING

# WACE HUMAN BIOLOGY

## Unit 4 Human Variation and Evolution

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# Introduction

Each book in the *Surfing* series contains a summary, with occasional more detailed sections, of all the mandatory parts of the syllabus, along with questions and answers.

All types of questions – multiple choice, short response, structured response and free response – are provided. Questions are written in exam style so that you will become familiar with the concepts of the topic and answering questions in the required way.

Answers to all questions are included.

A topic test at the end of the book contains an extensive set of summary questions. These cover every aspect of the topic, and are useful for revision and exam practice.

# Words To Watch

**account, account for** State reasons for, report on, give an account of, narrate a series of events or transactions.

**analyse** Interpret data to reach conclusions.

**annotate** Add brief notes to a diagram or graph.

**apply** Put to use in a particular situation.

**assess** Make a judgement about the value of something.

**calculate** Find a numerical answer.

**clarify** Make clear or plain.

**classify** Arrange into classes, groups or categories.

**comment** Give a judgement based on a given statement or result of a calculation.

**compare** Estimate, measure or note how things are similar or different.

**construct** Represent or develop in graphical form.

**contrast** Show how things are different or opposite.

**create** Originate or bring into existence.

**deduce** Reach a conclusion from given information.

**define** Give the precise meaning of a word, phrase or physical quantity.

**demonstrate** Show by example.

**derive** Manipulate a mathematical relationship(s) to give a new equation or relationship.

**describe** Give a detailed account.

**design** Produce a plan, simulation or model.

**determine** Find the only possible answer.

**discuss** Talk or write about a topic, taking into account different issues or ideas.

**distinguish** Give differences between two or more different items.

**draw** Represent by means of pencil lines.

**estimate** Find an approximate value for an unknown quantity.

**evaluate** Assess the implications and limitations.

**examine** Inquire into.

**explain** Make something clear or easy to understand.

**extract** Choose relevant and/or appropriate details.

**extrapolate** Infer from what is known.

**hypothesise** Suggest an explanation for a group of facts or phenomena.

**identify** Recognise and name.

**interpret** Draw meaning from.

**investigate** Plan, inquire into and draw conclusions about.

**justify** Support an argument or conclusion.

**label** Add labels to a diagram.

**list** Give a sequence of names or other brief answers.

**measure** Find a value for a quantity.

**outline** Give a brief account or summary.

**plan** Use strategies to develop a series of steps or processes.

**predict** Give an expected result.

**propose** Put forward a plan or suggestion for consideration or action.

**recall** Present remembered ideas, facts or experiences.

**relate** Tell or report about happenings, events or circumstances.

**represent** Use words, images or symbols to convey meaning.

**select** Choose in preference to another or others.

**sequence** Arrange in order.

**show** Give the steps in a calculation or derivation.

**sketch** Make a quick, rough drawing of something.

**solve** Work out the answer to a problem.

**state** Give a specific name, value or other brief answer.

**suggest** Put forward an idea for consideration.

**summarise** Give a brief statement of the main points.

**synthesise** Combine various elements to make a whole.

# 1 Assumed Knowledge

## QUESTIONS

1. The diagram shows a chromosome.

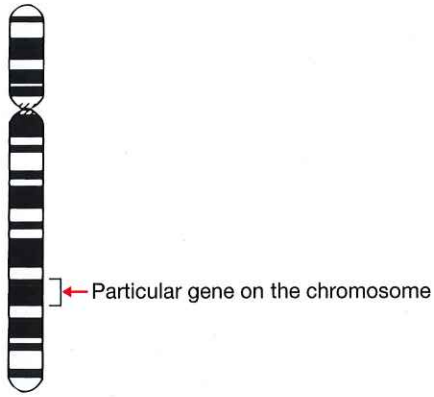



**Figure 1.1** Chromosome.

   (a) Define a chromosome.
   (b) What is the relationship between gene and chromosome?
2. Distinguish between a gene and an allele.
3. Distinguish between autosomes and sex chromosomes.
4. The diagram shows the karyogram for person A.

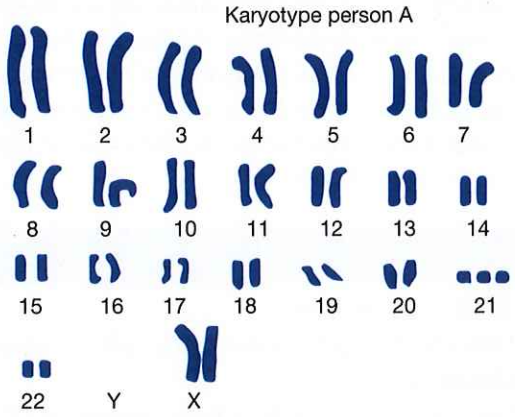



**Figure 1.2** Karyogram.

   (a) Is this person male or female?
   (b) Define trisomy.
   (c) Person A has a trisomy disorder. Name this disorder.
5. What is meant by the genome sequence?
6. Define a gene pool.
7. What is meant by allele frequency?
8. Define polyploidy.
9. What is a mutation?
10. Distinguish between genotype and phenotype.
11. What features classify humans as primates?
12. What is comparative genomics?
13. Define biodiversity.
14. What is a fossil?
15. What is an index fossil?
16. Define a species.
17. Where is most DNA in a human (eukaryote) cell?
18. The diagram shows a biological process.

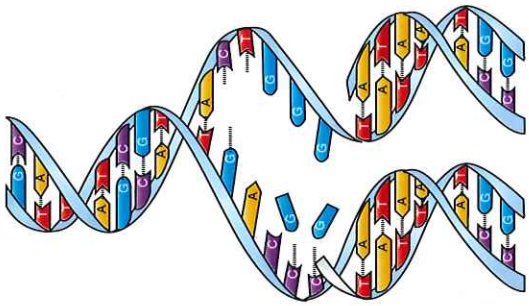


**Figure 1.3** Biological process.

    Name this process.
19. Name the four nitrogenous bases in DNA.
20. What is a nucleotide?
21. State the law of superposition.
22. Distinguish between absolute dating and relative dating.
23. List the levels of classification from kingdom to species.
24. (a) What is meant by binomial nomenclature?
    (b) In this system how are humans classified?
25. Humans are classified in the order 'Primates'. Name some other animals that are primates.
26. Define natural selection.
27. When does evolution occur?
28. List the major stages in the evolution of living things from organic molecules to multicellular organisms.
29. The photograph shows a model of the human head and brain.

**Figure 1.4** Human head and brain.

    (a) Identify structure A.
    (b) Outline the function of structure A.
    (c) Suggest why the size of structure A is important in human evolution.
30. Define biotechnology.

# 2 Mutation

**Mutation** is the change in the DNA genetic make-up of a cell. A mutation may be neutral and have no observable effect on the organism, it may be harmful and decrease the lifespan of the organism or it may be beneficial, giving it an evolutionary advantage over other individuals in the population.

## Causes of mutation

Mutations can occur spontaneously and randomly, e.g. due to errors in DNA replication or cell division, or they can be induced by mutagens (environmental factors) such as radiation (e.g. X-rays and UV rays) or chemical agents (e.g. benzene, formalin or Agent Orange).

If the mutation is in a gonad cell, it may be inherited with the sperm or ova, while mutations in body cells are not inherited. In the 1920s, Hermann Muller used X-rays to obtain mutant fruit flies, *Drosophila melanogaster*, so he could carry out his experiments on inheritance, showing radiation caused mutation. The atomic bomb on Hiroshima caused many mutations, showing gamma radiation is mutagenic.

## Types of mutation

There are two basic types of mutation – point mutation and chromosome mutation.

A **point mutation** changes the base sequences of a single gene and may form a new allele. For example, during DNA replication, sometimes one base is replaced by another nitrogenous base which in turn changes the mRNA. Since the code on the mRNA is translated into amino acids, the activity of the cell and the proteins it produces will be affected, e.g. on mRNA the codon GGG = glycine but if there is a base-pair substitution so that the mRNA reads CGG = arginine, the cell will now make arginine instead of glycine.

There are two types of point mutations – base pair substitutions and base pair insertions or deletions. Insertions and deletions can have a massive impact as they cause a **frameshift** and the triplet code is not read correctly, unless the shift is a multiple of three. To show the impact consider the sentence – *The dog ran up the road.* If you delete the 'e' from 'the', the sentence becomes – *thd ogr anu pth ero ad.* As the new sentence makes no sense, a deletion in a triplet code makes no sense and the polypeptide cannot be synthesised.

A **chromosome mutation** can involve either a change in the number of chromosomes or a rearrangement in the structure of a chromosome. Sometimes there are not 46 chromosomes present; Down syndrome, or trisomy 21, involves the presence of an extra copy of chromosome 21, so that the person has 47 chromosomes rather than 46. People with Klinefelter syndrome have 44 autosomes and the sex chromosomes XXY.

The rearrangement of a chromosome can involve inversion, when chromosome segments are upside down so the genes appear in the reverse order; translocation, when pieces are removed and inserted somewhere else; or deletion, when pieces of chromosome are lost.

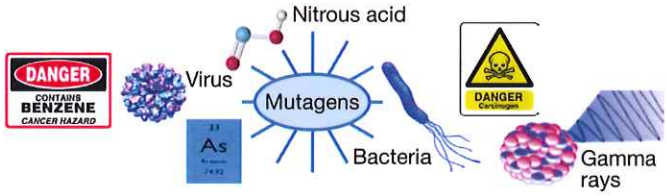



Figure 2.1 Mutagens.

## Mutation and variation

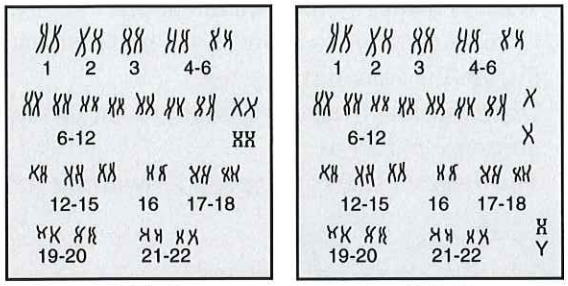
The discovery that mutations can be caused by radiation, certain chemicals and so on is most important to Darwin's theory of evolution. When Darwin proposed his theory he did not know how variations arose in populations. Mutation explains one source of variation in a population.

Variation can arise several ways. Variation occurs due to random segregation during meiosis – each chromosome of the homologous pair sorts independently and randomly so that the gametes can have different combinations of chromosomes. Variation occurs due to crossing over during meiosis as new combinations of characteristics arise. Variation occurs due to sexual reproduction when the union of a sperm with an ovum forms a new combination of genes, not identical to the parents. Variation is mainly important when the environment is changing. If one variant has features more suited to the new environment, natural selection will lead to evolution and survival of the population.

## QUESTIONS

1. What is a mutation?
2. Describe two 20th century examples of causes of mutation.
3. Use an example to show how changes in DNA sequences can result in changes in proteins and thus in cell activity.
4. What is a mutagen? Give an example.
5. Describe the cause of mutations.
6. Describe the circumstances needed for a mutation to be inherited.
7. Distinguish a point mutation from a chromosome mutation.
8. (a) What are the two types of point mutation?
   (b) Explain why base pair deletions of insertions can have a dramatic effect on polypeptide synthesis.
9. Explain how an understanding of the source of variation in organisms has provided support for Darwin's theory of evolution by natural selection.

10. (a) Describe three ways in which variation can arise in a population.
    (b) Explain why variation in a population is important.
11. The diagram shows the karyotype from a pair of human twins.




Twin A Twin B

Figure 2.2 Karyotype of twins.

    (a) What is a karyotype?
    (b) Are the twins identical? Explain your reasoning.
    (c) Karyotypes are often used to show genetic malfunctions and mutations. Describe one type of mutation detected by karyotypes and explain its significance.
12. Natural selection requires that there is variation within a population. Which of the following is a source of variation in both sexual reproduction and asexual reproduction?
    (A) Crossing over.
    (B) Mutation.
    (C) Random segregation.
    (D) Union of gametes.
13. The diagram shows the eye colour of three members of one family.

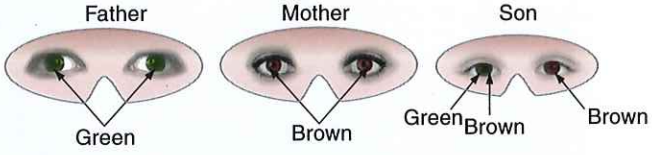



Figure 2.3 Eye colour of three family members.

The appearance of the son's right eye is probably due to a mutation in which of the following?
    (A) Sperm. (B) Ova.
    (C) Zygote. (D) Embryo.
14. Why are harmful mutations not always eliminated from the gene pool?
    (A) They are needed in case the environment changes.
    (B) They all provide immunity against one disease.
    (C) They are recessive and carried by heterozygous forms.
    (D) They are dominant and prevent their removal.
15. Mutations are a source of new alleles in organisms. What causes mutation?
    (A) Environmental agents such as X-rays, ultraviolet radiation and gamma rays.
    (B) Chemicals such as mustard gas which react with DNA.
    (C) Spontaneous occurrence during cell division.
    (D) All of the above.
16. When insecticides, such as pyrethrum, are first introduced, the recommended concentration kills nearly all of the targeted insect pests. After widespread use and several years, the same concentration is only partially effective. What is the development of resistance in insect pests due to?
    (A) The insects realised they needed to acquire a resistance for survival of their species.
    (B) The insects built up a resistance to the insecticide over several generations.
    (C) Some insects were naturally immune, survived and reproduced.
    (D) The insecticide caused a mutation to make the insect immune.
17. Chromosome 11 consists of 438 bases. If there is a change on this chromosome in the gene coding for β-chain haemoglobin for the sixth amino acid, the haemoglobin produced has different properties, i.e. the haemoglobins cluster together to form fibres to deform the red blood cell into a sickle shape. The person has sickle cell anaemia. Identify this type of mutation.
    (A) Point mutation.
    (B) Chromosome mutation.
    (C) Translocation.
    (D) Deletion.
18. The table shows the mutation rate per million gametes per generation.

| Disease | Dominant or recessive gene | Mutation per million gametes per generation |
|---|---|---|
| Haemophilia | Sex-linked recessive | 25 to 32 |
| Tay-Sachs disease | Recessive | 11 |
| Retinoblastoma | Dominant | 15 to 23 |
| Albinism | Recessive | 28 |

From this data, what can you conclude about mutation within the population?
    (A) A high proportion of people have these diseases.
    (B) The incidence of these diseases is low.
    (C) Sex-linked genes have a higher mutation rate than autosomal genes.
    (D) Everyone has a mutation, somewhere.
19. About 10 000 chemicals in common use in industrialised countries have been identified as mutagens. What could be one effect of these chemicals?
    (A) Increased birth rate.
    (B) Increased incidence of cancer.
    (C) Decreased water purity.
    (D) Decreased incidence of melanomas.

# 3 Mutation and Polypeptide Synthesis

A **mutation** is a permanent change in the genetic information. Mutations can occur in somatic cells or in germ line cells. Most mutations are in somatic cells, usually have a localised influence, e.g. the descendants of that cell and do not affect the next generation. A **somatic cell** is any cell in a multicellular organism except gametes. A mutation in a **germ line cell** occurs in any germ cell, will involve all cells and can be inherited by offspring. A germ cell is any cell that gives rise to the gametes.

Most mutations are recessive and harmful; some are lethal.

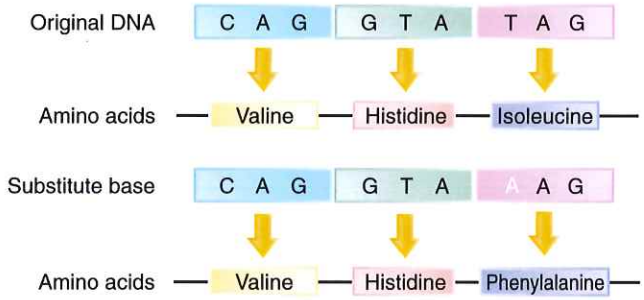



Figure 3.1 A change in code changes the amino acid sequence.

If a mutation causes a change in the DNA code, this will influence the protein produced. The activities of the cell can be disrupted if a particular protein cannot be produced. This can even lead to disease. For example, if a mutation on the β-haemoglobin gene puts thymine instead of guanine at one particular location, the person will suffer from the disease thalassemia major, which results in abnormalities such as severe anaemia and growth retardation.

## Spontaneous and induced mutations

**Spontaneous mutations** occur naturally with about one in every million to one in every billion divisions and the causative agent cannot be identified. Mutations can also be induced. In an **induced mutation** the causative agent can be identified, e.g. exposure to radiation such as X-rays, gamma rays or certain chemical substances.

## Missense and nonsense mutation

Sometimes a codon is changed by point mutation and the base pair substitution makes sense and still codes for an amino acid. However, it is not the right amino acid and it is a missense mutation. Sometimes the codon is changed by a point mutation and the code becomes a stop codon. This is a nonsense mutation and stops the polypeptide production at that point making a shorter polypeptide. The short polypeptide usually becomes a non-functional protein and the activities of the cell are interrupted.

## QUESTIONS

1. Define mutation.
2. Distinguish between a somatic cell and a germ line cell.
3. What is a spontaneous mutation?
4. What is meant by induced mutation?
5. Give an example to show how a mutation can disrupt the activities of a cell.
6. Distinguish between a missense mutation and a nonsense mutation.
7. The diagram shows polypeptide synthesis in a cell.

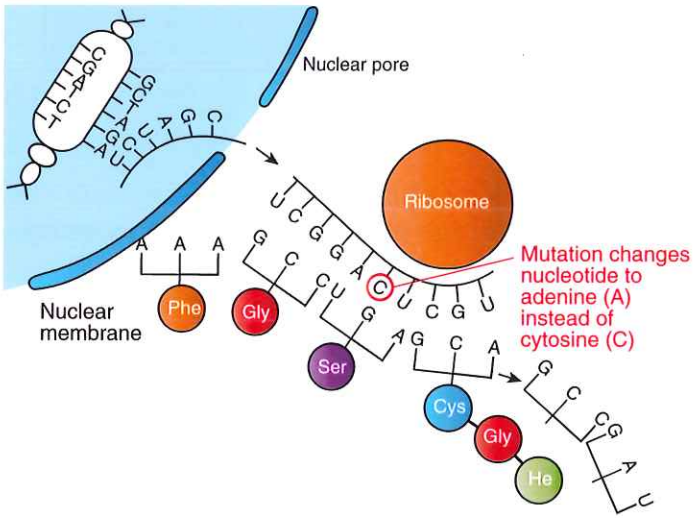



Figure 3.2 Polypeptide synthesis in a cell.

If a mutation occurred at the position marked in the diagram, use the table of the triplet codes to explain how the change would affect the synthesis of the polypeptide and the activities of the cell.

| | | | |
|---|---|---|---|
| UUU Phe | UGU Cys | GGU Gly | U – uracil (thymine) |
| UUC Phe | UGC Cys | GGC Gly | C – cytosine |
| | | GGA Gly | A – adenine |
| UCU Ser | AUU He | GGG Gly | G – guanine |
| UCC Ser | AUC He | | Cys – cysteine |
| UCA Ser | | AGU Ser | Gly – glycine |
| UCG Ser | ACU Thr | AGC Ser | He – isoleucine |
| | ACC Thr | | Phe – phenylalanine |
| UAA stop | ACA Thr | | Ser – serine |
| UAG stop | ACG Thr | | Thr – threonine |
| UGA stop | | | |

Figure 3.3 Some of the triplet genetic codes found in mRNA.

8. Which of the following mutations would have the most harmful impact on an organism?
   (A) A substitution mutation near the end of a coding sequence in a somatic cell.
   (B) Deletion of three nucleotides near the end of a coding sequence in a somatic cell.
   (C) Substitution of three nucleotides in a row.
   (D) A deletion of a single nucleotide close to the start of a coding sequence.

# 4 Point Mutations

A mutation is a permanent change in the genetic information and is a cause in genetic diversity. A gene mutation is a permanent change in the genetic information in a gene. The mutation can involve one or more base pairs and can be anywhere in the gene.

A point mutation is a change in one base in a single nucleotide in a gene. If the point mutation occurs in a gamete or zygote the change will affect every cell in the developing organism and will be passed to future generations. If the point mutation occurs in the developing embryo or foetus the change will affect tissues and cells that descend from this cell and may be passed to future offspring depending on the location of the mutation, e.g. in reproductive organs. If the point mutation occurs in an adult somatic cell the change will not be inherited by future generations and the effect on the person will depend on the specific mutation and how the body detects and responds to the error.

There are four basic types of point mutation – base pair substitution, insertion, deletion and inversion.

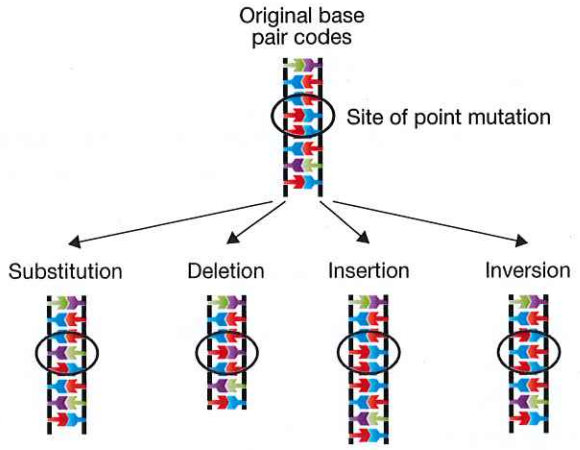



**Figure 4.1** Point mutations.

## Base pair substitution

In a base pair substitution one nucleotide and its complementary partner replace another pair of nucleotides. Sometimes this change is a **silent mutation** as the change has no effect on the protein being produced. For example, using the mRNA codes from Table 4.1 a base pair substitution from CUU to CUC will be a silent mutation as both code for the amino acid leucine. Other silent mutations may change the amino acid but have little effect on the protein. This can occur when the new amino acid has properties similar to the amino acid it replaced or the substitution has occurred in an area of the protein that is not an important determinant of the shape of the protein and its function.

Silent mutations occur fairly frequently but are hard to detect as they are not easily observed. Occasionally the change can be beneficial and can increase variation.

If the base pair substitution affects the shape of the protein or is at the active site of an enzyme, the change can seriously affect the functioning of the protein. For example, disulfide bridges are formed when two cysteine amino acids are brought close together with their sulfhydryl groups (–SH) on their side chains in close proximity due to the folding of the protein and a change in the code from UGC (cysteine a polar amino) to UUC (phenylalanine a non-polar amino acid) can affect the structure and functioning of the protein. Disulfide bridges are important in maintaining the structure of immunoglobulins (antibodies) and the antigen receptor sites on lymphocytes.

**Table 4.1** mRNA codes.

| | | | |
|---|---|---|---|
| UUU Phe | UCU Ser | UAU Tyr | UGU Cys |
| UUC Phe | UCC Ser | UAC Tyr | UGC Cys |
| UUA Leu | UCA Ser | UAA stop | UGA stop |
| UUG Leu | UCG Ser | UAG stop | UGG Trp |
| CUU Leu | CCU Pro | CAU His | CGU Arg |
| CUC Leu | CCC Pro | CAC His | CGC Arg |
| CUA Leu | CCA Pro | CAA Gln | CGA Arg |
| CUG Leu | CCG Pro | CAG Gln | CGG Arg |
| AUU He | ACU Thr | AAU Asn | AGU Ser |
| AUC He | ACC Thr | AAC Asn | AGC Ser |
| AUA He | ACA Thr | AAA Lys | AGA Arg |
| AUG Met | ACG Thr | AAG Lys | AGG Arg |
| GUU Val | GCU Ala | GAU Asp | GGU Gly |
| GUC Val | GCC Ala | GAC Asp | GGC Gly |
| GUA Val | GCA Ala | GAA Glu | GGA Gly |
| GUG Val | GCG Ala | GAG Glu | GGG Gly |

| | | |
|---|---|---|
| U – uracil (thymine) | Cys – cysteine | Met – methionine |
| C – cytosine | Gln – glutamine | Phe – phenylalanine |
| A – adenine | Glu – glutamine | Pro – proline |
| G – guanine | Gly – glycine | Ser – serine |
| Ala – alanine | His – histidine | Thr – threonine |
| Arg – arginine | He – isoleucine | Trp – tryptophan |
| Asn – asparagine | Leu – leucine | Tyr – tyrosine |
| Asp – aspartic acid | Lys – lysine | Val – valine |

Most base pair substitution mutations cause a missense mutation. A **missense mutation** is a changed codon that codes for an amino acid but does not necessarily make the correct sense. A **nonsense mutation** is a point mutation that changes a codon for an amino acid into a stop codon. The stop codon causes translation to stop shortening the polypeptide chain that is being synthesised. In most cases a nonsense mutation creates a non-functional protein. For example, a change in the code UCA for serine to UAA will create a stop at this point in translation and process of protein synthesis.

## Deletion

Deletion is a loss of nucleotide pairs in a gene sequence. This causes a **frameshift mutation**. The nucleotide sequence is read in multiples of three (the code for each amino acid) so the deletion of one nucleotide will change the reading sequence. A frameshift mutation will not occur if the change was an insertion or deletion of three nucleotides. The example in Figure 4.2 shows how the reading frame is changed with a deletion.

The old dog sat on a hot day on a red mat.

As a triplet code this would be read

The/ old/ dog/ sat/ on a/ hot/ day/ on a/ red/ mat/

If the 'e' from 'The' is deleted, the code is read

Th o/ ldd/ ogs/ ato/ nah/ otd/ ayo/ nar/ edm/ at

Figure 4.2 Frameshift mutation due to deletion.

## Insertion

Insertion is a point mutation where a nucleotide is added into the code. This will also cause a frameshift mutation. A frameshift mutation can be a nonsense mutation. Depending on the location of the insertion a number of different effects can occur.

Point mutations can occur spontaneously, e.g. during replication or can be induced by mutagens including ionising radiation, e.g. X-rays, cosmic rays, non-ionising radiation, e.g. UV light, various chemicals, e.g. benzene and some biological agents.

## Inversion

In an inversion point mutation a nucleotide pair reverse positions with the next pair of nucleotides.

## QUESTIONS

1. Define a mutation.
2. What is a gene mutation?
3. Define a point mutation.
4. Name the four basic types of point mutation.
5. What is a silent mutation?
6. Use Table 4.1 with the mRNA codes for amino acids to determine which of the following would be silent mutations.
   (a) GUU to GUG (b) ACG to AAG
   (c) GGU to GAU (d) CGG to UGG
   (e) GCU to GCC
7. Distinguish between a missense mutation and a nonsense mutation.
8. What is a frameshift mutation?
9. Use an example to show how a point mutation can have serious consequences.
10. What are the mRNA stop codes?
11. The diagram shows a section of DNA of a gene which undergoes a point mutation.

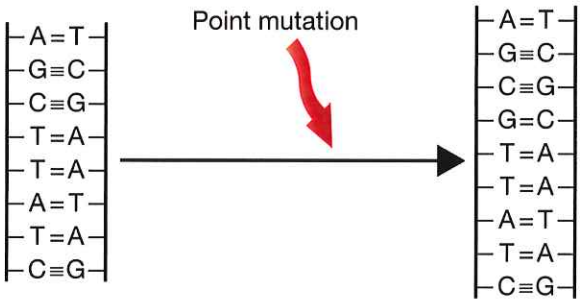



Figure 4.3 Point mutation in a gene.

Copy the diagram and label the location of the point mutation and which type of point mutation has occurred.

Use the diagram which shows a deletion of two nucleotides for the next THREE questions.

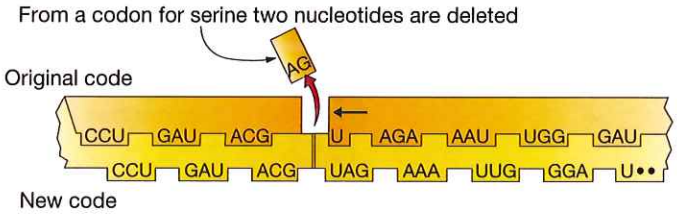



Figure 4.4 Deletion of two nucleotides.

12. What is the best description of this mutation?
    (A) Frameshift.
    (B) Chromosomal.
    (C) Insertion.
    (D) Base pair substitution.
13. Why is this mutation a nonsense mutation?
    (A) The new code does not read correctly.
    (B) The right proteins will not be produced.
    (C) There is no codon with only one nucleotide.
    (D) The new code is UAG which is a stop codon.
14. What was the code for serine that was involved in the mutation?
    (A) AGC (B) AGU
    (C) UGA (D) UAG
15. What is the best definition of a point mutation?
    (A) A permanent change in the genetic information.
    (B) An agent that interacts with DNA to cause a change.
    (C) A change in one base in a single nucleotide in a gene.
    (D) A change in an amino acid that alters protein structure.

# 5 Block Mutations

A block mutation is a permanent change to a segment of a chromosome that rearranges, deletes or disrupts many loci. Many block mutations are harmful changing the structure of the chromosome, though some are neutral especially if genes remain intact and sometimes the change can be beneficial, e.g. the change links genes that together now produce a positive effect.

Block mutations can be caused by **transposons** (transposable genetic elements) which are DNA segments that can move from one position to another in the chromosome. Transposons are sometimes called 'jumping genes' which is not how they move as they do not 'jump'. When the folding of the DNA molecule brings segments near each other transposons follow a 'cut and paste' mechanism to move to a new location or follow a 'copy and paste' mechanism replicating a section of DNA and adding it to another area. Transposons were discovered by Barbara McClintock when studying corn in the 1940s and she was awarded the Nobel Prize in Physiology or Medicine in 1983 for her discovery of 'mobile genetic elements'. She was the first female to receive this prize unshared. It is estimated that about 44% of the human genome is **repetitive DNA** which is multiple copies of DNA sections.

Types of block mutations include duplications, inversions, deletions, insertions and translocations.

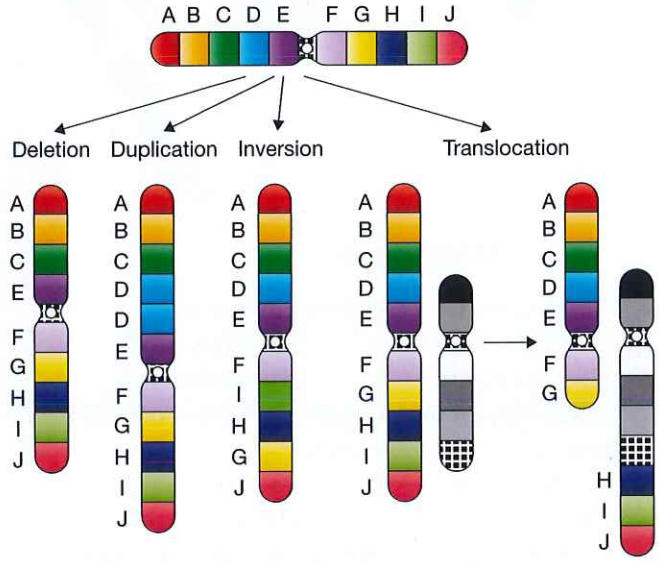



Figure 5.1 Types of block mutations.

Some sections of DNA are called **hotspots** as they are places that are more likely to undergo mutation than other places with an observable higher mutation frequency. Hotspots can be single nucleotides or short stretches of repeated nucleotides that have some basic instability or chemical tendency for nucleotide substitution.

## Deletions

Deletions occur when a section of DNA is removed from a chromosome. The effect of a deletion depends on the size and location of the removed block sequence. Large deletions can involve several genes and have greater effect on phenotype and the health of the individual. Cri du chat syndrome is a deletion disorder that is caused by a deletion on the short arm of chromosome 5.

## Duplications

Duplications occur when sections of DNA are replicated making the chromosome longer. Repetitive DNA can occur during DNA replication or recombination in crossing over and segregation during mitosis or meiosis. If the copies of a repeat sequence lie adjacent to each other they are called **tandem repeats**. Tandem repeats can vary in length with the large repeat units called satellites. Satellite DNA was first discovered when DNA was centrifuged and the repetitive units appeared as a distinct band in the tube. **Satellite DNA** (also now called simple sequence DNA) is a section of tandem, non-coding DNA that can be thousands of base pairs long. Microsatellite DNA is a short region of repeats that are used as genetic markers in DNA fingerprinting. In humans some microsatellites have 20 or more alleles which provides the variation to assist in identifying particular individuals by their DNA. **Trinucleotide disorders** occur when there are too many trinucleotide repeats in a gene, e.g. Huntington's disease occurs when there are more than 35 CAG repeats on the gene coding for the protein HTT. A genetic disorder due to duplication is Charcot-Marie-Tooth disease type 1 with duplication of 17p12 – a large section on the short arm of chromosome 17.

## Inversions

Inversions occur when a section of DNA breaks and is reattached in the reverse orientation and order. This changes chromosome structure. Inversion is a cause for haemophilia A with an inversion in the factor VIII gene on X chromosome. The inversion within this gene stops protein production which means that testing for this disorder often involves measuring protein activity rather than a genetic test for the inversion.

## Translocation

In translocation a section of one chromosome moves to a non-homologous chromosome. In a **reciprocal translocation** the non-homologous chromosomes exchange segments. Myeloproliferative syndrome is a genetic disorder caused by translocation of genetic material from chromosome 8 to other chromosomes, e.g. t(8;13)(p11;q12) which involves translocation of chromosomes 8 and 13 in lymphoma cells.

## QUESTIONS

1. Define a mutation.
2. What is a transposon?
3. Explain why the term 'jumping gene' is not an accurate description of transposons.
4. What is repetitive DNA?
5. Outline Barbara McClintock's contribution to our understanding of genetics.
6. List the main types of block mutations.
7. Describe hotspots.
8. The diagram shows a type of block mutation.

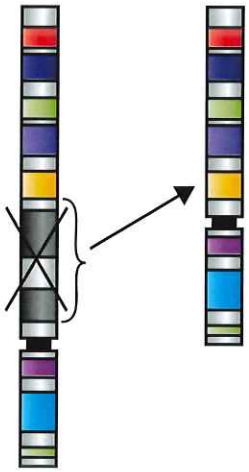


Figure 5.2 Type of block mutation.

Identify this type of block mutation and explain what has happened,

9. Construct a table to summarise the main types of block mutation – deletion, duplication, inversion and translocation explaining what is happening and giving a genetic example.
10. What are tandem repeats?
11. What is satellite DNA and how was it discovered?
12. Outline a use for microsatellite DNA.
13. What is a trinucleotide disorder and give an example.
14. The diagram shows an example of translocation.

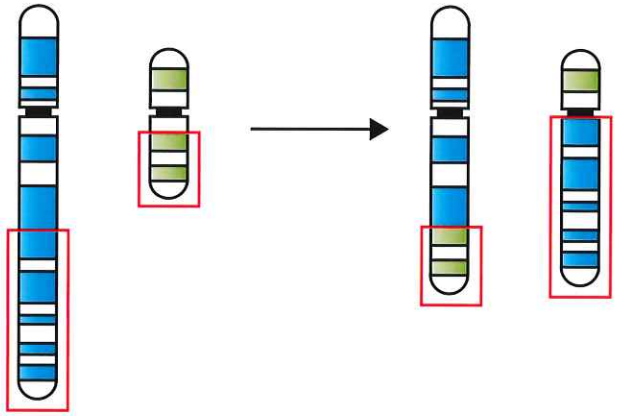


Figure 5.3 Translocation.

Explain why this is an example of reciprocal translocation.

15. If DNA duplications occur that involve one or more genes to make a gene pair and both copies stay in the genome and are inherited by future generations then a multigene family can be created.
    (a) Explain why the genes in a multigene family code for proteins with similar sequences.
    (b) Explain why the genes in a multigene are usually involved in the same body functioning.
    (c) Discuss why comparative genomics is interested in multigene families such as the globin genes for haemoglobin.

Use the following diagram for the next TWO questions.

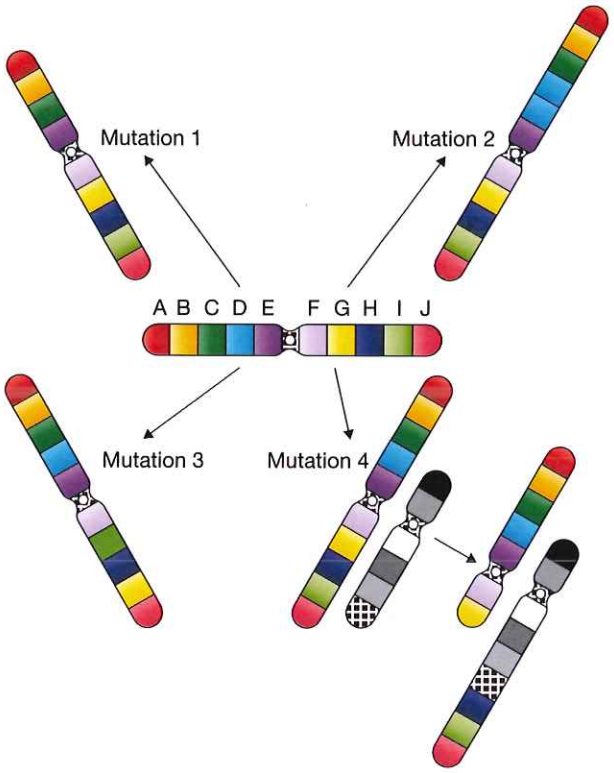



Figure 5.4 Mutation types.

16. Which mutation type shows inversion?
    (A) Mutation 1 (B) Mutation 2
    (C) Mutation 3 (D) Mutation 4
17. Which type of mutation would be involved in the creation of repetitive DNA?
    (A) Mutation 1 (B) Mutation 2
    (C) Mutation 3 (D) Mutation 4
18. The following code shows a section of a simple sequence DNA showing only one strand of the DNA.

    .GCTTAGCTTAGCTTAGCTTAGCTTAGCTTA.

    How many nucleotides are in this repeat unit?
    (A) 3 (B) 5 (C) 15 (D) 30
19. Which disease can be caused by translocation?
    (A) Trisomy 21. (B) Measles.
    (C) Cri du chat. (D) Malaria.

# 6 Transposable Genetic Elements

**Transposable genetic elements** or transposons are DNA segments that can move from one position to another in chromosomes. The movement is called **transposition**. Transposons have been found in both prokaryotes and eukaryotes and were first found in plants. They can cause mutations and can increase/decrease the amount of DNA in the genome. In humans, jumping genes have been linked with antibody production by the immune system and some of the rearrangements caused by jumping genes are believed to be the cause of some forms of cancer. Some people call transposons 'junk' DNA or 'selfish' DNA as they do not seem to benefit the host and often just make copies of themselves.

When the enzyme transposase is activated a 'jumping gene' can move from one locus on one chromosome to another chromosome in a 'cut and paste' action. Retrotransposons also use a 'cut and paste' method but they copy RNA rather than DNA. About 40% of the human genome consists of retrotransposons. Some transposons use a 'copy and paste' method.

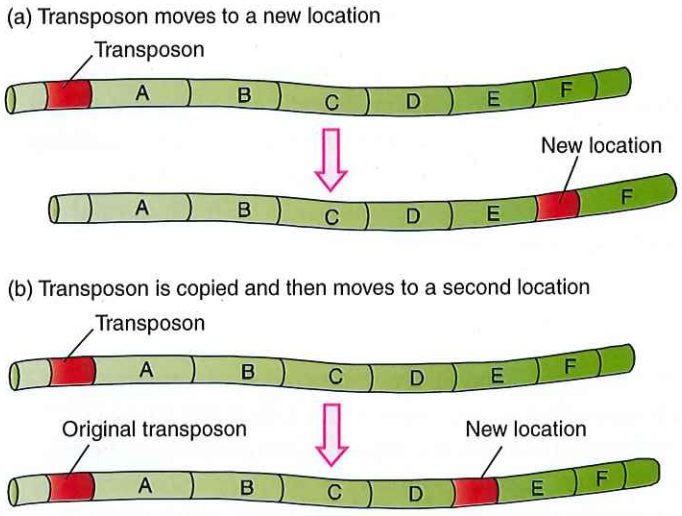



**Figure 6.1** Transposons.

Transposons have been used to study gene function and to cause mutagenesis in *Drosophila melanogaster*. In the fruit fly the transposons are called P elements and they seem to have only appeared 50 years ago. In bacteria some transposons not only carry the gene for transposase but also one or more genes for proteins that give antibiotic resistance. The transposons can jump from the DNA chromosome to the plasmid and back again. It is believed these transposons have caused the rapid spread of multidrug antibiotic resistance in bacteria.

## Transposons and mutation

Transposons are mutagens. If a transposon is inserted into a functional gene, it will most likely stop the function of that gene, e.g. if an L1 retrotransposon moves to a specific location on the X chromosome it causes the most common form of haemophilia. If the gap caused by the removal of a transposon is not correctly repaired, that site becomes a mutation. And if there are a string of inserts on a chromosome, identical homologous pairing becomes difficult in meiosis. Human diseases caused by transposons include haemophilia A and B, Duchenne muscular dystrophy, SCID and porphyria.

## QUESTIONS

1. Define transposable genetic elements.
2. What is transposition?
3. Distinguish between a prokaryote and a eukaryote.
4. In which organisms have transposons been found?
5. In the 1940s Barbara McClintock discovered the first transposons in maize, *Zea mays* (called corn in USA). She found transposons were responsible for gene mutations caused by insertions, deletions and translocations. Transposons cause the different colours, e.g. purple, yellow, white and streaking on 'Indian corn'.

**Figure 6.2** Indian corn.

Barbara McClintock received the Nobel Prize in Physiology or Medicine in 1983 for her work on corn genetics. Suggest one reason why her work was not recognised until 1983.

6. Outline how jumping genes affect humans.
7. HIV-1 is a retrovirus that causes AIDS. It acts like a retrotransposon. What is a retrotransposon?
8. Explain why transposons have been called by some people 'junk' DNA and 'selfish' DNA.
9. Transposons have been found in all branches of life. This has led to an interest in when transposons evolved and their effect on the evolution of genomes. Suggest a possible evolutionary history for transposons.
10. Construct a table to compare the effects of transposable genetic elements and germ line mutation on an organism.
11. What is a transposon?
    (A) DNA sequence that changes its position within the genome.
    (B) A permanent change in the genetic information.
    (C) The movement of information from one cell to another.
    (D) DNA sequence that is always expressed.

# 7 Chromosome Abnormalities

There are several types of chromosome abnormalities. Some individuals can have extra or missing chromosomes and some individuals have a different number of whole sets of chromosomes.

**Polyploidy** is the possession of more than two sets of chromosomes per nucleus, e.g. hexaploid ($6n$).

Monoploidy is the loss of an entire set of chromosomes. It is the haploid number of chromosomes.

Aneuploidy is the addition of all or part of a chromosome, e.g. in humans there could be 45 or 47 chromosomes instead of the normal 46 chromosomes in the nucleus.

Monosomy is the lack of one chromosome from the normal number of chromosomes.

Trisomy means there are three copies instead of the normal two copies of a particular chromosome.

## Nondisjunction

The change in the number of chromosomes can be due to nondisjunction during cell division. Nondisjunction is an error during mitosis or meiosis when both members of a pair of homologous chromosomes or both sister chromatids fail to separate properly.

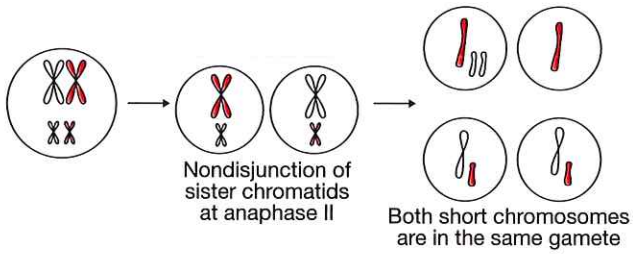



Figure 7.1 Nondisjunction in anaphase II in meiosis.

Nondisjunction leads to an abnormal number of chromosomes in a gamete. If it occurs in anaphase II there will be two affected daughter cells – one with two copies of the chromosome and the other with no copies of that chromosome.

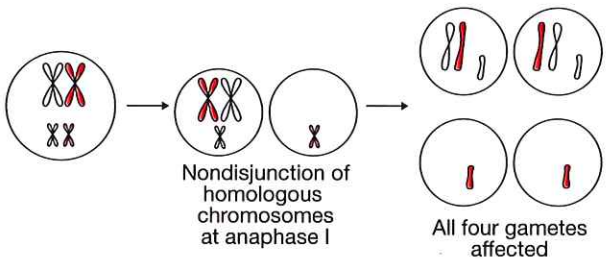



Figure 7.2 Nondisjunction in anaphase I in meiosis.

If the homologous chromosomes do not separate during anaphase I then all four daughter cells are affected – two daughter cells will have an additional chromosome and two cells will be missing one chromosome.

During fertilisation the fusion of an abnormal gamete with a normal gamete will create a zygote with an incorrect number of chromosomes, e.g. missing a chromosome or having an additional chromosome. Many cases of trisomy 21 (Down syndrome) are due to nondisjunction.

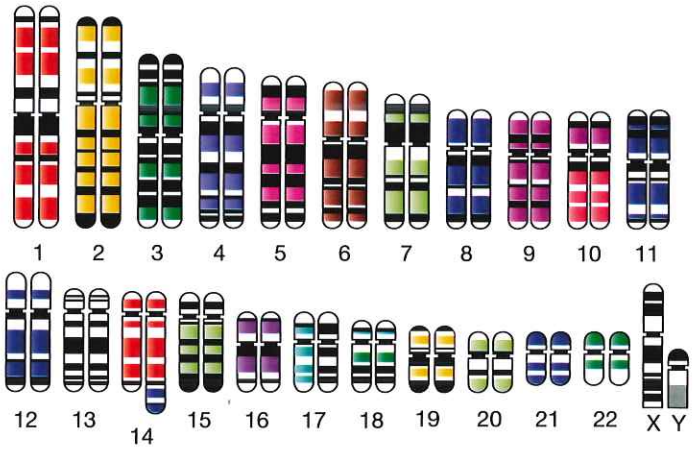



Figure 7.3 Trisomy 21 with translocation to chromosome 14.

Nondisjunction can also occur during mitosis. If the error occurs during early embryonic development then a large number of cells will have the aneuploidy condition. In mosaic Down syndrome the person has some cells with trisomy 21 and some cells with the normal 46 chromosomes in a cell.

## Unequal crossing over in meiosis

Chromosome abnormalities can occur when crossing over between adjacent non-sister chromatids in the tetrad in prophase I if meiosis is unequal. In the resulting gametes one cell may have a chromosome with a gene deletion and another cell will have a chromosome with a gene duplication. Transposable elements in the chromosome provides the sites for crossing over in non-sister chromatids.

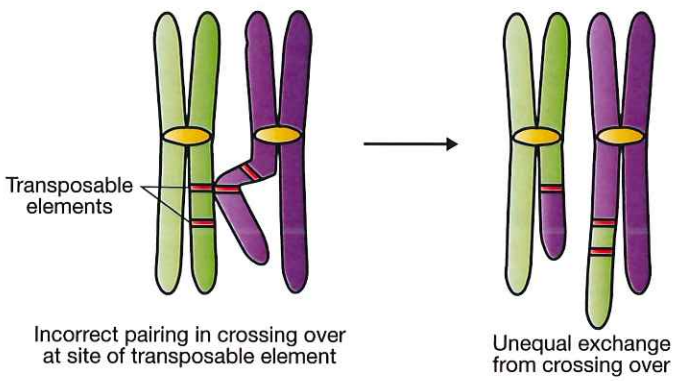



Figure 7.4 Unequal crossing over.

## Sex chromosomes and chromosome abnormalities

There are several genetic disorders that occur due to an incorrect number of sex chromosomes. A changed number of sex chromosomes does not seem to upset the genetic balance as much as an aneuploidy autosome condition. The Y chromosome has few genes and in somatic cells the extra copies of the X chromosome are inactivated as Barr bodies.

- **Klinefelter syndrome** is a chromosomal disorder that occurs when males have two or more X chromosomes, e.g. a male has XXY. It is a congenital disorder detected by karyotyping. Males have abnormally small testes and the man is sterile. It is not an inherited disorder and occurs in about 1 in 500 men to 1 in 1000 men in Australia with many not being diagnosed.
- **Turner syndrome** (TS) is a chromosomal disorder that occurs when females are missing part or a whole X chromosome, e.g. a female has XO. It is the only known viable monosomy in humans. The female will have 45 chromosomes instead of 46 chromosomes. The person may show mosaicism with some cells normal and some cells lacking the second X chromosome. Mosaic Turner syndrome occurs in about 20% of cases. Turner syndrome affects about 1 in every 2000 girls born in Australia with about 50% having XO in all their cells. They are sterile as the sex organs do not mature. Oestrogen therapy can aid the development of secondary sex characteristics.
- **XYY syndrome** has an extra Y chromosome in each cell in a human male. Individuals seem to have an increased risk for learning difficulties, often have heavy acne and have increased height. It is estimated to occur in about 1 in 1000 live births. The males have normal fertility.
- **Triple X syndrome** (XXX) has an X chromosome in each cell in a human female with a mosaic form being present in only some cells. Most cases are not diagnosed and since only one X chromosome is only activated at a time due to the formation of Barr bodies, there are few obvious signs of the condition and most are usually fertile.

## Cancer and chromosome abnormalities

Some translocations are associated with certain cancers. In chronic myelogenous leukaemia (CML) there is a reciprocal translocation between chromosome 22 and chromosome 9 with a large amount of chromosome 22 exchanging for a small tip of chromosome 9 – this forms the Philadelphia (Ph) chromosome. This rare cancer causes the overproduction of granulocytes.

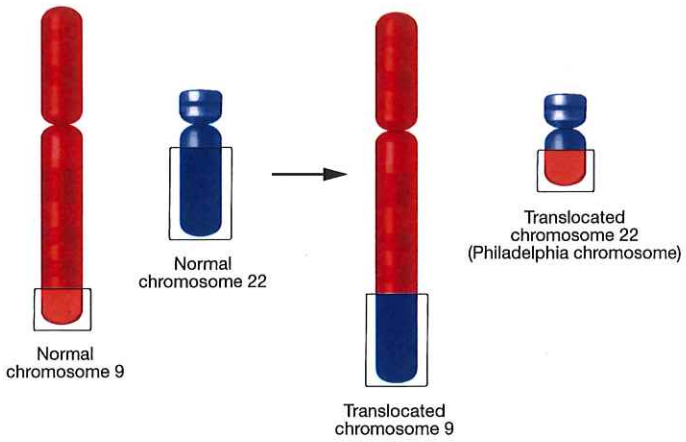



**Figure 7.5** Translocation forming the shortened chromosome 22 known as the Philadelphia chromosome.

## QUESTIONS

1. Distinguish between polyploidy, monoploidy and aneuploidy.
2. Distinguish between monosomy and trisomy.
3. Define nondisjunction.
4. Explain how nondisjunction leads to abnormal cells.
5. Explain how mosaic trisomy conditions arise.
6. Suggest why a number of anomalies involving the sex chromosomes are in the human population and do not have as great a genetic effect as autosome abnormalities.
7. Construct a table to summarise some genetic disorders caused by abnormalities involving the sex chromosomes.
8. What is the only known viable monosomy in humans?
   (A) Klinefelter syndrome.
   (B) Turner syndrome.
   (C) Down syndrome.
   (D) Edwards syndrome.
9. What will cause cell division to form four daughter cells that are all affected with chromosome abnormalities?
   (A) Nondisjunction in anaphase I of homologues in the tetrad.
   (B) Nondisjunction in anaphase II of sister chromatids.
   (C) Nondisjunction in anaphase I of sister chromatids.
   (D) Nondisjunction in anaphase II of homologues in the tetrad.
10. What is the best term to describe chromosomal abnormalities where there are either extra or missing copies of certain chromosomes?
    (A) Polyploidy.
    (B) Monoploidy.
    (C) Euploidy.
    (D) Aneuploidy.

# 8 Gene Expression Affects Phenotype

**Gene expression** refers to a gene being 'turned on'. When a gene is 'turned on' the coded information in the gene is used to produce a polypeptide. These polypeptides control the functioning of cells and the formation of the structures in a cell.

At any one time only some genes are expressed.

Genes can be switched on and off by various mechanisms. In many instances gene expression is controlled at the transcription stage. Gene expression causes cells to differentiate into particular types of cells.

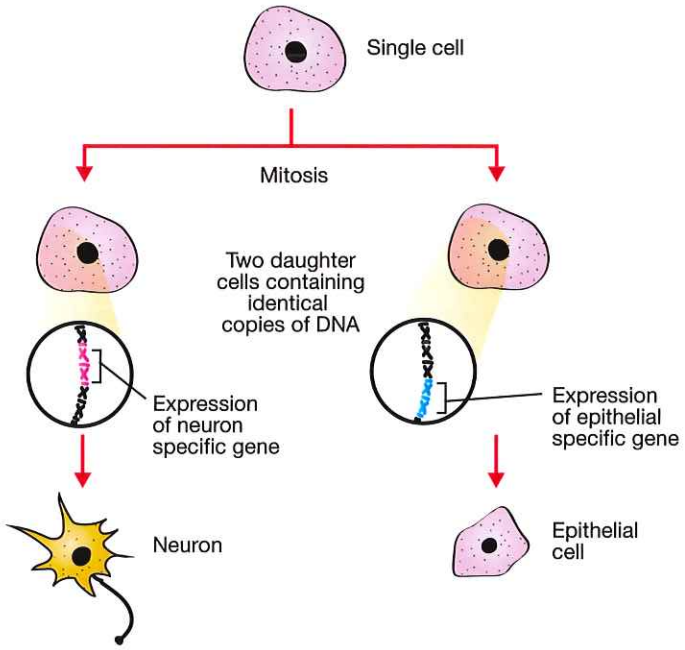



Figure 8.1 Gene expression.

If there is some problem in gene expression, the health of the individual can be at risk.

The interruption or an alteration in the process of producing a particular polypeptide means a particular chemical reaction may be unable to occur, which in turn may lead to a particular disease. For example, if there is a problem with the promoter there will be a problem in starting transcription and thus the polypeptide will not be synthesised and the gene will not be expressed.

A **promoter** is a regulatory area at the start of a gene that acts as the binding site for RNA polymerase.

It has been found that the 'switching on' and 'switching off' of particular genes which is the basis of gene expression has a great impact on the health of an individual.

For example, in the February 2005 issue of the Journal of Allergy and Clinical Immunology, Dr Gurjit Hershey reported that an analysis of children who were having an acute asthma attack had a different gene expression profile to children in a stable (controlled) asthma state. They found one gene cluster that was active during an acute asthma attack but was not active in stable asthma and other gene clusters that were active in stable asthma and not during acute asthma attacks. This research focused on genes concerned with epithelial cells, mucosa and the underlying cells.

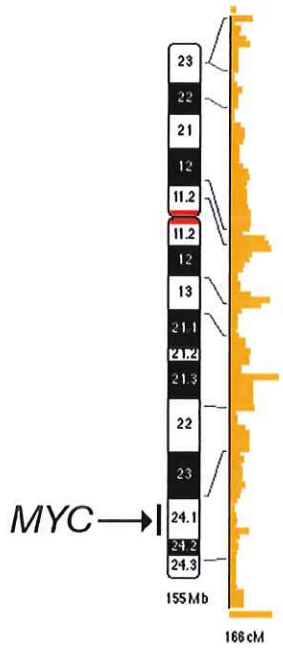



Figure 8.2 *MYC* gene on chromosome 8.

**Cancer** involves rapid cell division without cell differentiation so that a mass of cells accumulates in one place forming a tumour.

The protein *MYC* controls cell division and a study of the gene that produces *MYC* shows that if this gene is 'switched on' too much, a cancer develops. Researchers have found that injecting the antibiotic doxycycline into transgenic mice which had this *MYC* gene permanently 'switched on' caused the liver cancer cells in the mice to become harmless cells. They 'switched off' the gene and cancer cells became normal cells.

## QUESTIONS

1. Define gene expression.
2. What is a promoter?
3. Outline a way which will cause a gene not to be expressed.
4. Use an example to show how a knowledge of gene expression could aid suffers of a particular disease.
5. What is cancer?
6. Outline the function of the *MYC* gene.
7. Explain how a study of the *MYC* gene has aided cancer research.

# 9 Allele Frequency

An **allele** is an alternative for a particular trait, e.g. there are two alleles for height in pea plants – tall (T) or short (t). Alleles are the alternative forms of a gene and occupy a particular locus on a chromosome. The **frequency** of an allele in a population refers to the proportion of the population that have that allele.

## The Hardy-Weinberg principle

Population genetics studies how populations change genetically over time and the Hardy-Weinberg principle proposes that frequencies of alleles and genotypes remain constant from generation to generation provided that only Mendelian segregation and recombination of alleles occur. This is known as the **Hardy-Weinberg equilibrium**. This means that there is no evolution occurring in the population.

The Hardy-Weinberg principle gives the mathematical formula to show how the frequencies of two alleles (A and a) represented by symbols $p$ and $q$ maintains constant gene frequencies if the following conditions exist. 1. The population is large. 2. There is no mutation or the rate of mutation of a specific allele equals the reverse mutation. 3. There is no selective mating and reproduction is purely random. 4. There is no migration of individuals into or out of the population. The Hardy-Weinberg equation is as follows.

$$p^2 + 2pq + q^2 = 1 \text{ or } 100\%$$

This means that there is 100% probability that the offspring are AA ($p^2$), Aa ($2pq$), aa ($q^2$).

## Gene pool

A **gene pool** is the total aggregate of genes in a population at any one time. When studying a gene pool it is important to first define the population. A **population** is a group of organisms which can freely interbreed to produce fertile offspring living in a particular area at a particular time.

The gene pool of a localised population living in a remote area, e.g. Inuit in Greenland will have a different gene pool to the people living in a distant area, e.g. people living in Arnhem Land, Northern Territory. Allele frequencies of particular genes will differ if these two populations are compared.

## Genome

A **genome** is the complete complement of an organism's genes and the **genome sequence** gives the order of the As, Ts, Cs and Gs in the DNA code. **Genomics** is the study of whole sets of genes of their interactions.

Genes are coded DNA which is transcribed into RNA that is used in protein formation. Controlling proteins means that when differentiation occurs the final cell structure and functioning is controlled by which genes are switched on or off in gene expression, e.g. the switching on of genes that control specific proteins such as keratin and collagen and the switching on of particular enzymes that regulate specific reactions will determine the shape, composition and functioning ability of the cell. Allele frequency of particular traits will then show the frequency of particular phenotypes in a population.

**Phenotype** is the observable physical and physiological traits of an organism and the outward appearance of an organism. Phenotype depends not only on which alleles are present but also on environmental influences, e.g. temperature during development can affect gene expression in some organisms.

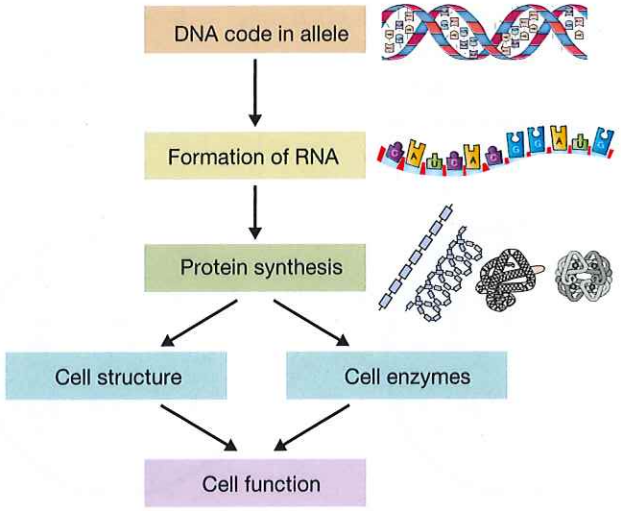



Figure 9.1 Alleles of a gene control cell functions.

## QUESTIONS

1. Define an allele.
2. What is meant by allele frequency?
3. How is the Hardy-Weinberg equilibrium used in population genetics?
4. Define a gene pool.
5. What is a population?
6. Define a genome and genomics.
7. Explain how genes determine the structure and functioning of a cell.
8. How does allele frequency affect phenotype in a population?
9. What determines phenotype?
10. What is allele frequency?
    (A) The number of people with a certain trait.
    (B) The number with an alternate trait.
    (C) The proportion of a population with that allele.
    (D) The ratio of different alleles in different groups.

# 10 Gene Flow

Gene pools are **dynamic** with changes in allele frequency caused by mutations, differing selection pressures, random genetic drift and changes in gene flow between adjoining groups. Gene pools can be used to compare populations at different times or locations.

**Gene flow** occurs when alleles are added to a gene pool or subtracted from the gene pool due to the movement of fertile individuals or gametes, e.g. due to migration of individuals from one population to another. Gene flow may involve natural selection or can occur due to chance.

Gene flow will change allele frequencies and in many instances will reduce the number of differences between populations. The higher the gene flow between two populations the less likely the two populations will evolve into two species. For example, two populations living geographically near each other without major travelling difficulties are likely to have high gene flow. If a particular allele did not previously exist in a population then its introduction can have a dramatic effect on natural selection.

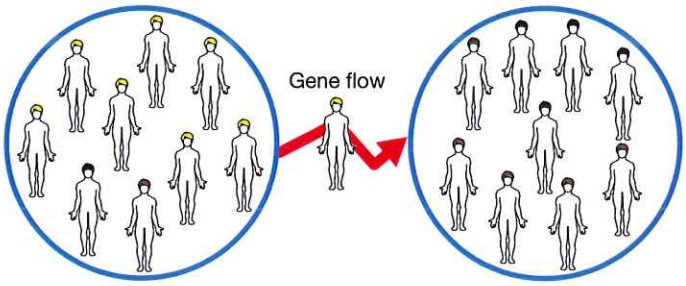



Figure 10.1 Gene flow.

Gene flow does not always require individuals to permanently migrate. For example, during the Vietnam War some US soldiers had children with Vietnamese women which altered the allele frequency of the Vietnamese gene pool.

## Factors affecting gene flow

There are several factors that can affect the rate of gene flow between different populations.

- Mobility of fertile individuals or gametes – high mobility increases the chances of gene flow from one population to another.
- Presence of barriers, e.g. mountain ranges, oceans, deserts.
- Cultural barriers, e.g. some groups prefer to find partners within their own group.

## Gene flow and natural selection

Gene flow often reduces the number of differences between populations with the higher the gene flow between two populations the fewer the differences and the less likelihood that the two populations will evolve into two species, e.g. in the evolutionary history of the genus *Homo* there were several species living at the same time showing divergence of groups leading to speciation.

A change in the allele frequencies in a population's gene pool over a number of generations means evolution is occurring. **Evolution** is a change in the genetic make-up of a population. Scientists are interested in changes in allele frequencies as it provides information about how populations with different size gene pools cope with changing conditions or how populations can change due to genetic drift or gene flow over time.

The size of the gene pool is most important when considering the survival chances of a population in a changing environment. A small gene pool means there is little genetic diversity and if conditions alter and there are no variants with features suited to the new conditions the population becomes endangered and may face extinction. A large gene pool makes a population more robust and able to withstand changing conditions and natural selection.

## Geographical isolation

Geographical isolation occurs when two populations occupy different geographical areas or different habitats within the same area so that they may not meet and cannot mate to produce offspring. Primary geographical barriers occur with water or mountain ranges separating two sections of a population.

**Hominin evolution** began in Africa and several species have migrated out of Africa and spread across the world, e.g. *Homo erectus* remains have been found in China and Java. Different groups evolved in different areas but gene flow still occurred as people travelled to other areas by walking, sailing or rowing boats or using animals such as horses. Since the development of planes and faster transport there is greater international travel and gene flow between different human populations. Thus geographical isolation which is a primary cause of speciation for other species is less important for humans.

## QUESTIONS

1. List some factors that make gene pools dynamic.
2. How can gene pools be used?
3. Define gene flow.
4. Outline how migration can cause gene flow.
5. Explain why gene flow can reduce the number of differences between populations.
6. Outline some factors that can affect gene flow.
7. Explain why geographical isolation does not restrict gene flow for humans compared to other species.

# 11 Genetic Drift

**Genetic drift** occurs when chance events cause unpredictable changes in allele frequencies from one generation to the next. Genetic drift occurs more easily in small populations than in large populations. Genetic drift is due to random events and since natural selection is rarely involved it does not necessarily lead to adaptations to specific environmental conditions.

Genetic drift is most noticeable if a population stays small over many generations. An allele may be eliminated by chance from a population even if it has an adaptive advantage.

Chance events that can affect survival and reproduction leading to genetic drift include the following.

- Natural disasters, e.g. an area may be flooded in a flash flood and those that make it to high ground survive.
- Some are accidentally killed by another species that is not a predator, competitor or pathogen.
- Chance events in fertilisation, e.g. by chance most offspring die and the only survivors have a particular genotype.
- Soils in one area become waterlogged leading to only some people, plants and animals surviving due to their chance location.

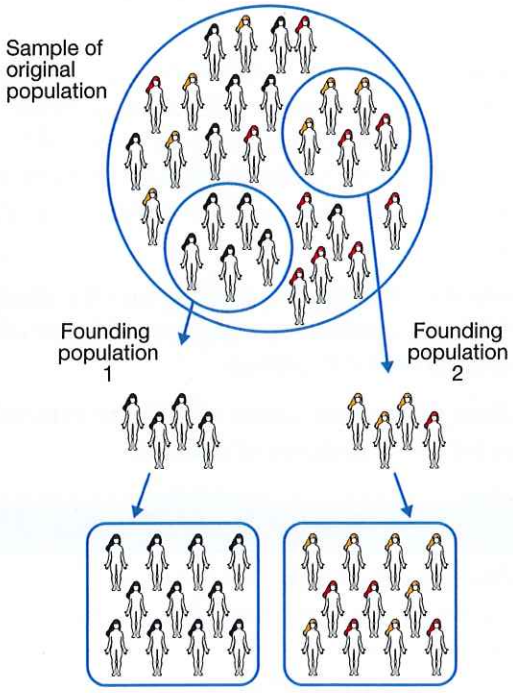



Figure 11.1 Founder effect.

## Founder effect

The founder effect occurs when a few individuals become isolated from a larger population and form a new population whose gene pool composition is not reflective of the original source population.

In isolated human populations the founder effect can account for the high frequency of certain genetic disorders in these populations. On the island of Tristan da Cunha in the Atlantic Ocean there is a high frequency of the recessive allele that causes the degenerative eye disease retinitis pigmentosa. One of the early colonists had this recessive allele and over many generations many descendants have the allele.

## Bottleneck effect

The bottleneck effect occurs when a large population is reduced in size, e.g. by a natural disaster or human activities and the gene pool of the surviving population is much smaller than the original gene pool. It is random which alleles are present in the survivors and which alleles are absent. With a small population there is a greater likelihood of further genetic drift which will continue until the population is large enough that chance events have less impact.

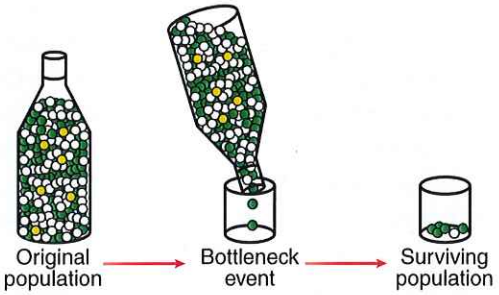



Figure 11.2 Bottleneck effect.

Around 10 000 years ago a bottleneck occurred with the cheetah population leading to the current low genetic variability of cheetahs.

## QUESTIONS

1. Define genetic drift.
2. In what way is genetic drift different to natural selection?
3. Identify some chance events that can lead to genetic drift.
4. In the Old Order Amish of Lancaster, Pennsylvania which was established in the early 1770s there is a high proportion of the recessive allele that leads to a combination of dwarfism and polydactylism (extra fingers). Suggest how this high allele frequency could have occurred.
5. Explain why a population that stays small and isolated from other populations is likely to show genetic drift.
6. When studying ABO blood groups ($I^A$, $I^B$ and i) it was found that Australian Aborigines lacked the $I^B$ allele. This means that the Australian Aborigines do not have blood groups A or AB unless partners have come from other groups. Suggest how the lack of allele $I^B$ could have occurred.
7. Explain why genetic drift can decrease genetic variation within a species and increase the amount of variation between local populations.

# 12 Microevolution

**Microevolution** refers to evolutionary change below the species level with changes in the genetic composition of a population from generation to generation.

## Causes of microevolution

There are several mechanisms that can cause the genetic composition of a population to change from generation to generation.

**Mutation** – mutations introduce new alleles and new genes into the gene pool. This could be a change in the nucleotide sequence of DNA, a point mutation that affects one base in a gene or a chromosomal mutation where there is deletion, duplication or rearrangement of many loci. Gene duplication and an increase in gene number have been important in the evolution of many organisms. For example, there was one common ancestral globin gene which duplicated and diverged into alpha globin and beta globin around 450 to 500 million years ago. Most mutations are harmful and do not stay in the population. Some mutations are recessive and stay in the population and can give an advantage under certain conditions, e.g. sickle cell anaemia allele gives protection against malaria in heterozygotes.

**Natural selection** – individuals with favourable characteristics survive to reproduce and their offspring inherit the favourable characteristics. Natural selection leads to evolution and adaptations to particular environments or lifestyles. Humans have used artificial selection to change the allele frequencies of different traits of many plants and animals, e.g. animal breeders producing different varieties of dogs or farmers producing different varieties of wheat and rice.

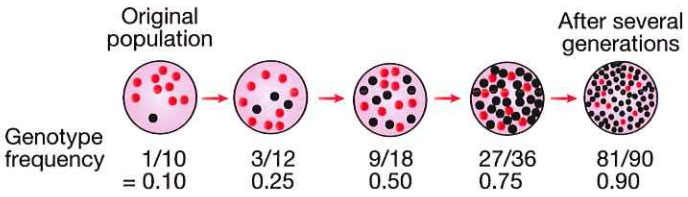



Figure 12.1 Genetic drift.

**Genetic drift** – chance events cause unpredictable changes in allele frequencies. In a small, isolated population some alleles may increase in frequency while others decrease in frequency due to random events that prevent/allow certain individuals to produce offspring.

**Gene flow** – alleles are added to or subtracted from a gene pool due to the movement of fertile individuals or gametes. Gene flow can lead to hybridisation. Some hybrids are sterile, e.g. mule while other hybrids have 'hybrid vigour' with 'improved' qualities compared to either parent and is still fertile, e.g. wheat varieties.

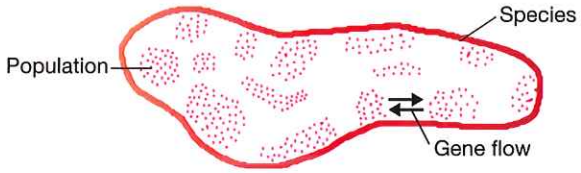



Figure 12.2 Gene flow.

## Examples of microevolution

There are many examples of microoevolution. Humans use many chemicals to keep areas clean, e.g. bleach, detergents and antibacterial soaps and use pesticides to protect crops, livestock and reduce the transmission of pathogens that cause infectious diseases. These chemicals act as selecting agents in natural selection.

**Insect** resistance to pesticides – insects can destroy crops, act as vectors for several human diseases and affect livestock. There are many pesticides on the market and many pesticides that have been used for a while are less effective. The pesticide acts as a selecting agent and those with resistance to the pesticide survive and produce offspring that are also resistant to the pesticide, e.g. DDT is no longer effective in malaria areas against mosquitoes, fruit flies are becoming resistant to malathion. The allele that provides resistance in the insects has a higher frequency in current generations than in previous generations.

**Bacteria** resistance to antibiotics – there are new strains of bacteria that are resistant to multiple antibiotics (often called superbugs), e.g. methicillin resistant *Staphylococcus aureus* (MRSA) and vancomycin resistant *Enterococcus* (VRE). Many of these bacteria are present in hospitals.

**Viruses** resistant to drugs – e.g. there are HIV strains that have at least some resistance to the antiretroviral (ARV) being used to treat AIDS patients.

***Plasmodium falciparum*** strains resistant to antimalarial drugs has led to a resurgence of malaria.

## QUESTIONS

1. Define microevolution.
2. List the four main processes that lead to microevolution.
3. Use an example to explain why many bacteria are now resistant to antibiotics.
4. During the ice ages sea level dropped and land bridges formed between mainland Australia, Tasmania, New Guinea, and the Indo-Malaysian peninsula. How would these land bridges have affected gene flow?

# 13 Tay-Sachs Disease

**Tay-Sachs disease** shows how the incidence of a genetic disease in a particular population can be related to the different factors acting on the dynamics of a gene pool. It is named after **Warren Tay** (1843-1927) and **Bernard Sachs** (1858-1944) who both described the disorder.

**Gene flow** is the movement of alleles into or out of a population, e.g. due to immigration or emigration. If gene flow is restricted and a new population in an area is isolated either geographically or culturally for many generations the **founder effect** can lead to the amplification of a disease-causing allele.

Tay-Sachs disease is an **autosomal recessive** disease that is very rare in the general population. Frequency of the disease shows the founder effect with the Ashkenazi Jews who lived for many centuries in Eastern and Central Europe, e.g. Germany, Poland, Russia, Lithuania and Ukraine. They lived in isolated groups due to political and religious reasons with minimal interbreeding with other local populations. The disease is also more common in some French-Canadian groups in Quebec, in the Old Order Amish community in Pennsylvania and in the Cajun population in Louisiana.

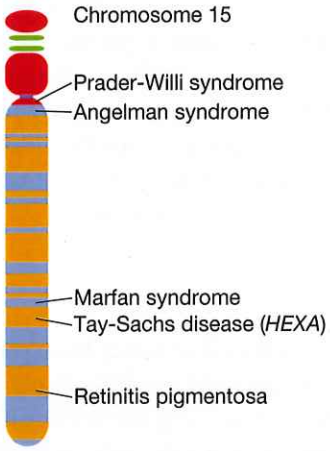



**Figure 13.1** Tay-Sachs disease on chromosome 15.

There are three types of Tay-Sachs disease – infantile (classic), juvenile onset (develops between ages 2 to 10) and late onset (begins between adolescence and mid 30s).

## Cause and transmission

Tay-Sachs disease is caused by a recessive allele on chromosome 15 for the *HEXA* gene. This gene codes for the vital enzyme hexosaminidase A (hex-A) which is found in lysosomes. Lysosomes are single membrane cellular organelles that break down and digest complex molecules for recycling. The recessive allele prevents this enzyme being produced. The child inherits a recessive gene from each parent. Offspring are homozygous with the recessive allele.

## Symptoms

The enzyme hex-A is involved in lipid metabolism. Without the enzyme lipids (GM2) accumulate in cells, especially nerve cells of the brain. Lipid metabolism and lipid storage are disrupted leading to a progressive deterioration of neurons in the nervous system. By two years of age most children have recurring seizures and reduced mental abilities. The child becomes blind, deaf, unable to swallow and eventually becomes paralysed. Severe forms lead to infant death at about age 3 or 4.

## Treatment and cure

There is no cure for Tay-Sachs disease.

**Diagnostic tests** are available to identify carriers and to detect an affected foetus prenatally. Test can involve an amniocentesis, biopsy and/or chorionic villi sampling. Palliative care to give support and comfort can involve medication to ease pain and control seizures, physical therapy, feeding tubes and respiratory assistance.

There are clinical trials of **stem cell transplantation** for people with late onset Tay-Sachs disease and **gene therapy** trials.

## QUESTIONS

1. How was the disease named?
2. Define gene flow.
3. What causes the founder effect?
4. Identify some populations that show the founder effect with Tay-Sachs disease.
5. Identify the inheritance pattern of Tay-Sachs disease.
6. What is the cause of Tay-Sachs disease?
7. How is Tay-Sachs disease related to lysosomes?
8. Why must a person with Tay-Sachs disease be homozygous genotype?
9. What happens when a person produces the defective enzyme in this disease?
10. Identify the three types of Tay-Sachs disease.
11. The diagram shows a normal neuron and a neuron in Tay-Sachs disease.

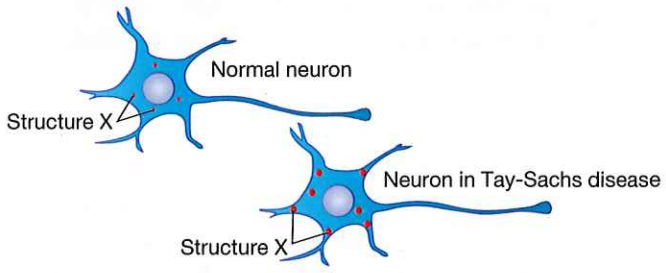



**Figure 13.2** Tay-Sachs disease neuron.

What is structure X?

(A) Nucleus (B) Mitochondria
(C) Golgi apparatus (D) Lysosomes

# 14 Sickle Cell Anaemia

**Sickle cell anaemia** (sickle cell disease – SCD) shows how the incidence of a genetic disease in a particular population can be related to the different factors acting on the dynamics of a gene pool.

Sickle cell disease is a group of disorders that have misshapen red blood cells due to an atypical haemoglobin called **haemoglobin S** that makes the red blood cells more rigid and likely to stick together to block small blood vessels, thus reducing life expectancy.

People **with sickle cell anaemia** are homozygous for the mutant allele. **Carriers** of the sickle cell allele (heterozygotes) have an increased protection against malaria.

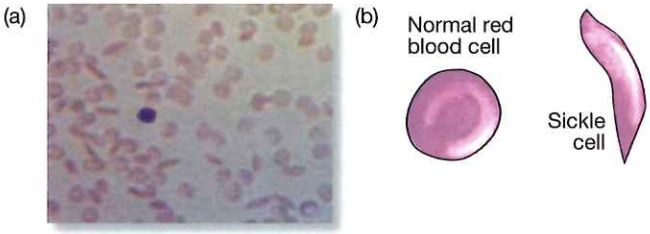



Figure 14.1 (a) HP sickle cell anaemia blood. (b) Normal red blood cells and sickle cell.

## Cause and transmission

Sickle cell anaemia is caused by a point mutation on chromosome 11 to the gene for beta haemoglobin. The gene codes for 287 amino acids and the affected allele has the code GTG (valine) rather than GAG (glutamic acid).

The terms recessive and dominant refer to the phenotype and the appearance caused by the genotype. Thus the abnormal form is a **recessive allele** when looking at sickle cell anaemia as a person will have the disease if they are homozygous with two copies of the allele. However, when looking at blood cell shape the allele is **codominant** as a person who is heterozygous will have a mixture of normal red blood cells and sickle shaped cells. The heterozygous form is called **sickle cell trait**. If considering malaria resistance, the allele could be considered dominant with only one copy of the allele needed to provide resistance.

## Effect on red blood cells

Red blood cells are distorted to the sickle cell shape (crescent moon shape) and although sickle cells can easily pick up oxygen like normal red blood cells there is a loss in elasticity of the red blood cells. The red blood cells particularly take on the sickle shape when deoxygenated. When the sickle haemoglobin releases the oxygen, the molecules 'stick' together and form long polymer chains. The polymers cause the cells to change shape and bend. When the red blood cells travel through the lungs again and pick up oxygen they resume their normal shape. A high proportion of abnormal haemoglobin molecules in a red blood cell will make the sickling permanent.

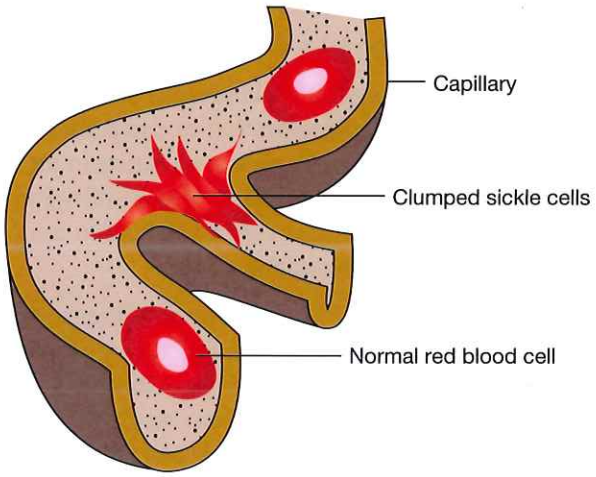



Figure 14.2 Sickle cells clump causing vessel occlusion.

The cyclic changing in the haemoglobin causes damage to the cell membranes making them more rigid. The rigid red blood cells are unable to flow through narrow capillaries and increases the viscosity of the blood. The red blood cells can also clump together causing vessel occlusion. This deprives the body's cells of oxygen producing hypoxia or low oxygen supply. The sickle cells can break in the capillaries and they break down after 10 to 20 days causing anaemia while normal red blood cells last about 120 days in the bloodstream.

## Symptoms

Individuals with sickle cell anaemia have normal periods followed by periodic painful attacks when sickling occurs. Symptoms include fever, chest pain, hard breathing, fatigue, episodic joint pain, pallor and acute painful enlargement of the spleen. Blocked vessels can lead to serious infections and organ damage and life expectancy is usually shorter. Infections include an increased risk of disease such as pneumococcus, influenza, meningocossus and SCD complications.

## Treatment and cure

Most treatment involves relieving symptoms and preventing infection, e.g. children should be fully immunised including annual influenza vaccinations and immunisation for meningococcus and pneumococcus. There are medications for chronic pain and in episodes of anaemia blood transfusions are needed.

**Hydroxyurea** is an oral medication that inhibits DNA synthesis causing immature red blood cells to produce foetal haemoglobin instead of adult haemoglobin giving some protection against haemoglobin S and reducing some SCD complications such as inflammation.

## Frequency

The disease is common among people with ancestors from Africa and Mediterranean countries, e.g. Greece, Turkey, Italy, Middle East, India and Spanish-speaking areas of South America and Central America.

Since the sickle cell disease is usually fatal or significantly reduces life expectancy, it would be expected that the frequency for the gene is low. However, the countries with a high allele frequency also have a high incidence of malaria and abundance of the *Anopheles* mosquito which transmits the malarial parasite. Since the heterozygotes (carriers of the sickle cell allele) have some protection against malaria, the lethal allele stays in the population. In these countries, people homozygous for the normal haemoglobin can die of malaria. Thus the frequency of the sickle allele varies worldwide depending on the selective pressure of malaria.

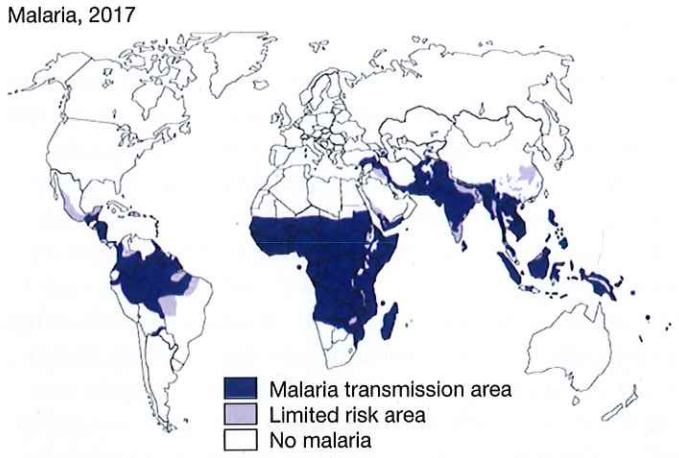



**Figure 14.3** Malaria areas.

## Sickle cell trait

Sickle cell trait is the heterozygous condition ($Hb^AHb^S$). The normal allele ($Hb^A$) is codominant to the sickle cell allele ($Hb^S$). Heterozygotes are usually healthy and do not show the severe symptoms of sickle cell disease. Some may suffer some sickle cell symptoms during extended periods of reduced blood oxygen, e.g. strenuous exercise or at high altitudes. The trait provides some immunity against malaria. The protozoan that causes malaria spends part of its life cycle in red blood cells and the presence of heterozygous amounts of sickle cell haemoglobin lowers the density of the parasite and reduces the symptoms of malaria.

## QUESTIONS

1. What is sickle cell disease?
2. If $Hb^A$ is the allele for normal red blood cells and $Hb^S$ is the allele for sickle cell:
   (a) Identify the genotype of a person with sickle cell disease.
   (b) Identify the genotype of a carrier for sickle cell anaemia.
3. Outline the cause of sickle cell anaemia.
4. To what do the terms dominant and recessive refer?
5. What is the inheritance pattern for:
   (a) Sickle cell anaemia?
   (b) Red blood cell shape – sickle cell or normal shape?
   (c) Malaria resistance?
6. How does the abnormal haemoglobin affect red blood cells?
7. What can cause permanency in the sickle cell shape?
8. Why do people with sickle cell anaemia often suffer from vessel occlusion?
9. Identify some of the treatments for SCD.
10. Why is hydroxyurea used as a treatment option?
11. Among which groups of people is sickle cell anaemia more common?
12. (a) Why would you expect a low allele frequency of sickle cell?
    (b) Explain why there is a high allele frequency for sickle cell in some areas of the world.
13. (a) Describe sickle cell trait.
    (b) Outline why this condition provides some immunity against malaria.
    (c) In which areas of the world would you expect to find a high allele frequency for sickle cell?
14. Explain why the frequency of sickle cell is lower in African Americans than in Africa.
15. The map shows the malaria areas of the world 2017.

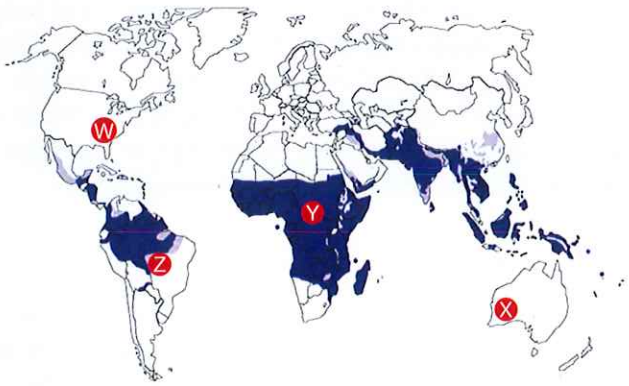



**Figure 14.4** Sickle cell allele frequency.

What would be the expected order from highest to lowest frequency for sickle cell for areas W, X, Y and Z?

(A) W, X, Y, Z (B) X, W, Y, Z
(C) Y, Z, W, X (D) Z, Y, W, X

# 15 Thalassemia

**Thalassemia** shows how the incidence of a genetic disease in a particular population can be related to the different factors acting on the dynamics of a gene pool.

Thalassemia is an autosomal recessive disease caused by a mutation either on chromosome 16 and/or chromosome 11. The disease affects the production of **haemoglobin**. Haemoglobin consists of haem, alpha globulins and beta globulins. There are several forms which depend on which of the four protein chains in haemoglobin has the mutation. All cause the formation of an abnormal form of haemoglobin. **Alpha (α) thalassemia** occurs when there is a mutation in the gene for the alpha globulin (two genes on each chromosome 16) and **beta (β) thalassemia** occurs when there is a change in the beta globulin gene (one gene on each chromosome 11). Both alpha and beta thalassemia have major or minor forms.

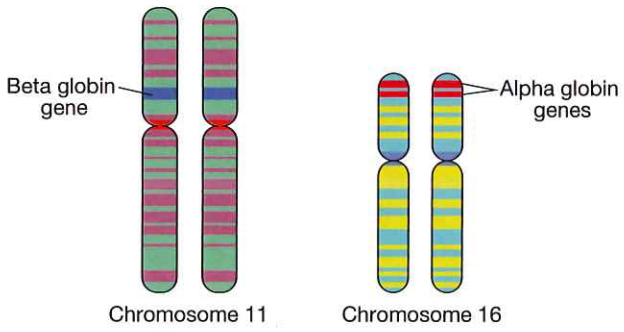



Figure 15.1 Genes causing thalassemia.

## Frequency

Thalassemia is most common in people from the Middle East, Mediterranean countries, e.g. Greece and Turkey, Africa and Asia. Alpha thalassemia is common in people of African, Middle Eastern, Chinese or South-East Asian descent, while beta thalassemia is common in people from Mediterranean areas, e.g. Africa, Greece, Malta and Italy. Immigration into Australia from these areas has increased the allele frequency for thalassemia.

## Cause and transmission

It is an inherited autosomal recessive disease with a number of thalassemia mutations for either the beta globin protein (on chromosome 11) or the genes for the alpha globin protein (on chromosome 16). People with thalassemia are homozygous genotype. Prenatal testing by amniocentesis checking the amniotic fluid or chorionic villus sampling of the placenta can check a growing embryo.

## Symptoms

The abnormal genes for haemoglobin cause the body to reduce globin synthesis and form less haemoglobin. There is increased friability of the cell membrane of the red blood cells leading to an excessive destruction of red blood cells and early haemolysis. This leads to mild or severe anaemia fatigue, jaundice (yellow skin), bone deformities, slow growth rates and cardiovascular problems.

Each type of thalassemia has distinctive symptoms. **Alpha thalassemia** causes bone problems with cheeks, forehead and jaw overgrowing, malnourishment, extremely enlarged spleen and jaundice. Severe alpha thalassemia can cause stillbirth or early death due to heart failure. **Beta thalassemia** usually appearing before 2 years of age causes paleness, frequent infections, poor appetite, jaundice, enlarged organs and require regular blood transfusions

Iron overload can damage the heart, liver and endocrine system. When the bone marrow expands the bones widen causing changes to the bone structure, e.g. in face and skull. The spleen becomes larger to aid fighting infection and destroys old red blood cells.

Thalassemia can give some protection against malaria, e.g. against *Plasmodium falciparum.*

## Treatment and cure

Treatment depends on the type and severity of thalassemia with minimal treatment needed for people who are carriers or who have mild forms to blood transfusion programs with folate supplements and additional chelation therapy for people with moderate to severe forms of thalassemia. Blood tests check the complete blood count and there are special haemoglobin tests. **Blood transfusions** can take 1 to 4 hours and are given when the person is anaemic or has an infection or illness. Some people require a transfusion every 2 to 4 weeks to maintain normal haemoglobin and red blood cell levels. **Iron chelation therapy** is needed to reduce the iron overload caused by the blood transfusions.

Other treatments include **bone marrow transplants** to replace faulty stem cells and researchers are investigating **gene therapy** to insert the normal haemoglobin gene into stem cells in bone marrow.

Lack of treatment can make the person more likely to develop infections.

## QUESTIONS

1. Identify the inheritance pattern for thalassemia.
2. Which chromosomes are affected in thalassemia?
3. Outline why there are the two types of thalassemia.
4. Where is thalassemia most common?
5. Describe the symptoms of thalassemia.
6. Describe the main treatment for thalassemia.
7. Explain why some people with thalassemia need iron chelation therapy.

# 16 Natural Selection, Speciation and Evolution

**Natural selection** occurs when factors in the environment confer a selective advantage on specific phenotypes to enhance survival and reproduction.

**Speciation** is the evolution of a new species. Speciation involves the reproductive isolation of a group from the rest of its species with no genetic interchange.

**Allopatric speciation** involves geographical separation, e.g. mountain ranges form, rivers change course, land bridges form/disappear, deserts form or dynamic Earth movements such as earthquakes and volcanic eruptions occur.

**Sympatric speciation** occurs when the new species develops within the same geographical region as the parent species.

Speciation is likely to occur if the isolated group is a small population. Genetic drift and the founder effect will lead to changes in the gene pool of the separated group so that allele frequencies will differ from the original parent population and there will be genetic divergence.

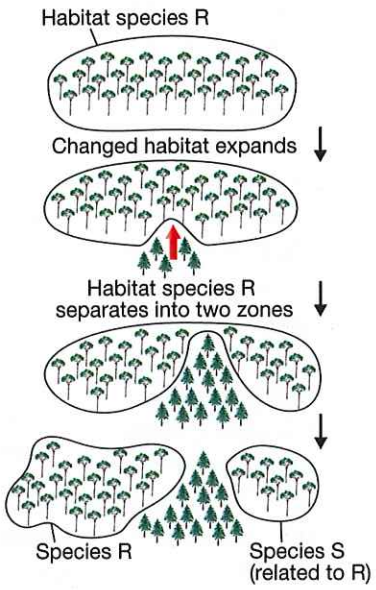



Figure 16.1 Speciation by geographic isolation.

## If the barrier disappears in allopatric speciation

If the barrier that separated the parent population and the small population disappears then:

- Speciation had led to sufficient changes that the two groups can no longer interbreed to produce fertile offspring and there are two distinct species.
- Speciation had not occurred and the two groups can interbreed to produce fertile offspring making them one species.
- The two groups have distinct differences and there is limited gene flow which can lead to sympatric speciation or a cline can develop when the differences change with location.

## Mechanisms in evolution

**Evolution** is the change in the gene pool from one generation to the next due to processes such as natural selection, mutation and genetic drift. The main mechanisms underpinning the theory of evolution by natural selection are as follows.

- **Inherited variation** – the ultimate source of variation within a population is mutation. Mutations produce random changes with mutation rates influenced by the chemical composition of the gene and the arrangement of the gene in the chromosome. Most new alleles due to mutation are neutral and do not aid the individual during their lifetime. Mutagens are chemical substances or radiation that cause mutation.
- **Struggle for existence** – in many populations more are born than can survive to reproduce causing 'survival of the fittest'. Those with traits that best cope with pressures in a particular environment will survive and produce offspring. The offspring that inherit these features will survive and these features will increase in frequency in the population.
- **Isolation and differential selection** – isolation is very important in the processes of evolution and speciation. Isolating mechanisms reduce gene flow between populations and over time the cumulative changes in the gene pools of different groups can lead to speciation where interbreeding to produce fertile offspring is no longer possible.
- **Changes in gene pools** – an allele becomes 'fixed' in a gene pool when all the individuals in the population are homozygous for that allele. Gene flow between populations, e.g. due to immigration or emigration helps maintain relatively similar allele frequencies between different populations and reduces the chance of speciation. If a new allele arises on one population, gene flow allows this allele to spread to other populations. This means the gene pool of the species can change over time due to mutation, changing environmental selective pressures and/or genetic drift.

There is evolution as the gene pool changes.

## QUESTIONS

1. When does natural selection occur?
2. Define speciation.
3. Distinguish between allopatric speciation and sympatric speciation.
4. If the barrier in allopatric speciation is removed, what possible outcomes could occur?
5. Define evolution.
6. Describe the main mechanisms for evolution.

# 17 Allele Frequencies and Natural Selection

The **gene pool** is the sum total of all the alleles of all the genes of all the individuals in a population. A population can be defined by its gene pool.

**Allele frequency** is the proportion or percentage of a particular allele among all alleles at a given locus in a population gene pool. In natural populations allele frequency will vary from generation to generation. If surviving offspring have a favourable combination of alleles then these alleles are likely to be present with an increased frequency in the next generation.

Some populations are 'closed' with effective barriers preventing migration into the population, while other populations are 'open' with overlapping boundaries that allow interchange with other populations.

Evolution occurs as a result of accumulated changes in the gene pool over time.

The **Hardy-Weinberg principle** gives a mathematical formula to show how the frequencies of two alleles (A and a) represented by symbols $p$ and $q$ maintains constant gene frequencies if the above conditions exist. If the conditions are not met then there will be **genetic drift** and the gene pool will change.

The Hardy-Weinberg equation is as follows.

$$p^2 + 2pq + q^2 = 1$$
or 100%

This means that there is 100% probability that the offspring are AA ($p^2$), Aa ($2pq$), aa ($q^2$).

The Hardy-Weinberg equation can also be used if there are more than two alleles for the same gene.

$(p + q + r)^2$ with $r$ representing the third allele.

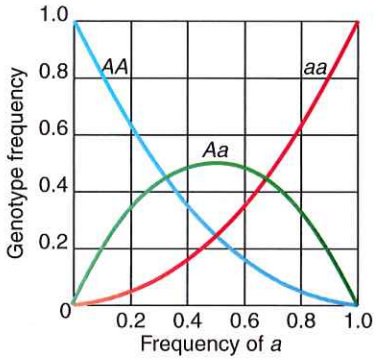



Figure 17.1 The relationship between the frequency of allele a in a population and the frequency of genotypes AA, Aa and aa.

## Constancy of gene frequencies

In large populations of sexually reproducing individuals with random matings the gene frequencies remain constant. Disturbances such as mutations, migrations, natural selection and random genetic drift can lead to significant changes in gene frequencies.

## QUESTIONS

1. What is a gene pool?
2. What is allele frequency?
3. Outline the relationship between evolution and allele frequency.
4. Outline the difference between a 'closed' population and an 'open' population.
5. In a population of 650 individuals it was found that 195 individuals had a genotype of AA, 97 had a genotype of aa and 358 had a genotype of Aa.
   (a) What is the frequency per cent for each of these genotypes?
   (b) What is the frequency of allele A and allele a?
6. In a small population of experimental animals it was found that 52 had the genotype BB, 158 had the genotype Bb and 49 had the genotype bb. What was the frequency of the alleles B and b in this population?
7. The graph shows two alleles A and a that are present in a population with allele frequencies 0.8 A and 0.2 a. There is a large number of random matings with 0.8 + 0.2 male gametes randomly fertilising 0.8 + 0.2 female gametes.

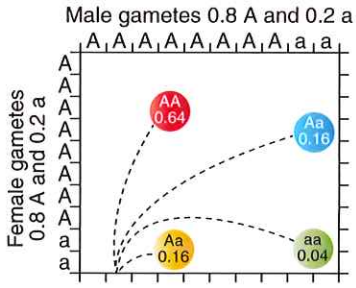



Figure 17.2 Allele frequency for a large number of random matings.

   (a) What are the frequencies of the genotypes of the first generation?
   (b) If these offspring grow and develop and in turn become parents of the next generation, what will be the frequencies of the alleles for this mating?
   (c) From your result in part (b), what can you conclude about gene frequencies if it is a large population that sexually reproduces and has random matings?
8. Identify some conditions that will lead to changes in gene frequencies.
9. Explain why a large number of recessive alleles stay in a population and why this can be an evolutionary advantage.
10. What is a closed population?
    (A) An isolated population with no migration into the population.
    (B) A large population on an island.
    (C) Populations within certain physical boundaries.
    (D) Populations with fixed allele frequencies.

# 18 Activity – Model for Natural Selection

Some Human Biology students decided to perform a first-hand investigation to model natural selection. They used green, red, yellow and orange cards to represent different hominins and used a dice with one side orange, two sides red and three sides green to represent selective pressure which was based on cranial capacity ($cm^3$) and tool making ability.

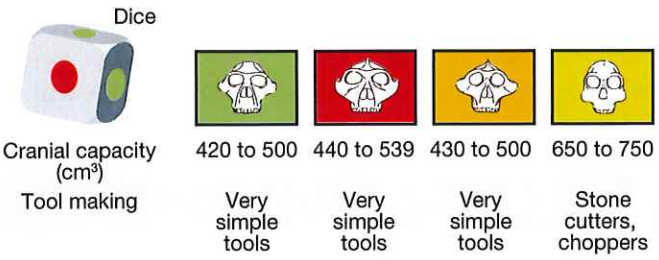



**Figure 18.1** Hominin cards.

They started with 40 cards representing 40 hominins in the original population, which consisted of 10 green cards, 10 yellow cards, 10 red cards and 10 orange cards. They shuffled the cards and then dealt them out in pairs. Each pair reproduced one offspring according to rules constructed by the students, so that 20 offspring were added to the population. The students then rolled the dice, simulating individuals not surviving selection pressures in the environment, and 20 cards were removed as 'non-survivor'. This method kept the hominin population constant at 40 individuals for every year as the carrying capacity for the environment. The numbers were recorded for year 1.

They repeated this method, accumulating results for '7 years' for this hominin population. Their results are in the table below.

**Table 18.1** Student results for hominin population.

| Year | Red | Green | Orange | Yellow | Total |
|---|---|---|---|---|---|
| 1 | 10 | 10 | 10 | 10 | 40 |
| 2 | 6 | 10 | 12 | 12 | 40 |
| 3 | 2 | 11 | 13 | 14 | 40 |
| 4 | 1 | 8 | 12 | 19 | 40 |
| 5 | 0 | 3 | 15 | 22 | 40 |
| 6 | 0 | 0 | 12 | 28 | 40 |
| 7 | 0 | 0 | 7 | 33 | 40 |

## QUESTIONS

1. In this experiment by the students, what is the 'selecting agent'?
2. How did the hominin population change over time?
3. (a) Outline how the students kept the population total constant over each year.
   (b) Explain why it was important to keep the population total constant.
4. For this population what was the 'favourable characteristic for survival'?
5. If the students decided to simulate a model for large jaw muscles to eat tough fibrous bark and leaves, how could they change the method so that the results would show natural selection for hominins?
6. Explain how changing the colours on the sides of the dice to 1 green : 1 red : 1 yellow : 1 orange would influence evolution.
7. (a) Describe an experiment you could perform to model natural selection that differs from the one described above.
   (b) Explain how this model of natural selection could be applied to human evolution.
8. Explain why isolation is another important factor that leads to the evolution of a new species.
9. Explain why scientists use models.
10. The cards in this experiment have pictures of the hominins *Australopithecus africanus, Paranthropus robustus, Paranthropus boisei* and *Homo habilis* who all lived around 2 million years ago.
    (a) Identify each hominin with its colour card.
    (b) Explain how this example shows hominin evolution.
11. Which of the following best describes natural selection?
    (A) The mechanism for evolution.
    (B) Survival of the strongest.
    (C) Producing mutations to survive in a changing environment.
    (D) Changing to fit in with the environment.
12. In a population, a certain gene is found to have a frequency of 25%. When would a decrease in the frequency of this gene occur?
    (A) When the gene became recessive.
    (B) When individuals with the gene had fewer offspring.
    (C) When individuals with this gene were immune to a lethal disease.
    (D) When the population size decreased.
13. How could students increase the reliability of their results in an experiment to model natural selection?
    (A) Repeat the experiment several times to see if they get the same results each time.
    (B) Use more accurate measuring devices.
    (C) Make sure their method will give results that answer the aim.
    (D) Check secondary sources to find out what other people have done on the same subject.
14. Which scientist proposed the theory of natural selection?
    (A) Samuel Wilberforce (B) Charles Darwin
    (C) Gregor Mendel (D) Thomas Huxley

# 19 Endonucleases

An **endonuclease** is an enzyme that breaks the phosphodiester bond in the sugar phosphate backbone within a polynucleotide chain.

**Deoxyribonuclease** cuts DNA non-specifically at any nucleotide sequence.

## Restriction endonucleases

**Restriction endonucleases** are degradative enzymes that recognise and cut up DNA, including DNA of certain phages that is foreign to a bacterium. The substrates for the enzymes are the **recognition sequences**.

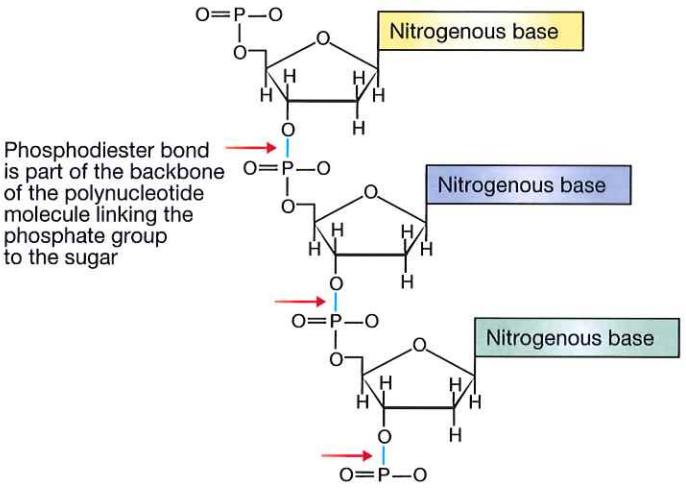



Figure 19.1 Phosphodiester bond that is broken by endonucleases.

Found in bacteria and archaea the restriction endonucleases provide a form of protection for the prokaryotes against viral infection. When a virus injects DNA into the bacterium the restriction endonucleases cut up the viral DNA so that the virus cannot take over the bacterium and use the host for its own reproduction.

Restriction endonucleases are extremely useful in DNA manipulation as the ability to cut DNA at specific sites allows researchers to determine genetic diversity and to remove DNA from specific sources. Useful restriction endonucleases cut DNA at very specific nucleotide sequences. The nucleotide sequence is called the **restriction site** and is usually about four to eight nucleotide sequences long and is where the restriction endonuclease cuts the sugar phosphate backbone by hydrolysing the phosphodiester bond between two nucleotides.

## Types of restriction endonucleases

Restriction endonucleases can be divided into four main types based on their composition, where they break the polynucleotide chain and need for a cofactor.

**Type I enzymes** – cut DNA at a random distance from the recognition sequence.

**Type II enzymes** – cut DNA at specific positions close to their recognition sequence.

**Type III enzymes** – cut DNA away from their recognition sequence and need two of the recognition sequences in opposite orientations within the same DNA molecule.

**Type IV enzymes** – recognise modified, usually methylated DNA.

Type II restriction enzymes are the main type used in laboratories for DNA analysis and gene cloning as they produce specific discrete restriction fragments that will form distinct gel banding patterns.

## QUESTIONS

1. Define an endonuclease.
2. Outline the action of deoxyribonuclease.
3. What is a restriction endonuclease?
4. Enzymes act on a substrate. What is the substrate for restriction endonucleases?
5. Outline the host defence role of restriction endonucleases.
6. Explain why restriction endonucleases are useful in DNA manipulation.
7. What is the restriction site?
8. Construct a table to summarise the four types of restriction endonucleases.
9. Which type of restriction endonucleases are mainly used in laboratories for DNA manipulation and explain why this type is the most useful.
10. The diagram shows stages of a virus entering a bacterium.

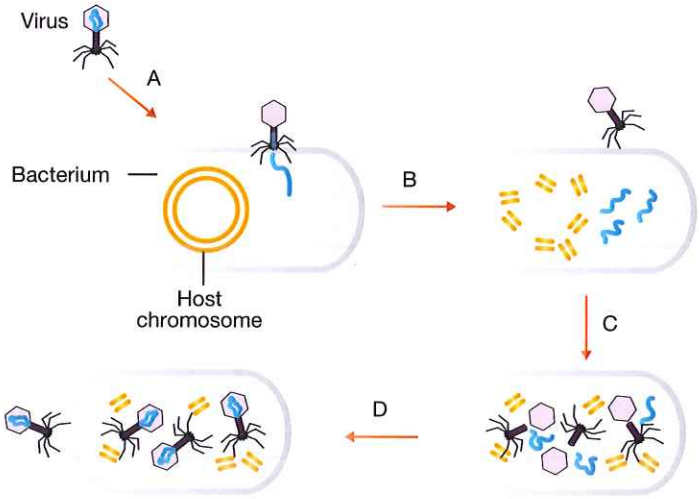



Figure 19.2 Viral infection of a bacterium.

At which step would the bacterium release an restriction endonuclease?

(A) Step A (B) Step B
(C) Step C (D) Step C

# 20 DNA Ligases

**DNA ligases** are linking enzymes that are needed in DNA replication. They catalyse the bonding of the 3′ end of a new DNA fragment to the 5′ end of a growing chain.

In **DNA replication** DNA ligases join Okazaki fragments by forming a bond between their free ends to make a long continuous strand on the lagging strand.

DNA ligase will cause the formation of a phosphodiester bond between the phosphate and sugar to form the backbone of a DNA molecule. This can **repair** a single strand break using the complementary strand of the double helix as a template. It can also repair breaks in double stranded DNA.

**DNA ligation** is the process when linear fragments of DNA are joined together with covalent bonds.

Like all enzymes each DNA ligase works best at an optimal temperature, optimal pH and can require cofactors and/or ATP.

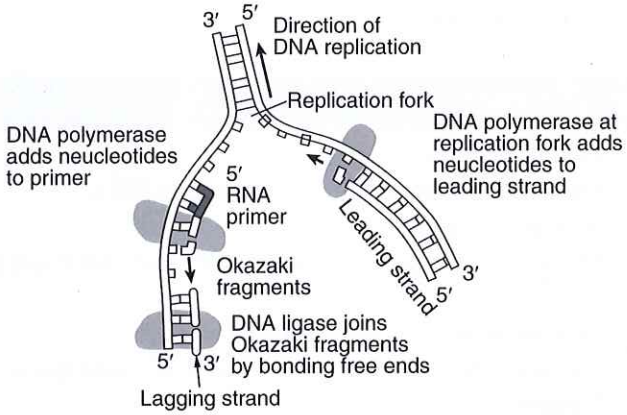



Figure 20.1 DNA ligase joins Okazaki fragments.

## Use in DNA manipulation

Since DNA ligase is involved in DNA replication and DNA repair, the enzyme can be used in **genetic engineering** to form recombinant DNA. After a restriction enzyme has cut a specific gene or nucleotide sequence from a DNA molecule and cut the receiving DNA molecule, DNA ligase is used to insert the DNA fragment. For example, it is often used to insert DNA fragments into bacterial plasmids that act as vector backbones.

The DNA ligase from **bacteriophage T4** is often used in laboratory research, e.g. in making DNA libraries. T4 DNA ligase works best at 16°C. Bacteriophage T4 infects *Escherichia coli* bacteria which are used in research. T4 only undergoes the lytic life cycle which destroys the host cell. *E. coli* is frequently used in biotechnology, e.g. *E. coli* plasmids are used as vectors and used to produce 'designer' proteins which can then be mass produced in industrial fermentation, e.g. producing human insulin.

Ligases from **thermophilic bacteria** are enzymes that are stable at high temperatures and are often used in ligase amplification reactions as they do not denature at high temperatures.

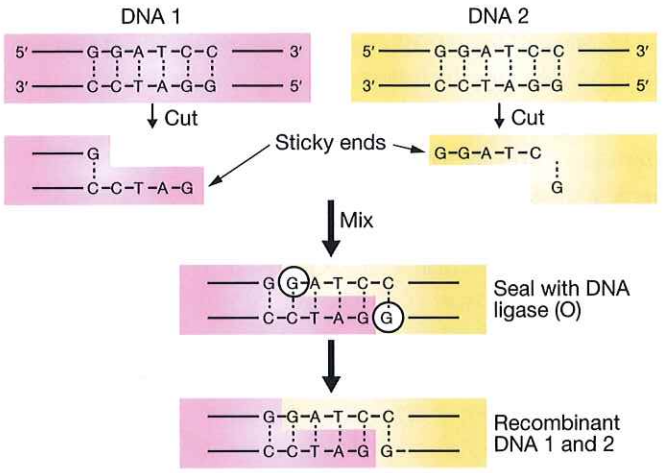



Figure 20.2 DNA ligase joins sticky ends.

## QUESTIONS

1. Define DNA ligase.
2. How is DNA ligase involved in DNA replication?
3. Define DNA ligation.
4. Describe the bond formed by DNA ligation.
5. Explain why DNA ligase is useful in DNA manipulation.
6. (a) What is bacteriophage T4?
   (b) Explain why T4 DNA ligase is often used in laboratory research.
7. Explain why DNA ligases from thermophilic bacteria are used in laboratory research.
8. Explain why DNA ligation in DNA manipulation requires each of the following.
   (a) Two or more DNA fragments with either blunt or compatible sticky ends.
   (b) A buffer containing ATP.
   (c) T4 DNA ligase.
9. The diagram shows a DNA fragment being inserted into a bacterial plasmid.

Figure 20.3 DNA manipulation using bacterial plasmid.

Which enzymes cut the plasmid and attach the inserted gene respectively?

(A) Restriction enzyme and DNA polymerase.
(B) DNA polymerase and endonuclease.
(C) Restriction enzyme and DNA ligase.
(D) DNA ligase and restriction enzyme.

# 21 DNA Polymerases

**DNA polymerases** are enzymes that catalyse the elongation of new DNA at a replication fork by the addition of nucleotides to the existing chain. The DNA polymerase adds nucleotides one by one and the rate of elongation depends on the species, e.g. in human cells the rate is around 50 nucleotides per second while in bacteria it can be 500 nucleotides per second.

There are different DNA polymerases and different species have different numbers of DNA polymerases. For example, *E. coli* has two different DNA polymerases involved in replication – DNA polymerase III and DNA polymerase I. However, the structure of DNA polymerase is **highly conserved** so that the subunits and the functions they perform are similar from one species to another.

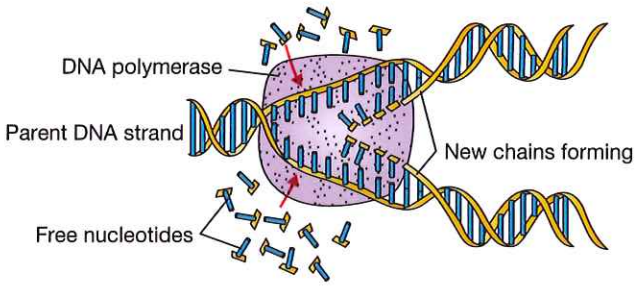



Figure 21.1 DNA polymerase is involved in DNA replication.

## How it works

DNA polymerase 'reads' the existing DNA as a template code and adds free nucleotides to the 3′ end of a newly formed strand. It cannot begin the formation of a new chain and needs a **primer** to initiate replication. DNA polymerase usually works in pairs to create the two new DNA strands. It can only work in the 5′ → 3′ direction which means the two strands of the replication fork are copied in different ways. The **leading strand** is copied continuously in a direction towards the replication fork while the **lagging strand** is synthesised in Okazaki fragments which are short sections in a direction away from the replication fork.

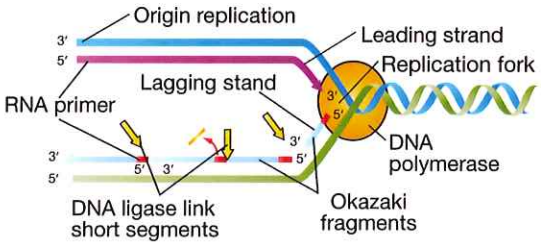



Figure 21.2 The two strands elongate in 5′ → 3′ direction.

As soon as a nucleotide is added to a strand, DNA polymerase will 'proofread' the code against the template checking for a mistake so that misplaced pairs can be corrected. If a mistake is found the enzyme will remove the nucleotide and then continue synthesis.

Because DNA polymerase only adds to the 3′ end of a pre-existing polynucleotide it means it cannot complete the 5′ ends of daughter strands where the primer was attached. This means that DNA in daughter cells gets shorter each time the DNA replicates and the cell divides.

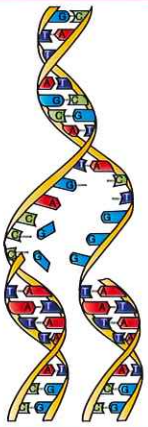
| Diagram of DNA replication | What is happening at each stage |
|---|---|
|  | DNA double helix. |
| | Double helix unwinds. |
| | Double helix unzips and two strands separate catalysed by the enzyme helicase. |
| | Free nucleotides from the cytoplasm join at the free sites as the helix unwinds. Adenine always pairs with thymine and guanine pairs with cytosine to maintain the sequence of nucleotides. |
| | DNA polymerase catalyses the formation of bonds to join the nucleotides to the DNA strand. |
| | Two identical strands of DNA are formed from the original strand. |

Figure 21.3 DNA replication.

## QUESTIONS

1. Define DNA polymerase.
2. How can the rate of elongation due to DNA polymerase vary?
3. DNA polymerases are highly conserved. What does this mean?
4. Outline how DNA polymerases work.
5. Explain why there has to be a leading strand and a lagging strand.
6. Explain why a primer is needed in DNA replication.
7. The diagram shows a problem with DNA polymerase and DNA replication.

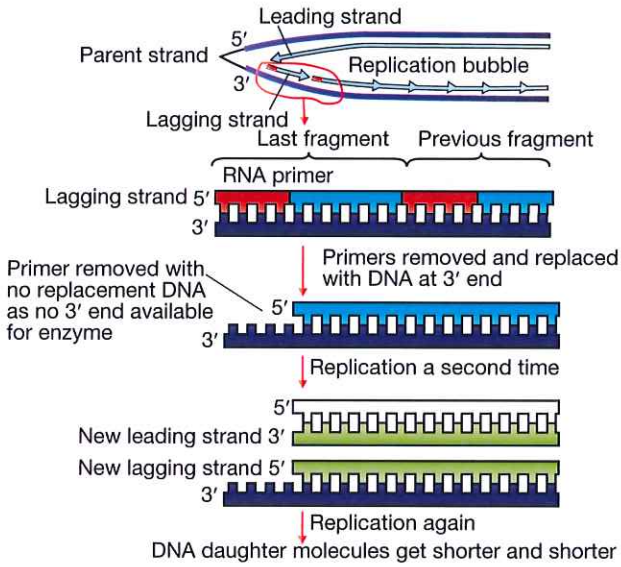



Figure 21.4 DNA polymerase and replication.

Outline the problem that is shown in this diagram.

# 22 The Polymerase Chain Reaction

The **polymerase chain reaction** (PCR) is a technique for amplifying DNA in vitro by incubating nucleotides with special primers and DNA polymerase. It allows any chosen segment of DNA to be copied many times in a test tube in a relatively short time span.

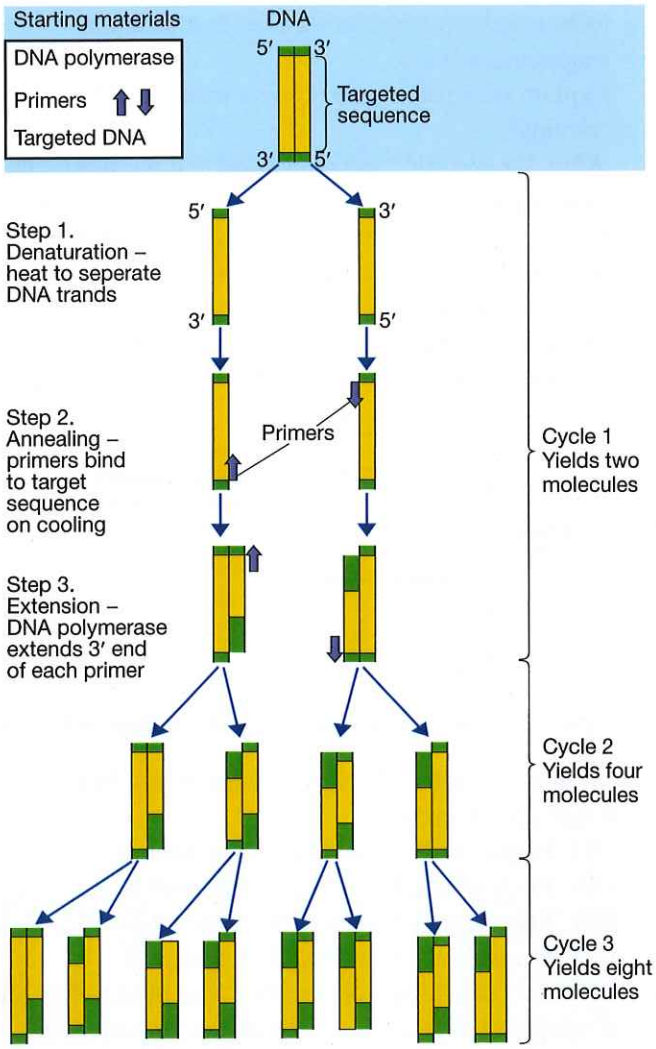



Figure 22.1 DNA polymerase is involved in DNA replication.

## How it works

There are three basic steps in PCR which will exponentially increase the number of molecules each cycle.

**Step 1 – Denaturation** – the DNA strands are separated by heating.

**Step 2 – Annealing** – the two primers – each respectively complementary to the end of one of the strands of the target sequence bind by hydrogen bonding to its target DNA sequence during cooling.

**Step 3 – Extension** – a heat stable DNA polymerase extends the primers in the 5′ → 3′ direction.

The ability to amplify very small amounts of DNA has been invaluable for researchers, e.g. when studying the DNA sequences from archaeological remains and forensic evidence as these samples are often in very small amounts and can be partially degraded.

## QUESTIONS

1. What is the polymerase chain reaction?
2. Outline why PCR is important.
3. What are the three main steps in PCR?
4. Taq DNA polymerase is named after the thermophilic bacterium *Thermus aquaticus* from which it was extracted. This bacteria lives in hot springs. Discuss why the discovery of Taq DNA polymerase greatly aided research investigations.
5. The diagram shows a type of DNA manipulation.

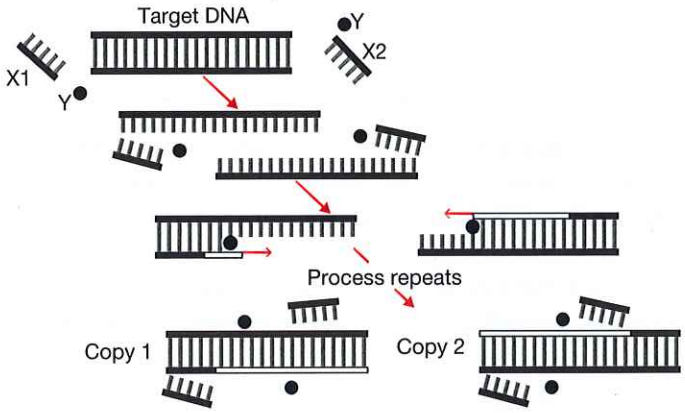



Figure 22.2 DNA manipulation.

   (a) Identify X1, X2 and Y.
   (b) Explain what is happening in this diagram.
6. Explain why PCR is invaluable when studying DNA samples that come from crime scenes or archaeological remains.
7. Explain why PCR is important in prenatal diagnosis.
8. What is the correct order of the polymerase chain reaction?
   (A) Extension, annealing, denaturation.
   (B) Annealing, extension, denaturation.
   (C) Annealing, denaturation, extension.
   (D) Denaturation, annealing, extension.
9. Which of the following DNA manipulation techniques amplifies a DNA sample?
   (A) Polymerase chain reaction.
   (B) DNA ligation.
   (C) Gel electrophoresis.
   (D) Southern blotting.
10. In the polymerase chain reaction, what is the function of DNA polymerase?
   (A) Cause denaturing of parent DNA strand.
   (B) Attach to end of target DNA to initiate process.
   (C) Extend primers in 5′ → 3′ direction.
   (D) Extend new DNA in 3′ → 5′ direction.

# 23 Gel Electrophoresis

**Gel electrophoresis** is a technique that separates nucleic acids or proteins, on the basis of their size and electrical charge, by measuring their rate of movement through an electrical field in a gel. It can also be used to separate nanoparticles.

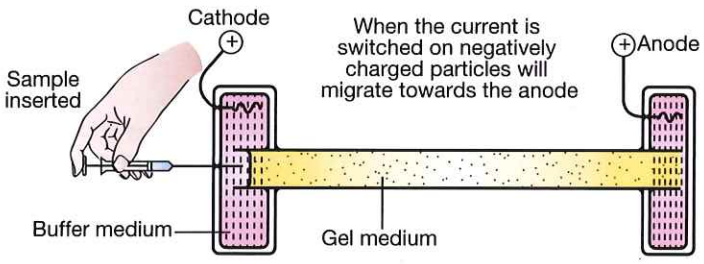



Figure 23.1 Gel electrophoresis.

## How it works

The gel medium acts as a sieve with the polymer fibres in the gel hindering the movement of larger and/or longer molecules. Nucleic acids have a negative charge and many proteins have a negative charge and when the current is turned on negatively charged particles travel towards the anode from where they are inserted near the cathode.

## Restriction fragment analysis

Restriction enzymes are used to cut DNA into fragments that can be sorted by gel electrophoresis. The gel needs to be an anticonvective medium, e.g. agarose or polyacrylamide gel to stop thermal convection when the electric field is applied. As the fragments travel in the gel a characteristic band pattern is produced dependent on the gel used and the size of the target molecules.

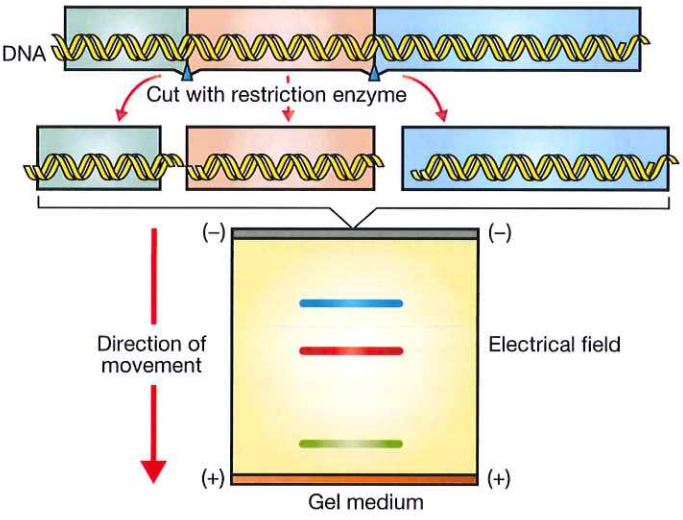



Figure 23.2 Gel electrophoresis with different DNA fragments.

As the process separates DNA fragments, gel electrophoresis is often used as a way to obtain pure samples of particular DNA sequences.

## QUESTIONS

1. What is gel electrophoresis?
2. Identify some substances that can be separated using gel electrophoresis.
3. What properties are needed for a suitable medium?
4. How is gel electrophoresis used in restriction fragment analysis?
5. Explain why part of the process is often called 'sieving'.
6. When the electric field is stopped and the fragments have separated a post electrophoresis stain is often applied, e.g. ethidium bromide which binds to the DNA and is fluorescent under UV light. Explain why a stain is applied.
7. The diagram shows a band pattern for three restriction fragments.

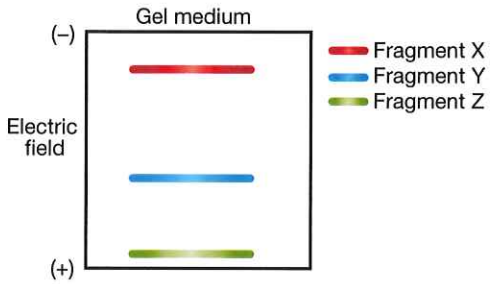



Figure 23.3 Gel electrophoresis and three fragments.

What is the order of restriction fragments from longest to shortest?

(A) Fragment X, fragment Y, fragment Z
(B) Fragment Y, fragment Z, fragment X
(C) Fragment Z, fragment Y, fragment X
(D) Fragment Z, fragment X, fragment Y

8. DNA was taken from young child and tested to find out if a mutated allele for a genetic disease was present.

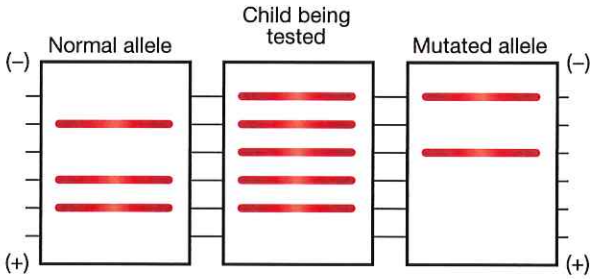



Figure 23.4 Gel electrophoresis for a child's DNA.

What is the genotype for this child for this gene?

(A) Homozygous normal for gene.
(B) Heterozygous for the gene.
(C) Homozygous for mutated phenotype.
(D) Genotype cannot be determined for this gene.

# 24 Developmental Biology

There are many areas of study that provide evidence for evolution including developmental biology.

**Developmental biology** is the study of the process by which organisms grow and develop. Studies investigate the processes that develop a zygote into a fully grown adult, e.g. genetic control of cell growth, differentiation and morphogenesis. **Morphogenesis** is the development of body shape and organisation. Developmental biology includes the study of classical embryology with molecular genetics.

During development there are changes in shape, form, size and degree of complexity. Comparing these stages between different species and groups of organisms and the genes that control each development shows the interrelatedness of each species and enables the construction of evolutionary trees.

Development involves differentiation where cells become specialised in structure and function. This involves biochemical as well as structural changes.

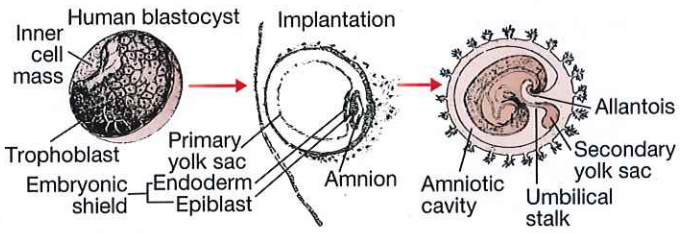



Figure 24.1 Early stages of human development.

## Embryology

Early stage development in animals from zygote (fertilised egg) to multicellular embryo follows a basic pattern that is studied in embryology.

**Embryology** is the study of the development of an embryo from fertilisation to the foetus stage. Comparative embryology compares the embryos of different species and shows the relationship between different species of animals.

Embryonic development of animals, both vertebrate and invertebrate, goes through three basic stages:

- **Cleavage** – where the zygote divides into a mass of daughter cells with the progressive subdivisions occurring in particular geometric patterns depending on the species.
- **Gastrulation** – the mechanical evolution forming the germ layers where the cells are rearranged into distinct layers and the developmental stage is called a gastrula.
- **Organogeny** – where differentiation leads to the formation of tissues and organs.

## Molecular systematics

**Molecular systematics** uses genetics to study the evolution of relationships among individuals and species to determine the history and processes involved in the history of life on Earth. To increase knowledge of why organisms grow and develop modern researchers are using genomics, biochemical techniques, bioinformatics and computer data analysis to investigate the molecular mechanisms of growth and development in different organisms.

Studies look at the nucleic acid sequence and the proteome to work out the systematic positions of different nucleotide sequences and the changes that have occurred over time. Comparisons are then made between different species and evolutionary relationships determined.

The recent discovery of horizontal gene transfer has shown that different genes within one organism may have different histories and different ancestors – different phylogenies. For example, the **R plasmid** in bacteria has genes for resistance to two antibiotics (ampicillin and tetracycline) and horizontal transfer between bacteria has led to the proliferation of antibiotic resistant bacteria. The ability of plasmids to insert themselves into DNA of host cells is used in genetic engineering to make recombinant DNA.

## QUESTIONS

1. Define developmental biology.
2. What is morphogenesis?
3. What happens during development?
4. Define embryology.
5. Identify a function of comparative embryology.
6. The following diagram shows a series of diagrams collected by a Human Biology student.

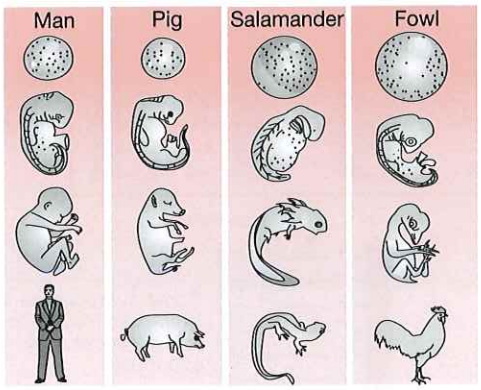



Figure 24.2 Series of diagrams for evolution.

   (a) What area of evidence for evolution was this student investigating?
   (b) Explain how this area of evidence supports the theory of evolution.
7. Outline the three basic stages of embryonic development that occurs in all animals.
8. What is involved in molecular systematics?

# 25 Comparative Genomics

**Genomics** is the study of whole sets of genes and their interactions, e.g. it includes how particular genes are expressed, how they are regulated, their interactions with other genes and how gene expression affects growth, development and evolution.

Comparative studies of DNA of both genomic and mitochondrial DNA and proteins provides additional evidence for evolution. This information can then be used to construct phylogenetic trees to show evolutionary relationships.

## Bacterial enzymes

The discovery of bacterial enzymes that can cut foreign DNA (**restriction enzymes**) opened up new research areas that enabled the sequencing of genomes.

## DNA hybridisation

DNA hybridisation was one of the first techniques used to study DNA and RNA molecules. It uses the complementary base pair properties to compare nucleotide sequences from different sources. The process is often called **nucleic acid hybridisation** using base pairing between a gene and a complementary sequence on another nucleic acid molecule.

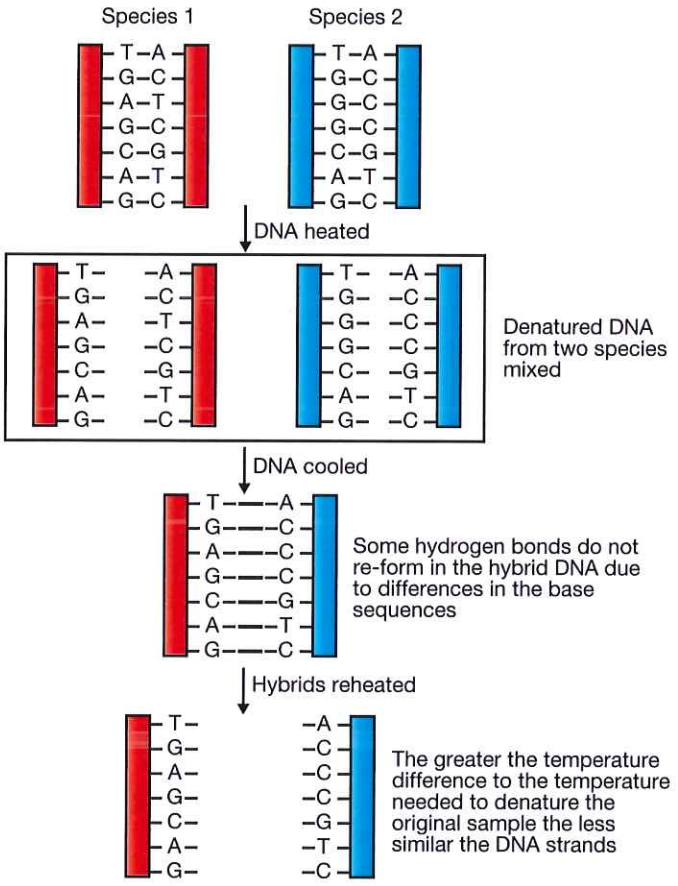



Figure 25.1 DNA hybridisation.

When DNA is heated the hydrogen bonds holding the two strands of the double helix together are broken in **denaturation**. The two strands separate. Restriction enzymes are used to cut the strands into smaller sections. Comparable segments from two different species are mixed and when they cool the hydrogen bonds re-form to make a double helix with random collisions causing some hybrid double helix molecules to form. Similar regions will form strong bonds. The DNA is reheated and the temperature needed to cause separation indicates the degree of similarity between the two species. The higher the temperature the greater the number of hydrogen bonds and the more genetically similar the species.

DNA hybridisation has been used to work out the relationships between many species. Differences in mitochondrial DNA have been used to work out the evolutionary tree of humans and other primates and when different groups shared a common ancestor.

## QUESTIONS

1. Define genomics.
2. What property of DNA is the basis for DNA hybridisation?
3. Outline what happen in the denaturation of DNA.
4. Describe the process of DNA hybridisation.
5. What is a probe?
6. How can probes be labelled?
7. The diagram shows stages in DNA hybridisation.

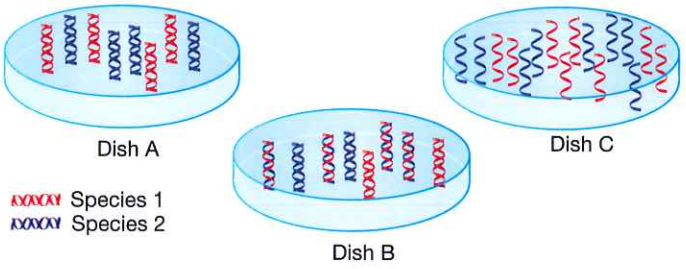



Figure 25.2 Stages of DNA hybridisation.

Place these dishes in the correct order and explain what is happening at each stage.

8. Give an example of the use of DNA hybridisation.
9. Distinguish between genetic screening and genetic testing.
10. What do you call a labelled single stranded nucleic acid molecule that tags a particular nucleotide sequence in the nucleic acid sample?
    (A) Primer
    (B) Probe
    (C) Operon
    (D) DNA hybrid

# 26 Molecular Homology

**Homology** is the study of similarities in characteristics due to shared ancestry.

**Molecular homology** is the study of similarities in molecules such as DNA and proteins due to shared ancestry.

When comparing amino acid sequences of two species it is important to correctly align the nucleotide sequences. Closely related species will have few differences and a recent common ancestor. If there has been a long time period since the last common ancestor point mutations such as deletions, base-pair substitutions and insertions or block mutations will accumulate leading to a large number of differences in the nucleotide sequence. These mutations are usually due to rare and spontaneous changes.

| Organism | Human | Monkey | Dog | Horse | Pig | Rabbit | Kangaroo | Duck | Chicken | Turtle | Tuna | Moth | *Neurospora* | *Candida* |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Human | _ | 1 | 11 | 17 | 0 | 12 | 12 | 17 | 18 | 19 | 21 | 36 | 63 | 66 |
| Monkey | | _ | 12 | 16 | 12 | 11 | 13 | 16 | 17 | 18 | 32 | 35 | 62 | 65 |
| Dog | | | _ | 10 | 3 | 6 | 7 | 12 | 14 | 13 | 18 | 28 | 64 | 66 |
| Horse | | | | _ | 5 | 11 | 11 | 16 | 16 | 16 | 27 | 33 | 64 | 68 |
| Pig | | | | | _ | 6 | 7 | 13 | 13 | 13 | 17 | 31 | 64 | 67 |
| Rabbit | | | | | | _ | 7 | 10 | 11 | 11 | 26 | 29 | 62 | 67 |
| Kangaroo | | | | | | | _ | 14 | 15 | 14 | 27 | 31 | 66 | 68 |
| Duck | | | | | | | | _ | 3 | 7 | 26 | 29 | 61 | 66 |
| Chicken | | | | | | | | | _ | 8 | 26 | 31 | 61 | 66 |
| Turtle | | | | | | | | | | _ | 27 | 33 | 65 | 67 |
| Tuna | | | | | | | | | | | _ | 41 | 72 | 69 |
| Moth | | | | | | | | | | | | _ | 59 | 61 |
| *Neurospora* (bread mould) | | | | | | | | | | | | | _ | 61 |
| *Candida* (fungus) | | | | | | | | | | | | | | _ |

**Figure 26.1** Minimum number of nucleotide differences in cytochrome c molecules of different organisms. Cytochrome c is involved in the electron transport pathway in respiration.

## Universality of genetic code

The **genetic code** is made from different combinations of three bases (triplet code) from a choice of four nitrogenous bases. There are 64 possible codes ($4^3$) with about 20 amino acids to code which means some codes are redundant. The genetic code is shared by nearly all species and is universal from simple viruses and bacteria to complex animals, e.g. RNA codon CCG is translated as the amino acid proline in all organisms. The universality of the genetic code and protein structures shows the interrelatedness of living organisms. Humans, with the same standard genetic code, are part of the evolutionary tree and share genetic history with other organisms.

## Protein evidence

The composition of key proteins such as cytochrome c and haemoglobin are compared to determine the relationship of two species. Cytochrome c is found on the inner membrane of mitochondria and is part of the electron transfer chain in aerobic respiration. It is present in organisms that use aerobic respiration An analysis of cytochrome c shows the similarities and differences between species and has been used to determine a molecular clock to work out the history of evolutionary relationships.

The genetic data has been used to infer all present-day organisms must have had a common ancestor and systematics has been used to determine a tree of life. Since it is found in so many organisms it is useful for working out the major branches of the evolutionary tree, but it is not particularly suitable for finer divisions, e.g. chickens and turkeys have identical molecules for cytochrome c while ducks differ by one amino acid, or humans and chimpanzees have the identical molecule while rhesus monkeys differ by one amino acid.

## Immunological evidence

The production of antibodies against the introduction of specific blood proteins can be used to work out the relationship between species. For example, if human serum is introduced into a rabbit the rabbit will produce antibodies against the human (anti-human antibodies). If the anti-human antibodies are tested against other species the degree of precipitation shows the degree of closeness to humans, e.g. the anti-human antibodies have 100% precipitation with human serum and 8% precipitation with pig serum.

## QUESTIONS

1. Define homology.
2. What is molecular homology?
3. What is meant by the universality of the genetic code?
4. Describe a protein that is used to determine evolutionary relationships.
5. How is immunological evidence used to determine evolutionary relationships?
6. The table shows the degree of precipitation when 'anti-human' antibodies was added to serum from each animal.

| Animal | Precipitation (%) |
|---|---|
| Pig | 8 |
| Lemur | 37 |
| Spider monkey | 58 |
| Baboon | 75 |
| Gorilla | 97 |
| Chimpanzee | 97 |

Discuss the usefulness of using immunological evidence such as this to determine evolutionary trees.

7. Which of the following would *not* be studied in molecular homology?
   (A) Structure of haemoglobin.
   (B) Composition of cytochrome c.
   (C) Nucleotide sequence.
   (D) Pentadactyl limb structure

# 27 Ancestral Gene Homologues

**Homologue genes** or homeotic genes are DNA sequences that are similar in different organisms. Molecular phylogenetics is a new field of study that uses DNA sequencing technology to study evolutionary relationships. The presence of similar DNA sequences suggests a common ancestor – the greater the similarity in the molecules, the more recent in time the common ancestor.

One of the first genes to be investigated in molecular phylogenetics as a molecular marker was the 'ssu rDNA' gene. The similarities and differences in this gene in different species were used to construct an evolutionary tree of the eucaryotes showing the divergence of protists, fungi, plants and animals. The development of better analysis techniques showed that there were problems with the initial evolutionary tree. The problem was called long branch attraction and is caused by incorrectly relating genes that evolve very quickly. When working out specific relationships for new organisms, the ssu rDNA gene is often used as one of the initial studies but it cannot provide complete answers.

The **homeobox genes** controlling embryo development are another example of homologue genes found in different species. The gene cascades for limb development are similar in chickens, humans and fruit flies.

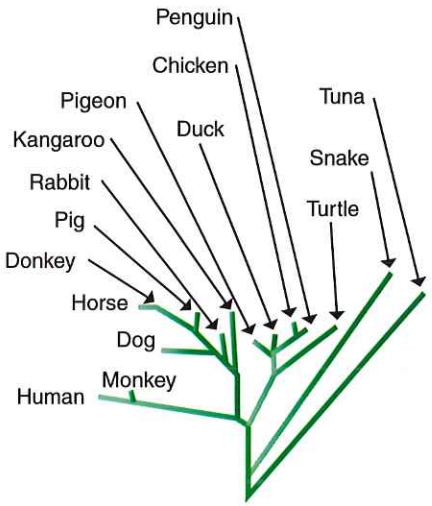



Figure 27.1 Vertebrate evolutionary tree based on cytochrome c sequences.

**Cytochrome c** is an essential protein that performs a key step in cellular respiration. It has changed little in millions of years and is found in every aerobic eukaryote. The genes producing cytochrome c are homologues. Human cytochrome c consists of 104 amino acids and 37 of these amino acids have been found at equivalent positions in every sequenced cytochrome c. It is estimated that the primitive common ancestor that had the precursor for cytochrome c existed over two billion years ago and thus comparing the DNA sequences for cytochrome c is useful for comparing the lower animal classes. There is no difference in the DNA sequences for cytochrome c for humans and chimpanzees and thus it cannot be used for working out the evolutionary relationship between these species.

To compare DNA sequences of homologue genes, the first step requires an alignment of the two sequences. There are several procedures used to get the best alignment for the two sequences. Computer programs are essential and gaps can be introduced to maximise the number of positions that have the same amino acid. The gaps correspond to insertions or deletions that occurred during evolution. The collected data is then used to construct a phylogenetic tree.

Table 27.1 Comparison of number of different amino acids in cytochrome c.

| | Human | Pig | Dog | Turkey | Tuna | Moth |
|---|---|---|---|---|---|---|
| Human | 0 | 10 | 11 | 13 | 21 | 31 |
| Pig | | 0 | 3 | 9 | 17 | 27 |
| Dog | | | 0 | 10 | 18 | 25 |
| Turkey | | | | 0 | 17 | 28 |
| Tuna | | | | | 0 | 32 |
| Moth | | | | | | 0 |

## QUESTIONS

1. What is a homologue gene?
2. Outline the area of study of molecular phylogenetics.
3. How is molecular phylogenetics used to show relationships between species?
4. Identify a gene that is a homologue and name the groups it is able to link in an evolutionary tree.
5. What is cytochrome c?
6. Name the major groups of organisms that have cytochrome c.
7. Explain why the genes for cytochrome c are homologues.
8. Explain why the gene homologues of cytochrome c are useful for working out evolutionary relationships for the lower animal classes.
9. Homologue genes have been used to create 'molecular clocks' as mutations occur at a certain rate in most genes. Outline why homologue genes are suitable to construct molecular clocks.
10. The gene cascade for skeletal and neurological development in limbs is similar in humans, chickens, rodents, nematodes and molluscs. What does this indicate about the gene cascade?
    (A) It is an isologue.
    (B) It is a homologue.
    (C) It is transgenic.
    (D) It is a translocation.

# 28 Molecular Clocks

A **molecular clock** (or sometimes called an evolutionary clock) is an evolutionary timing method based on the observation that the code for some genes and particular regions of the genome seem to change at constant rates. If the rate of nucleotide change is constant then the number of changes is proportional to the time that has elapsed since the species branched from a common ancestor.

Molecular clocks need to be calibrated against other independent evidence, e.g. using the fossil record.

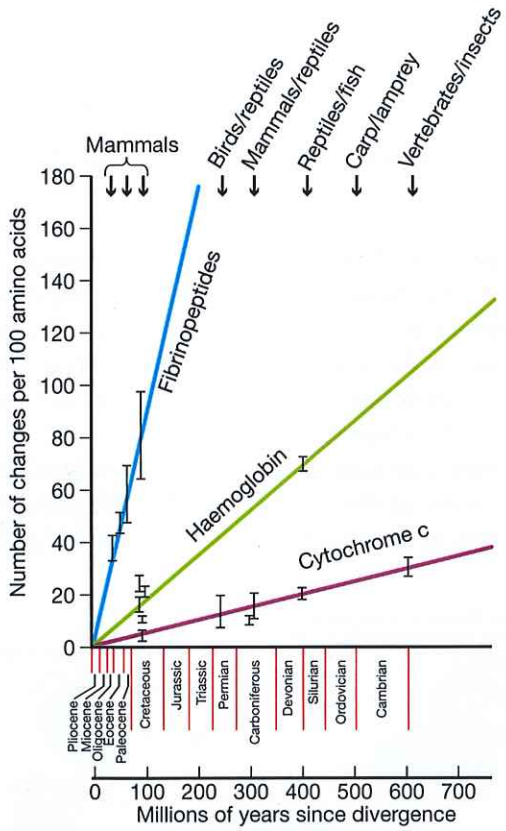



**Figure 28.1** Molecular clocks.

Molecular clocks have been constructed using mtDNA, haemoglobin, cytochrome c, calmodulin, myoglobin and immunoglobulins. Calmodulin is a protein found in eukaryotes that controls calcium levels in cells and provides a molecular clock that dates back 1500 million years to the first eukaryotic organisms. Human and chicken calmodulins have identical amino acid sequences. This means that calmodulin is not particularly useful for determining primate evolution, rather it allows an estimate of 1000 million years for the divergence of protozoans, plants and multicellular animals.

The use of molecular clocks can be limited if very large time scales are involved due to a large number of differences and the problem of saturation. There are also problems with very short time scales as differences could be alternate alleles in polymorphism and not different species.

## Immunological evidence

There are five major types of immunoglobulins (antibodies) found in mammals – IgM, IgA, IgG, IgE and IgD. IgA is found in birds and mammals in mucous secretions, e.g. to protect epithelial surfaces in the respiratory tract and urogenital tract. In 2001 researcher Kathy Belov at the Australian Museum found evidence for a massive immunoglobulin evolution before the divergence of the monotremes (prototheria), marsupials (metatherian) and placental mammals (eutheria). Comparisons of the structure of immunoglobulins of lower vertebrates and vertebrate groups besides mammals (e.g. birds and reptiles) show the relatedness of different groups and can be used to determine divergence from a common ancestor. For example, IgM, IgA, IgG and IgE in monotremes are structurally identical to other mammals but are different to birds and reptiles.

A comparison of the heavy chain constant region of IgA between birds, therians and the echidna indicates that monotremes diverged prior to the separation of marsupials and eutherians and contradicts mitochondrial data. And the immunoglobulin profile of short-beaked echidnas differs to that of platypuses.

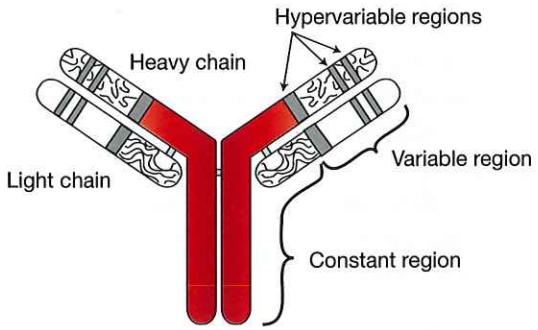



**Figure 28.2** Structure of immunoglobulin and heavy chain of constant region.

IgM has been used to estimate the times of mammalian divergences. Comparisons of the heavy chain constant region of IgM of the short-beaked echidna with birds and eutherians suggests monotremes and eutherians separated around 170 million years ago (mya) and marsupials and eutherians separated around 130 mya. Australian and American marsupials separated around 65 mya and the two monotremes (platypus and echidna) separated 22 mya.

## QUESTIONS

1. What is a molecular clock?
2. Describe a use of molecular clocks and outline the scientific principle behind this use.
3. Identify some organic chemicals that have been used to construct molecular clocks.
4. How is the timing of molecular clocks calculated?
5. Explain how molecular clocks have helped increase understanding of the divergence of the mammals.

# 29 Evolution of Genes

The biotechnological revolution caused by the advance of genetic engineering techniques has meant many scientists are now investigating gene similarities between different species to show evolutionary relationships. An analysis of homologues, homeotic or *Hox* genes of different species shows similar DNA sequences suggesting these genes evolved in a common ancestor. Mutations in homeotic genes can cause significant changes in the development of an organism. An analysis of differences in these gene cascades provides information about divergence and the evolution of new body structures which can be used to construct evolutionary trees.

Previously, evolutionary trees had been constructed using anatomy, physiology and embryology, with time lines based on fossil evidence to determine relationships between species. If there is conflict between fossil evidence and molecular analysis, further research is needed as each technique has both advantages and disadvantages and more data is needed before final decisions can be made about the evolutionary tree.

## Phylogenetic trees

A **phylogenetic tree** is a graphical representation of an evolution to show how different groups diverged over time. If the end of each branch reaches the top, it is an extant species, while a branch ending lower down indicates extinction. Branch lengths can indicate time frames. Usually the collected data can be interpreted in several ways which means that several evolutionary trees can be drawn.

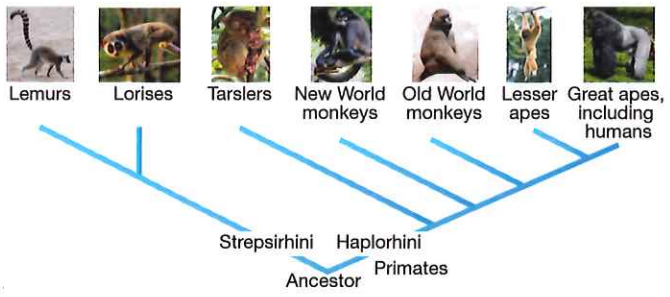



Figure 29.1 Phylogenetic tree for primates using cross-hybridisation techniques.

The 'serial endosymbiotic' theory proposes that mitochondria and chloroplasts were once free-living bacteria but were captured by primitive cells and turned into organelles for aerobic respiration and photosynthesis respectively. This theory was strongly supported by the molecular phylogeny of the mitochondrial and chloroplast genes. With further study the evolutionary relationships will be more clearly defined, e.g. analysis of chloroplast genes and the nuclear DNA of red algae, green algae, and plant groups to construct an accurate evolutionary tree for multicellular photosynthetic organisms.

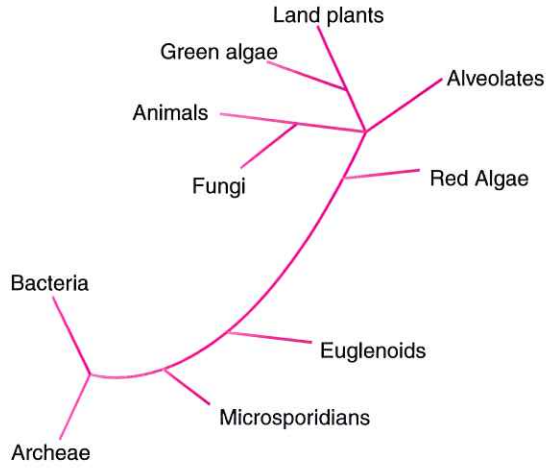



Figure 29.2 Evolutionary tree based on ssu rDNA gene.

Haemoglobin is an iron protein able to transport oxygen. X-ray crystallography has been used to study haemoglobins from several vertebrates, e.g. horse, pig, bar-headed goose and three species of fish. Fish haemoglobins have more differences than the air-breathing vertebrates with insertions and deletions. Mammals have different haemoglobin at different life stages, e.g. embryo, foetus and adult. Myoglobin is found in the muscles of many vertebrates, including humans and is often used for oxygen storage rather than transportation. Thus a study of the evolution of the globin genes provides information about the evolution of the vertebrates and other animal groups.

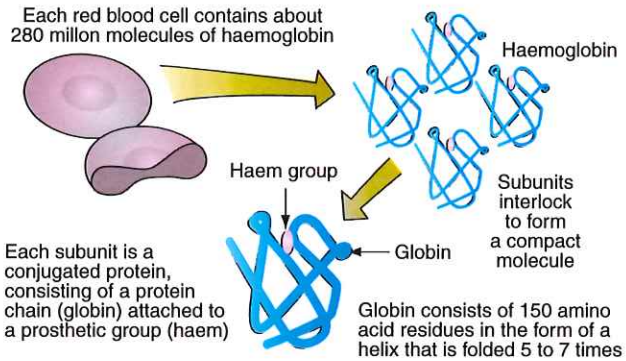



Figure 29.3 Haemoglobin.

Techniques used to compare the total genome of two species include DNA-DNA hybridisation and chromosome painting. The DNA sequences are compared and differences in genes show evolutionary relationships. Some of the most informative data has used rDNA genes and genes on mitochondrial DNA (mtDNA). As the amount of data about the genomes of different species is collected, there has been a corresponding need to improve computer capacity to store and analyse this data. Knowledge of the total genome and an analysis of the genes 'switched on' in particular types of cells provides data so that particular genes can be linked with particular activities.

## Problems in determining gene evolution

Several areas cause problems when determining gene evolution and working out phylogenetic trees. One of the biggest problems in determining the relationship of the different groups in the animal kingdom is the short time frame in which they appeared. All major groups appeared before and during the Cambrian period thus the base of the evolutionary tree is very clustered requiring more evidence to help clarify relationships. This also means that many of the branches on the tree are very long and different branches have different mutation rates. Thus branches based on quickly evolving molecules could appear superficially long. Back mutations can also be a source of error in tree construction. A back mutation occurs when there has been a change in an amino acid over time, but a later second mutation returns the code to its original state. Thus no change is recorded, but it should have counted as two changes.

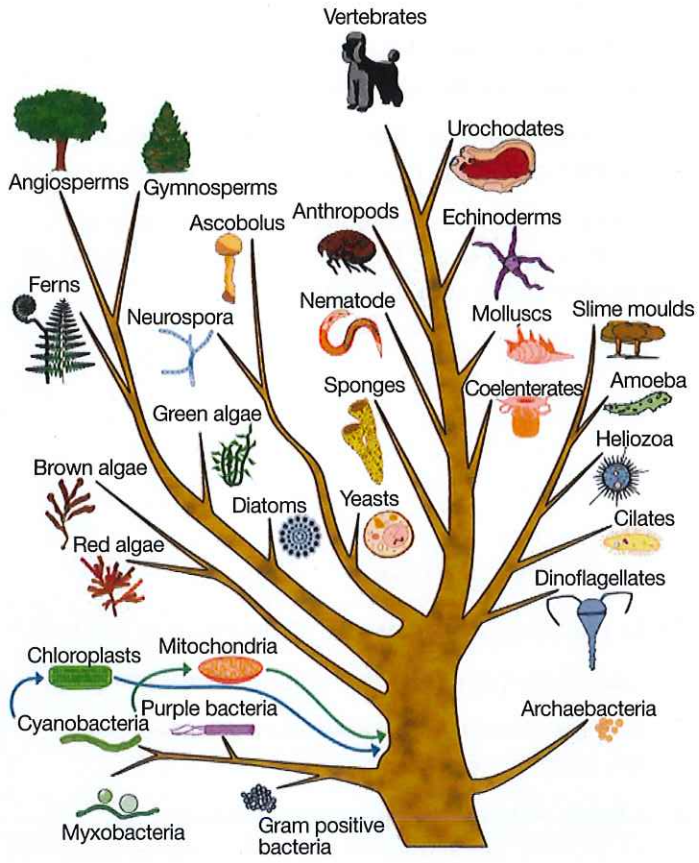



**Figure 29.4** Phylogenetic tree for all living things.

## QUESTIONS

1. Discuss how research into evolutionary relationships has changed in recent years.
2. How were evolutionary trees constructed before the biotechnological revolution?
3. Describe what happens if there is conflict in data between fossil evidence and molecular analysis when constructing evolutionary trees.
4. What is meant by 'extant species'?
5. Discuss why several possible evolutionary trees can be drawn for a group of organisms.
6. Outline the serial endosymbiotic theory.
7. Identify some techniques used to compare the total genome of two different species.
8. Discuss how the impact of the collection of data about the DNA sequences of different species has impacted on the development of technologies.
9. Explain how the fossil record of the Cambrian period has caused problems in the construction of phylogenetic trees.
10. Define back mutation and explain how back mutations cause problems when determining the evolution of genes.
11. Outline how particular genes are linked with particular actions.
12. DNA sequence analysis of different bacteria has shown that genes can pass from one group to another in a 'horizontal' transfer. This means that the genes in different species of bacteria may not necessarily descend from a common ancestor. For example, it appears that the bacteria *Mycobacterium tuberculosis* has picked up eight genes from its human host. Discuss how this 'horizontal transfer' of genes could cause problems with working out the evolution of genes.
13. Convergent evolution can occur not only on a macroscopic level but also on a microscopic level, e.g. a comparison of the amino acid sequences of cows and langur monkeys show that both synthesise a lysozyme that share the same activity but have evolved from different ancestral molecules. Define convergent evolution and use examples to show how both macroscopic and microscopic levels of convergent evolution need to be considered when looking at the evolution of genes.
14. It is very hard for school students to collect first-hand data to show the evolution of genes. Explain how you could gather secondary evidence to process, analyse and draw conclusions about evolutionary relationships from the analysis of genes.
15. When collecting secondary data, how can you ensure the reliability of the source?
16. Study Figure 29.4, the phylogenetic tree for all living things.
    From this tree which group of organisms is most closely related to the vertebrates?
    (A) Gymnosperms.
    (B) Angiosperms.
    (C) Arthropods.
    (D) Urochordates.

# 30 Cladistics

**Cladistics** is a type of classification that places species into clades based on descent from a single common ancestor. A **clade** is a group of species that belong to a single branch of the evolutionary tree. It includes the single common ancestor and all descendants. It is usually constructed by considering a single homologous characteristic, e.g. base sequences of a gene or the corresponding amino acid sequences of a protein. Using genetic information forms a **molecular clade**.

A **cladogram** is a diagram used in cladistics to show ancestral relationships between species. It is an hypothesis of branching sequences and shows the most probable sequence of divergence in clades. A **node** on a cladogram is a branch point indicating the evolution of **shared derived characteristics**. Derived characters are the traits that tie the members of a clade together, e.g. the possession of feathers can be used to form a clade of birds. A clade using feathers is a **morphological clade**.

A **shared ancestral character** is a feature found in an ancestor of the taxon. For example, all primates have a backbone (a shared ancestral characteristic that was present in the ancestral vertebrate) and all primates have an opposable toe/thumb (not a feature found in their ancestors).

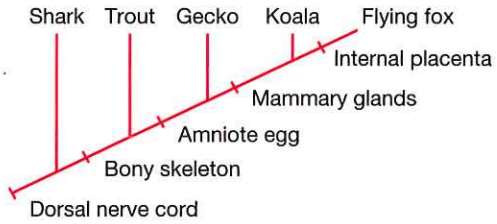



Figure 30.1 Cladogram.

Unlike phylogenetic trees, a node in a cladogram can be placed at any point and there are no varying lengths of the branches to show the timing of divergence. Many cladograms do not follow the traditional taxa used for classification and can clash with conventional terminology. For example, birds, crocodiles and lizards can be placed into a clade from their most recent common ancestor which puts the crocodiles and birds in the same group and does not recognise the class Reptilia.

## Groupings for cladograms

Clades can be nested within clades.

A clade is a **monophyletic group** as it gives the ancestor and all of its descendants.

In a **paraphyletic group** the common ancestor is given and only some of the descendants given. In a **polyphyletic group** all of the descendants are given but there is an unknown ancestor(s).

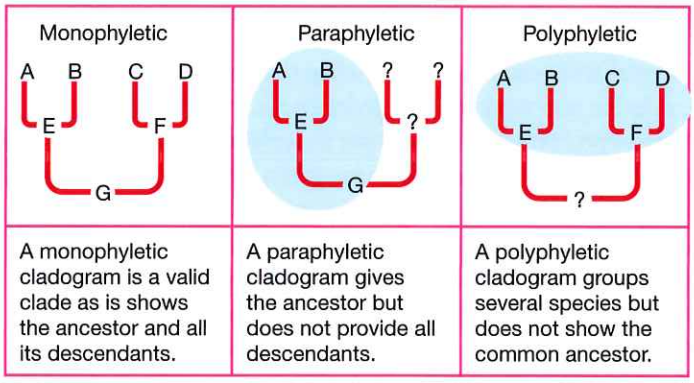



Figure 30.2 Cladogram.

## Constructing cladograms

To construct a cladogram you need to follow these rules:

- The original lineage should split into two when new species arise.
- All species in the clade can be traced back to the ancestral species, which no longer exists.
- All species derived from the common ancestor must be included in the taxon (must be monophyletic).

The cladogram can be constructed as a sloping linear line or branching lines. The two cladograms in the figure below give identical information.

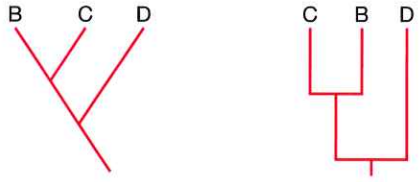



Figure 30.3 Cladogram types.

In the process of constructing a cladogram you need to:

1. List all the organisms that will be in the cladogram.
2. List the characteristics to be compared and tabulate for each organism.
3. Decide which trait is ancestral (primitive – plesiomorphic) and which trait is derived (apomorphic)
4. Convert the information into a cladogram so that nodes show a common ancestor and branches show divergence.
5. The resulting cladogram needs to show each branch as an evolutionary step related to the loss/gain/ modification of a feature.

If there are a large number of structural differences between the organisms you wish to put into a cladogram, e.g. a cladogram including a snail, a mushroom and a frog there are many possible features that could be used and decisions need to be made about what weight is given to each feature.

(a)

| Feature | Chordates | | | | |
|---|---|---|---|---|---|
| | Tunicate | Shark | Toad | Goanna | Human |
| Notochord | ✓ | ✓ | ✓ | ✓ | ✓ |
| Hinged jaw | | ✓ | ✓ | ✓ | ✓ |
| Legs | | | ✓ | ✓ | ✓ |
| Amniotic egg | | | | ✓ | ✓ |
| Mammary glands | | | | | ✓ |

(b)

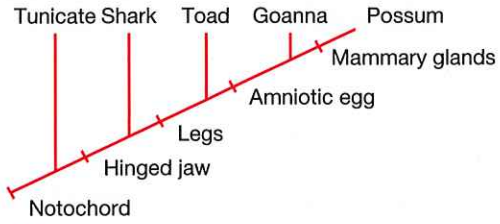



**Figure 30.4** (a) Data for a cladogram construction. (b) The resulting cladogram.

## Scaled cladograms

A scaled cladogram has a time axis so that each node indicates the approximate date of divergence of the two groups.

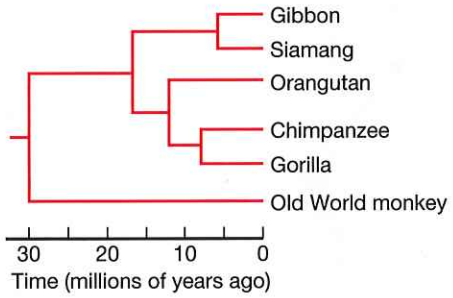



**Figure 30.5** Scaled cladogram.

## Molecular cladograms

Molecular cladograms using the base sequence of a gene are very useful when constructing cladograms as the data is objectively quantifiable and the data can be used to compare very diverse organisms. For example, the protein cytochrome c has been used to construct clades of highly diverse organisms. Cytochrome c is a carrier of the electron transport chain and is found in mitochondria of eukaryotes. This means it can be used to construct a clade for any eukaryotes.

## QUESTIONS

1. What is cladistics?
2. Define a clade.
3. What is a cladogram?
4. What is a node on a cladogram?
5. Distinguish between a phylogenetic tree and a cladogram.
6. Distinguish between a shared derived character and a shared ancestral character.
7. Distinguish between a molecular clade and a morphological clade.
8. Explain why a clade is a monophyletic group.
9. Distinguish between a monophyletic group, a paraphyletic group and a polyphyletic group.
10. Outline the rules that need to be followed to construct a cladogram.
11. What is a scaled cladogram?
12. Identify some benefits of using molecular cladograms.
13. Use the following information to construct a cladogram.

| Feature | Chordates | | | | | |
|---|---|---|---|---|---|---|
| | Lancelet | Lamprey | Eel | Frog | Snake | Human |
| Notochord | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Backbone | | ✓ | ✓ | ✓ | ✓ | ✓ |
| Hinged jaw | | | ✓ | ✓ | ✓ | ✓ |
| Lung | | | | ✓ | ✓ | ✓ |
| Amniotic egg | | | | | ✓ | ✓ |
| Hair | | | | | | ✓ |

14. The table gives a comparison of a certain protein's structure found in four animals.

| Organism | Number of different amino acids in the cytochrome c relative to humans |
|---|---|
| Human | 0 |
| Monkey | 1 |
| Turtle | 15 |
| Moth | 31 |

What is shown by this data?

(A) Humans evolved from moths.
(B) Turtles and moths are more closely linked on the evolutionary tree than turtles and monkeys.
(C) Mammals evolved from reptiles.
(D) Monkeys and humans are closely related.

15. According to studies of DNA sequencing, there is a 97.6% similarity between humans and chimpanzees and a 91.1% similarity between humans and rhesus monkeys. What can be concluded from this information?

(A) 0.5% of the human gene pool evolved recently.
(B) In geological terms humans and chimpanzees had a fairly recent common ancestor.
(C) Humans, chimpanzees and rhesus monkeys need to be classified in the same family.
(D) Evolution favoured upright stance.

# 31 Homologous and Analogous Traits

**Analogous structures** perform the same or similar function often by a similar mechanism but have evolved separately.

**Homologous structures** are similar structural or biochemical features due to a common ancestor.

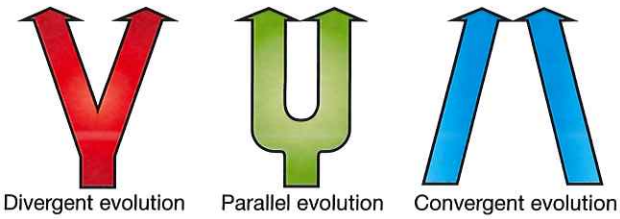



Figure 31.1 Patterns of evolution.

When reconstructing phylogenies it is very important to distinguish between homology and analogy. The wing of the bird and wing of the bat are homologous as they are both descended from the forelimb of vertebrates due to adaptive radiation from a common ancestor that had a pentadactyl limb. The two wings are also an example of analogy as they independently evolved as a means of flying.

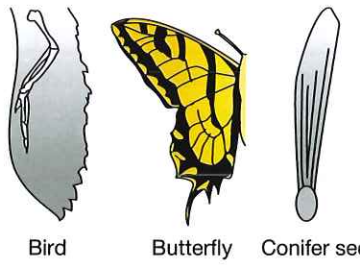



Figure 31.2 Analogous traits.

## Examples of analogous traits

Analogous traits are due to convergent evolution with a similar function and superficial similar appearance:

- Wings of different groups.
- Streamlined body shape of fast aquatic swimmers.
- Elongated body, large front claws for burrowing, e.g. moles.

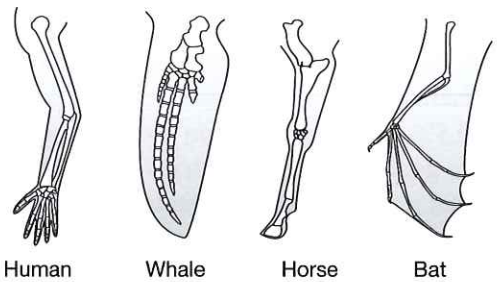



Figure 31.3 Homologous traits.

## Examples of homologous traits

Homologous traits are due to divergent evolution with a common origin for the structure but not necessarily a common function:

- Pentadactyl limb.
- DNA sequences.
- Vertebrate heart.

## QUESTIONS

1. Distinguish between analogous structures and homologous structures.
2. Construct a table to compare analogous and homologous structures for structure, function and ancestry.
3. The diagram shows two types of vertebral columns.

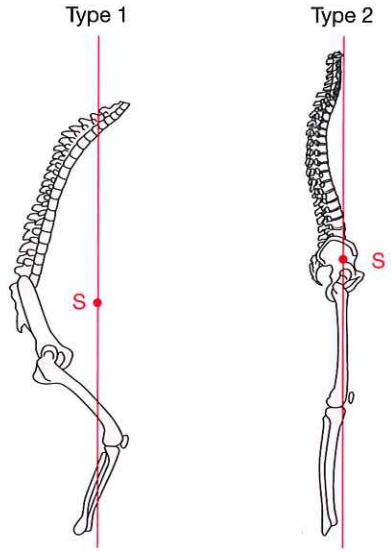



Figure 31.4 Two types of vertebral columns.

The vertebral columns belong to a human and an ape. Identify each type and discuss why these are homologous structures.

4. All mammals have seven cervical vertebrae. In a giraffe these vertebrae can have a combined length of over two metres, while in a human these vertebrae extend a much shorter distance from the skull to the thoracic (rib) cage. Explain how this relationship would have evolved.
5. Which of the following are analogous structures?
   (A) Wing of parrot and wing of eagle.
   (B) Giraffe neck vertebrae and human neck vertebrae.
   (C) Shape of dolphin and shape of shark.
   (D) Gorilla hand shape and human hand shape.
6. What is the best definition of the pentadactyl limb?
   (A) A five-digit limb.
   (B) A leg with five toes.
   (C) An arm with five fingers.
   (D) An arm or leg found on land animals.

# 32 Activity – Vertebrate Forelimbs

Some Human Biology students wished to analyse and compare the structure of a range of vertebrate forelimbs as part of their study of comparative anatomy. Comparative anatomy is the study of the structures, e.g. organs or bones, found in different groups.

They decided to study the pentadactyl limb. The **pentadactyl limb** is the five-digit limb found in some amphibians and reptiles, birds and mammals. The pentadactyl limb is an example of an homologous structure. An homologous structure is a similar structure found in different organisms due to common ancestry but used for different purposes.

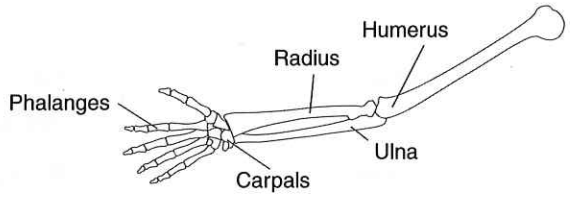



Figure 32.1 Parts of the human forelimb (arm).

The students collected diagrams and/or prepared specimens of vertebrate forelimbs and drew an outline of the limb of each animal.

The students decided that there are many underlying similarities between the forelimbs of these groups, showing that these groups had a common ancestor, and since that time natural selection has led to different modifications of the limb.

## QUESTIONS

1. (a) What is comparative anatomy?
   (b) Choose one feature studied in comparative anatomy and explain how it provides evidence for the theory of evolution.
2. The diagram shows the forelimb of several different animals.

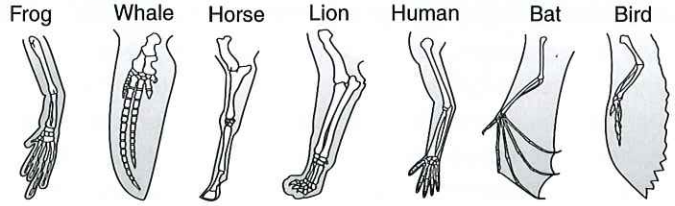



Figure 32.2 Forelimb of several different animals.

   (a) What is an homologous structure?
   (b) Describe how each of these animals has adapted the structure for its own purpose.
   (c) Explain how homologous structures are used to support the theory of evolution.

3. The diagram shows the skeleton of a horse and the skeleton of a whale.

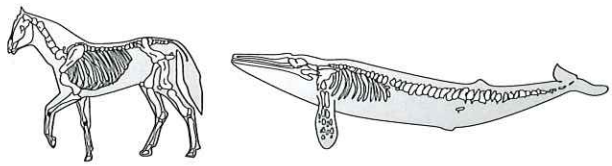


Figure 32.3 Skeletons of a horse and whale.

What does a comparison of their anatomy show?
(A) Differences can be explained by convergent evolution from a common ancestor.
(B) Differences can be explained by inheritance of acquired characteristics.
(C) Similarities can be explained by adaptive radiation from a common ancestor.
(D) Similarities can be explained by convergent evolution.

4. The diagram shows the forefeet of four different ancestors of the modern horse, *Equus*.

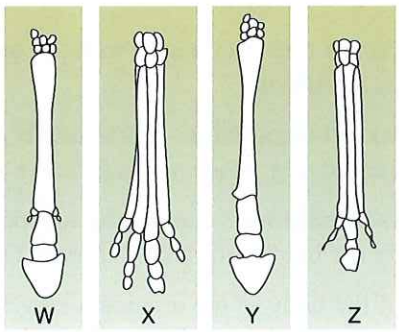



Figure 32.4 Forefeet of four ancestors of modern horse.

What is the correct order in which they occurred?
(A) Y, W, Z, X
(B) W, Y, X, Z
(C) X, Z, W, Y
(D) X, Z, Y, W

5. The diagram shows the forelimbs of four different vertebrates.

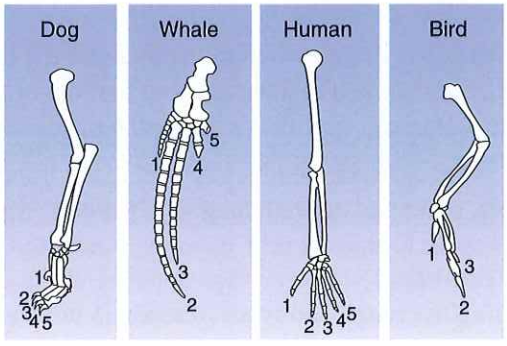



Figure 32.5 Forelimb of four different vertebrates.

An analysis of these forelimbs shows that they:
(A) Evolved from a common ancestor.
(B) Adapted to similar environments.
(C) All belong to the same class.
(D) Use the pentadactyl limb for the same function.

# 33 Fossils

A **fossil** is anything that indicates the existence of past life. The word comes from the Latin 'fossilis' which means 'something dug out of the ground'. A fossil can be part of or a whole organism or a trace left behind. The study of fossils is called **palaeontology** and fossils provide us with information about the history of life of Earth.

## Using fossils

The fossil record is incomplete and cannot represent the entire biodiversity of a time or a location due to many factors.

- Remains need to become fossilised.
- The fossil needs to be preserved over a long time and not destroyed in the rock cycle.
- The fossil needs to be found.

## Types of fossilisation

Fossilisation is a rare occurrence with the organism dying in particular situations.

**Fully preserved organisms** – occasionally whole bodies can be preserved, e.g. in ice, mummified or in amber.

**Hard parts remaining** – usually the soft parts decay, leaving only the hard parts such as bones, teeth and shells.

**Moulds** – if the body of the organism leaves an impression, the cavity it made may form a mould. An **external mould** is an impression of the outside of the body while an **internal mould** is an impression of the inside, e.g. of a shell made by mud or sand.

**Cast** – if the mould cavity is filled with material, a cast is formed. Casts can be formed from dissolved minerals, e.g. quartz or calcite and form a replica. Casts can also be trace fossils, e.g. formed by footprints.

**Petrification** – occurs when minerals enter the tissues of a dead organism and replace them so that the organism turns to stone, e.g. petrified wood and opal shells.

**Trace fossils** provide indirect evidence, e.g. footprint, burrows, trails and animal dung, e.g. Kalbarri trace fossils and dinosaur footprints near Broome. Fossilised dung is called **coprolites** with the shape and size of the coprolites indicating the range and types of animals in an area.

## Index fossils

If the organism lives for a short time and is found over a wide area it can be used as an **index fossil**, e.g. trilobites in the Palaeozoic era. Index fossils are used to correlate different rock layers so that the age of each layer can be determined.

## Geological time scale

Scientists have used the appearance and disappearance of different fossil species to divide geological time into eras, periods and epochs. The sudden appearance of many organisms with shells around 540 million years ago was used by scientists to mark the beginning of the Palaeozoic (ancient life) era. The mass extinction of many invertebrates marks the end of the Palaeozoic era and the beginning of the Mesozoic era, just as the mass extinction of the dinosaurs marks the end of the Mesozoic era and the beginning of the Cainozoic era. The eras were named as 'ancient life', 'middle life' and 'recent life'.

Table 33.1 Geological time scale.

| Era | Period | Epoch | Time began |
|---|---|---|---|
| Cainozoic | Quaternary | Holocene | 11 000 ya |
| | | Pleistocene | 1.75 mya |
| | Neogene | Pliocene | 5.3 mya |
| | | Miocene | 23.5 mya |
| | Palaeogene | Oligocene | 33.7 mya |
| | | Eocene | 56 mya |
| | | Palaeocene | 65 mya |
| Mesozoic | Cretaceous | | 145 mya |
| | Jurassic | | 201 mya |
| | Triassic | | 252 mya |
| Palaeozoic | Permian | | 298 mya |
| | Carboniferous | | 360 mya |
| | Devonian | | 420 mya |
| | Silurian | | 445 mya |
| | Ordovician | | 485 mya |
| | Cambrian | | 540 mya |
| Precambrian | | | 4.6 bya |

The eras are subdivided into periods. The early geologists who investigated the rocks from these different layers named the periods after the place or the type of rock found. For example, the first rocks of the Mesozoic era that were studied were limestones from the Jura Mountains and hence the age was given as 'Jurassic' and the chalk layers, called in French *Terrain Crétacé*, were called the 'Cretaceous'. Since the Mesozoic seemed to have three divisions, the earliest layer was then called the 'Triassic'.

The oldest fossils are bacteria and the fossils are around 3500 million years old. These bacteria formed stromatolites which are banded domes of sediment consisting of bacteria and cyanobacteria colonies.

## Stratigraphic correlation

Stratigraphic correlation looks at the relationship between the geological age of different rock layers in different areas. Layers with similar index fossils are the same geological age.

Stratigraphic correlation uses the **law of superposition** which states that the oldest layers are found at the bottom and the youngest layers are found at the top, unless there has been dynamic movement of the strata, e.g. by folding and/or faulting.

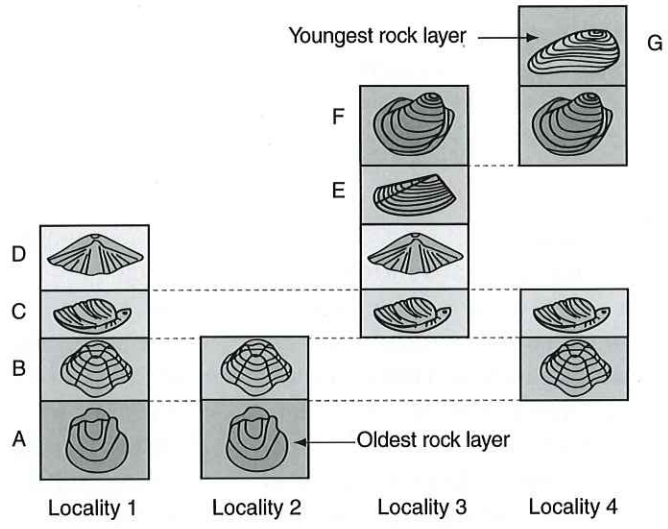



**Figure 33.1** Stratigraphic correlation using index fossils.

Comparing fossils with each other and with present-day species provides important evidence to show how organisms have changed throughout the history of the Earth. The data can also be used to reconstruct ancient climates, environments and geography.

The study of fossil evidence shows several trends in the origin of life on Earth.

- A trend from simple to more complex structures.
- A trend from aquatic to terrestrial life forms.
- A trend from asexual to sexual reproduction.
- An increase in the degree of protection or care of the embryo as it develops.

## QUESTIONS

1. What is a fossil?
2. Define palaeontology.
3. Explain why the fossil record is incomplete.
4. Describe how a mould and a cast are formed.
5. How can organisms leave a trace of their presence?
6. What is an index fossil?
7. Use an example to show how index fossils are used in geology.
8. How did scientists determine the geological time scale, dividing time into the Palaeozoic, Mesozoic and Cainozoic eras?
9. Use examples to show the difference in the naming of the geological eras and the geological periods.
10. How can palaeontological evidence provide information about life on Earth?
11. Use palaeontological and geological evidence to discuss when life is believed to have begun on Earth.
12. A study of fossils shows that life has changed throughout the history of the Earth. Identify the trends and patterns in life on Earth.
13. The diagram shows three stratigraphic columns from nearby areas.

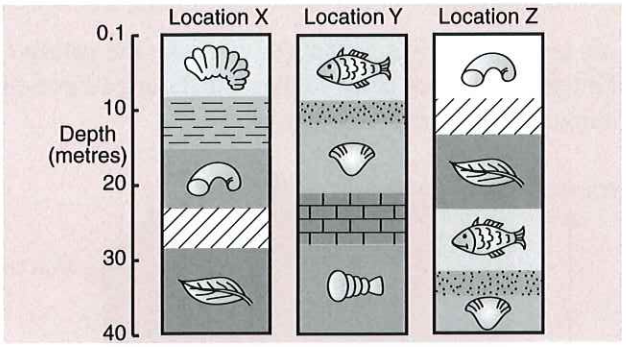



**Figure 33.2** Three stratigraphic columns.

(a) Which locality has the oldest stratum? Explain your answer.
(b) Explain how index fossils are used to date rock layers.

14. The photograph shows living stromatolites at Shark Bay, Western Australia.
Why is the study of stromatolites important in increasing our understanding of life on Earth?

**Figure 33.3** Stromatolites.

(A) Stromatolites are made by cyanobacteria and cyanobacteria fossils are the oldest fossils currently found.
(B) Stromatolites are made by coral and show the location of ancient shallow seas.
(C) Stromatolites are formed by masses of trilobite exoskeletons and show the mass extinction at the end of the Permian.
(D) Stromatolites are formed by primitive eukaryotes and give a date for the evolution of eukaryotes from prokaryotes.

15. Approximately how long has life existed on Earth?
(A) 3500 years (B) 4600 years
(C) 3.5 million years (D) 3500 million years

# 34 Absolute Age and Relative Age

Once fossils have been found it is necessary to date them. Dating techniques for sequences in the fossil record requires a combination of relative and absolute dating techniques.

## Relative dating

In 1669 **Nicolas Steno** proposed the basic principles of stratigraphy and suggested the **law of superposition** which states that the oldest layers are found at the bottom and the youngest layers are found at the top, unless there has been dynamic movement of the strata, e.g. folding and faulting.

This law is used by scientists to work out the **relative age** of a fossil or a rock layer so that it is described as being younger or older than another layer.

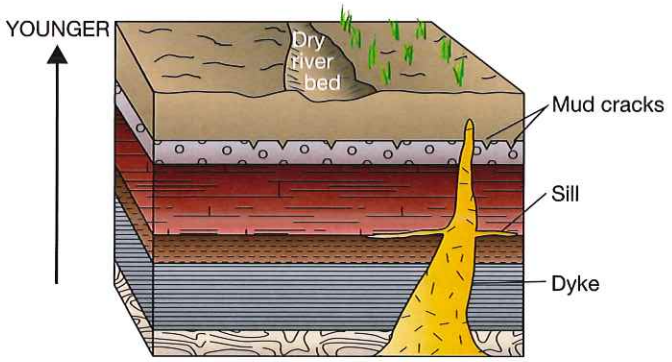



Figure 34.1 Relative age.

In 1815 **William Smith** (1769-1838), an English surveyor, published the first geological map of its kind showing the distribution of rock strata in England and Wales. He had noticed that there were different kinds of fossils in different layers and that certain fossils were always found in the same strata. From this work, the concept of an index fossil was introduced. An **index fossil** is widely distributed but only lived for a short time which means it clearly indicates the age of the rock layer.

By comparing fossils and working out relative ages the **geological time scale** was established. Current divisions of geological time are eon, era, period, epoch and age. For example the last three eras are the **Palaeozoic** (ancient life), the **Mesozoic** (middle life) and the **Cainozoic** (recent life) with the Cainozoic era divided into the Palaeogene, Neogene and Quaternary periods. The time frame for each division is not equal, the divisions were based on the appearance of specific fossils.

Some people use the supereon division **Precambrian** for the time before the massive abundance of fossils of several different types, e.g. trilobites, graptolites and brachiopods of the Cambrian period (540 million years ago).

## Fluorine absorption analysis

Fluorine absorption analysis can only be used on bone and is a method of relative dating as fluctuations in the amount of fluoride in ground water means that bones from the same area can be compared. When bone is in the ground, fluoride ions from the soil water replace some of the ions in the bone. The longer the bone is in the ground, the more fluorine present. Problems arise as there are many natural variables, e.g. rainfall, soil permeability and temperature so that bones being compared need to have been exposed to the same changing conditions over time. Different bones absorb fluorine at different rates which means that bones of similar density need to be compared. Differences can be due to age, e.g. adult bone/juvenile bone, part of the body, e.g. femur/jawbone or when comparing different animals.

## Absolute dating

Whereas relative dating gives the age in terms of younger than or older than, **absolute dating** gives the age in **years**. Radioactive isotopes (radioisotopes) are used to determine the age of rocks or fossils giving the actual age in years plus or minus the experimental error involved in the particular method used. Each radioactive isotope decays at a fixed, known rate called its **half-life**. The half-life is the time it takes for half the atoms to decay and if the half-life is plotted onto a graph, the decay forms an exponential curve. The atom that undergoes radioactive decay is known as the **parent nuclide** and the atom which is formed is known as the **daughter nuclide**. The rate of radioactive decay is *not* affected by changes in temperature, pressure or the chemical environment.

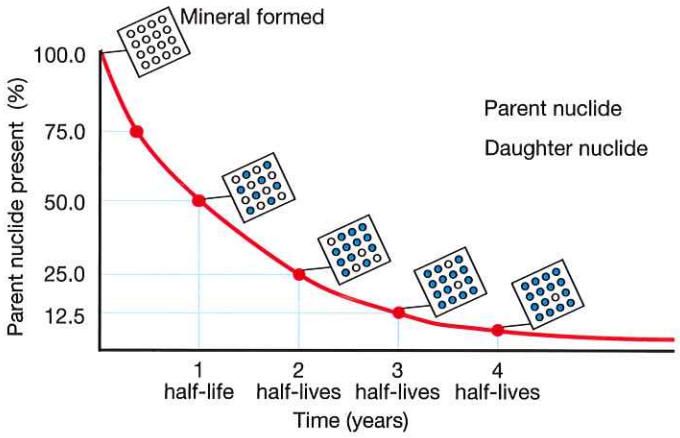



Figure 34.2 Radioactive decay.

Radiometric dating is only useful for a limited number of half-lives as the amount of radioactive isotope remaining after a certain time is too small to be measured. This means that different radioisotopes are used for different time frames and for different purposes. Geological absolute dating requires long half-lives while medical applications use radioisotopes with short half-lives.

Useful geological radioisotopes include carbon-14, potassium-40, rubidium-87 and samarium-147. There are two decay chains for uranium-lead decay with uranium-238 ($^{238}U$ to $^{206}Pb$) with a half-life of 4.47 billion years and uranium-235 ($^{235}U$ to $^{207}Pb$) with a half-life of 710 million years. Uranium is very useful for dating rocks as almost all igneous and metamorphic rocks contain enough uranium and lead for accurate dating and dating methods can be used on powdered whole rocks, mineral concentrates or single grains.

The oldest rocks of terrestrial origin dated so far come from Jack Hills, Western Australia dated at 4.4 billion years old using uranium isotopes in zircon minerals. The rock was extracted in 2001 and the results of the study were published in 2014.

**Table 34.1** Useful radioisotopes.

| Dating method | Half life | Used to date |
|---|---|---|
| Carbon-14 ($^{14}C$) | 5730 years | Bones, teeth, charcoal, linen |
| Rubidium-Strontium (Rb-Sr) ($^{87}Rb$) | $48.8 \times 10^9$ years | Minerals containing iron/ magnesium, e.g. mica, limestone |
| Potassium-argon (K-Ar) ($^{40}K$) | $1.26 \times 10^9$ years | Rocks contains potassium, e.g. feldspar, clay and igneous rocks |
| Samarium-neodymium (Sm-Nd) ($^{147}Sm$) | $1.06 \times 10^{11}$ years | Old igneous and metamorphic rocks, meteorites and cosmic fragments |
| Uranium-lead (U-Pb) ($^{238}U$) | $4.47 \times 10^9$ years | Igneous and metamorphic rocks |

Carbon is part of all the main building blocks of life and is thus present in all living organisms. As carbon-14 has a half-life of 5730 years the oldest date reliably measured by radiocarbon is around 50 000 years. This means that the absolute age of many fossils is calculated by the age of the rocks in which they are found. Rocks that form at a particular time, e.g. volcanic rocks such as tuff and basalt are very useful for dating fossils. Sedimentary rocks often contain fossils with the bodies of organisms being covered by sediment to become part of the forming rock layer.

## Fission tracks

**Fission tracking** occurs when uranium-238 decays (in fission process) and the released high speed atoms cause damage to nearby rocks and minerals (fission tracks). In a rock containing uranium-238 the number of fission tracks will increase over time at a rate that depends on the uranium content. The age of the rock can be calculated by measuring the uranium content and the density of the fission tracks.

## QUESTIONS

1. State the law of superposition.
2. The following diagram shows a stratigraphic column.

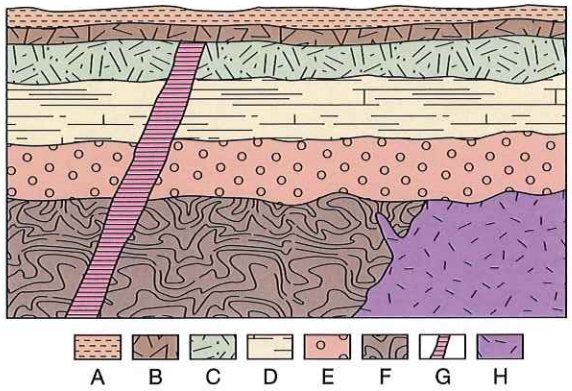



**Figure 34.3** Stratigraphic column.

   (a) Put the rock layers in order from oldest to youngest.
   (b) What type of dating is shown by your answer in part (a)?
3. Define an index fossil.
4. What is the Precambrian?
5. Identify the periods of the Cainozoic era.
6. Describe fluorine absorption dating.
7. List some of the variables that need to be considered with fluorine absorption dating.
8. Define absolute dating.
9. What is radiometric dating?
10. What is meant by the half-life of a radioisotope?
11. Distinguish between the parent nuclide and the daughter nuclide.
12. Identify some conditions that would influence fluorine absorption dating but not affect radiometric dating.
13. Explain why different radioisotopes are used for different purposes.
14. List some useful geological radioisotopes.
15. Explain why uranium is very useful for absolute dating.
16. Where are the oldest known rocks of terrestrial origin in the world?
17. Explain why carbon-14 has limited use in dating fossils.
18. Describe fission tracking.
19. A newspaper report claims a person has found ancient human bones dated 1.2 million years old using carbon-14 dating. What is the likely scientific response to this announcement?
   (A) Request samples to confirm data.
   (B) Change theories to accept new data.
   (C) Disbelief as humans did not exist 1.2 million years ago.
   (D) Rejection as carbon-14 cannot date that age.

# 35 Radiometric Dating

**Radiometric dating** uses radioisotopes to determine the absolute age of fossils or rocks plus or minus the experimental error involved in the particular method used.

## Carbon-14

Radiocarbon dating is very important for dating fossils of organic material such as bone, shell, charcoal, linen, wood, paper, and cave paintings which have pigments from honey, milk, or oil seed. All living things contain carbon in a known ratio and when the organism dies, the carbon-14 decays and the ratio of carbon-14 to carbon-12 changes. Scientists can measure this ratio and then estimate the age of the sample. The half-life of carbon-14 is 5730 ±30 years. The use of particle accelerators as mass spectrometers means it is possible to measure the number of carbon-14 atoms directly rather than waiting for their decay allowing dating up to 70 000 years ago.

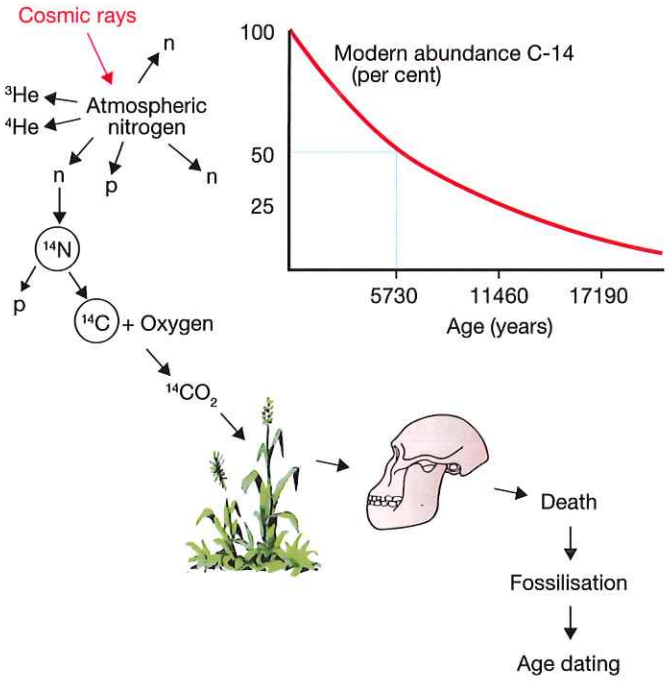



Figure 35.1 Carbon-14 decay is used to date fossils.

When using radiocarbon dating for artefacts made by people the date of the use needs to be distinguished from the date the material being used was harvested or gathered. For example, with writing on paper such as the Dead Sea Scrolls, the radiocarbon date gives the time the flax was cut and the leaves died which is not necessarily the exact age of when the scrolls were written. In this case the writing style aids the dating process.

It has been discovered that the ratio in the atmosphere of one atom of carbon-14 to every million million ($10^{12}$) carbon-12 atoms is not a totally fixed ratio. The study of growth rings on trees and gas analysis of ice cores from glaciers and Antarctica has required corrections to be made to allow for fluctuations in the carbon-14 content of the atmosphere. The amount of carbon as carbon dioxide in the atmosphere has also changed, especially in the last 200 years.

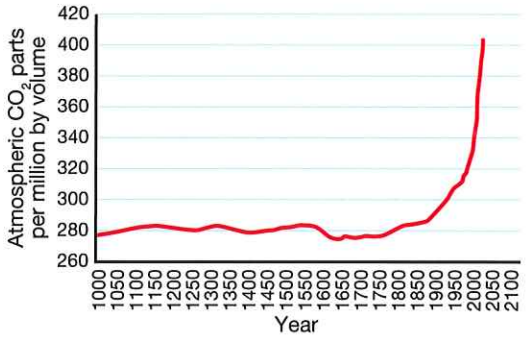



Figure 35.2 Changing amount of carbon dioxide in the atmosphere.

## Potassium-argon dating

Potassium-40 is a naturally occurring radioisotope that decays to form argon-40 and calcium-40. Potassium-40 has a half-life of $1.26 \times 10^9$ years. This long half-life means it is used in dating volcanic rocks and minerals older than 200 000 to 300 000 years and is limited to a small range of rocks past a certain age.

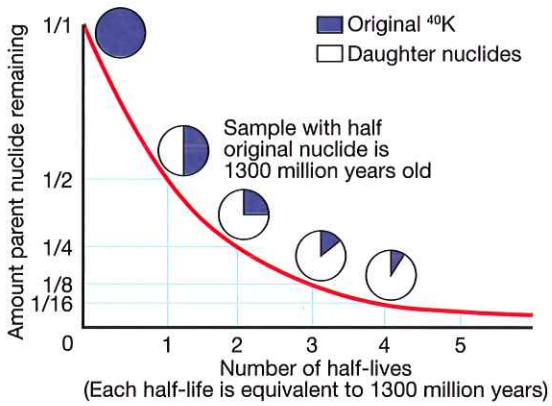



Figure 35.3 Carbon-14 decay is used to date fossils.

Potassium-argon dating has been very important in dating and understanding primate fossil records. Many of the hominid fossils in East Africa have been found in layers associated with volcanism and sedimentation and potassium-argon dating has been used to date the volcanic ash deposits and thus the fossils in these deposits. For example, the footsteps at Laetoli were made by three hominin individuals walking in powdery volcanic ash from a nearby eruption and the tuff was dated using potassium-argon dating and stratigraphy.

Figure 35.4 Footsteps at Laetoli dated 3.6 million years.

## Fission tracking

Fission tracking dating is based on the spontaneous fission of uranium-238. Uranium-238 is the most abundant isotope of natural uranium and it splits into two halves that rapidly speed off in different directions. The moving fission fragments affect the other minerals, e.g. the mineral zircon ($ZrSiO_4$) causing a trail to form through the rock or crystal. Volcanic zircons have no tracks when they form and as the uranium undergoes fission the number of tracks increase with time at a rate that depends on the uranium content. Each fission tack is formed by the fission of a single atom of uranium-238. A measurement of the density of the tracks and the amount of uranium can give an accurate estimate of the age of the rock. Fission tracking can be used on minerals and glass in a range from 100 years to 4500 million years before the present. Detrital zircons are used to give the age of sediments. Zircon grains from Jack Hills, Western Australia have been dated at 4.4 million years old and are the oldest known rocks of terrestrial origin.

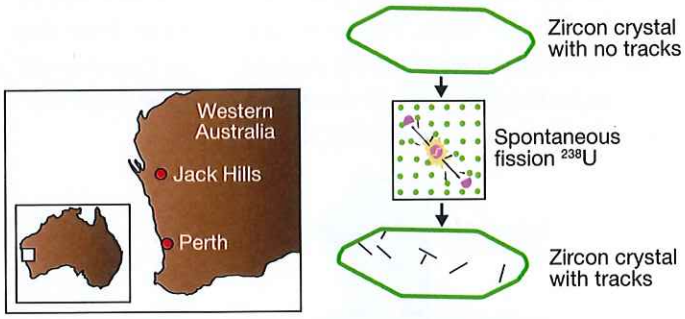



Figure 35.5 (a) Jack Hills, Western Australia. (b) Formation of fission tracks in zircon crystal.

## Rubidium-strontium dating

Rubidium-87 has a half-life of $48.8 \times 10^9$ years and is used to determine the time of crystallisation of igneous intrusions, the time of metamorphism and collect information about the mineral crystals in sedimentary rocks. With a long half-life it is usually only used to date rocks older than about 100 million years old. It has been used to date meteorites, Moon rocks and rocks in the Pilbara region and Marble Bar.

## QUESTIONS

1. What is radiometric dating?
2. Identify some materials dated using radiocarbon.
3. State the half-life of carbon-14.
4. Why has the use of mass spectrometers aided absolute dating?
5. Use an example to explain why radiocarbon dating of artefacts made by humans may not give the exact time of use.
6. Explain why scientists have had to develop an adjustment curve for radiocarbon analyses.
7. State the half-life of potassium-40.
8. Outline a limitation on the use of potassium-argon dating.
9. Use an example to show how potassium-argon dating has assisted the interpretation of fossils in hominid evolution.
10. (a) Explain why zircon crystals are often used when determining the age of a rock.
    (b) Give an example of the use of zircon fission tracks.
11. Explain why rubidium-strontium dating is used to date Moon rocks but not rocks from the last ice age which occurred around 12 000 years ago.
12. The data shows the radioactive decay of carbon-14 over time.

| Time (years) | Fraction of original radioactivity (%) |
|---|---|
| 0 | 100 |
| 5000 | 54 |
| 10 000 | 29 |
| 15 000 | 16 |
| 20 000 | 8 |
| 25 000 | 5 |
| 30 000 | 2 |
| 35 000 | 1 |

    (a) Graph the data for the decay of carbon-14.
    (b) A new fossil sample was found to contain 30% of its original radioactivity. What is the estimated age of this fossil?
13. For what age of rocks can fission tracking be used?
14. Uranium-238 has a half life of approximately $4.5 \times 10^9$ years. How many years are needed for there to be only one eighth of the parent nuclide remaining?
    (A) 9.0 million years (B) 4.5 billion years
    (C) 9.0 billion years (D) 13.5 billion years
15. Approximately how much carbon-14 remains after an organism has been dead around 2000 years?
    (A) 20% carbon-14 (B) 40% carbon-14
    (C) 60% carbon-14 (D) 80% carbon-14

# 36 Changes in Technology

Acquiring further evidence for evolutionary relationships is directly related to the development of new technologies and instruments. New technologies provide new data that requires our models and theories to be modified and extended. There has been significant developments in the fields of dating techniques, comparative genomics, comparative biochemistry and bioinformatics.

## Radiometric dating

The use of radioactive isotopes has enabled scientists to work out the absolute age of rocks and fossils. Before the use of radioisotopes scientists used the law of superposition, which gives relative ages to indicate time frames, e.g. if fossil A was in a layer below fossil B, then fossil A is older than fossil B. To find out the ages for the formation of the Earth and the origin of life on Earth, radioisotopes with long half-lives are needed. For example, uranium-238 has a half-life of 4.5 billion years and so can be used to date rocks many hundreds of thousands of years old. In 1993, uranium-238 was used to date the oldest fossil-containing rocks of the Cambrian period at 554 million years.

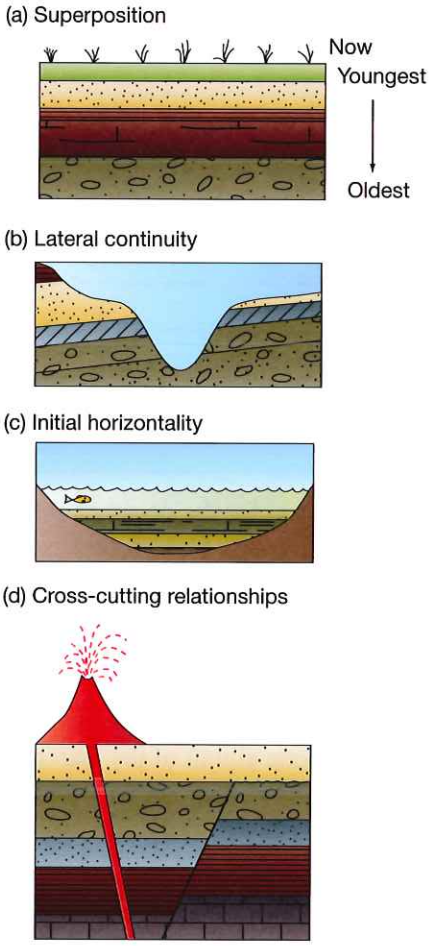



Figure 36.1 Stratigraphic correlation.

The estimated error involved in dating when using this method is less than ten per cent. A **mass spectrometer** is an instrument used to measure the ratio of isotopes present in a sample. This ratio can then be used, knowing the half-life of the radioisotope, to work out the age of the rock. Thus the absolute age of different fossils and rocks can be determined and the history of the evolution of life on Earth can be worked out.

Carbon-14 is constantly being produced in the atmosphere when cosmic rays interact with atmospheric nitrogen. The ratio of carbon-12 to carbon-14 can be measured. When a living organism dies the ratio of carbon-12 to carbon-14 decreases with a half-life of carbon-14 at 5730 years.

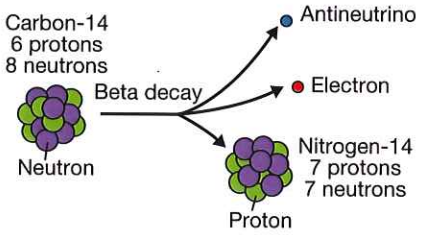



Figure 36.2 Carbon-14 undergoes beta decay to form nitrogen-14.

## Electron microscopy

The invention of the light microscope led to the cell theory when people were able to view the structure of cells. This changed scientific thinking and scientists quickly began investigating all aspects of cell structure. The invention of the electron microscope provided resolution about 1000 times better than that of light microscopes and magnification can be as high as 300 000 times. This increase in ability to detect fine details has helped interpret fossil evidence. Very thin slices of rocks can be viewed under an electron microscope and if the rock contains fossils, fine details can be examined.

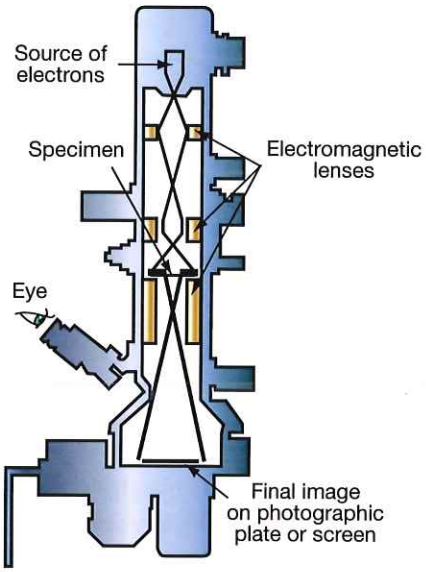



Figure 36.3 Electron microscope.

The use of the electron microscope is especially important when looking at fine details of hominid evolutionary trends. Dental development shows age at death, scratch marks on the teeth provides information about diet and defects in enamel and structure gives information about disease and life events.

**Scanning electron microscopes** are used to study the cut marks on bones showing how early hominids butchered their prey and the range of tools that were used in the process.

## Comparative genomics

**Genomics** is the study of whole sets of genes and their interactions. The development of new technologies has enabled the deciphering of the genomes of many different species, e.g. PCR amplification increases the amount of sample for testing and the ability to compare different genomes, e.g. using DNA-DNA hybridisation has provided valuable data about the degree of relationship between different species. For example, analyses of fossil Neanderthal samples shows that the Neanderthals are the closest known relative of modern humans. Genomic data is now combined with morphological evidence to adjust evolutionary trees with more accurate times of divergence.

## Biochemical analysis

New techniques in biochemical analysis have increased our understanding of the processes that occur in cells. New staining techniques use various coloured dyes that react with different organic compounds, and gas and liquid chromatography identify specific compounds present in a sample. Radioactive isotopes can be used as tracers to show particular metabolic pathways and DNA sequencing techniques allow mapping of the genome. Once the biochemical data is collected, comparisons can be made between different species, and similarities and differences provide information about common molecules and chemical reactions in cells. The discovery that RNA not only stores genetic information but can catalyse certain chemical reactions led to the hypothesis that first life used RNA for basic cellular processes.

## Computer technology

The recent advances in our understanding of DNA, the genetic code and amino acid sequences, e.g. in the Human Genome Project required the storage and analysis ability of massive amounts of data. This was possible with the invention of computer databases that can hold and manipulate incredible amounts of information. Table 36.1 shows the size of the genomes for several different species. **Bioinformatics** is the use of computers, software and mathematical models to process and analyse large data sets and biological information.

**Table 36.1** Genome projects.

| Organism | Size of genome (Mb DNA) |
|---|---|
| Bacterium (*E. coli*) | 4.2 |
| Yeast (*Saccharomyces cerevisiae*) | 150 |
| Fruit fly (*Drosophila melanogaster*) | 120 |
| Mouse (*Mus musculus*) | 3000 |
| Human (*Homo sapiens*) | 3000 |

## QUESTIONS

1. What is the difference between relative age and absolute age?
2. Explain how stratigraphic correlation is used to work out the ages of different rock layers.
3. For many years stratigraphic correlation was used to date fossils and rocks. Discuss a new technology that has made the dating of rocks and fossils more precise.
4. The diagram shows radioactive decay used in potassium-argon dating.

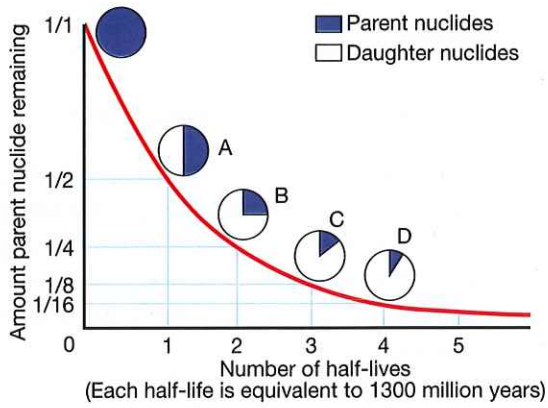



**Figure 36.4** Potassium-argon dating.

   (a) Identify the parent nuclide used in potassium-argon dating.
   (b) What would be the ages of rocks found at points A, B, C and D?
5. Explain why the electron microscope is used to look for the earliest fossils of life on Earth.
6. What is comparative genomics?
7. Explain how the invention of computer databases has assisted our understanding of the origin of life and the evolution of living things.
8. Discuss how the development of a named biochemical technique has increased our knowledge of the origin of life on Earth.

9. The homeobox genes are the developmental genes in foetuses that control the anterior to posterior regions and control the development from just behind the brain to the tail. Scientists have compared the homeobox genes in many animals, e.g. insects, nematode worms, molluscs, fish, frogs, birds and mammals (including humans) and have found that there are many similarities.
   (a) If all these different groups have similar homeobox DNA sequences, when would this DNA sequence be expected to have evolved in the history of life on Earth?
   (b) If the homeobox DNA sequence has changed little for millions of years, what can be inferred about its importance for survival?
10. The diagram shows the forming and assimilating of carbon-14 into living organisms.

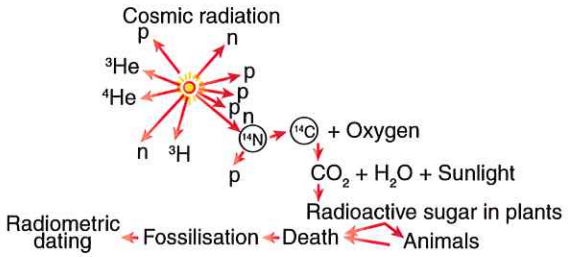



Figure 36.5 Forming and assimilating carbon-14.

Explain how carbon-14 is formed, enters living things and can be used to determine the age of a fossil.

11. Discuss why scientists use the latest technologies to compare the amino acid sequences in DNA in different species such as bacteria, yeast, fruit flies and humans.
12. (a) What does PCR stand for?
    (b) What is needed for PCR?
    (c) Outline the function of PCR.
13. Identify two functions of RNA.
14. Define bioinformatics.
15. The table shows the half-life of several radioisotopes. Which of these radioisotopes would be useful in dating rocks believed to have formed very early in the history of the Earth?

| | Radioisotope | Half-life |
|---|---|---|
| (A) | Iodine-131 | 8.05 days |
| (B) | Carbon-14 | 5700 years |
| (C) | Uranium-238 | $4.51 \times 10^9$ years |
| (D) | Strontium-90 | 28 years |

16. Which instrument would be most useful for seeing the earliest fossils of life on Earth?
    (A) Light microscope.
    (B) Electron microscope.
    (C) Mass spectrometer.
    (D) Deep sea submersible.
17. The invention of which instrument led to the formation of the cell theory as the basis of life on Earth?
    (A) Light microscope.
    (B) Electron microscope.
    (C) Mass spectrometer.
    (D) Deep sea submersible.
18. Figure 36.6 shows the decay curve of carbon-14.

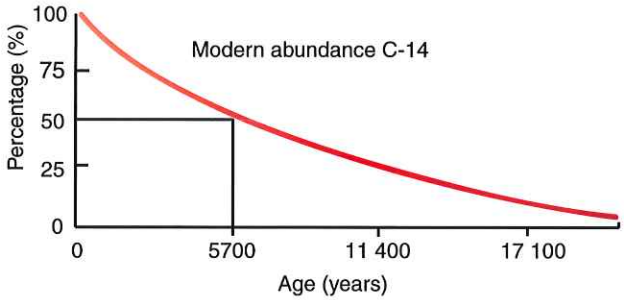



Figure 36.5 Carbon-14.

If a fossil contains only 50% of its original amount of carbon-14, how old is the fossil?
(A) 50 years old (B) 5700 years old
(C) 114 000 years old (D) 17 100 years old

19. Figure 36.7 below shows three stratigraphic columns.

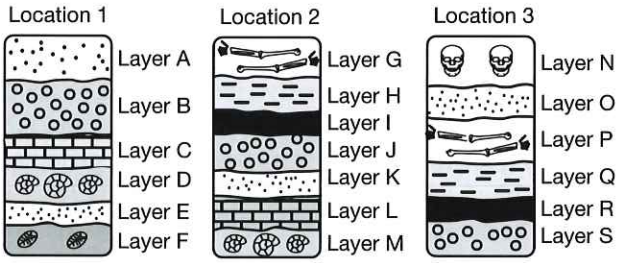



Figure 36.7 Three stratigraphic columns.

What is the oldest layer?
(A) Layer F (B) Layer M
(C) Layer S (D) Layer N

20. The development of which technology allowed a more precise dating of rocks and fossils?
    (A) Light microscope.
    (B) Electron microscope.
    (C) Mass spectrometer.
    (D) Deep sea submersible.
21. If a fossil was dated 430 million years old, what technology is likely to have been used?
    (A) Electron microscope.
    (B) Uranium-238 dating.
    (C) Carbon-14 dating.
    (D) Liquid chromatography and mass spectrometer.

# 37 Activity – Fossils Give a Timeline for Life on Earth

Using dates of known fossils, it is possible to construct a timeline to show how life on Earth has changed and the relative times at which this occurred. Unfortunately, fossils only show about 15% of the history of the Earth; the other 85% is Precambrian when organisms either did not exist or were soft bodied and left few fossils.

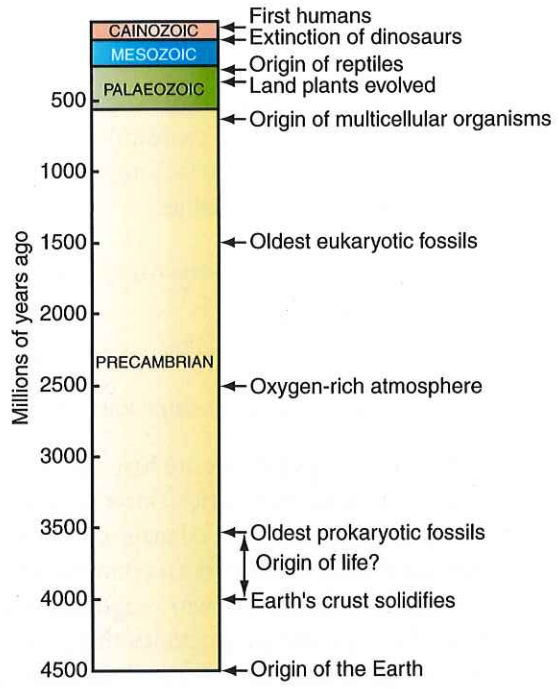



**Figure 37.1** Events in the Earth's history.

## QUESTIONS

1. Why do we have little record of life on Earth for most of the history of the Earth?
2. Use the following data to construct a scaled timeline for these events.

| Time | Event |
|---|---|
| 4.6 billion years ago | Formation of Earth |
| 300 million years ago | Early reptiles |
| 3.5 billion years ago | Early cyanobacteria |
| 700 million years ago | Worms and jellyfish |
| 600 million years ago | Early trilobites |
| 350 million years ago | Early amphibians |
| 120 million years ago | Early birds |
| 1.2 billion years ago | First simple animals |
| 200 million years ago | First mammals |
| 10 000 years ago | Woolly mammoth disappeared |
| 3 million years ago | Early people-like animals |
| ? | When you were born |

3. A student brings three fossils to school and tells his class that he believes they all came from the same sedimentary rock layer.

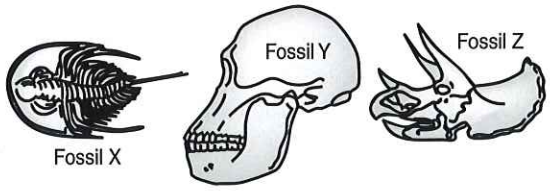



**Figure 37.2** Three fossils.

Discuss why his classmates disagreed with his interpretation.

4. Read the following information about the Cainozoic Era and draw a scaled timeline to show changes in Australia and its plants and animals.

The Cainozoic era began 65 million years ago and Australia was attached to Antarctica and South America with a warm and wet climate. Fossil evidence enabled geologists to divide the era into two periods – the Palaeogene (65 to 23 million years ago and the Neogene (23 million years ago to present). Each of these periods has then been subdivided into epochs.

The Gondwanan vegetation 65 mya was dominated by broad leafed rainforest, e.g. *Nothofagus*. The oldest marsupial fossils are dated 55 mya and are found at Murgon in southern Queensland.

Australia separated from Antarctica 45 mya and moved north. The first Acacias appear in the Eucla Basin (Nullabor Plain) 45 mya and in the Murray Basin 25 mya. Cliamte fluctuations occur 35 mya with inland dry areas and coastal warm, wet areas. The first fossil Eucalypts appear around 25 mya with their distinctive capped fruit (gum nuts). Lake Eyre fossils include pelicans, flamingoes, freshwater dolphins, lungfish and many marsupials.

Fossil coal of south-eastern Australia dated 23 mya contains fossil pollen of plants that only grow in cool, temperate regions in large conifer swamps.

Australia became drier and the vegetation changed as the centre became desert 20 mya and the rainforests contracted. Grasslands spread and were inhabited by large emus, diprotodons, meat-eating kangaroos and large tortoises.

Australia moved close to South-East Asia 15 mya forming the highlands of New Guinea and many animal groups arrived on Australia, e.g. rodents, goannas and elapid snakes (fanged venomous snakes that gave rise to death adders, copperheads, taipans, brown snakes, bandy-bandies). Inland areas were dry and hot with frequent bushfires.

By 5 mya sand deserts and large inland salt lakes form.

By 2 mya sclerophyll forests and dry rainforests had replaced the rainforests west of the Great Dividing Range. The dry rainforests were dominated by Araucaria (southern pine) and the sclerophyll forests contained casuarinas (she-oaks), Podocarpus (conifers) and Cyathea (tree ferns). There were several glaciations of ice and sea level dropped. Only a small area around Mt Kosciuszko and a larger area of Tasmania had permanent snow. Land bridges allowed migration for some groups leading to the characteristic flora and fauna of Australia.

# 38 Historical Development of Evolution Theories

During the 18th and early 19th centuries several people put forward ideas suggesting evolution. This challenged the accepted view of the times which held that each organism had been specially created and had not changed since that time. This was a time of political and social unrest, with the French Revolution and the Industrial Revolution changing not only the monarchy and the class system, but also the workplace with the growth of larger cities. People were becoming more interested in science and in explaining the structure of matter and why it behaved as it did. They were ready to investigate the scientific basis of all phenomena.

**Georges Cuvier** (1769-1832) believed the fossils in the strata below Paris showed catastrophism, special creation followed by a catastrophe like Noah's flood, followed by repopulation from other areas. **Jean Baptiste de Lamarck** (1744-1829) believed evolution occurred through the inheritance of acquired characteristics. **Charles Lyell** (1797-1875) suggested uniformitarianism, the idea that geological processes have not changed and are the same today as in the past. **Thomas Malthus**, in 1798, wrote an 'Essay on the Principle of Population', which proposed that there was a 'struggle for existence' as human populations increased geometrically while food supplies increased arithmetically.

Figure 38.1 Charles Darwin.

In 1858, **Charles Darwin** (1809-1882) and **Alfred Wallace** (1823-1913) both published articles in the *Journal of the Proceedings of the Linnaean Society* in England. Both suggested the mechanism for evolution using evidence from their individual extensive travels around the world. Alfred Wallace spent over four years in South America, particularly Brazil, exploring the Amazon and Rio Negro rivers. He also visited Indonesia and identified the Wallace Line, between the islands of Bali and Lombok, which separates the Asian animals on the eastern side from the Australian animals on the western side.

Charles Darwin sailed on the survey ship HMS Beagle for a five year journey on which he visited many countries including Africa, South America, Galapagos Islands, Australia and New Zealand. In 1859 Darwin published his book, *The Origin of Species*, which presents quantities of evidence for natural selection. However, the book also created great controversy and also led some people to give Darwin more credit than Wallace for the explanation of evolution.

## Divergent evolution

Divergent evolution occurs when different groups arise from a common ancestor due to adaptive radiation. This leads to populations that look quite different from each other. For example, adaptive radiation from the ancestral kangaroo has led to the musky rat kangaroo, the rat kangaroos, the red kangaroo, tree kangaroo and the wallabies.

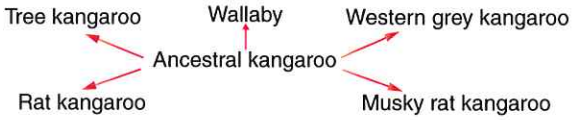



Figure 38.2 Adaptive radiation from ancestral kangaroo.

The finches of the Galapagos Islands are historically important as they were used by Charles Darwin as an example of adaptive radiation. The Galapagos Islands are volcanic islands that have never been connected to each other or to South America. Darwin suggested that the finches arose from one ancestral species that probably came from the mainland. Over time, natural selection and isolation led to the evolution of the different finch species, e.g. tree finches, ground-dwelling finches and insect eaters.

## Convergent evolution

Convergent evolution leads to superficial similarities due to organisms living in the same habitat or having the same lifestyle although they are not closely related. For example, convergent evolution has led to a dolphin, which is a mammal, and a shark, which is a cartilaginous fish, looking superficially similar, although reproduction, body temperature control etc are quite different.

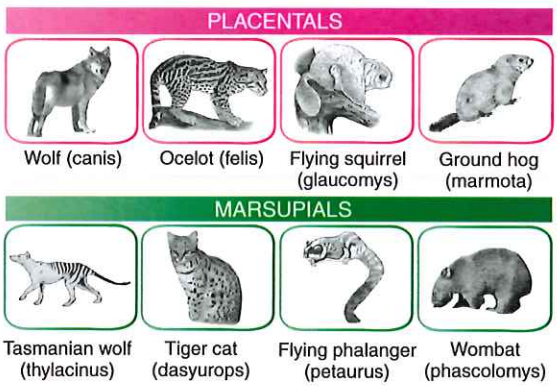



Figure 38.3 Convergent evolution.

## QUESTIONS

1. Many modern hospitals are the breeding ground of a multiple drug-resistant strain of the golden staph bacteria, *Staphylococcus aureus* or MRSA. A national study in 1988 found that 6% of patients, on average, acquired an infection in hospital. Using Darwin's theory of natural selection, explain why these bacteria are found in hospitals.
2. The diagram shows the extinct glyptodon and the modern armadillo, which were observed by Charles Darwin when he visited South America.

**Figure 38.4** Extinct glyptodon and modern armadillo.

Explain how Darwin used the extinct glyptodon and the modern armadillo to support his theory of evolution.

3. Compare divergent and convergent evolution. Use examples to illustrate your answer.
4. Fifteen million years ago the largest kangaroo resembled a possum and was the size of a large rat. At present there are sixty-two different species in the kangaroo family. How does the Darwin/Wallace theory of evolution explain this change in kangaroos?
5. How can the mechanism of evolution proposed by Darwin and Wallace be best explained?
   (A) Inheritance of characteristics acquired during an organism's lifetime.
   (B) Ability of an organism to adapt to environmental change.
   (C) Survival and reproduction of those with features most suited to the environment.
   (D) Forms with mutations survive and reproduce as they are more suited to the changed environment.
6. Alfred Russel Wallace (1823-1913) travelled throughout the East Indies and noticed that the Australian fauna and Asian fauna divided along a line between Bali and Lombok and between Borneo and Sulawesi (Celebes). This is called the Wallace line.

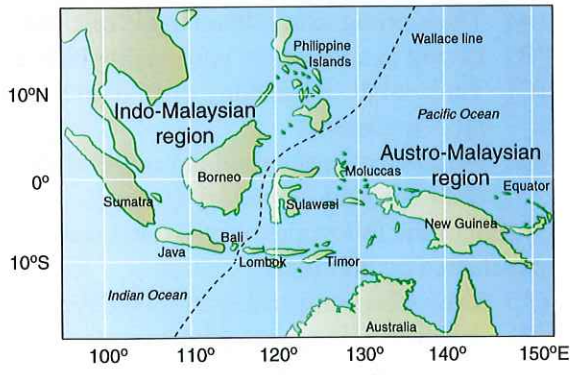



**Figure 38.5** Wallace line.

What caused the biogeography of the Wallace line?
   (A) Migration from Asia to Australia and then separation with natural selection in isolation.
   (B) Migration from Australia to Asia and then separation with natural selection in isolation.
   (C) Collision of the Australian plate with the Asian plate.
   (D) Constant interchange from Asia to Australia and back again.
7. Darwin's theory of natural selection has often been summarised as the *'survival of the fittest'*. What does the term 'fittest' refer to?
   (A) Organisms that are the strongest in the population.
   (B) Organisms that live the longest.
   (C) Organisms that produce the most healthy offspring.
   (D) Organisms that dominate over all others in the population.
8. On the Galapagos Islands there are two types of tortoise. On the moister islands the tortoises have short necks and domed shells, while islands with less fresh water and with tall shrubs and bushes have tortoises with longer necks with a ridge on the shell that allows the tortoise to raise its long neck.

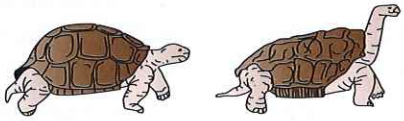


**Figure 38.6** Two types of tortoise found on the Galapagos Islands.

Identify the most likely selection pressure for the evolution of the long neck in tortoises.
   (A) Height of the bushes.
   (B) Weight of the tortoiseshell.
   (C) Width of pools of water.
   (D) Abundance of leaf litter and ground vegetation.
9. The development of the theory of evolution was originally used by Darwin to explain the change in different species over time. His book *The Origin of Species* caused great controversy and the idea of 'survival of the fittest' has been used by many people in varying fields of enterprise. For example, Hitler wished to create an Aryan super-race and Karl Marx envisaged a class struggle for survival. One of the areas still under debate is:
   (A) How variations arise in the gene pool.
   (B) How offspring inherit the favourable characteristics.
   (C) Why some species change and others do not.
   (D) Whether change is gradual or by punctuated equilibrium.
10. Historically, what is the correct order of proposed theories of evolution?
   (A) Lamarck, then Darwin and Wallace.
   (B) Lamarck, then Wallace, then Darwin.
   (C) Lamarck, then Darwin, then Wallace.
   (D) Wallace, then Lamarck, then Darwin.

# 39 Punctuated Equilibrium

In 1972, Niles Eldredge and Stephen Jay Gould published a paper titled '**Punctuated Equilibrium**' that suggested that smooth transitional sequences are not as common as Darwin's theory predicted and that species remain unchanged for long lengths of time, a period of stasis, to be replaced very quickly by a new species which is a variation of the old one. For example, trilobites, brachiopods and some dinosaurs, including Tyrannosaurus, show punctuated equilibrium. Natural selection as proposed by Charles Darwin is often called '**gradualism**'.

There is great debate about which mechanism is 'better', with each side quoting specific examples. It seems as through gradualism is more common among the microscopic protistans while more complex organisms show punctuated equilibrium.

## QUESTIONS

1. The diagram shows two theories about the mechanism of natural selection. Use these two examples of butterfly evolution to show how the concept of punctuated equilibrium differs from the gradual evolutionary process proposed by Darwin.

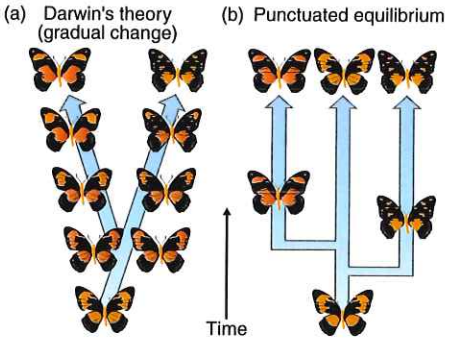



Figure 39.1 Two theories of evolution.

2. There is much debate about whether evolution occurs gradually or by punctuated equilibrium. Which mechanism is correct?
3. Figure 39.2 shows a series of events.

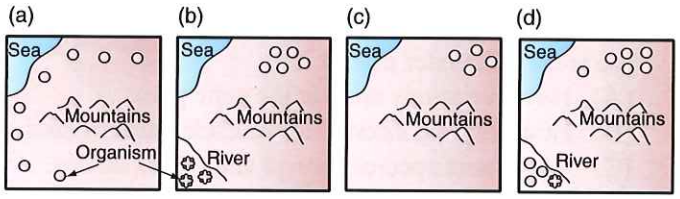



Figure 39.2 Formation of a new species.

   (a) Place them in their correct order and explain how the new species evolved.
   (b) Relate these diagrams to biogeography.
4. The diagram shows the changes in a single species of snail found in successive layers of rock from a freshwater deposit 10 million to 3 million years ago.

Figure 39.3 Pliocene snail fossils.

   Name this type of change.
   (A) Punctuated equilibrium.
   (B) Gradualism.
   (C) Cloning.
   (D) Transgenic isolation.
5. What is the method of natural selection as proposed by Charles Darwin which states that change occurs by small degrees every generation?
   (A) Punctuated equilibrium.
   (B) Gradualism.
   (C) Cloning.
   (D) Transgenic isolation.
6. In the theory of punctuated equilibria, what is meant by the period of stasis?
   (A) Long time periods when the species stays the same.
   (B) Abrupt appearance of new species.
   (C) Gradual changes occurring in small increments each generation.
   (D) The species stays in the same geographic area.
7. In Western and eastern Australia, there are underground orchids of the genus *Rhizanthella*, which obtain all nutrients from mycorrhizal fungi. What is shown by the evolution of orchids with no leaves and no chlorophyll?
   (A) Two species can form a unique symbiotic relationship.
   (B) Natural selection will lead to leafless plants in Australia.
   (C) Orchid mycorrhizae inhibit chlorophyll production in orchids.
   (D) Geographic isolation in Australia favours underground plants.
8. By the end of the Palaeozoic era, many of the modern orders of Insecta had appeared. Some modern insect species appear very similar to their ancestral form. What is this lack of change most likely due to?
   (A) Their strong exoskeleton made of chitin.
   (B) Living in a relatively unchanging habitat.
   (C) Their small size making them hard to catch by most predators.
   (D) Presence of specialised digestive, circulatory, respiratory and excretory systems.
9. Which of the following events would lead to punctuated equilibrium?
   (A) Slow increasing aridity of inland Australia.
   (B) A slow rise in global temperatures.
   (C) Cretaceous extinction of the dinosaurs.
   (D) Plate tectonics causing plate divergence.

# 40 Human Evolution

Humans are classified as a **primate** as they have an opposable thumb, shoulder joint with high rotating ability, forward facing eyes with stereoscopic vision, reduced snout and olfactory centre of the brain, enlarged skull with large cerebrum, five digits on limbs, nails (not claws) and bicuspid teeth. The primates include lemurs, lorises, monkeys, gibbons, baboons, chimpanzees, orangutans, gorillas and humans.

The **hominoids** is a broad term that refers to both the primate superfamily that includes humans and the greater and lesser apes – the gibbons, gorillas, chimpanzees. The hominoids diverged from the Old World monkeys around 20 to 25 million years ago. The **hominins** refers to species on the human branch of the evolutionary tree – a member of the family Hominidae which include the genera *Homo* and *Australopithecus*. Hominins are bipedal and have relatively large brains.

## Physical evolution

The fossil record shows many different hominins with physical differences in brain size, jaw structure, teeth, arm length in proportion to body length and degree of sexual dimorphism. The oldest fossils of *Homo sapiens* are from Africa and date nearly 200 000 years old. All other species of hominins are extinct.

## Cultural evolution

Humans are unique in their ability to manufacture and use complex tools. Other animals can make and use tools but not with the same complexity, e.g. a chimpanzee will strip the leaves off a thin branch and then use the stick to get termites out of a termite mound to eat the extracted termites. Theoretical models of natural selection do not account for culture and technology. The ability to make and use tools alters the selection pressures so that it is not necessarily the 'fittest' that can survive to reproduce.

## Are humans still evolving?

There is some debate about whether or not human physical evolution is still occurring. The agricultural revolution, Industrial Revolution, modern medicine and global mass transport have changed human lifestyles, the way food is collected and distributed, the structures that are used as homes, the survival rate of infants, the ability to overcome diseases and injuries, life expectancy and the ability to travel to distant continents. The gene pool for the human population is less influenced by environmental conditions. In many Western societies infertile couples can use a variety of reproductive technologies which means their genes will stay in the gene pool and be passed onto the next generation.

There are some localised studies that link human evolution with fertility related natural selection. Some research projects have found a link between a woman's fertility and particular features, e.g. a woman's height, blood pressure and amount of body fat and cholesterol is linked to the number of children she has. Using these results they have predicted how allele frequency will change very slowly over time for these traits.

## Human genetic variation

There are genetic variations between human populations and between individuals within a population. Many human genes only have one form of allele and are not polymorphic. These genes are 'fixed' and each human is around 99.9% biochemically similar to other humans. The differences between humans allows DNA fingerprinting to identify individuals. Analyses of human genomes has provided information about genetic difference between different groups. **Ancestry information markers** (AIMs) can be used to put people into broad, geographically based groups. Some of these markers relate to health risks associated with people from certain geographical areas. There is debate about the usefulness of this data as human migration and interbreeding between 'races' makes it difficult to place many individuals into one particular group. The **1000 Genomes Project** started in 2008 as an international research project to catalogue human genetic variations, e.g. looking for disease variants that have severe effects such as cystic fibrosis and Huntington's disease and disease variants that have mild effects such as diabetes, cognition traits and heart disease. The research found that the average person has around 250 to 300 loss of function variants in annotated genes. In 2010 it was announced that the project was to be expanded into the 10 000 Genomes Project.

### QUESTIONS

1. List the feature that classifies humans as a primate.
2. Distinguish between the hominoids and the hominins.
3. Identify some areas of physical evolution that distinguish each different type of hominin.
4. How does culture distinguish humans from other species?
5. Explain why there is debate about the current physical evolution of humans.
6. What are 'fixed' genes?
7. What is the 1000 Genomes Project?
8. The human population has increased from around five million at the end of the last ice age to more than seven billion today. How can ancestry information markers provide information about today's human population?

# 41 The Primates

The primates are an order of placental mammals. Primates include the prosimians – lemurs, indri, aye-aye, loris, bush-baby and tarsiers, and the anthropoids – monkeys, capuchin, macaques, baboon, mandrills, gibbons, orangutan, chimpanzee, gorilla and humans. The earliest primates were probably tree dwellers with many primate features related to an arboreal lifestyle.

Primates are identified by a combination of features related to fingers/toes, arms/legs, eyes/ears, teeth/snout and particular behaviour.

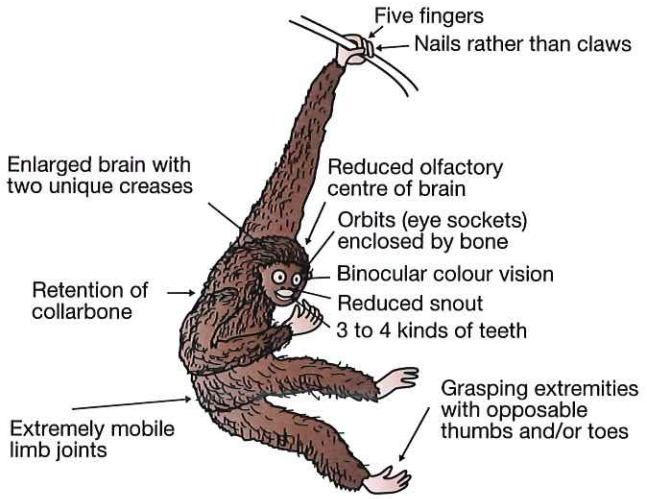



**Figure 41.1** Primate features.

## Primate features

The criteria used to classify the order Primates have changed over the last 300 years. Linnaeus in 1758 used the presence of four incisors (cutting front teeth), two clavicles (collarbones), two mammary glands on the chest and at least two prehensile extremities to identify primates. Robert Martin in *Primates: A Definition* (1986) included a wider range of features where subdivision depends on the degree or grade of a feature, e.g. compared with most other mammals primates have an increased brain size relative to their body size. Martin considered means of location, relative brain size, reproductive biology, dental patterns, geographic distribution, influence of major sense organs on the shape of the skull and habitat.

Primates have the following features.

- An opposable thumb and/or toe with hands/feet adapted for grasping.
- Shoulder joint with high rotating ability.
- Forward facing eyes with stereoscopic vision.
- Reduced snout and olfactory centre of the brain.
- Enlarged skull with large cerebrum.
- Five digits on limbs, nails (not claws).
- Bicuspid teeth.
- Most primates have cones as well as rods and have colour vision.
- Primates typically produce few young each gestation with a long period of development after birth.

## Types of primates

Traditionally the primates were divided into the prosimians and the anthropoids. They are now subdivided into three groups – with the prosimians separated into the lemurs of Madagascar and the lorises and pottos of tropical Africa and southern Asia (wet noses) and the tarsiers (dry noses).

Lemur

Loris

**Figure 41.2** Wet nosed primates.

The earliest primates were most likely similar to the lemurs, lorises and pottos. The eyes of lemurs are widely spaced and directed to the side giving them little binocular vision and a lack of cones means a lack of colour vision. The rhinarium is the moist, flashy pad at the end of the nose of lemurs and lorises and aids their sense of smell. They mark their territories with scent and many have a grooming claw on their second toe.

**Figure 41.3** Tarsier.

Tarsiers have short stocky bodies, long tail, long hind legs and are unusual as they can turn their heads 180 degrees in both directions. They do not have a rhinarium and thus have a reduced sense of smell.

## Anthropoids

The anthropoids are divided into the catarrhine (narrow nosed with nostrils that face downwards) monkeys, apes and humans and the platyrrhine (flat nosed with nostrils that face sideways) monkeys.

Compared to the other primates anthropoids have the following features.

- Flatter face with a dry nose (no rhinarium) and a complete upper lip.
- Larger body size relative to body weight.
- More rounded skull.
- Relatively larger brain.
- Binocular, colour vision.
- Mainly diurnal.
- Prolonged parental care, gestation and maturation.
- Increased mutual grooming.

**New World monkeys** – are platyrrhine monkeys that live in Central and South America. Their thumbs are relatively unopposable and many have a long prehensile tail which acts as a 'fifth' limb to grasp objects, e.g. spider monkeys, squirrel monkeys, howler monkeys and capuchins.

**Old World monkeys** – are catarrhines that live in Africa and Asia. None have a prehensile tail. They have hard patches of bare skin (ischial callosities) on the buttocks that support their weight when sitting, most are quadrupeds and either walk on the ground like baboons or live in trees like guenons and many species show sexual dimorphism. They include the baboons, leaf monkeys, Rhesus monkeys and vervets. The terrestrial baboons also have longer muzzles and larger canine teeth than the New World monkeys.

**Hominoids** – include humans and the great apes – the gibbons, gorillas, chimpanzees, orangutan and bonobo. Apes have no tail and their arms are longer than their legs. They do not have external tails, have an increased brain size relative to their body size and have an extended maturation period with greater reliance on learnt and complex behaviour.

## QUESTIONS

1. Discuss how primate classification has changed over the years.
2. List some of the physical features that are used to classify primates.
3. (a) How were primates traditionally subdivided?
   (b) How are primates now subdivided?
4. What is the most probable appearance of the earliest primates?
5. Identify features that would support the statement that the prosimii are more primitive that the anthropoidea.
6. Distinguish between platyrrhine and catarrhine.
7. Construct a table to compare the features of New World monkeys, Old World monkeys and apes.
8. If you found a fossil, describe two structural features that would identify it as a primate.
9. Orangutans are anthropoids and the world's largest tree-climbing mammal. They are currently restricted to the forests of western Borneo and the top of Sumatra although fossil evidence shows they once lived across Sumatra, Borneo, java and South-East Asia. They are an endangered species mainly due to forest destruction and illegal poaching and smuggling. Outline the importance of captive breeding programs for orangutans.
10. The diagram shows two types of noses found in the anthropoidea.

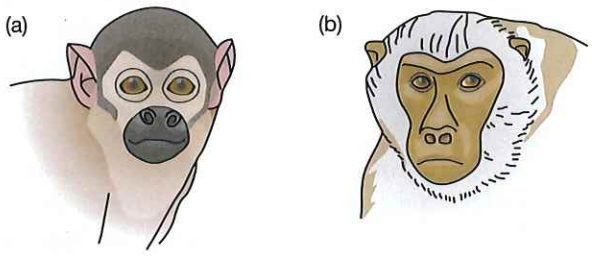



Figure 41.4 Nose types of the anthropoids.

Identify each type of nose and the animal groups that possess that feature.

11. The map shows the distribution of New World monkeys, Old World monkeys and apes.

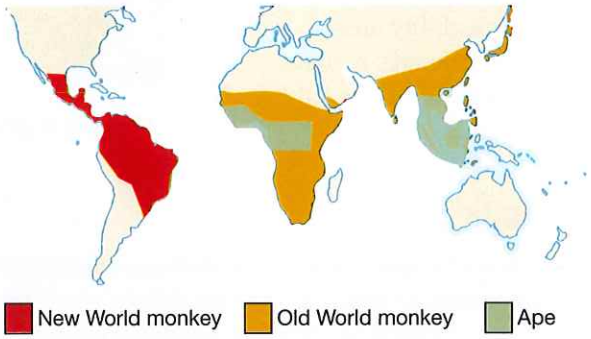



Figure 41.5 Distribution of anthropoids.

Which geographical areas are the native ranges of:
(a) Spider monkeys? (b) Baboons?
(c) Orangutan? (d) Gorilla?

12. Most primates live in groups with a permanent adult male. Infants depend on their mothers for all of the food for the first 2 to 6 months of life and most are carried by their mother for another 6 to 12 months. Discuss how social grouping has assisted primates in their evolution.
13. Outline the benefit of having an opposable big toe and opposable thumb.
14. Spider monkeys have the most prehensile tail of any primate and their thumb is reduced in size to a small tubercule. Discuss the advantage of these features.

# 42 Primate Hands and Feet

The shape and structure of primate hands and feet are important characteristic features of the order.

## Opposable thumb and/or toe

An opposable and mobile thumb gives primates the ability to grasp objects. Some primates also have an opposable big toe. To be able to grasp and manipulate objects certain features are needed. Most primates can move their thumb so that it can touch the second digit or the palm of their hand. All non-human primates have a divergent or partially opposable big toe.

An opposable thumb allows primates to have a **power grip** holding an object between the fingers and the palm The thumb reinforces the fingers. The power grip aids an arboreal lifestyle enabling branches to be grasped. Spider monkeys, which have the most prehensile tail of any primate have only four fingers with a small tubercle for a thumb. This increases the ability of the spider monkey to swing through the trees more easily as a thumb grasping a branch in the opposite direction to the fingers could be broken and would get in the way of the swinging motion of brachiation.

The **precision grip** relates to the ability of the index finger to touch the thumb. The arrangement and length of the human thumb gives increased manipulative ability and unlike other primates humans have the unique capacity to thread a needle.

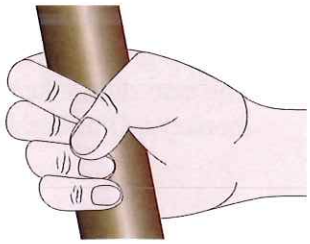


Figure 42.1 Power grip.

## Five digits

Many orders of mammals have become specialised and have reduced numbers of digits, e.g. the horse hoof is one toe. Primates are relatively unspecialised with extremities similar to primitive mammals. Primates still retain the pentadactyl limb with the five digits, although some species show a reduction in the size of the thumb or the second finger.

## Nails not claws

Most primates have nails which are a protective shield covering sensitive tactile pads. The fingertips have sensory nerve fibres to enhance the sense of touch. The cutaneous ridges on the inner surfaces of fingertips and very little oil secretion make the grip firmer and reduce fingers slipping when grabbing branches. Exceptions include some New World monkeys which have claws and some prosimians have a claw on one digit.

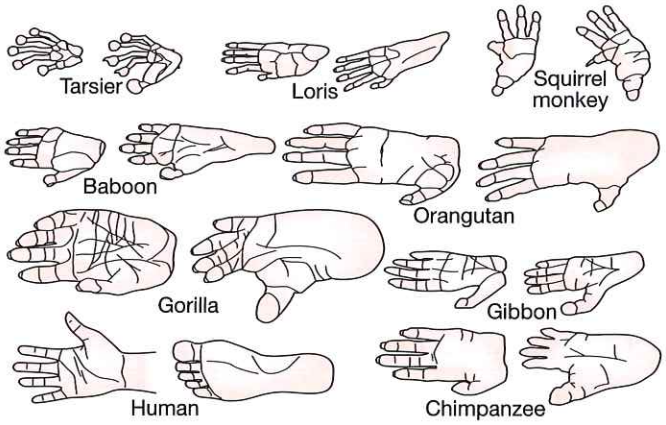



Figure 42.3 Primate hands and feet.

## QUESTIONS

1. What is an opposable thumb?
2. What is the benefit of having an opposable thumb?
3. How is the big toe of humans different to the big toe of other primates?
4. What is the power grip?
5. Identify some of the types of power grips.
6. What is the precision grip and which primates have this grip?
7. Identify some different types of precision grips.
8. From the diagram compare the hands of the orangutan, gorilla and human relating structure to use.
9. Evidence suggests that the hand of the last common ancestor of chimpanzees and humans had a hand relatively similar to modern humans and since then the chimpanzee hand has changed. What does this suggest about the structure of human hands?

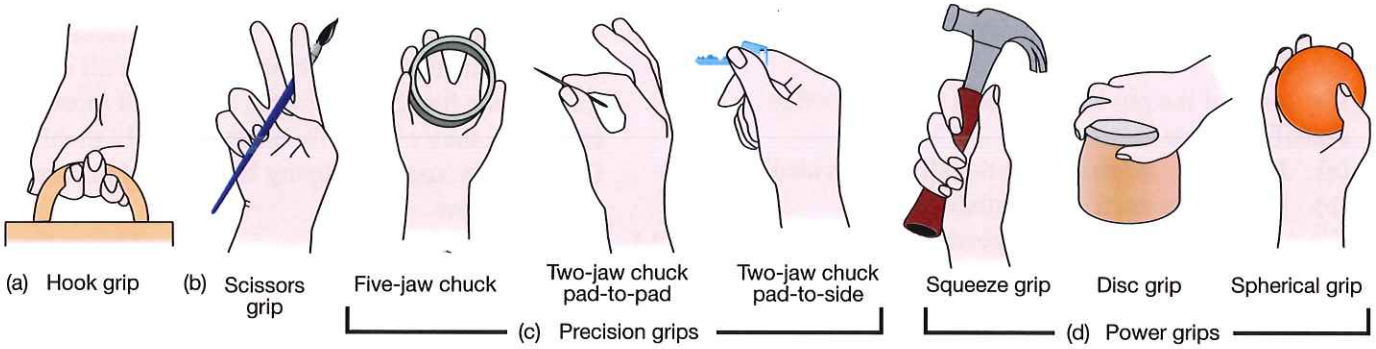



Figure 42.2 Human hand grips.

# 43 Primate Head and Skull

Most primates rely on their vision more than their sense of smell. There is also a trend towards forward directed eyes and stereoscopic vision with a flatter face than most other mammals. The brain of primates is larger relative to their body size compared with other mammals and this also has influenced the shape of the skull.

## Forward facing eyes

Primate eyes are typically close together and forward facing so that the field of view from each eye overlaps and gives **binocular vision**. Both eyes can focus on the target object. The information from each eye is transmitted by the optic nerve to the brain. The enlarged brain of the primates then superimposes the two images over each other which then interprets the messages as three-dimensional – **stereoscopic vision** – giving a perception of depth.

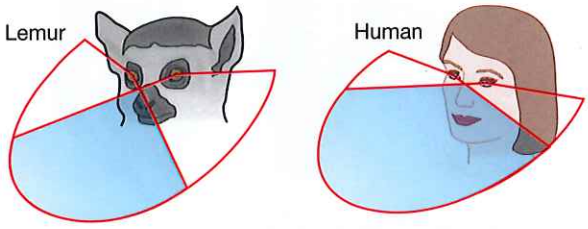



Figure 43.1 Binocular vision of lemur and human.

The overlapping binocular vision means that there is a reduced field of view. Many primates overcome this problem by having a mobile head and neck.

## Colour vision

Primates are one of the few groups of mammals to see in colour. Primate retinas have cones as well as rods and have a fovea which is an area of closely packed cones that produce sharp visual images. Cones give colour vision and fine visual discrimination while rods are important when there is little light available. Other animals that have colour vision are most fish, amphibians, reptiles and birds. Most diurnal primates see in three colour vision (red, green and blue); nocturnal primates do not. Colour vision also requires specialised areas in the brain in the cerebral cortex to interpret the messages. Colour vision is an advantage for fruit eaters so they can tell when the fruit is ripe. Some primates, especially the monkeys, use the ability to see in colour for social communication. For example, the bright red, white and blue skin on the face of the male mandrills and the female baboon will show the male her pink, swollen 'sexual skin' on her hindquarters when she is ovulating and in oestrus.

## Enlarged brain

Mammals tend to have a larger brain size relative to body size compared to other animals, and primates tend to have a larger brain relative to body size compared to other mammals and humans have an even larger brain relative to body size compared to other primates. The cerebral cortex which is the outer region of the cerebrum is involved in sensory and motor responses as well as the higher nervous functions. In primates a larger cerebral cortex gives increased manipulative and grasping ability of the hand, colour vision and stereoscopic vision.

Table 43.1 Comparative brain sizes.

| | Chimpanzee | Orangutan | Gorilla | Human |
|---|---|---|---|---|
| Brain size ($cm^3$) | 300 to 480 | 300 to 550 | 400 to 670 | 1000 to 2000 |

## Teeth

Primate teeth are unspecialised which means that diet is varied. The shape and size of the four different kinds of teeth found in primates are used to differentiate between the different groups, e.g. canine teeth are large in monkeys and apes. Teeth consist of a central pulp surrounded by dentine surrounded by enamel. Enamel is the hardest known biological substance. The thickness of the enamel indicates the 'toughness' of the diet. Enamel also preserves growth lines so it is possible to calibrate events from teeth sections.

## Reduced snout

There is a trend for a reduction in the importance in the sense of smell in the primates which leads to a flatter face and reduced snout. The area of the brain needed for olfaction is also correspondingly smaller. Some primates, e.g. baboons have a large muzzle to accommodate their large canines and not for olfactory receptors and nervous tissue.

## QUESTIONS

1. What is binocular vision and stereoscopic vision?
2. How does forward facing eyes affect the field of view and how do primates overcome this problem?
3. Most primates have good colour vision being able to distinguish blue and yellow-green and many can see red. What structures give colour vision?
4. How does brain size affect the skull?
5. When a male mandrill becomes the dominant alpha male of the group his face and rump become particularly bright red and blue. What is a benefit of this colouring?
6. Explain why primates have less need for a long snout than other mammals.

# 44 Primate Skeletons and Locomotion

The vertebral column is the backbone. It consists of a series of vertebrae (bony or cartilaginous segments) going from the head to the tail, enclosing the spinal cord.

The arrangement of the primate vertebral column gives flexibility and allows a variety of forms of location, e.g. an upright posture holding the upper body erect allows the primate to walk, to sit, to leap and in humans to stand with bipedal walking.

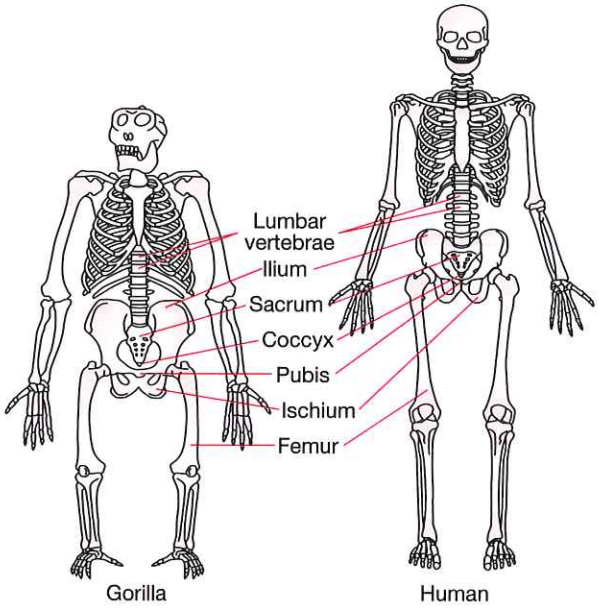



Figure 44.1 A human skeleton and a gorilla skeleton.

The clavicle (collarbone) in primates allows rotation of the forearm and certain aspects of hip structure means primates exhibit a range of different forms of locomotion. Rotating the arm in the shoulder socket also means the arm can reach behind the body. This movement is important in an arboreal lifestyle and enables some primates to swing from branch to branch through the trees.

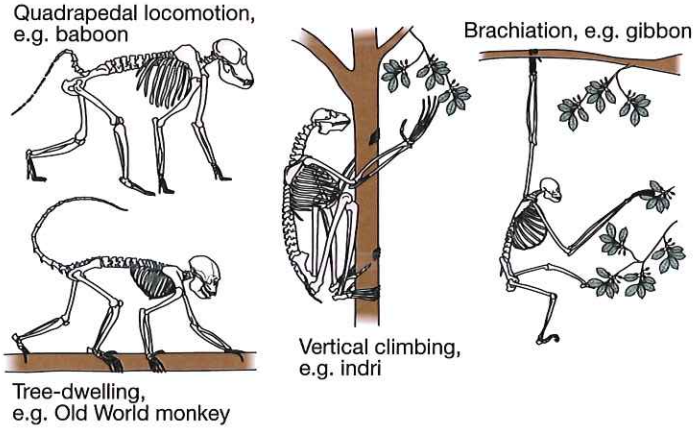



Figure 44.2 Different types of primate locomotion.

The earliest primates and most prosimians and some monkeys are arboreal quadrupeds, that is, they walk or run on all fours along branches, e.g. loris, squirrel monkeys. An arboreal lifestyle requires an opposable thumb/big toe for gripping, mobile shoulders, hip and elbow joints and stereoscopic vision.

Some arboreal primates are 'leapers', e.g. lemurs and tarsiers. They have long flexible backs, long hind limbs and long hands and feet. Long, strong legs allow them to hang suspended from branches.

Some primates are terrestrial quadrupeds, e.g. Old World monkeys and some of these have developed **knuckle walking**, e.g. gorillas and chimpanzees. They extend their arms and support their weight on their knuckles holding their body in a semi-upright position with their shoulders higher than their hips.

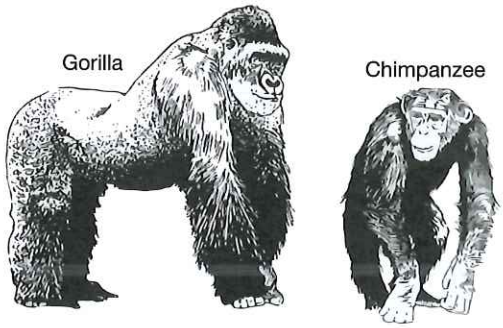



Figure 44.3 Primate knuckle walking.

The larger primates often use a suspensory method of hanging from branches by their arms using **brachiation**, a form of swinging movement, to get from tree to tree.

Species that run on the ground have longer limbs and some loss of gripping feet. Humans are the only truly upright primate. In humans the S shaped curve of the backbone, the lower, broader pelvis, the position of the foramen magnum and the arrangement of the knee joint are structural changes that show upright stance.

## QUESTIONS

1. What is the vertebral column?
2. (a) What is the clavicle?
   (b) How does the clavicle aid locomotion in primates?
3. Which type of locomotion was most likely used by the earliest primates?
4. Identify features that are needed for an arboreal lifestyle.
5. Describe knuckle walking.
6. Describe brachiation.
7. Identify features needed for running on the ground.
8. Which of these features best indicates that a fossil primate lived mainly on the ground?
   (A) Long prehensile tail. (B) Large head.
   (C) No opposable toe. (D) Mobile shoulder.

# 45 Primate Reproductive Features

Primate reproduction and care for the young are characteristic of the order. They are among the slowest breeders, relative to body size among the mammals.

## Gestation and parental care

Modern primates have long gestation periods (length of pregnancy) relative to maternal body size. Usually a long gestation period for mammalian species is related to maternal body size with larger mammals having longer gestation periods. In primates the longer gestation is related to foetal development especially brain development. Primates also usually have reduced numbers of offspring per birth with single live births being the normal.

Primates have a long period of parental care with slow postnatal growth of the baby and the entire life span is usually extended. Sexual maturation is also later in life.

The increased length of time spent in childhood means that there is a greater dependency on learnt behaviour rather than innate or instinctive behaviour. Learnt behaviour can involve improving motor skills, social skills and cognitive skills, e.g. by observing the actions of others in their group, imitating these behaviours, practice and trial and error.

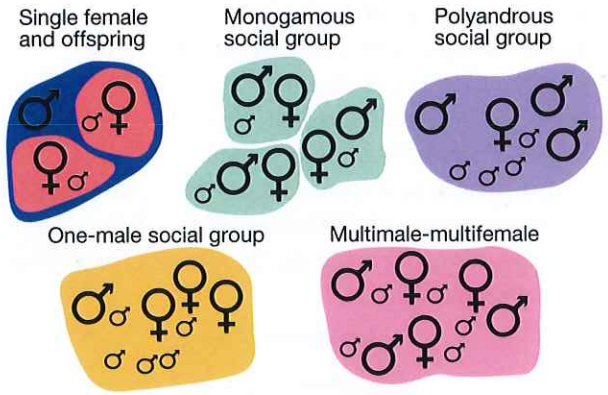



Figure 45.1 Types of primate social organisation.

## Social organisation

Most primates live in groups with a permanent adult male though social structures can change with changing activities. Infant primates depend on their mothers for all food requirements for the first 2 to 6 months after birth and most are carried by their mother for another 6 to 12 months. The social grouping of primates provides protection for the mother and child. The size of the group depends on the habitat. For example, in forests there are often small groups with only one male, while on the open grasslands where there are more predators the groups tend to be larger groups with males in a dominance hierarchy.

The simplest social organisation is a **female with her offspring,** e.g. nocturnal primates, orangutans. The range of the adult male and female overlap and the adults come together for reproduction. The range of the male may be large overlapping the ranges of several females.

**Monogamous social groups** consist of an adult female-male pair and their immature offspring, e.g. gibbons, tarsiers. The sizes of both sexes are relatively the same.

**Polyandrous social groups** consist of one mature female and her offspring with several sexually active males, e.g. in tamarins, though this type of grouping is a relatively rare organisation.

**One-male social group** consists of several females and their young and a single adult male, e.g. baboons, howler monkeys and gorillas. There are several different forms of this grouping with some being dominated by a stable core of females that choose the male who may join the group for a limited time and aid defence of the territory. Or it can be dominated by a silverback male who drives off males when they reach maturity.

**Multimale-multifemale group** consists of several adult males and many females and their offspring forming a large group, e.g. macaques. There is usually a dominance hierarchy among the males and females.

**Fission-fusion groups** change size and composition throughout the year. Individuals leave or join groups and some separate to form a subgroup, e.g. while foraging for food or they all join together if a rich food source is found or to occupy a particular sleeping site.

## QUESTIONS

1. What is gestation?
2. The table shows gestation periods and female weight.

Table 45.1 Gestation periods.

| Mammal | Female weight (kg) | Gestation time (days) |
|---|---|---|
| Human | 60 | 266 |
| Chimpanzee | 40 | 227 |
| Gorilla | 80 | 260 |
| Sheep | 60 | 148 |
| Cow | 700 | 284 |
| Rabbit | 1 | 30 |

Compare the trend of female weight and gestation for primates and other mammals.

3. Outline a benefit of longer parental care.
4. Construct a table to compare different types of primate social organisation.
5. Explain why the size of a social grouping usually depends on the habitat.

# 46 The Hominidae

The Hominidae is a family in the primate order. By convention all 'family' names end in '-idae'. Superfamilies end in '-oidea' and subfamilies end in '-inae'.

The definition of 'hominid' has changed over time with the term now having a broader meaning than it did in the past. **Hominids** now include the great apes and their ancestors which means it includes the modern humans, chimpanzees, gorillas and orangutans.

**Hominins** refers to bipedal humans and relatives, e.g. other species in the genus *Homo* and the australopithecines.

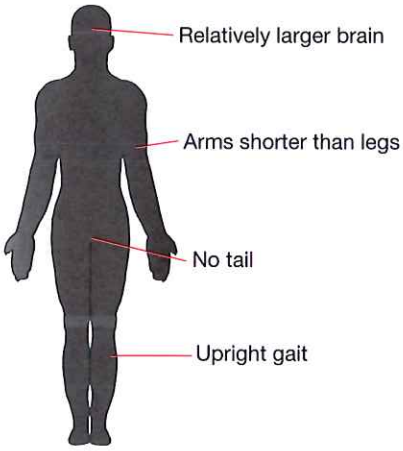



Figure 46.1 Hominidae features.

## New technologies and hominids

Changes in classification are due to improved technologies and increased knowledge and understanding about evolutionary relationships. Technological advances include the use of **DNA-DNA hybridisation** to compare the DNA of different species, **DNA sequencing** to find differences in either nuclear or mitochondrial DNA (mtDNA) between two species to determine their likely time of divergence from a common ancestor and **karyotype analysis** to compare the number, shape and banding of chromosomes of different species to determine their degree of similarity. Classification systems will thus change over time as they are a human construct used to show how we believe organisms should be grouped based on current knowledge and understanding.

For example, until recently the great apes were classified in the family Pongidae. This taxon is now obsolete as increased knowledge about molecular genetics shows that humans, chimpanzees and gorillas form a clade of closely related species that share more than 97% of their DNA with the human genome. The great apes and humans are now all placed in the family Hominidae.

## Hominid features

Hominids are the largest primates and have the following features.

- No tail and no ischial callosities.
- Upright or semi-upright stance.
- Same dental formula $\frac{2\,1\,2\,3}{2\,1\,2\,3}$ with broad incisors.
- Prominent face and prognathous jaw.
- Arms shorter than legs.
- Only the orangutan has an arboreal lifestyle.
- Sexual dimorphism with males larger than females.
- Large brain relative to body size.

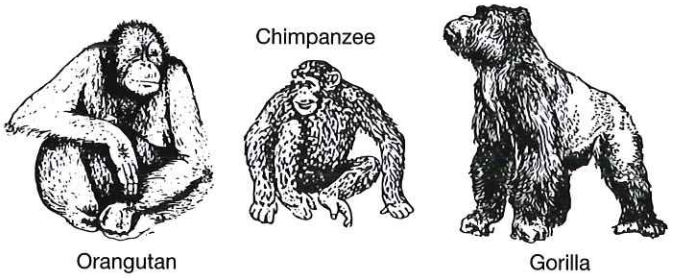



Figure 46.2 Members of the Hominidae family.

## QUESTIONS

1. What is the Hominidae?
2. What ending is given by convention for the classification taxon 'family'?
3. How has the term 'hominid' changed over time?
4. Why does classification change?
5. Outline some new technologies that are being used to determine the classification of the Hominidae.
6. The diagram shows the dental formula for a New World monkey.

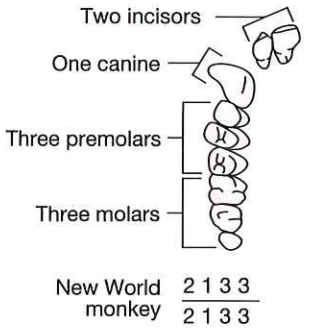



Figure 46.3 Dental formula for New World monkey.

How do the teeth of New World monkeys differ from the hominids?

7. List the main features of the hominids.
8. Hominids have flattened nails and humans have broader fingertips than other primates. Outline a benefit of flattened nails.

# 47 The Hominins

The **hominins** refers to the human branch of the evolutionary tree. It includes bipedal humans and extinct ancestors and close relatives, e.g. other species in the genus *Homo* and the australopithecines.

Evidence suggests that the earliest hominins were small in size, had brains similar in size to modern apes and had a set of characteristics that allowed both upright bipedal walking and climbing.

The earliest possible hominin, *Sahelanthropus tchadensis*, lived 7 to 6 million years ago though its position on the evolutionary tree is debated. Fossils of *S. tchadensis*, e.g. skull, jawbones and teeth found in Chad in 2002 show several hominin features, e.g. flatter face than apes and position of the foramen magnum, the opening for the spinal cord is oval, downward pointing and near the front of the skull suggesting an erect spine.

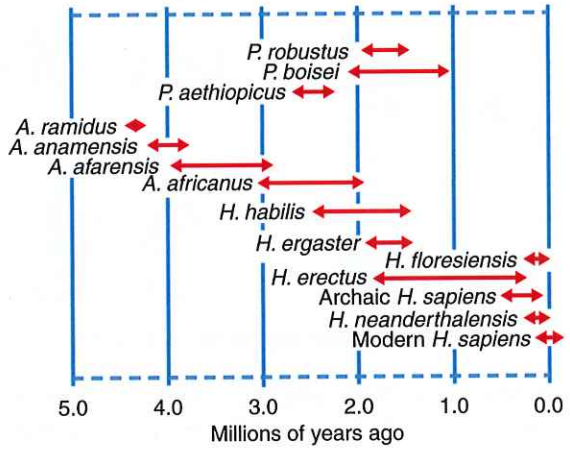



Figure 47.1 Timeline for some hominins.

## Features of hominins

Hominins share derived characters with humans. **Derived characters** are features that evolved in the lineage leading up to a clade that sets the members of that clade apart from other individuals. Shared derived characters of hominins include:

- Reduced canine teeth.
- Relatively flat face.
- Bipedalism and more upright stance.

## Bipedalism

Bipedalism is advantageous in grassland and woodland ecosystems and requires several features dealing with the following.

- Position of the foramen magnum.
- Shape of pelvis.
- S shaped spine.
- Carrying angle of the femur.
- The foot.
- Shape of the rib cage.

**Position of the foramen magnum**, the entrance of the spine to the cranium, needs to be well forward for upright posture so that it is vertically beneath the top of the skull.

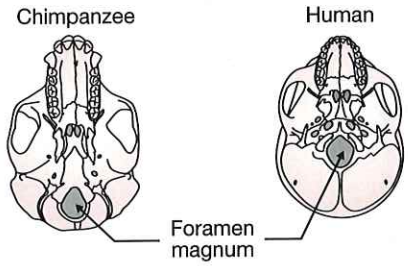



Figure 47.2 Foramen magnum is more forward in hominins than in apes such as the chimpanzee.

**Shape of pelvis** – the human pelvis is short and broad compared to the long narrow pelvis of apes. The shorter and bowl shaped pelvis is more stable for an upright stance. The change in shape allows muscle attachment for bipedal locomotion and there has been a consequent modification of the associated muscles, e.g. development of the human gluteus maximus muscle (flesh of the buttocks).

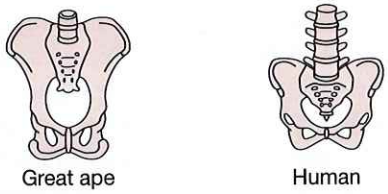



Figure 47.3 Shortened bowl shaped pelvis is suited for an upright stance and bipedal locomotion while longer pelvis is more suited for quadrapedal locomotion.

**S shaped spine** of humans allows the centre of gravity to fall over the hips while the C shaped spine of the chimpanzee has a forward centre of gravity. The S shape is able to withstand more stresses and function as a spring.

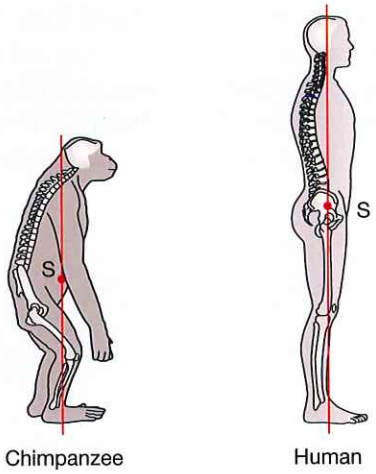



Figure 47.4 S shaped spine aids bipedalism rather than C shaped spine of apes.

**Carrying angle** of the femur, the thigh bone, is angled out from the knee so that weight distribution remains closer to the central axis giving more stability when walking upright. This changes the arrangement of the knee joint.

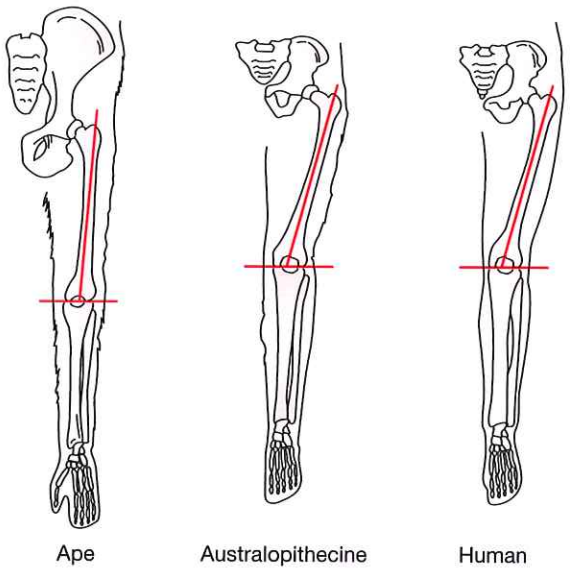



Figure 47.5 The structure and angle of the femur with the angle of the knee joint shows bipedalism.

**Foot** – the human big toe is not opposable and is rotated to be in line with the rest of the foot. The big toe is the last point of contact with the ground providing additional leverage when walking and giving the human footprint distinctive features. The human foot has two arches – a longitudinal arch and a transverse arch which help absorb shocks. The transverse arch is unique to humans.

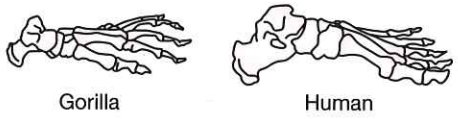



Figure 47.6 Human foot has two arches.

**Shape of rib cage** – the human head balances on the backbone and humans have a more barrel shaped chest so the torso can flex and the arms are free to swing and manipulate objects. Apes have powerful neck muscles with a cone shaped chest and a large gut below the diaphragm. They can reach above their heads for climbing and food gathering.

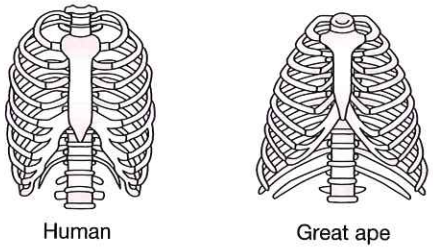



Figure 47.7 Rib cage.

## QUESTIONS

1. Who are the hominins?
2. Describe the earliest hominins.
3. Explain why some people believe *Sahelanthropus tchadensis*, was a hominin.
4. (a) What are derived characters?
   (b) Identify some shared derived characters of hominins.
5. (a) What is bipedalism?
   (b) Why is bipedalism an advantage in grassland and woodland ecosystems?
6. The diagram shows some suggestions why bipedalism was favoured in natural selection and evolution.

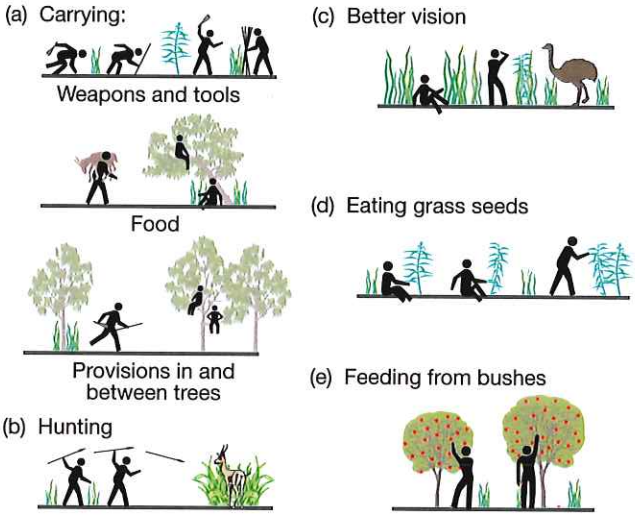



Figure 47.8 Selective pressures for bipedalism.

Construct a table to summarise these suggestions.

7. Construct a table to summarise the features that indicate bipedalism and explain why each feature is needed for bipedalism.
8. The diagram shows the top of the femur of three primates – a human, a chimpanzee and an australopithecine hominin.

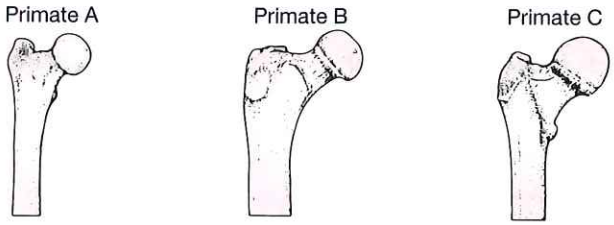



Figure 47.9 Upper section of the femur of three primates.

Identify each primate and explain why you made this choice.

9. Which of the following is *not* needed for bipedalism?
   (A) Large brain.
   (B) S shaped spine.
   (C) Short broad pelvis.
   (D) Foramen magnum towards front of skull.

# 48 The Australopithecines

Around 4 to 2 million years ago there was an increase in hominin diversity. Many of these hominins are classified as australopithecines. Australopithecines are bipedal with human-like hands and feet but have smaller brains (about one third of modern human).

The genus was named by **Raymond Dart**, an Australian scientist who was professor of anatomy at the Medical School of the University of Witwatersrand, South Africa. He discovered a fossil skull near Taung in 1924 which he named ***Australopithecus africanus*** (southern ape of Africa). Dart's fossil showed a small brain with rounded skull, a more forward position of the foramen magnum, and human-like teeth especially the first molars. Dart published his findings in *Nature* in 1925 causing much scepticism and disbelief.

Some of the criticism was due to the evidence of the **Piltdown Man** which was found in 1912 and later proved to be a forgery. The Piltdown Man forgery had a large brain and large protruding jaw inferring intelligence evolved before other features. This was the common belief at the time. *A. africanus* clearly showed that bipedal locomotion existed before the human brain reached its current proportions. In 1953, using new chemical evidence of fluorine dating, the Piltdown Man was shown to be a forgery with an ancient human skull and the jaw of a modern orangutan.

## *Ardipithecus ramidus*

Fossils of ***Ardipithecus ramidus*** were discovered in Aramis, Ethiopia in 1992 and dated at 4.4 mya. Fossils that have since been found include jaws, teeth, crania, and limb bones of over 100 individuals. One skeleton has 40% of an adult hominin with cranial and post cranial elements. The hominin is quite primitive with large canines; an ape-like base of the cranium; and the cranium is quite small. The original classification was *Australopithecus ramidus* but the fossils were reclassified as *Ardipithecus ramidus* in 1995 as there are sufficient differences for the fossils to be placed into their own genus, e.g. thin enamel caps on the molars compared to other later australopithecines and the brain size is quite small. Analyses of male and female teeth suggest that there was little difference in body size of males and females and there was low competition between males. This could mean food gathering and child care was done by both sexes in a family group.

## *Australopithecus anamensis*

In 1994 specimens where discovered in Kanapoi and Allia Bay and described by Leakey et al in 1995. Maeve Leakey and her colleagues assigned the name *Australopithecus anamensis*. Skull fragments show primitive teeth with thick enamel on the molars like other australopithecines. Fossils show a bipedal leg though the forearm and finger bone show arm weight bearing leading to debate about the extent of bipedalism. The fossils have been dated as 4.2 to 3.9 mya.

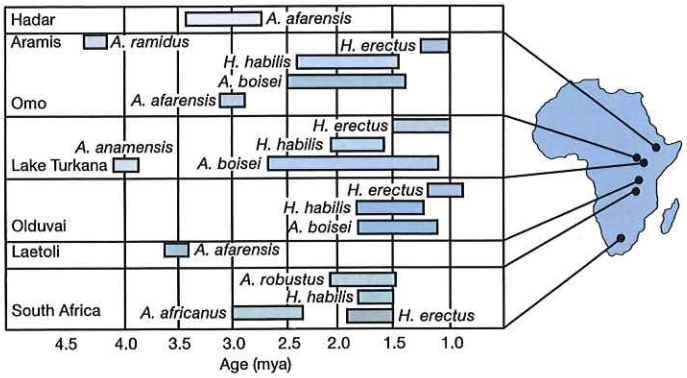
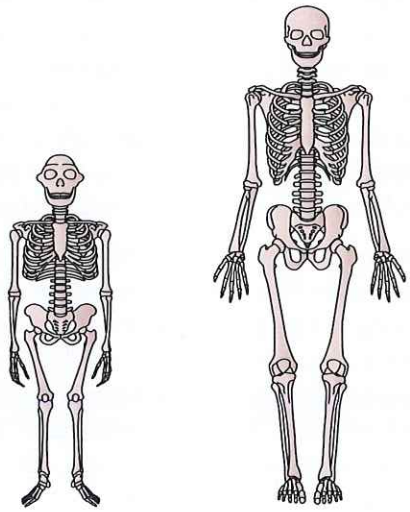



**Figure 48.1** Australopithecine fossils.

## *Australopithecus afarensis*

*A. afarensis* were small hominids with a small brain, ape-like face, bony ridge over the eyes, flat nose, large teeth (front and back) with large canines and back teeth with thick enamel, sexual dimorphism and a bipedal gait. They lived 4.2 to 3.0 mya. Most fossils have been found at Laetoli and Hadar in Eastern Africa. In Hadar in 1974 40% of a skeleton nicknamed '**Lucy**' was found and publicity about Lucy make *A. afarensis* one of the best known hominin. Lucy was small, about one metre tall with a head the size of a baseball and lived 3 mya. Lucy was bipedal as shown by her pelvis which is shorter and broader than that of an ape although the angle is not yet fully human. Ape-like features include long arms and short legs, and sexual dimorphism. The teeth show features of both apes and humans.



(a) *A. afarensis* (b) Modern human

**Figure 48.2** Comparing modern human with *A. afarensis*.

In 1978, Mary Leakey announced fossilised footprints at Laetoli, in rocks more than 3.5 mya, including a long trail, approximately 30 metres long, made by at least two, probably three, hominids walking with bipedal steps. This is very important evidence as it shows the fully bipedal locomotion as well as the stature of the hominids, approximately 1.4 metres tall.

## *Australopithecus bahrelghazali*

In 1995 a fossil hominid was found in Chad, Central Africa dated 3.5 to 3.0 mya. The find is important as it is 2500 km west of the Rift Valley and the previously established range of early hominids and lived at roughly the same time as Lucy. It closely resembles *A. afarensis* in size and proportions and more fossils will need to show if it is a separate species or a regional type of *A. afarensis*.

## *Australopithecus africanus*

First found in 1924 by Dart, *A. africanus* has relatively large teeth compared to humans, especially molars and premolars. *A. africanus* is slighter larger than *A. afarensis* and brain size was probably also slightly larger. Fossils come from Southern Africa and pelvic bones, femur and foot bones show *A. africanus* was bipedal.

## Gracile and robust australopithecines

Australopithecines have only been verified as being found in Africa and are conventionally separated into two groups – the gracile and the robust. Some people put the robust forms into their own genera ***Paranthropus*** (alongside man) which then creates *P. boisei* from Eastern Africa, *P. robustus* from Southern Africa and *P. aethiopicus* from Kenya and Ethiopia. In this classification system the gracile forms keep the name *Australopithecus*, e.g. *A. africanus* from Southern Africa and *A. afarensis* from Eastern Africa.

The robust forms are heavier and larger toothed and do not appear until 2.6 mya. The robust forms follow a separate evolutionary path to humans and disappear around 1.0 mya with later forms living at the same time as *Homo erectus* and *Homo habilis*. The gracile forms are lighter and smaller toothed and appear before 2.6 mya. The dates of the gracile and robust forms influence the construction of evolutionary trees.

The body size and average weight of robust and gracile australopithecines are relatively similar and overlap with most differences being in their face and dentition. Differences also are due to sexual dimorphism. *A. africanus* has a more dish shaped face compared to the vertical form in the robust. The gracile forms have larger front teeth compared to the back while the robust forms have huge back teeth and small incisors and canines which infers different diets.

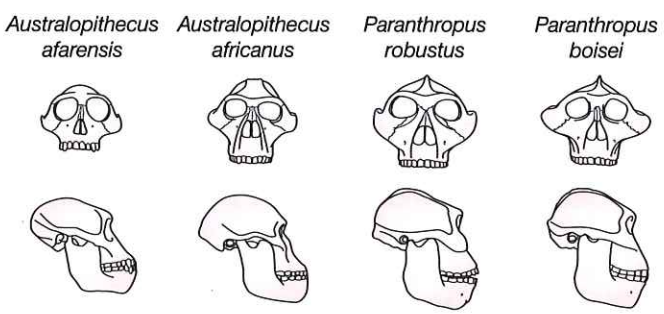



Figure 48.3 Skulls of different australopithecines.

The different robust forms show a difference in height with *P. boisei* taller but not heavier than *P. robustus* and the teeth suggest a diet processing large amounts of tough, fibrous vegetation, e.g. seeds and nuts while the gracile forms primarily ate leaves and fleshy fruits. *P. boisei* is a more robust extreme than *P. robustus* with larger sagittal and nuchal crests on the skull, larger brow ridges and a large jaw.

## *Paranthropus robustus*

In 1938, Robert Broom announced the discovery of a new genus he called *Paranthropus robustus*. Specimens have been found in Southern Africa, e.g. Swartkrans *P. robustus* is greater in size than *A. africanus* with a larger and more robust skull and teeth. Brow ridges are well developed, a crest of bone runs along the top of the skull, the forehead is low with a broad massive face that is flat or dished. Diet would have been tough, coarse food that required prolonged chewing. *P. robustus* lived 2 to 1 mya.

## *Paranthropus boisei*

*Paranthropus boisei* (originally *Zinjanthropus boisei)* was first discovered by Mary Leakey in 1959 at Olduvai Gorge, Eastern Africa and given the name 'Nutcracker Man'. It is more robust than *P. robustus*, with massive face and cheek teeth and lived 2 to 1.6 mya. Living at the same time as *H. erectus*, an accomplished hunter has led to the suggestion that the robust forms might have been prey and hunted to extinction by early *Homo* forms.

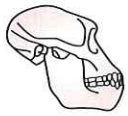
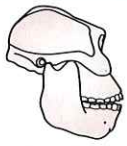
## *Paranthropus aethiopicus*

In 1985, Alan Walker discovered the '**Black Skull**' in Turkana dated 2.6 mya. It is one of the oldest robust forms with a small cranial capacity of 410 cc and other primitive traits, like *A. afarensis*, such as a joined crest in the back of the skull, the upper face projects considerably and the upper dental row converges at the back. It also has robust features such as a broad face, a very large palate and a large area for back teeth. There is debate over the evolutionary pathway with some suggesting the similarity to *A. afarensis* means that *A. africanus* is not its ancestor.

## QUESTIONS

1. Who were the australopithecines?
2. Who named the genus *Australopithecus*?
3. Explain why the find of the Piltdown Man was believed at the time and led to disbelief when Dart found the first australopithecine fossils.
4. The table shows two robust and two gracile forms of the australopithecines – *A. africanus, A. afarensis, P. robustus* and *P. boisei*.

**Table 48.1** Skulls of different australopithecines.

| Species | Front view of skull | Side view of skull |
|---|---|---|
| Species A | |  |
| Species B | | |
| Species C | |  |
| Species D | | |

Identify each species and explain why you made each choice.

5. Sexual dimorphism occurs when there are differences between the secondary sex characteristics of males and females and has been related to social organisation. The table shows the comparative body mass of some australopithecines that lived in grassland or woodland ecosystems.

**Table 48.2** Sexual dimorphism in some gracile and robust australopithecines.

| Australopithecine | Body mass (kg) | |
|---|---|---|
| | Female | Male |
| *A. afarensis* | 29 | 45 |
| *A. africanus* | 30 | 41 |
| *P. robustus* | 32 | 40 |
| *P. boisei* | 34 | 49 |

(a) Which of these species shows the greatest sexual dimorphism?

(b) Discuss how sexual dimorphism could be used to infer different social organisation patterns in the different species of australopithecines.

6. The relative size, shape, wear patterns and enamel of teeth combined with the jaw size and shape and musculature are used to infer prehistoric diets. What can you infer from the following data?

(a) *Paranthropus boisei* had a massive, robust jaw with very large chewing muscles attached to the skull. The molar teeth were extremely large and flat with very thick enamel but the incisors and canines were very small.

(b) *A. afarensis* had small pointed canines larger than human canines with wide front incisors and ape-like premolars with bumps on the chewing surfaces. They have a gap (diastema) between the canine and the premolars.

7. The diagram shows the dental arch of an ape, an *Australopithecus afarensis* and a human.

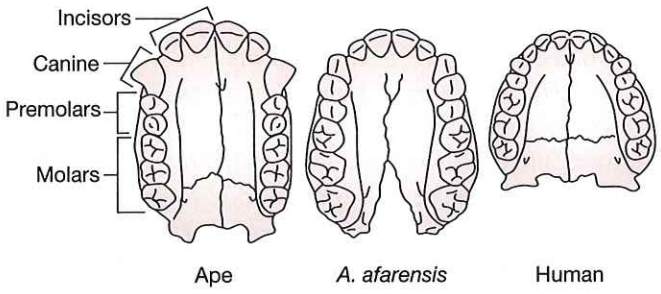



**Figure 48.4** Dental arch of ape, A. afarensis and human.

Compare the *A. afarensis* dental arch to humans and apes.

8. Explain why the robust forms of australopithecines cannot be on the human lineage branch of the evolutionary tree.

9. In 2008 a nine year old boy in Malapa, South Africa found the collarbone of an immature australopithecine that was dated 1.95 to 1.78 mya. Investigation at the site by his palaeoanthropologist father Lee Berger uncovered the partial skeletons of a juvenile male and an adult female. The skeletons show small body size, long upper limbs, curved finger bones, narrow face with modest brow ridge, pronounced lower jaw and small premolars and molars. The thighbone is strong and indicates bipedal locomotion. Sexual dimorphism is similar to modern humans. Suggest why these fossils were classified as a separate species – *Australopithecus sediba*.

10. Potassium/argon (K/Ar) dating of volcanic material from Olduvai Gorge was used by Louis Leakey to date the fossil (nicknamed Zinj) classified at that time as *Zinjanthropus boisei* (it is now classified as *Paranthropus boisei*). Potassium-40 is a radioisotope with a half-life of 1.25 billion years and this was the first time that K/Ar dating was used at a hominin site. Explain why a date of 1.75 mya for Zinj was a significant surprise for researchers and the public.

11. In 1974 Mary Leakey found fossilised footsteps at Laetoli that were between rock layers of volcanic ash dated 3.6 mya. There were tracks of many animals, e.g. giraffes and elephants as well as the hominins that were made in the wet ashfall. Explain why the footsteps, attributed to *Australopithecus afarensis* are an important discovery and the subject of debate.

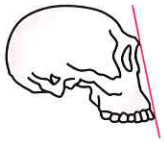
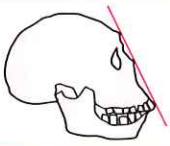
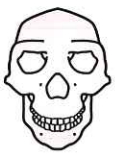
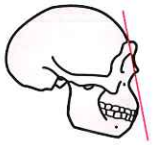
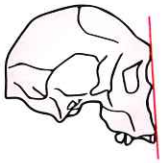
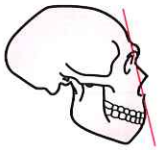
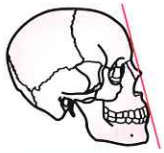
# 49 Genus – *Homo*

The genus *Homo* includes modern humans and many close ancestors. The earliest member of the genus *Homo* in the current fossil record is *Homo habilis* – the 'handy man' and other members include *H. erectus, H. ergaster, H. heidelbergensis, H. georgicus, H. floresiensis, H. neanderthalensis* and *H. sapiens*.

The genus evolved in Africa with only one living (extant) species – *Homo sapiens* – modern humans. There is great variation in phenotypic features of humans. **Polymorphic features** include adult height, skin colour, body shape, hair and eye colour, blood groups, nose and cheekbone structure, proportional finger length and fingerprint. **Sexual dimorphism** occurs when there are differences between the sexes beside the sexual organs, e.g. secondary sexual features, size and height.

It is not easy to identify the features that mark the genus *Homo* as many features are behavioural, which leave little fossil evidence.

Table 49.1 Some members of genus *Homo*.

| Species | Front view of skull | Side view of skull |
|---|---|---|
| *Homo habilis* | |  |
| *Homo ergaster* | |  |
| *Homo erectus* |  |  |
| *Homo heidelbergensis* | |  |
| *Homo neanderthalensis* | |  |
| *Homo sapiens* | |  |

## Features of *Homo*

Species in the genus *Homo* have the following features.

**Larger brains** – though this must be measured relative to body size. Most *Homo* species have a brain size larger than 700 cc with a large cranial vault above the face to hold the brain with increased cranial vault thickness.

**Prominent nose** – with raised bone surfaces around the nasal opening and a bony spine at the centre of the base of the nasal opening.

**Smaller teeth** – especially the cheek teeth and tooth eruption is slow.

**Flatter faces** – with thin bone of lower jaw and there may be a chin.

**Short base of skull** – can be folded and related to speech.

**Late sexual maturity** – with increased parental care.

## Appearance of genus *Homo*

The genus *Homo* is believed to have begun around 2.8 mya and modern humans *Homo sapiens* appeared around 200 000 years ago. Recent suggestions link a changing palaeoclimate to the survival, variation and spread of *Homo* from Africa to Europe and Asia around 1.8 mya. This suggestion proposes that the unstable climatic conditions, e.g. wet and dry seasons favoured adaptability rather than particular adaptations for particular conditions.

Analysis of skulls of different species in the genus *Homo* suggests that there were significant differences in facial features and that different species had distinct body shapes and forms.

## QUESTIONS

1. Who belong to the genus *Homo*?
2. Where did the genus originate and where are extant species living?
3. List some phenotypic features that vary in *H. sapiens*.
4. Identify some features that are used to identify the genus *Homo*.
5. Define polymorphism.
6. What is sexual dimorphism?
7. Identify the main features of the genus *Homo*.
8. When did the genus *Homo* first appear?
9. What is the earliest member of the genus *Homo*?
10. When is it believed that *H. sapiens* first appeared?
11. Which of the following appeared first in the fossil record?
    (A) *Homo habilis.*
    (B) *Homo erectus.*
    (C) *Homo ergaster.*
    (D) *Homo heidelbergensis.*

# 50 *Homo habilis*

*Homo habilis* ('handy man') was named by Louis Leakey in 1964 from fossils found at Olduvai Gorge dating 1.75 mya. Leakey named the fossil 'handy man' as stone tools were found near the remains. Fossil finds date *H. habilis* from about 2.3 to 1.5 mya.

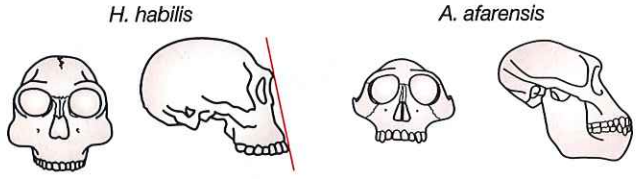



**Figure 50.1** Comparing *Homo habilis* and *Australopithecus afarensis.*

Compared to the australopithecines *H. habilis* has a larger brain, about 700 cc, smaller teeth, stood more erect and was taller. *H. habilis* was around 120 cm tall, with a light build and weighed around 40 kg.

## Classification of *H. habilis*

The classification of *H. habilis* caused controversy when the name was announced and the debate still continues. Some people feel *H. habilis* remains are too ape-like and show an advanced australopithecine while others feel it was an early *Homo erectus*.

Between 1969 and 1976 more *H. habilis* were found at Koobi Fora as well as simple tools, dating 2.0 and 1.6 mya. Like most hominin fossil finds, the fossils from Koobi Fora caused problems. Skull 1470, found in 1972 was dated at 1.9 mya and had a large brain, 750 cc, with a more human braincase shape but has the australopithecine broad cheekbones. While Skull 1813, found in 1973, was dated 1.8 mya, had a small brain, 510 cc, but looks more 'human' than Skull 1470. Skull 1805 has teeth like Skull 1813, a face like Skull 1470 and a braincase of 580 cc with a primitive crest along the midline like robust australopithecines.

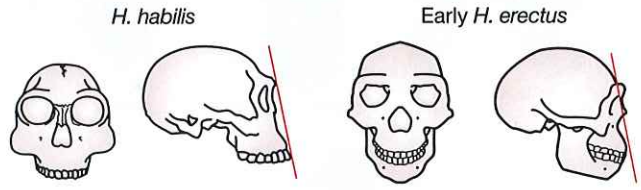



**Figure 50.2** Comparing *Homo habilis* and early *Homo erectus.*

This great variety in the fossils, with younger fossils having smaller brains, could mean that there were other human-like inhabitants in Eastern Africa between 2.4 to 1.6 mya.

In 1991, Bernard Wood suggested that the finds from Olduvai should remain *H. habilis* while the fossils from Koobi Fora should be split into two different species *H. habilis* and *H. rudolfensis*. Other suggestions soon followed and the debate continues.

## *H. habilis* lifestyle

The construction of tools to gather food shows a change in lifestyle. The earliest tools found at Olduvai, East Africa have been given the name Oldowan culture and relate to simple pebble tools such as stone flakes used as cutters and choppers to cut up a carcass and crush bones to get the marrow. A hammerstone is used to strike a core to remove flakes and rock fragments to create a chopping or cutting edge.

Habitat was mainly grassland with the climate becoming drier and colder. In the Pleistocene epoch (2.6 mya to 11 800 years ago) there were repeated glaciations. They were probably forager scavengers eating roots, berries and meat. They built shelters and lived in groups.

## QUESTIONS

1. Who named *Homo habilis*?
2. Explain why *H. habilis* was called the 'handy man'.
3. Compare *H. habilis* with australopithecines.
4. Discuss why there is controversy over the classification of *H. habilis*.
5. What classification was suggested by Bernard Wood?
6. The diagram compares *H. habilis* and *H. rudolfensis*.

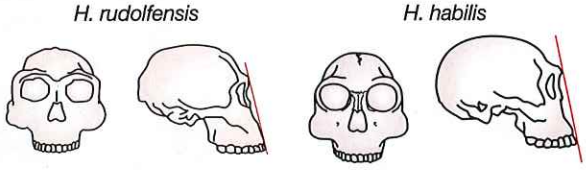



**Figure 50.3** Comparing *Homo habilis* and *Homo rudolfensis.*

Describe some of the differences between the skulls of *H. habilis* and *H. rudolfensis.*

7. In 2010 fossil bones with cut marks were found in Dikika, Ethiopia. The distinctive stone tool cut marks show that flesh had been removed from the bones and parts of the bone had been smashed to remove marrow, though no tools were found with the remains. The bones are dated 3.4 mya. The bones were most likely left behind by *A. afarensis* who lived at this time. Discuss why this was an important discovery and shows how knowledge of the human evolutionary tree is constantly being revised.
8. Why was *Homo habilis* given this name?
   (A) The bones were found at Habila, Ethiopia.
   (B) The expedition sponsor was Tony Habila.
   (C) The finger bones are long and handy.
   (D) Tools were found near the remains.

# 51 *Homo ergaster*

*Homo ergaster* is the 'working man' as many stone tools have been found with fossil remains. They manufactured hand axes which shows a certain level of thought and application using bifacial knapping of a stone nodule. This culture is known as the Acheulian stone tool industry and is also associated with *H. erectus*.

They had a human-like body shape with similar limb proportions and height which would have been quite different to the ape-like ancestors. The face projects forward but the lower jaw was receding with no external chin. *H. ergaster* probably walked and ran in a manner similar to modern humans and lived between 1.9 and 1.5 mya. Fossils have been found in East Africa and South Africa.

## Turkana Boy

The first specimen, found in Turkana, Kenya in 1971 was called 'Turkana Boy' and is a complete skeleton dated 1.5 mya. The fossil was the first to be classified as *Homo ergaster* as it is similar to modern humans with slender body frame, tall height, shorter arms and longer legs and was fully bipedal but had a modest brain size. Analysis of the bones of the skull which had not fully fused together and teeth place the boy between 7 and 15 years old when he died. He was around 160 cm tall and could have grown up to be 170 cm tall as an adult.

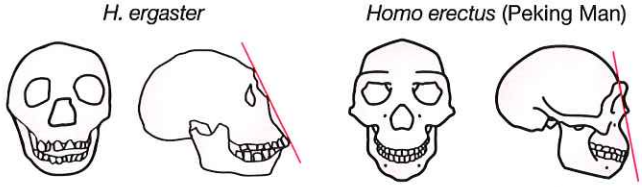



**Figure 51.1** *Homo ergaster* and *Homo erectus* (Peking Man).

## Classification of *H. ergaster*

In 1992, Colin Groves argued that there is insufficient differentiation between *H. ergaster* and *H. erectus* while Bernard Wood suggested that *H. ergaster* is the sister group of *H. sapiens* and is sufficiently different *to H. erectus*. He believes that the differences between the East African *H. erectus* and the Asian *H. erectus* should be made into separate species with the African forms being called *H. ergaster* so that the name *H. erectus* would be used solely for the Asian hominins. As with the australopithecines, the classification of *H. habilis*, *H. ergaster*, and *H. rudolfensis* has still not been completely resolved and the final determination will influence the construction of evolutionary trees. Each new find adds to the complexity of human lineage.

## Acheulian tools

*H. ergaster* and *H. erectus* are associated with the Acheulian tool industry which existed around 1.6 mya. Hand axes were made by bifacial knapping which involves hitting one side to flake off a piece and then turning the stone over to flake off another piece. This forms a hand axe tool with a sharp and fairly straight edge.

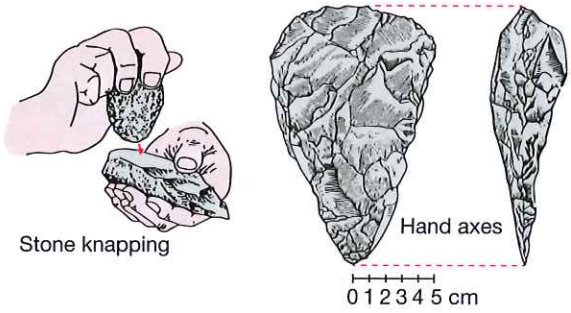



**Figure 51.2** Acheulian tool culture.

## QUESTIONS

1. How was *Homo ergaster* named?
2. Describe the appearance of *Homo ergaster.*
3. Where and when did *H. ergaster* live?
4. (a) Why is 'Turkana Boy' well known?
   (b) Describe the appearance of Turkana Boy.
5. Outline the debate about the classification of *H. ergaster*.
6. Describe stone knapping as used in Acheulian culture.
7. In a recent investigation neuroscientists recorded the brain activity of different humans as they made tools using stone knapping techniques that would have been used in Acheulian culture. The positron emission tomography (PET) scans showed brain activity in the premotor cortex during the tool making process. This area of the brain is also involved in understanding language. How could the results of this investigation affect our understanding of human evolution?
8. The diagram shows tools of the Oldowan culture as used by *Homo habilis*.

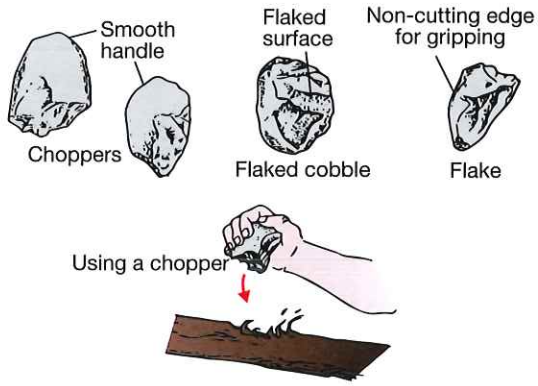



**Figure 51.3** Oldowan culture of *Homo habilis*.

Compare the tools of the Oldowan and Acheulian cultures.

# 52 *Homo erectus*

*H. erectus* meaning 'upright man' is the first hominin to be identified outside Africa and Europe with fossils found in Java and China. Many names were used in classifying these fossils, e.g. *Pithecanthropus* (Java remains) and *Sinanthropus* (Northern China remains). During the 1950s, it was decided to lump them all together into *Homo erectus*. However, there is still debate whether European and African fossils should be classified with the Asian fossils.

*H. erectus* existed between 1.8 mya and 250 000 years ago, though some place *H. erectus* as late as 30 000 years ago.

## Physical appearance

The many fossils of *H. erectus* from Asia show prominent brow ridges; protruding jaw although a less projecting face and smaller teeth than *H. habilis*; an increase in body size; sexually dimorphic (East African specimens); receding forehead; no chin and long, low, thick skull. Cranial capacity ranged from 750 to 1250 cc and they used tools. The incisors of some *H. erectus* have a scooped out appearance at the back – shovel shaped – which may be an adaptation in the primitive forms for processing foods in hunter gatherers. These shovel shaped incisors have been found in fossils from China and East Africa.

Adult *H. erectus* was about 1.6 to 1.8 metres tall and weighed 40 to 68 kg.

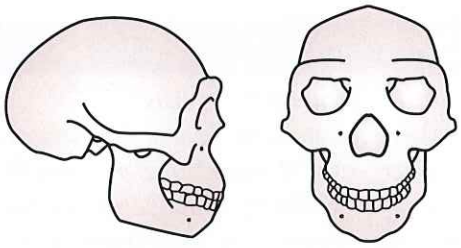


Figure 52.1 *Homo erectus.*

## *H. erectus* and the evolutionary tree

The origins of *H. erectus* are unclear with the oldest *H. erectus* fossils from East Africa dated at 1.8 mya and fossils from two sites in Java dated at 1.8 and 1.6 mya. The dates are important in understanding hominin development and in understanding the pattern and timing of human migration from Africa.

The fossil remains of *H. erectus* from different parts of the world are important with it being a long-lived, successful species.

Some evidence suggests that *H. erectus* changed over time, e.g. later *H. erectus* had a larger brain, more protrusive nose, the body was less robust and tool technology (*Acheulian*) became more sophisticated.

The development of hand axes and cleavers with a biface (worked both sides) with sharper and straighter edges were used for butchering and some believe this means *H. erectus* was a hunter while others believe he was a scavenger. The earliest Acheulian tools are dated at 1.4 mya, existed for over 1 million years and have never been found in Eastern Europe or East Asia.

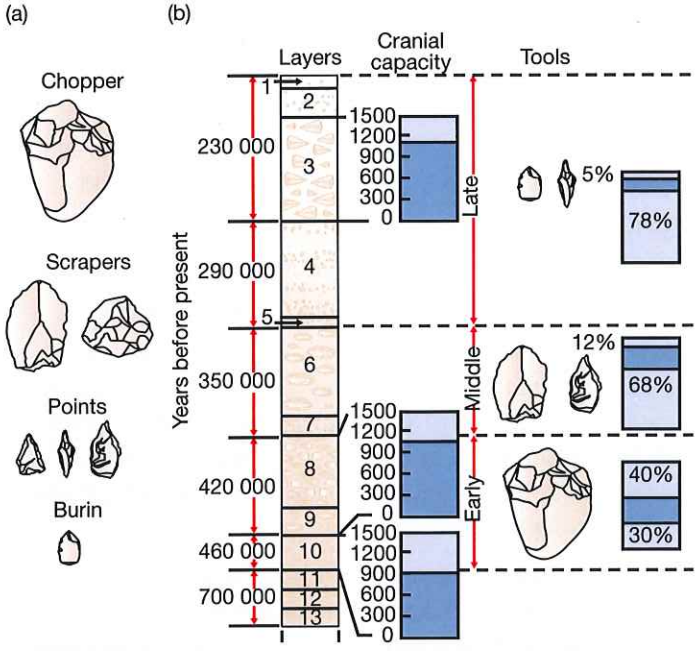



Figure 52.2 *Homo erectus* and Acheulian culture.

## QUESTIONS

1. How was *Homo erectus* named?
2. What other names have been given to fossils now classified as *H. erectus*?
3. When did *H. erectus* live?
4. Describe the physical appearance of *H. erectus*.
5. Why is *H. erectus* important?
6. What culture is associated with *H. erectus*?
7. The diagram shows four primate skulls.

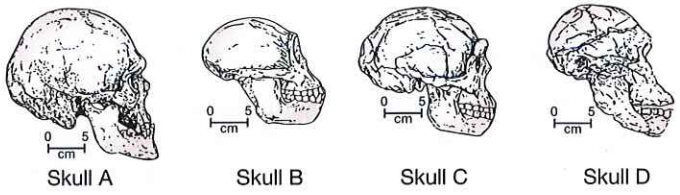



Figure 52.3 Four primate skulls.

(a) Place these skulls in order from youngest in geological time to the oldest.
(b) Describe two features you used in determining the order from youngest to oldest.
(c) Which one is likely to be *Homo erectus*? Explain your reasoning.

# 53 *Homo heidelbergensis* and *Homo floresiensis*

## *Homo heidelbergensis*

In 1907, Otto Schoetensack found the lower jaw of an early hominin south of Heidelberg, Germany. He placed it in the genus *Homo heidelbergensis* (Heidelberg Man) which caused immediate controversy. It has the receding chin and robust jaw of *H. erectus* but a larger brain (1100 to 1300 cc) with a taller braincase, more prominent face and nose, and smaller molars. The fossils are dated 500 000 years old.

It is believed *H. heidelbergensis* lived from 600 000 to 200 000 years ago with fossils found in East Africa, North Africa, South Africa and several parts of Europe.

Some people include these fossils with the archaic *Homo sapiens* while others believe they are the ancestor of the Neanderthals. Similar fossils have been found in Africa and across Europe. It is believed *H. heidelbergensis* is the last common ancestor of Neanderthals in Europe and modern humans in Africa.

*H. heidelbergensis* manufactured stone tools using the prepared-core technique in a manner similar to Neanderthals and early modern humans. Tools were made for specific functions, e.g. knapped hand axes and wooden spears for hunting and evidence suggests that tools were kept and reused rather than being thrown away after use.

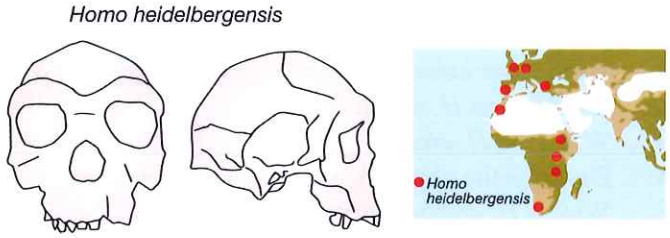



**Figure 53.1** *Homo heidelbergensis*.

## *Homo floresiensis*

Fossils of *H. floresiensis* were found in a cave on the island of Flores, Indonesia in 2003 and dated at 18 000 years old. It was a tiny hominin around 1.3 metres tall and weighing 25 kg. Digital analysis of the skull in 2009 showed that it had more similarities with *Homo* species that lived 1.5 mya than with modern humans. The cave also contained stone tools and the remains of a pygmy form of an extinct elephant called *Stegodon*.

Living in caves *Homo floresiensis* coexisted with modern humans living from 95 000 to 12 000 years ago.

Initially there was controversy over the finds, debating whether the remains belonged to a modern human with an underdeveloped brain and short stature, e.g. was microcephalic or was a small stature, dwarf sized human. Many questions also arose about how the species reached the island, their relationship to *H. erectus*, and their culture.

Investigators took measurements of many skulls of *H. floresiensis* and made comparisons with modern humans. The results indicated that the individuals belonged to a dwarfed sized but otherwise normal human.

Some researchers pointed out that the dwarfing process could have occurred early en route to Flores while others argued that many species of organisms on the islands tended towards dwarfism.

It was also suggested that a race consisting only of microcephalics could not have survived up to 95 000 years or 40 000 generations.

The presence of *H. floresiensis* until around 12 000 years ago raises questions about the evolution and extinction of the species. A massive volcanic eruption in the area around 12 000 years ago greatly affected the island of Flores but evidence in the ash layers suggests *H. floresiensis* survived this event and other eruptions.

Functional tools, many small and sophisticated with evidence of the use of fire suggests intellectual ability and manual dexterity.

It is possible the species was destroyed by modern humans.

## QUESTIONS

1. Describe *Homo heidelbergensis*.
2. What is believed about the relationship between *H. heidelbergensis, H. neanderthalensis* and *H. sapiens*?
3. When and where did *H. heidelbergensis* live?
4. Describe the culture of *H. heidelbergensis*.
5. The diagram shows one suggestion about the spread of different *Homo* species across the world.

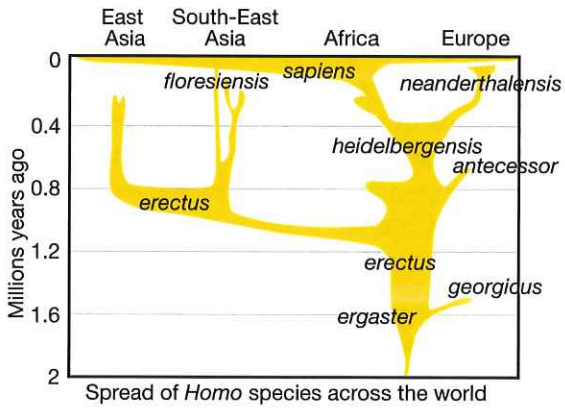



**Figure 53.2** Spread of *Homo* species around the world.

From this diagram when and where was the last common ancestor of *H. sapiens* and *H. floresiensis*?

# 54 *Homo neanderthalensis*

The first recognised Neanderthal fossils were found in 1856 in a cave in the Neander Valley, Germany. In 1908, at La Chapelle-aux-Saints, France, **Marcellin Boule** found the body of a 40 year old Neanderthal who was unusually robust and had osteoarthritis. Boule described the Neanderthal as brutish, bent over, bent kneed and this became the popular public image of the Neanderthal. The Neanderthals lived in Europe for around 300 000 years before the arrival of modern humans. Fossils have been found across Europe into Siberia and south-west Asia and they lived from 350 000 to 28 000 years ago.

## Physical appearance

Neanderthals had short stocky bodies (less than 170 cm tall); big faces; heavy brow ridges; large skulls with a distinctive 'bun' shape at the back; definite chin; cheekbones were swept back; and they had midfacial prognathism. The forward thrust of the face has been explained as an adaptation to cold, as a result of enlarged noses or as a response to increased anterior dental loads using the teeth as 'vice-like' tools.

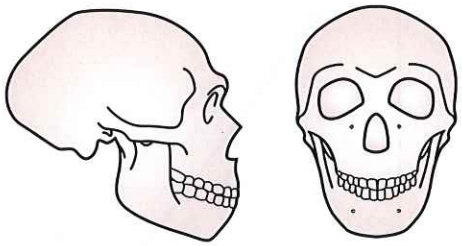


Figure 54.1 *Homo neanderthalensis*.

## Mousterian culture

The Neanderthals lived in Europe and Western Asia using a tool culture called **Mousterian**. They used a type of prepared core **stone tool knapping** trimming a piece of flint into a disc shaped core then chipping flakes from the edge thus collecting more flakes per core. The flakes were used as scrapers, points, knives and some were hafted to wooden shafts. The technique is known as the **Levallois** technique and appears to have started around 250 000 to 200 000 years ago. At this time there was a reduction in the number of large hand axes manufactured.

Neanderthals lived in open sites, rock shelters and caves and were successful hunters. Mousterian sites show a range of habitats and animal remains show the Neanderthals were seasonal hunters, e.g. killing reindeer in winter and red deer in summer. Head and neck injuries resemble rodeo performer injuries suggesting close contact with hoofed, clawed or taloned animals as well as care by more able members of the group for the wounded.

The Neanderthals in Eurasia deliberately **buried their dead,** e.g. at La Chapelle and Amud with the body positioned into a flexed posture. Burial first occurred around 100 000 years ago and infers a type of formal ritual as well as a possible religious belief in life after death. The presence in the graves of tools, animal bones and flowers infers symbolic behaviour but has been questioned due to poor excavation documentation.

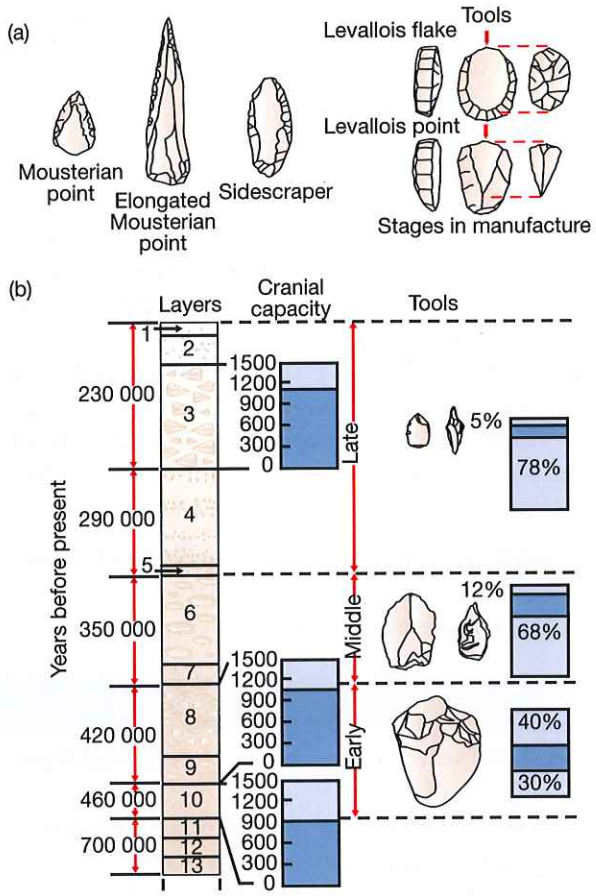



Figure 54.2 *Homo neanderthalensis* culture.

## Classification

There are many interpretations of these fossils with respect to their relationship to modern humans so that some have placed them as a subspecies *Homo sapiens neanderthalensis*; others have classified them as their own species *H. neanderthalensis*; and others have classified them as a form of archaic *H. sapiens*.

In 1997, **Svante Paabo** used genetic material extracted from the bones of the original Neanderthal specimen found in 1856 and tested the mtDNA. He reported that the Neanderthals could not be the ancestors of modern humans, they did not interbreed and the Neanderthal lineage is four times older than the human one meaning they split off much earlier than humans. Paabo found that there are 27 differences on one strand of Neanderthal DNA while the maximum genetic variation for the entire human race along the same strand has eight mutations.

Neanderthal nuclear DNA suggests they had red hair, though a different colour red to modern human red hair and pale skin. The pale skin would aid vitamin D synthesis.

The Neanderthals lived in areas at the same time as *H. sapiens* with an overlap of about 6000 years coexistence. There is currently great debate about the relationship of the Neanderthals and modern humans. Genome comparisons suggest divergence occurred 440 000 years ago but there is between 1 and 4 per cent of the DNA of modern Europeans and Asians, but not Africans, that comes from Neanderthals. This infers there was some assimilation with interbreeding between archaic populations.

## The Denisovans

The **Denisovans** were identified in 2010 from bones in the Denisova cave in Siberia with DNA analyses showing archaic human features, closely related to Neanderthals. Fossils from Spain show the Denisovans also lived in Western Europe. DNA analysis of a bone thought to be *H. heidelbergensis* or a Neanderthal was closer to the Denisovans than the other two groups. The Denisovans were neither Neanderthal nor modern humans and have some genetic similarities with current Melanesians.

Further study is needed before any absolute conclusions can be drawn about the relationship between the Neanderthals, Denisovans and *H. sapiens*.

## QUESTIONS

1. When and where were the first Neanderthal fossils found?
2. Why did people for many years always think of Neanderthals as brutish and bent over?
3. Where and when did Neanderthals live?
4. Describe the physical appearance of Neanderthals.
5. Which culture is associated with the Neanderthals?
6. Describe how Neanderthals made stone tools.
7. What types of habitats were occupied by Neanderthals?
8. What evidence suggests that Neanderthals had close contact with large animals?
9. More than a dozen Neanderthal grave sites show ritualised burial ranging from a simple pit to stone lined chambers. Discuss what ritualised burial indicates about *H. neanderthalensis*.

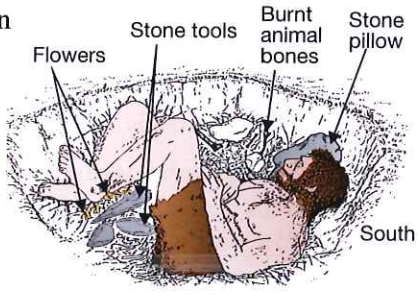



Figure 54.3 *Homo neanderthalensis* burial pit.

10. Name some ways that the Neanderthals have been classified.
11. Outline how the work of Svante Paabo helped distinguish *H. neanderthalensis* and *H. sapiens*.
12. How has nuclear DNA provided information about the Neanderthals?
13. What was the overlap in time of *H. sapiens* and *H. neanderthalensis*?
14. What is the current debate about the relationship of Neanderthals and modern humans?
15. Who were the Denisovans?
16. Why is it difficult to classify the Denisovans?
17. Use the Denisovans to explain why modern technologies have been able to dramatically increase our understanding of hominin relationships leading to modifications of existing theories of the human evolutionary tree.
18. The graph shows the relationship between the distance from the equator and the average human skin pigmentation.

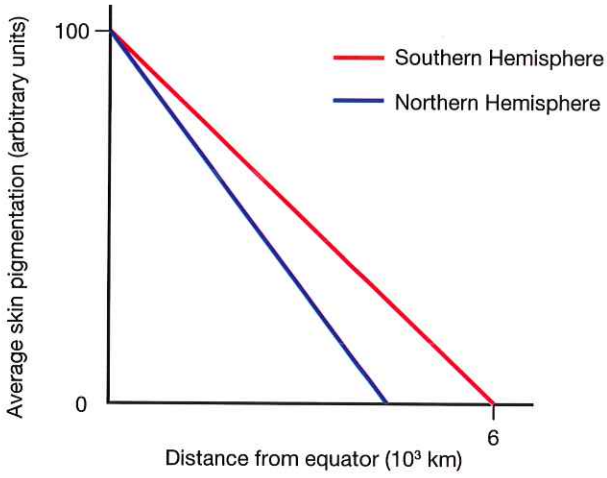



Figure 54.4 Distance from the equator and average skin pigmentation.

How does the information in this graph relate to our current knowledge about the skin colour of the Neanderthals?

19. The diagram shows the skulls of five primates – a chimpanzee, an australopithecine, *H. erectus*, *H. neanderthalensis* and *H. sapiens*.

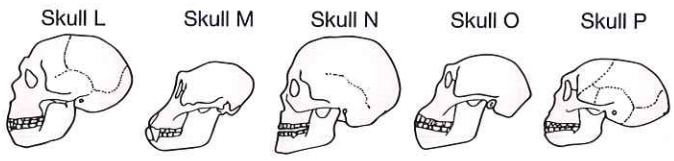



Figure 54.5 Five primates.

(a) Identify each skull.
(b) Explain how you identified *H. neanderthalensis*.

# 55 *Homo sapiens*

*Homo sapiens* are the only extant species of the genus *Homo*. Earliest modern *H. sapiens* probably came out of Africa, however there is dispute about the date(s) of this migration. The first people in Western Europe to be considered modern *H. sapiens* are called **Cro-Magnons** named after the site in France where eight individuals were found in 1868.

## Physical appearance

Modern humans are distinct from Neanderthals as they have a sharply rising forehead; less prominent brow ridges; prominent chin; face is not forward thrusting; more rounded skulls shorter from front to back; shorter lower jaws and a taller less robust skeleton. The rib cage is barrel shaped and the hands are dextrous.

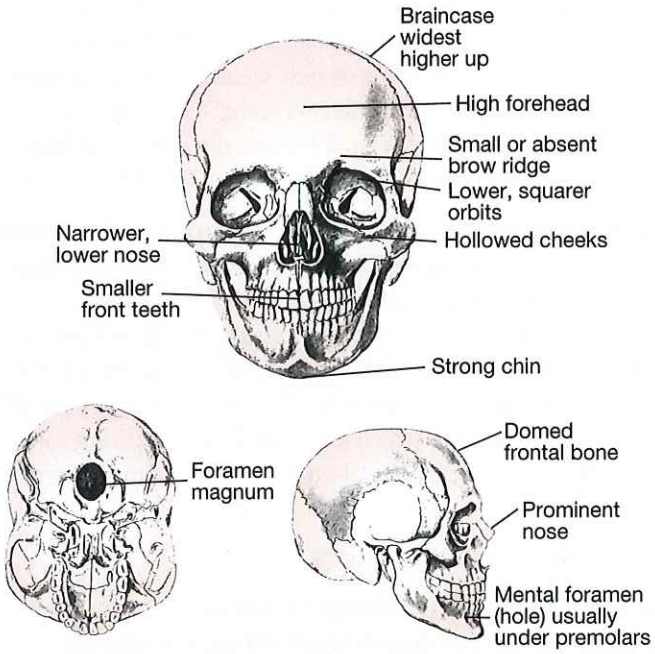



**Figure 55.1** *Homo sapiens*.

## Migration

Fossils of *H. sapiens* have been found from all around the world. The oldest African fossils of *H. sapiens* are dated 200 000 years old. In Israel modern skulls were found at Mt Carmel dated at 115 000 years ago and the Qafzeh Cave fossils are dated 100 000 years old. In China, modern humans were found in the Upper Cave at Zhoukoudian dated 18 000 to 10 000 years old. In 1984, in Jinniushan, China, a partial modern skeleton is dated at 200 000 years old. In Australia, modern human fossils at Lake Mungo are dated at 30 000 years and the first evidence of cremation at 25 000 years. Australian archaeological sites have been dated 55 000 years.

## Cultural developments

*H. sapiens* introduced a new tool industry making blade tools, e.g. for scraping, cutting, chiselling, and piercing. Palaeontologists have divided the time frame into the various stages of tool development. Early *H. sapiens* were hunter gatherers using not only stone tools but using bone, ivory and antler producing fish hooks and needles. Tools became more sophisticated and Cro-Magnons began engravings and sculptures. Burials were more elaborate with offerings in graves. By 30 000 years ago grave bodies were being decorated with necklaces and surrounded by clay figurines, bone tools and stone tools. Palaeolithic art was produced with engravings on stone and bone, e.g. cave drawings in Chauvet Cave, France dated 30 000 years ago. Small figurines called 'Venuses' have also been found across Europe.

**Figure 55.2** Representations of early art and clay figurines.

After the Stone Age there have been more recent cultural developments of *H. sapiens* including the agricultural revolution, the Bronze Age, the Iron Age, the Industrial Revolution and even more recently the technological revolution.

## QUESTIONS

1. What is the only extant species of the genus *Homo*?
2. How were the Cro-Magnons named?
3. Describe the physical appearance of *H. sapiens*.
4. What is the date of the oldest African *H. sapiens* fossils?
5. When studying the skull of *Homo sapiens*, identify a feature that shows an advancement compared to other *Homo* species and explain how this feature aided the evolution of modern humans.
6. A hominin skull W was found with some artefacts and dated at 100 000 years old. At another location a hominin skull Y was found with artefacts and dated at 20 000 years old. What artefacts were most likely discovered with skull Y but not skull W?
   (A) Pigments used in rock paintings.
   (B) Animal bones with cut marks.
   (C) Stone tools for cutting.
   (D) Remains of fire cooking.

# 56 Dating Hominin Fossils

Once fossils have been found they need to be dated using either relative dating or absolute dating techniques.

## Relative dating

Relative dating uses the **law of superposition** which states that the oldest layers are found at the bottom and the youngest layers are found at the top, unless there has been dynamic movement of the strata, e.g. by folding and/or faulting. This law gives the **relative age** of a fossil or a rock layer as being younger or older than another layer. Relative dating uses index fossils. An **index fossil** is widely distributed but only lived for a short time which means it clearly indicates the age of the rock layer.

The **geological time scale** was worked out using the relative ages of different rock layers dividing time since the formation of the Earth into five eras. The time frame for each division is not equal as the divisions were based on the appearance of specific fossils. Each era is subdivided into periods and the periods subdivided into epochs.

## Fluorine analysis

Fluorine analysis can only be used on bone and is another method of relative dating. When bone is in the ground, fluoride ions from the soil water replace some of the ions in the bone. The longer the bone is in the ground, the more fluorine present. Absolute age can only be determined if the numerical age of another item from the same area is known, otherwise the age is either 'younger than' or 'older than' the other object. The process is limited as not all bones absorb fluorine at the same rate so that bones of similar density must be compared and some natural variables may not have remained constant, e.g. rainfall, temperature and the bones being compared need to have been exposed to the same changing conditions. Bones from the same area need to be compared due to fluctuations of fluoride in ground water.

## Absolute dating

Absolute dating give the age in **years**. Radioactive isotopes (radioisotopes) are used to determine the age of rocks or fossils giving the actual age in years plus or minus the experimental error involved in the particular method used. Each radioactive isotope decays at a fixed, known rate called its **half-life**. The half-life is the time it takes for half the atoms to decay and if the half-life is plotted onto a graph, the decay forms an exponential curve with certain radioisotopes being useful for a certain time frame. Useful radioisotope dating methods include radiocarbon dating, potassium-argon dating, rubidium-strontium dating and uranium-235 dating.

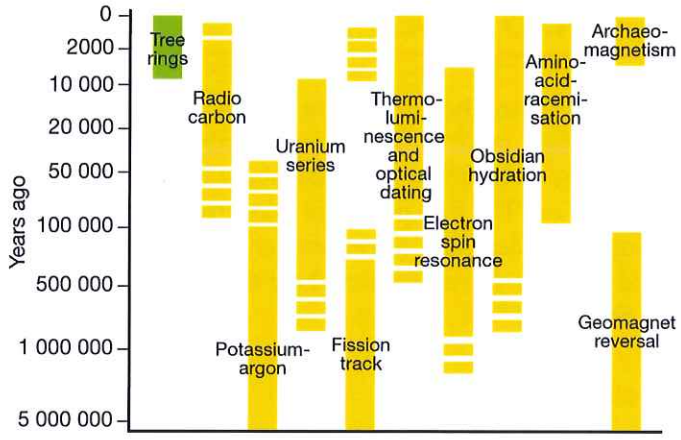



Figure 56.1 Dating techniques.

## Carbon-14

Radiocarbon dating is important in dating organic material, e.g. bone, shell, charcoal. All living things contain carbon in a known ratio and when the organism dies, the carbon-14 decays and the ratio changes. The half-life of carbon-14 is 5730 ± 30 years. A new technique using particle accelerators as mass spectrometers means it is possible to measure the number of carbon-14 atoms directly rather than waiting for their decay allowing dating up to 70 000 years ago.

Radiocarbon dating is also used on materials that are used by humans, for example, linen, paper, and cave paintings which have pigments from honey, milk, or oil seed. When it was discovered that the ratio in the atmosphere of one atom of carbon-14 to every million million ($10^{12}$) carbon-12 atoms is not a totally fixed ratio an analysis of growth rings on trees was used to make corrections for fluctuations in the carbon-14 content of the atmosphere.

## Potassium-argon dating

The naturally occurring radioactive isotope potassium-40 decays to form argon-40 and calcium-40. Potassium-40 has a half-life of 1.25 billion years. Potassium-argon dating is suitable for volcanic rocks and minerals older than 200 000 to 300 000 years which means it is limited to a small range of rocks past a certain age. However, it has been important in understanding primate fossil records. For example, many of the hominid fossils in East Africa are in layers associated with volcanism and sedimentation and this method is used to date the volcanic ash deposits.

## Fission tracking

Fission tracking dating is based on the spontaneous fission of uranium-238 and can be used on minerals and glass in a range from 100 years to 4500 million years before the present. Volcanic zircons have no tracks when they form and as the uranium undergoes fission the number of tracks increases with time at a rate that depends on the uranium content.

A measurement of the density of the tracks and the amount of uranium can give an accurate estimate of the age of the rock. It was used to date the rocks in the Olduvai Gorge in the Great Rift Valley in East Africa at sites where remains of australopithecines and *Homo erectus* have been found.

## Palaeodiet and stable isotopes

Carbon has two stable isotopes $^{12}C$ and $^{13}C$ and nitrogen has two stable isotopes $^{14}N$ and $^{15}N$. The ratios of stable carbon and nitrogen isotopes in human tissues show a relationship with diet. For example, bone collagen mainly comes from dietary protein and an analysis of samples from Neanderthals and early humans shows palaeodiet (food eaten in ancient times), e.g. it is estimated that Neanderthals ate around 80% meat and 20% vegetables.

Differences in carbon isotopes in tooth enamel can be used to provide information about diet.

## Strontium and oxygen isotopes

Ratios of strontium isotopes ($^{87}Sr/^{86}Sr$) vary with local geology and oxygen isotopes ($^{18}O/^{16}O$) in rainwater vary with local climate. Strontium and oxygen in food and drinking water become part of bones and teeth and thus the mineral component of bones and teeth can indicate if the hominin has migrated from its place of birth to its place of death.

## Carbon dioxide and temperature changes

The Pleistocene lasted from 1 800 000 to 10 000 years ago during which time there were many continental glaciers. Studies of gas bubbles from Antarctica glacial ice cores show changing levels of atmospheric carbon dioxide.

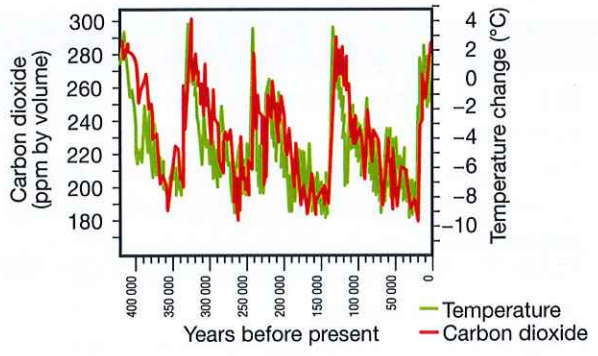



Figure 56.2 Changes in temperature and carbon dioxide levels in the last 400 000 years.

Climate changes affect the type and distribution of different ecosystems and the migration of hominins. Correlations can be made between dates, climate and the selective pressures acting on the different hominins, e.g. the Neanderthals were adapted to cold conditions and lived with mammoths, cave bears, cave lions and woolly rhinos. During ice ages sea level is low and land bridges allow greater migration, e.g. Indonesian islands were occupied by *H. erectus* in the early Pleistocene (1.5 mya) and Aboriginal Australians reached Australia at least 50 000 years ago.

## QUESTIONS

1. Distinguish between relative dating and absolute dating.
2. What is an index fossil?
3. What is the geological time scale?
4. Construct a table to summarise fluorine absorption dating including how absolute age is determined, processes involved in bone accumulating fluorine and limitations of the method.
5. Identify hominin dates that were determined using fission tracking.
6. The table shows geological time for the Cainozoic era.

Table 56.1 Geological time.

| Era | Period | Epoch | Years ago | Life on Earth |
|---|---|---|---|---|
| Cainozoic | Quaternary | Holocene | 12 000 | *H. sapiens* |
| | | Pleistocene | 18 000 000 | *H. ergaster, H. erectus, H. heidelbergensis, H. neanderthalensis, H. floresiensis, H. sapiens* |
| | Tertiary | Pliocene | 6 000 000 | *Ardipithecus ramidus, Australopithecus afarensis, Australopithecus africanus, Paranthropus boisei* |
| | | Miocene | 23 000 000 | Apes flourish, *Pliopithecus, Dryopithecus, Sivapithecus* |
| | | Oligocene | 34 000 000 | Monkeys, apes, mammal families evolve |
| | | Eocene | 54 000 000 | Prosimians, possible early monkeys |
| | | Palaeocene | 65 000 000 | Earliest primates |

In geological time, when have hominins existed?

7. What is meant by the half-life of a radioisotope?
8. Name some useful radioisotopes.
9. Explain why carbon-14 is a very useful radioisotope.
10. What is the oldest age that can be dated using current radiocarbon dating techniques?
11. Explain why adjustments had to be made about radiocarbon data.
12. Outline a use of potassium-argon dating.
13. What is fission tracking?
14. How can stable isotopes provide information about palaeodiet?
15. What information can be gained from strontium and oxygen isotopes?
16. Explain how dating climate changes provides information about hominin activities.

# 57 Models for Human Evolutionary Trees

There are several models for human evolutionary trees using current data. The human fossil record is incomplete and is open to different interpretations.

As new discoveries increase our knowledge, e.g. finding the remains of *H. floresiensis* on the island of Flores in Indonesia current theories need to be refined or replaced. In many instances the new theories are contested and there is debate about placement and divergence on the evolutionary tree.

Current evolutionary trees are produced using both anatomical and molecular evidence.

## Fossil evidence

The fossil evidence with both where it was found and with other remains and artefacts provides information about the environment, habitat, lifestyle, diet and use of tools.

**Digital calipers** are used to measure dimensions of bones to compare features of different fossils to help determine the range of measurements for each acknowledged species. **Scanning electron microscopes** (SEM) are used to study surface features of bones, e.g. microwear patterns on teeth to help determine diets of different species and cut marks left by tools. SEM scans of cut sections of bones and teeth can show the age at death and identify ancient diseases. **CT (computerised tomography) scans** are used to make internal images of fossils and can be transformed into solid replicas by **stereolithography**, e.g. to make whole skull reconstructions.

## Molecular evidence

DNA extracted from bones provides several kinds of information that is used to construct human evolutionary trees. Three types of DNA are studied.

**Nuclear DNA** – contains genome for the individual, e.g. studies of CD4 locus on chromosome 12 which suggests non-African populations descended from ancestors who left Africa about 90 000 years ago and studies of the beta-globin gene on chromosome 11 which suggests Asians have been evolving distinctive patterns for 200 000 years.

**Y chromosome DNA** – looks at amino acid sequences on the Y chromosome which determines male sex in humans to show inheritance along the male line without inheritance from two parents.

**Mitochondrial DNA** (mtDNA) which is found in mitochondria in the cytoplasm and is inherited from the egg and through the female line.

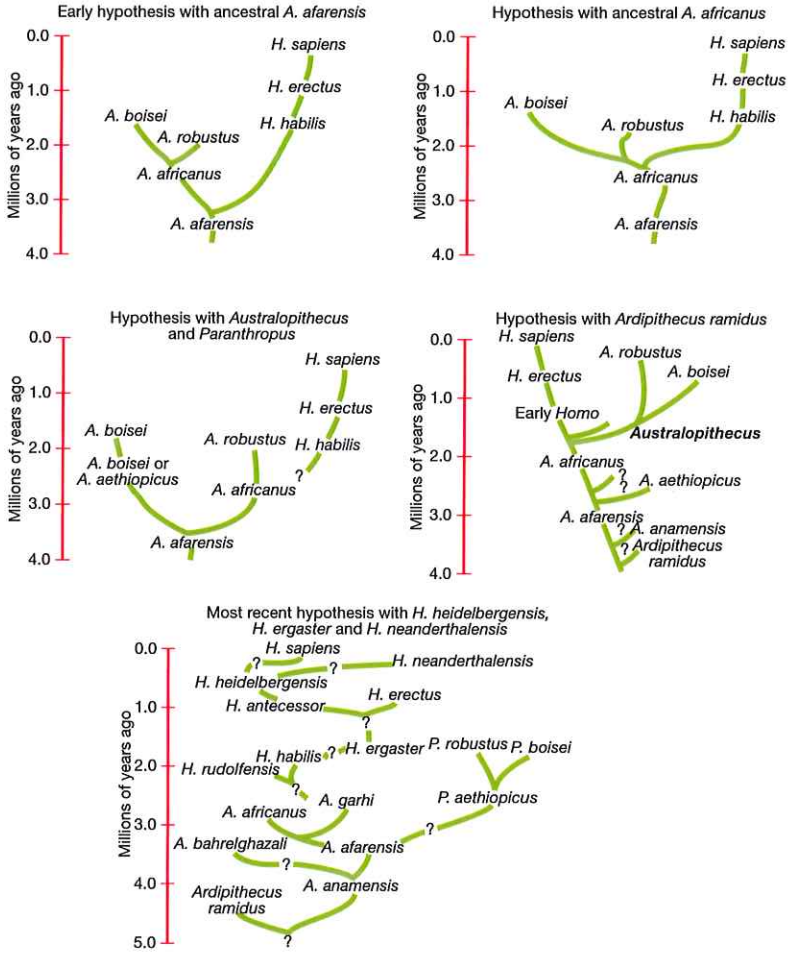



Figure 57.1 Possible evolutionary trees.

Studies of mtDNA of human populations around the world suggest that there was a common ancestor about 150 000 years ago in Africa (African Eve) and studies are investigating human migration patterns to each continent.

## QUESTIONS

1. Explain why there can be several models for human evolution.
2. Use an example to show why models of human evolutionary trees need to be constantly reviewed and revised.
3. Construct a table to summarise different ways fossil remains can be studied.
4. Construct a table to summarise the different types of DNA used in molecular studies of human evolutionary trees.
5. Studies of mtDNA in modern-day humans have shown that there is the highest level of genetic variation in mtDNA in African populations. What does this suggest?
   (A) African environments have high mutation rates.
   (B) *Homo sapiens* evolved in Africa.
   (C) Gene flow is limited in non-African areas.
   (D) Natural selection does not involve mtDNA.

# 58 Models for Evolution of *Homo sapiens*

There are two main theories to explain the migration of humans to a worldwide distribution – the theory of regional continuity and the 'out of Africa' model. The evidence for ancient migrations comes from fossil bones, artefacts, genetic material and the rock layers in which they are found.

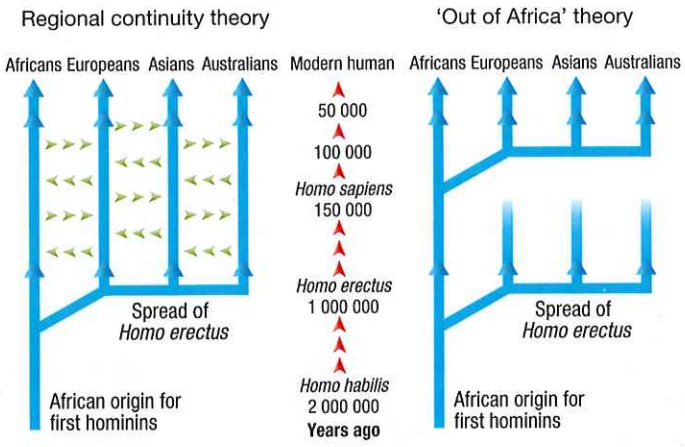



**Figure 58.1** Two models for human evolution.

## Theory of regional continuity

Sometimes this theory is called the **multiregional continuity model** and is most closely associated with Milford Wolpoff and Alan Thorne (Thorne and Wolpoff, 1992; Wolpoff et al, 1994). The regional continuity model explains the migration of *H. erectus* out of Africa into the different regions of the world where each region gradually evolved into modern populations with continued interbreeding and gene flow to allow the maintenance of one gene pool and one species. The migration of *H. erectus* occurred about 1 mya so that the change from archaic to modern *H. sapiens* occurred simultaneously around the world.

Supporting evidence for the regional continuity theory comes from the dating of fossil finds that are dated before the proposed second wave from Africa, e.g. the Jinniushan skeleton, dated at 200 000 years old suggests modern humans were in China well before the replacement model allows and the strong brow ridge found in Java Man is a common feature in modern populations in the area. mtDNA evidence of 'Mungo Man' in Australia dated around 60 000 years old shows the fossil is not descended from 'mitochondrial Eve' although the fossil is anatomically identical to modern *H. sapiens*.

Proponents of the theory of regional continuity feel that these features show a single species spreading worldwide and changing over time, as a whole, due to interbreeding and cultural exchange.

## 'Out of Africa' model

This model suggests that there have been more than one migration out of Africa with a recent common ancestral population migrating out of Africa with the features of modern humans. Mitochondrial DNA (mtDNA) studies support the 'out of Africa' model dating the migration of *H. sapiens* around 100 000 years ago and the studies also infer that all present-day humans have descended from this group. The 'out of Africa' model is supported by Christopher Stringer and Peter Andrews. Supporting evidence comes from the studies of mtDNA of all modern humans which some trace back to an 'African Eve'. The oldest fossil of *H. sapiens* has been found in Africa and the greatest variety in mtDNA which infers the greatest number of female lineages occurs in Africa.

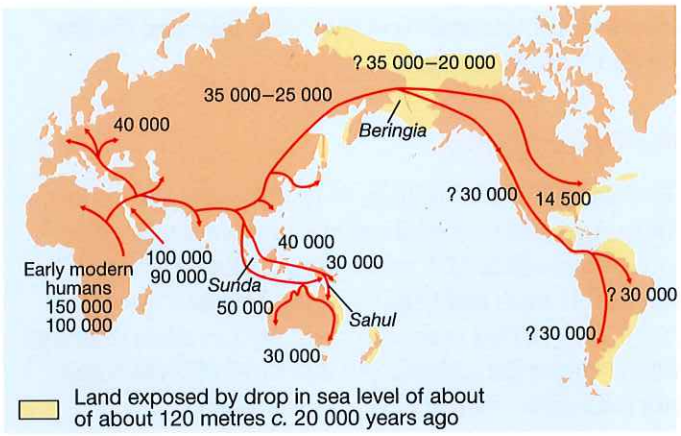



**Figure 58.2** Migration routes.

## Intermediate theories

Some people believe there is an intermediate version that is a combination of the above two theories and there are several ideas on how this intermediate version occurred.

Fred Smith and Erik Trinkaus suggest that Africa was predominant and modern features spread gradually from Africa with interbreeding between adjacent populations. This accounts for gene flow with the Neanderthals. Another intermediate version has continuity in some regions outside Africa, e.g. the Neanderthals became extinct but the descendants of archaic people in China and Indonesia could have interbred and evolved in modern people. Testing intermediate theories and identifying gene flow is more difficult than contrasting the two main theories.

## QUESTIONS

1. What type of evidence is used to determine human migration worldwide?
2. Construct a table to compare the two theories about the migration of humans to a worldwide distribution.
3. What are the intermediate theories?

# 59 Development of Culture

Culture has been a significant feature of human development with humans changing from being efficient hunter gatherers to living in large cooperative groups in cities involved in job specialisation. From *Homo habilis* to *Homo sapiens* there is a gradual sophistication in the use of tools.

Communication systems and tool making have greatly influenced the ability of humans to change their environment so that modern technologies will have a profound impact on the lifestyles of future human populations. Human ingenuity has caused such changes that there has been a dramatic increase in population numbers in the last 300 years.

Historians divide prehistory into ages, e.g. Stone Age, Bronze Age, Iron Age and in the late 19th century the terms Palaeolithic and Neolithic were invented for the Old and New Stone ages.

## The Palaeolithic

The Palaeolithic (2.5 million to 12 000 years ago) is the 'Stone Age' and is often divided into three stages – the Lower Palaeolithic (2.5 million to 300 000 years ago) with pebble tools and hand axes, the Middle Palaeolithic (300 000 to 40 000 years ago) with the Levallois technique and the Upper Palaeolithic (40 000 to 10 000 years ago) with microlithic industry (very small tools).

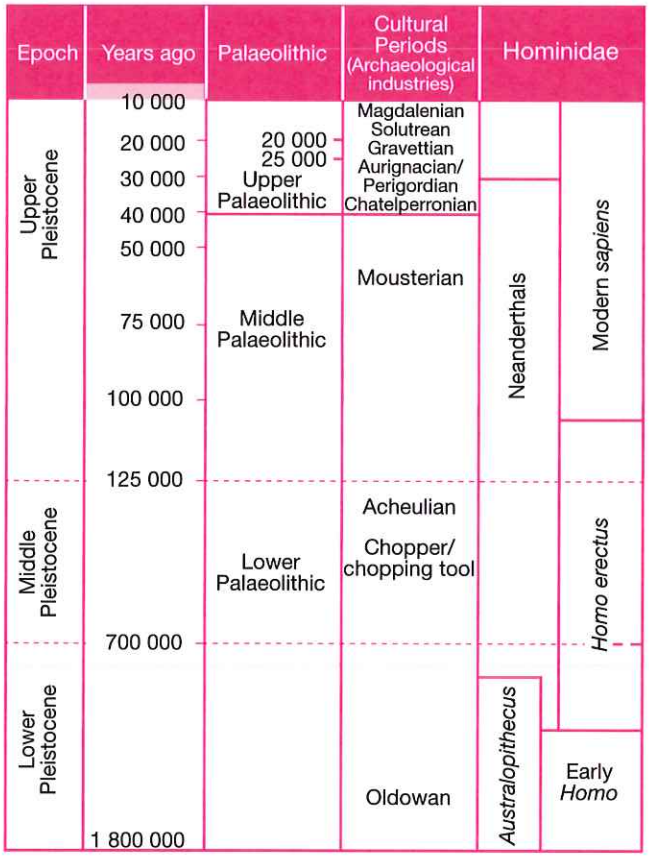



Figure 59.1 Hominin culture periods.

## Sequence of cultural development

The **australopithecines** are not credited with tool manufacture. *A. afarensis* probably lived in small family groups. It did not have a 'tool kit' but probably used tools like the modern chimpanzee. *Paranthropus boisei* had no known culture and tools found could belong to *Homo* found at the same site.

The first hominin identified as *Homo* is ***Homo habilis*** – the handy man – which shows the importance of tool making in our understanding of what it is to be human. The **Oldowan culture** used the earliest tools, e.g. pebble tools such as stone flakes used as cutters and choppers used for cutting meat, skinning animals, sawing wood or cutting plant material. They were hunter gatherers living on the grasslands of Africa with probable home bases.

Stone tools became more refined and advanced with the introduction of the flake industry. The **Acheulian culture** first appeared around 1.5 mya and is associated with *H. erectus* and *H. ergaster* with large hand axes made from a core by stone knapping. The hand axes were multipurpose tools used for chopping, digging up roots and bulbs, butchering carcasses and cutting bones. *H. erectus* used fire, hunting, had home bases and was the first hominin confirmed out of Africa. In 1948 Hallam Movius proposed the **Movius Line** dividing the Acheulian tool users of Europe and Africa from the chopping tool cultures of East Asia showing the most easterly extent of hand axes. In 2014 stone tools from north-eastern China dated 50 000 years old question the accuracy of the Movius Line.

*H. heidelbergensis* is associated with the Acheulian culture making tools with shaper edges. They were hunter gatherers and had home bases.

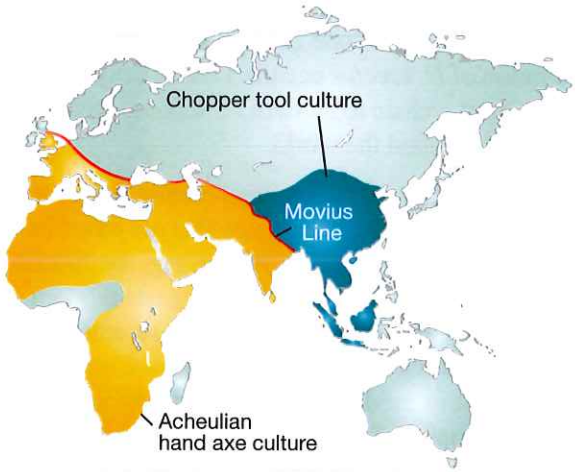



Figure 59.2 Distribution of choppers and hand axes.

The **Levallois technique** was introduced around 300 000 years ago and gave increased control over tool production. In this method the tool manufacturer mapped out the final shape of the flake that could be struck from the core with a single hit.

The **Mousterian culture** associated with *H. neanderthalensis* used stone knapping and the Levallois technique to make hand axes and other flake tools, e.g. to be used as scrapers, points, knives and some were hafted to wooden shafts. They lived in open sites, rock shelters and caves and were successful hunters, with close contact with hoofed, clawed or taloned animals as well as care by more able members of the group for the wounded.

At the beginning of the Upper Palaeozoic around 40 000 years ago a new blade industry and then the microlithic industry were introduced. Bone, antler and ivory were now used to make tools and composite tools using different materials were introduced, e.g. harpoons with detachable heads and spear throwers. Pigments were used as paint, e.g. on cave walls and on some bodies at burial and small figurines were made in ivory and moulded clay.

As camps became larger, more permanent and buildings more complex with stone lined fire hearths social organisation became more complex and sophisticated.

## Neolithic Revolution

Around 15 000 to 10 000 years ago the Neolithic Revolution (agricultural revolution) occurred with the development of agriculture and the change from nomadic to village life. The growing of crops and the domestication of animals led to the development of settlements and job specialisation.

## After the Neolithic

The discovery of metal technology caused the end of the Stone Age. Copper working began 8000 BCE and the Bronze Age (copper and tin) began about 2000 BCE.

By 5000 years ago the city state of Sumer existed which led to the development of writing, improved art and craft and an increase in the human population. The Egyptian, Greek and Roman civilisations flourished and declined, the Middle Ages began and natural events such as wars, famine, disease and drought influenced and limited human population growth until the Industrial Revolution which was the next major cultural change in human population history.

The Industrial Revolution began about 300 years ago with the invention of the steam engine by James Watt. The inventions of new machines led to the development of factories, larger cities, more people and new strains were placed on the Earth's resources. Investigations into disease, hygiene and medicine increased life expectancy, infant mortality rate dropped and people survived infectious diseases.

Improved transport such as ships, trains, planes increased the blending of cultures so that diet, clothing, language, music, art have become more cosmopolitan. Immigrants bring certain aspects of their native culture to their new land and these become incorporated into the culture of their adopted country. Increased population mobility has meant there is an increased rate of cultural development.

## QUESTIONS

1. (a) What is the Palaeolithic?
   (b) Describe the subdivisions of the Palaeolithic.
2. Construct a table to summarise the inferred culture of the hominins – *Australopithecus afarensis, Paranthropus boisei, H. habilis, H. erectus, H. heidelbergensis, H. neanderthalensis, H. sapiens.*
3. (a) What is the Movius Line?
   (b) How does the Movius Line show that models and interpretations of the human fossil record are often contested and need to be refined or replaced?
4. What is the Neolithic Revolution?
5. Outline what followed the Neolithic Revolution.
6. The diagram shows the Levallois technique.

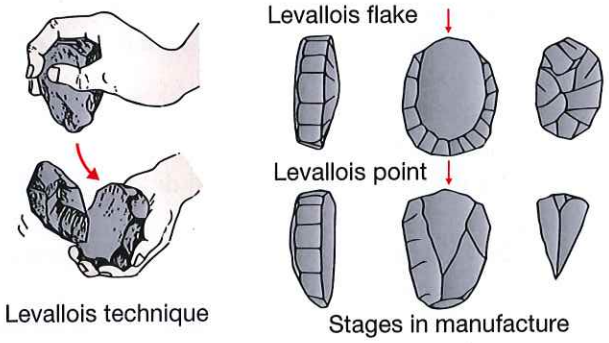



**Figure 59.3** Levallois technique.

Identify the hominin most associated with this stone working technique and describe the tools that were made.

7. Briefly assess the impact of cultural development on human evolution.
8. The graph shows the change in the human population over the last 10 000 years.

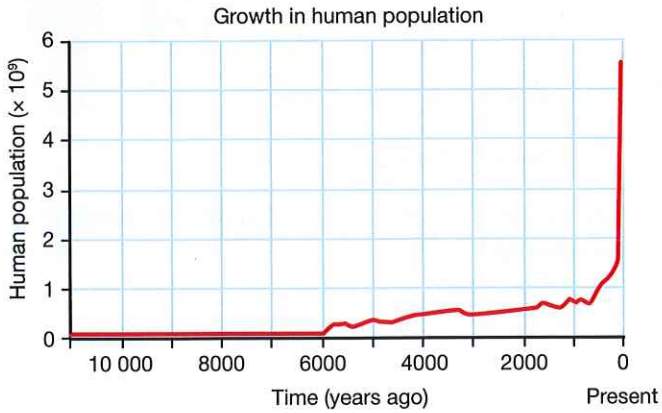



**Figure 59.4** Growth in human population.

Suggest why human population has changed this way in the last 10 000 years.

9. (a) Who is associated with Palaeolithic art?
   (b) Where is early art located?
   (c) What figurines are found in early art?

# Topic Test

## Section A – Multiple Choice (35 marks)

1. The order of bases in a section of DNA is TTAACG. When exposed to X-rays this section in some new cells was found to be TTACCG. What is the best term to describe what happened?
   (A) Replication error.
   (B) Block mutation.
   (C) Point mutation.
   (D) Deletion.
2. Why are mutations important?
   (A) They always introduce new favourable alleles.
   (B) They are the ultimate source of variation.
   (C) They occur in organisms that are not suited to their particular environment.
   (D) They help reduce population size as most mutations are lethal
3. Which of the following is the best description of a gene pool?
   (A) The frequency of all alleles from generation to the next in a population.
   (B) The total number of genes on all chromosomes in a particular population.
   (C) The genetic material found in all organisms within a certain community.
   (D) All copies of every type of allele at all loci in every individual in a population.
4. The diagram shows genotype frequency for a population over several generations.

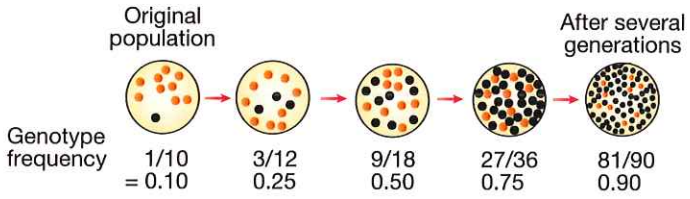



Figure TT.1 Genotype frequency.

What is the best term for the change over several generations?
   (A) Gene flow.
   (B) Founder effect.
   (C) Speciation.
   (D) Genetic drift.
5. Which of the following is *not* considered to be a mutagen?
   (A) X-rays.
   (B) UV rays.
   (C) Radio waves.
   (D) Gamma rays.
6. Tay-Sachs is a hereditary disease that occurs in 1 in every 2500 births in the Ashkenazi Jewish population while in other populations the incidence is about 1 in every 500 000 births.
   Which of the following could *not* explain the frequency of Tay-Sachs disease?
   (A) Heterozygotes are carriers and produce enough normal enzyme not to show disease symptoms.
   (B) It is a late onset disease so people have reproduced and already passed on the allele before symptoms appear.
   (C) Founder effect with original population having high incidence of gene which is maintained.
   (D) Genetic drift with small Jewish populations remaining isolated changing allele frequency.
7. What is pleiotropy?
   (A) The ability of a single gene to have multiple effects.
   (B) The ability of a cell to give rise to many different types of cells.
   (C) The addition of all or part of a chromosome, e.g. in humans there could be 45 or 47 chromosomes instead of the normal 46 chromosomes in the nucleus.
   (D) The lack of one chromosome from the normal number of chromosomes.
8. The diagram shows the change in phenotype as a population evolved over time.

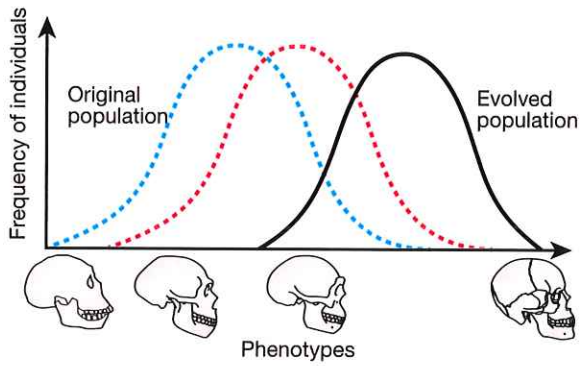



Figure TT.2 Change in phenotype over time.

What would be the best name for this type of change?
   (A) Convergent evolution.
   (B) Directional natural selection.
   (C) Stabilising natural selection.
   (D) Divergent evolution.
9. Which instrument would be most useful to see the earliest fossils of life on earth?
   (A) Light microscope.
   (B) Electron microscope.
   (C) Mass spectrometer.
   (D) Deep sea submersible.

10. The diagram shows two stratigraphic columns taken from nearby areas. Index fossils have been indicated in some of the layers.

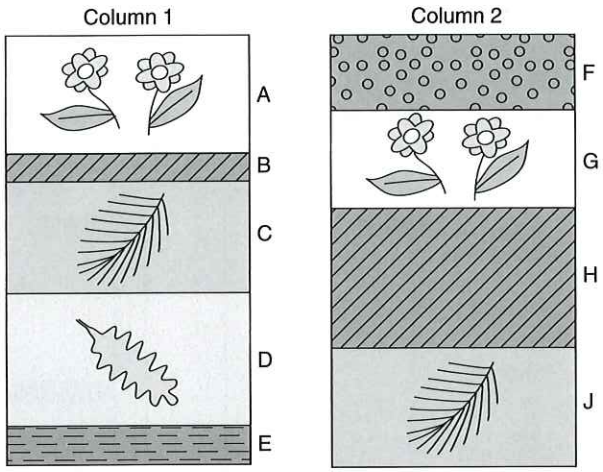



**Figure TT.3** Two stratigraphic columns.

What can be concluded from these columns?

(A) Column 2 location was more favourable to fossil formation than column 1 location.
(B) Layer C and layer J are the same age.
(C) Layer B is older than layer J.
(D) Layer E is the youngest layer in the two sequences.

11. One of the first discoveries of Neanderthals occurred in the Feldhofer Cave in the Neander Valley, Germany.

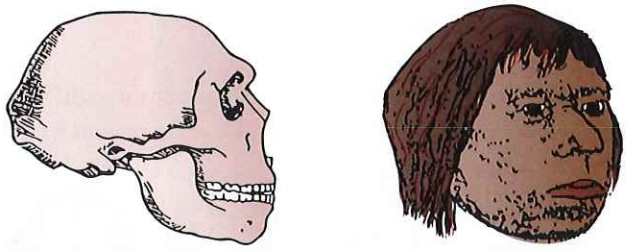


**Figure TT.4** Neanderthal man.

A later find in 1908 was a skeleton of an elderly Neanderthal man with chronic arthritis. This later find affected perceptions at the time about Neanderthals so that most people believed Neanderthals were:

(A) Hunched over and brutish in appearance.
(B) Tall and muscular with large intelligent brains.
(C) Capable of dangerous feats capturing large prey.
(D) Using language and symbols for communication.

12. What is an index fossil?

(A) A distinct organism that lived for a short time period over a wide area.
(B) A distinct organism that lived over a long time period in a restricted area.
(C) A large variety of closely related species with a worldwide distribution.
(D) Any trace or remain of an organism.

13. The diagram shows a blood smear from a person with a genetic disease.

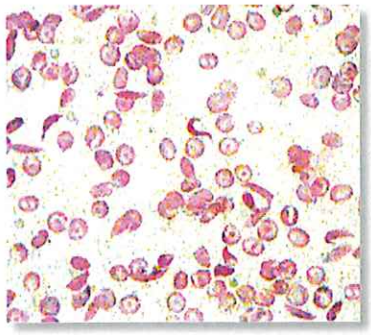


**Figure TT.5** Human blood smear under HP light microscope.

This disease is caused by an atypical haemoglobin called haemoglobin S. This abnormal haemoglobin differs from normal haemoglobin by only a single amino acid. What is this disease?

(A) Tay-Sachs disease.
(B) Thalassemia.
(C) Phenylketonuria.
(D) Sickle cell anaemia.

14. Which of the following provides evidence that supports the theory of evolution?

(A) Transgenic species.
(B) Punnett squares.
(C) Structural morphology.
(D) The DNA double helix.

15. The diagram shows steps in fossil formation.

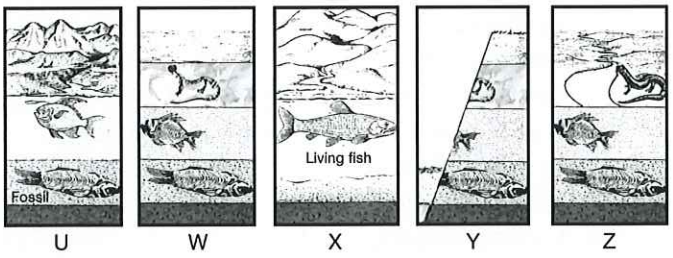



**Figure TT.6** Fossil formation.

What is the correct order of these diagrams from oldest to youngest?

(A) Y, Z, W, U, X
(B) X, Z, W, U, Y
(C) X, W, U, Z, Y
(D) X, U, W, Z, Y

16. Which of the following is the most likely outcome of gene flow?

(A) Reduction in genetic differences between populations over time.
(B) Increase in genetic differences between populations over time.
(C) Increase in the mutation rate within a population over time.
(D) Speciation with differences permanently separating populations.

Use the diagram which shows a type of primate locomotion for the next TWO questions.

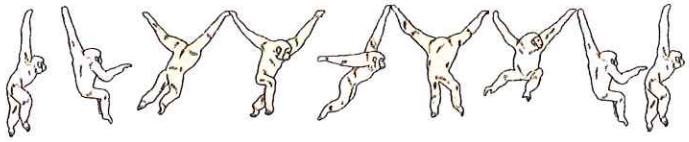


Figure TT.7 Type of primate locomotion.

17. What is the best name for this type of movement?
    (A) Brachiation.
    (B) Knuckle swinging.
    (C) Quadrapedal locomotion.
    (D) Arboreal climbing.
18. Which primates mainly use this type of locomotion?
    (A) Lemurs.
    (B) Baboons.
    (C) Gibbons.
    (D) Australopithecines.
19. Which of the following correctly lists the cultural advances of the hominin group in the most likely sequence of development from earliest to most recent?
    (A) Manufacture of simple stone tools, use of fire, painting, agriculture.
    (B) Manufacture of simple stone tools, painting, use of fire, agriculture.
    (C) Agriculture, manufacture of simple stone tools, painting, use of fire.
    (D) Agriculture, painting, use of fire, manufacture of simple stone tools.
20. A Neanderthal fossil was dated by DNA extracted from the bone at approximately 35 000 years old. Which of the following methods was most likely used to date this fossil?
    (A) Potassium-argon dating.
    (B) Carbon-14 dating.
    (C) Fluorine analysis.
    (D) Stratigraphy.
21. Although there are more than 60 species of potato and hundreds of different strains, the Irish potato famine of 1845 to 1850 caused the death of one million people. What important lesson should have been learnt from this example?
    (A) Genetic engineering is needed to create more transgenic species.
    (B) Biodiversity needs to be maintained in all crops.
    (C) Potato seed banks should store more potato seeds.
    (D) Farmers should have chosen a different species more suited to Irish conditions.
22. Which of the following can be used as an example of divergent evolution?
    (A) Wings of butterflies and bats.
    (B) Body shape of wombat and mole.
    (C) Sexual dimorphism in *P. robustus*.
    (D) Hominins and apes are primates.
23. The following diagram shows a possible relationship between some Australian marsupials.

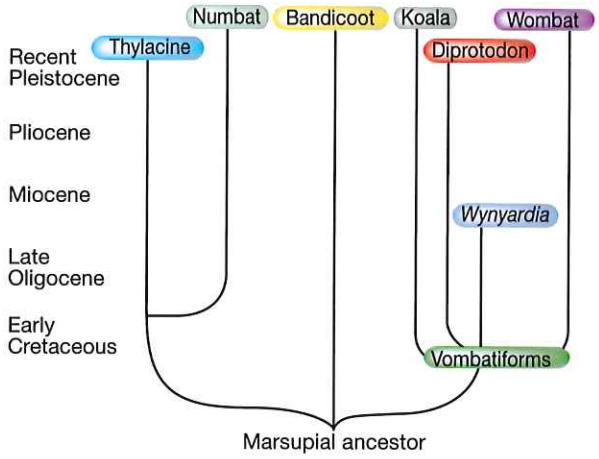



Figure TT.8 Possible evolutionary tree for some Australian marsupials.

Which extant animal is most closely related to the extinct thylacine?
(A) Numbat.
(B) Koala.
(C) Bandicoot.
(D) *Wynyardia*.

24. The diagram shows the lower jaws of three different primates – a gorilla, an australopithecine and a human.

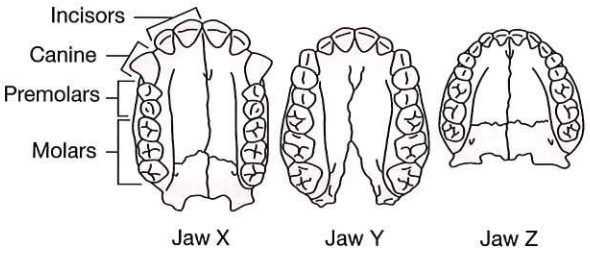



Figure TT.9 Primate jaws.

Which of the following would be the most likely match of jaw and species?

| | Gorilla | Australopithecine | Human |
|---|---|---|---|
| (A) | Jaw Z | Jaw Y | Jaw X |
| (B) | Jaw Z | Jaw X | Jaw Y |
| (C) | Jaw X | Jay Y | Jaw Z |
| (D) | Jaw Y | Jaw X | Jaw Z |

25. What is the name of the study of social behaviour based on evolutionary theory?
    (A) Sociobiology.
    (B) Anthropology.
    (C) Palaeontology.
    (D) Morphology.
26. In an investigation DNA and the enzyme DNA polymerase are placed together in a solution that is repeatedly heated and cooled in cycles to show the polymerase chain reaction (PCR). Which of the following is correct?
    (A) Cooling causes strands of DNA polymerase to denature.
    (B) Cooling binds the sugars to within DNA strand.
    (C) Heating breaks the bonds between the bases in the strands of DNA.
    (D) Heating provides the activation energy for the DNA polymerase.
27. Protein analysis can be used to determine when two species had a common ancestor. Which of the following would show species with the most recent common ancestor?
    Species with the most similar:
    (A) Number of amino acids in a protein.
    (B) Sequence of amino acids in a protein.
    (C) Shape of the protein molecule.
    (D) Sequence of nucleotides in a protein.
28. Plasmids are used to transfer selected genes from one species to another using enzymes at several steps in the process. What enzymes are needed to cut plasmids and to insert genes into the plasmid respectively?
    (A) Restriction enzyme, DNA ligase.
    (B) DNA ligase, restriction enzyme.
    (C) Restriction enzyme, DNA polymerase.
    (D) DNA polymerase, DNA ligase.

Use the diagram that shows some stone tools that distinguish a particular hominin culture for the next TWO questions.

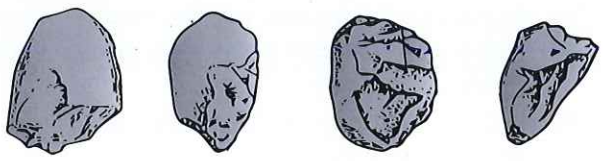


**Figure TT.10** Some stone tools.

29. Which hominins are identified with this culture?
    (A) Cro-Magnon man.
    (B) *Homo neanderthalensis.*
    (C) *Homo erectus.*
    (D) *Homo habilis.*
30. What is the name of this culture?
    (A) Oldowan culture.
    (B) Acheulian culture.
    (C) Mousterian culture.
    (D) Aurignacian culture.
31. Which feature does *not* correctly compare a feature of a primitive primate with a more advanced primate?
    (A) Shorter gestation period.
    (B) Greater number of teeth.
    (C) Greater reliance on vision than sense of smell.
    (D) Smaller cranial capacity relative to body size.
32. The diagram shows the pelvis of an ape (left) and a human (right).

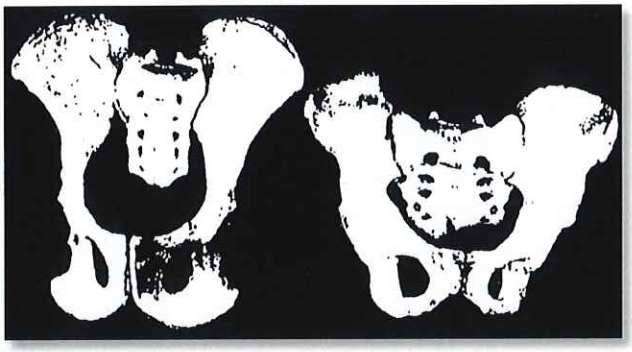


**Figure TT.11** Ape pelvis and human pelvis.

What trend in pelvis shape is linked with bipedal locomotion?
    (A) Long narrow pelvis.
    (B) Short, broad pelvis.
    (C) Long, broad pelvis.
    (D) Short, narrow pelvis.
33. The shape of the brain surface of many hominins has been determined by making endocasts while some endocasts occur naturally, e.g. the discovery by Raymond Dart of the skull of the Taung child. What information can be shown by an endocast?
    (A) The foramen magnum is centrally located in apes.
    (B) Cranial capacity achieved its maximum size with *Paranthropus robustus.*
    (C) The brains of australopithecines were more human-like than ape-like.
    (D) The frontal lobe was well developed with vertical forehead in australopithecines.

34. The fossil record shows that several hominins existed in the same time period.

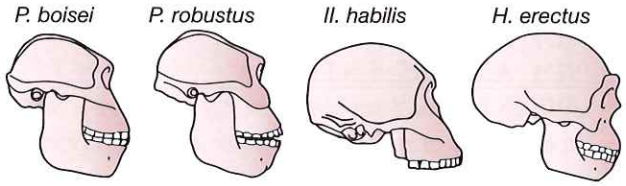



Figure TT.12 Some of the hominins that existed within the last 3 million years, some within overlapping time periods.

Which of the following lived at the same time?
(A) *H. habilis, P. robustus, A. ramidus.*
(B) *H. habilis, P. aethiopicus, A. africanus.*
(C) *H. erectus, P. aethiopicus, A. garhi.*
(D) *H. erectus, P. boisei, A. ramidus.*

35. Which feature does not aid bipedalism?
(A) Larger cranial capacity.
(B) Short, broad pelvis.
(C) S shaped spine.
(D) Foramen magnum more forward placed.

## Section B – Written Response (65 marks)

36. The diagram shows a code change in the amino acid sequence.

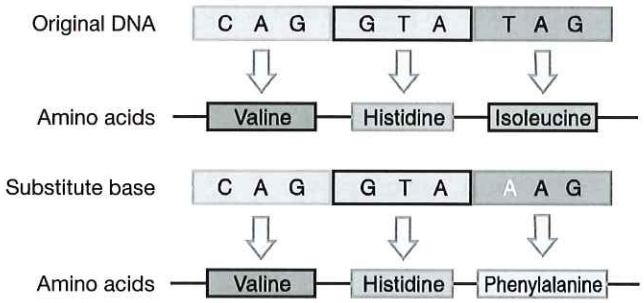



Figure TT.13 Change in code in amino acid sequence.

(a) What term is used to describe this type of change? (1 mark)
(b) Describe how this change could affect human health. (3 marks)

37. (a) What is a chromosome mutation and some different types of chromosome mutation? (3 marks)
(b) List some causes of chromosome mutation. (3 marks)

38. Explain how each of the following is important in DNA replication.
(a) Nucleotide composition and pairing. (2 marks)
(b) Weak bonds between strands of DNA. (2 marks)
(c) DNA polymerase. (2 marks)

39. (a) Explain how mutagens can affect genes. (2 marks)
(b) Construct a table to show how each of the following mutations affect chromosome number in an organism – base substitution, trisomy and polyploidy. (3 marks)

40. The diagram compares the skeleton of a human and a gorilla.

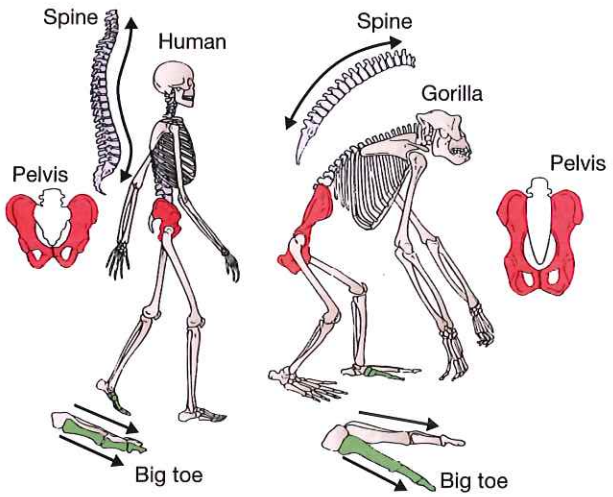



Figure TT.14 Skeletons of a human and a gorilla.

Construct a table to compare the three features – spine, pelvis and big toes of a gorilla and a human. (6 marks)

41. Iceland was settled by a small founder population and geographical isolation has kept the gene pool relatively small. Research at the Molecular and Cell Biology Laboratory at The Icelandic Cancer Society, Reykjavik, Iceland suggests that a founder mutation in the *BRCA2* gene (BReast CAncer 2 gene) accounts for 7% to 8% of female breast cancers and for 40% of male breast cancers in Iceland.
(a) Explain how a founder population can lead to a higher prevalence of a particular disease within a human population. (3 marks)
(b) Explain what is meant by a 'bottleneck' population. (2 marks)

42. Discuss why the fossil record is incomplete. (3 marks)
43. The diagram shows one suggested timeline for several different hominins.

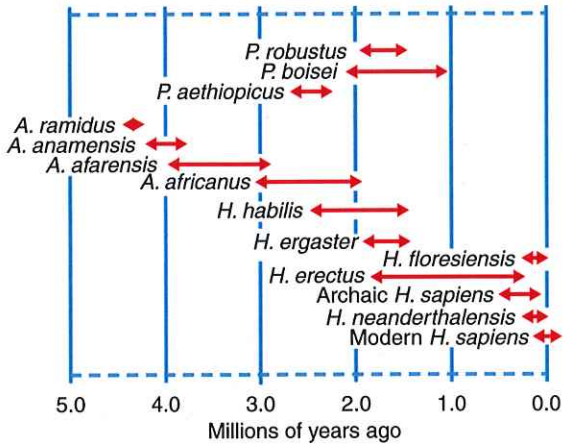



**Figure TT.15** Timeline for several hominins.

Explain how the construction of timelines shows the difficulty in determining the exact pathway of the evolution of humans. (6 marks)

44. (a) Outline some of the areas of evidence that show there has been a change in diet during the evolution of humans. (4 marks)
    (b) Relate the change in diet during the evolution of humans to increase in brain size. (3 marks)
45. (a) Define culture. (1 mark)
    (b) Discuss the relative importance of genetic and cultural evolution during the evolution of humans. (6 marks)
46. Use a named example to show how an advance in technology has changed scientific thinking about evolutionary relationships. (2 marks)

47. The diagram shows the field of vision for a domestic cat, a human, a rabbit and a horse.

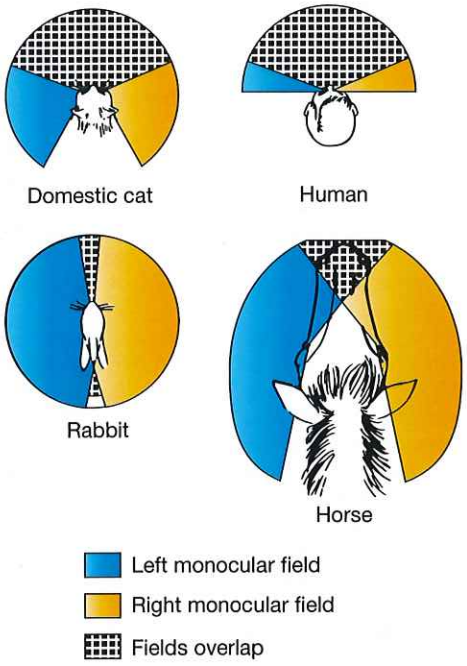



**Figure TT.16** Field of view for four animals.

(a) Distinguish between binocular vision and stereoscopic vision. (2 marks)
(b) Many prey species have laterally placed eyes. Use an example from the diagram to outline why the selective pressure for this field of view gave these animals an evolutionary adaptive advantage. (2 marks)
(c) Primates have forward facing eyes. Explain why this is an important feature for the lifestyle of a named primate. (2 marks)
(d) Identify another feature of vision that is associated with many primates and suggest why this feature gives an adaptive evolutionary advantage. (2 marks)

# Answers

## 1 Assumed Knowledge

1. (a) Chromosomes are long strands of hereditary information containing genes. Chromosomes are made up of DNA and histone proteins.
   (b) Each gene is found at a particular locus on a chromosome.
2. A gene is a section of DNA coding for proteins that expresses itself as the phenotype for that trait, e.g. gene for plant height whereas an allele is an alternative for a particular trait, e.g. there are two alleles for height in pea plants – tall (T) or short (t). Alleles are the alternative forms of a gene and occupy a particular locus on a chromosome.
3. Autosomes are chromosomes that are not directly involved in determining sex whereas sex chromosomes are responsible for determining the sex of an individual. In humans there are two sex chromosomes X and Y. Males are XY and females are XX.
4. (a) Person A is female as they have XX.
   (b) Trisomy is the presence of three copies of a homologous chromosome rather than the normal two copies.
   (c) Person A has trisomy 21 with three copies of chromosome 21 – this is Down syndrome.
5. A genome sequence is the order of the As, Ts, Cs and Gs in the DNA code.
6. A gene pool is the total aggregate of genes in a population at any one time.
7. The frequency of an allele in a population refers to the proportion of the population that has that allele.
8. Polyploidy occurs when cells contain more than two haploid ($n$) sets of chromosomes, e.g. triploid ($3n$), tetraploid ($4n$).
9. A mutation is a permanent change in the genetic information. This causes genetic diversity.
10. The genotype shows the genetic make-up or set of alleles of an organism that controls a characteristic whereas phenotype is the observable physical and physiological traits of an organism and the outward appearance of an organism. A dominant phenotype can have two dominant alleles, e.g. TT or one dominant allele and one recessive allele, e.g. Tt. The recessive phenotype has two recessive alleles, e.g. tt.
11. Humans are classified as a primate as they have – 1. An opposable thumb. 2. Shoulder joint with high rotating ability. 3. Forward facing eyes with stereoscopic vision. 4. Reduced snout and olfactory centre of the brain. 5. Enlarged skull with large cerebrum. 6. Five digits on limbs. 7. Nails (not claws). 8. Bicuspid teeth.
12. Comparative genomics studies compare genomic features of different organisms, e.g. looking for differences and similarities in DNA sequence, genes, gene order and regulatory sequences.
13. Biodiversity refers to the amount of variation within the group.
14. A fossil is a remain or trace of a pre-existing organism.
15. Index fossils are fossils of organisms that lived for a short time and were found over a wide area.
16. A species is a group of organisms that can interbreed to produce fertile offspring. They share a common gene pool.
17. Most DNA in eukaryote cells is in the nucleus. DNA is also found in mitochondria and chloroplasts.
18. Diagram shows DNA replication.
19. Four nitrogenous bases are adenine, thymine, cytosine and guanine.
20. A nucleotide is the basic unit of DNA consisting of a nitrogenous base, sugar and a phosphate group.
21. The law of superposition states that the oldest layers are on the bottom and the youngest layers are on top, unless there has been folding or faulting or another form of dynamic Earth movement.
22. Relative dating uses the law of superposition determining if something is 'older than' or 'younger than' whereas absolute dating gives a date in years with the experimental error of the method used in determining the date.
23. Levels are kingdom, phylum, class, order, family, genus, species.
24. (a) Binomial nomenclature is calling an organism by its genus species name.
    (b) Humans are *Homo sapiens*.
25. Primates include the lemurs, lorises, tarsiers, monkeys, macaques, baboons, gibbons, orangutans, chimpanzees and gorillas.
26. Natural selection is a process that leads to a change in a population over time due to some phenotypes having more success surviving and reproducing in particular environmental conditions.
27. Evolution occurs when natural selection causes changes in relative frequencies of alleles in the gene pool.
28. The order of evolution of living things was organic molecules, membranes, prokaryotic heterotrophic cells, prokaryotic autotrophic cells, eukaryotic cells, colonial organisms and multicellular organisms.
29. (a) Structure A is the cerebrum.
    (b) The cerebrum is concerned with receiving and responding to sensory signals and carrying out mental processes, e.g. memory, abstract thought.
    (c) The size of the cerebrum is an important indicator in human evolution as the cranial capacity relative to body size is related to thought processes, the ability to make complex tools, produce language with symbolic thinking and complex social interactions.
30. Biotechnology is the use of organisms or their parts to make useful products.

## 2 Mutation

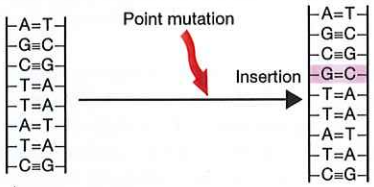
1. Mutation is the change in the DNA genetic make-up of a cell.
2. In the 1920s, Hermann Muller used X-rays to obtain mutant fruit flies, *Drosophila melanogaster*, so he could carry out his experiments on inheritance, showing radiation caused mutation. The atomic bomb on Hiroshima caused many mutations, showing gamma radiation is mutagenic.
3. During DNA replication, sometimes one base is replaced by another which in turn changes the mRNA. Since the code on the mRNA is translated into amino acids, the activity of the cell and the proteins it produces will be affected, e.g. on mRNA the codon GGG = glycine, but if there is a base pair substitution so that the mRNA reads CGG = arginine, the cell will now make arginine instead of glycine.
4. Mutagens are environmental factors such as radiation (e.g. X-rays and UV rays) or chemical agents (e.g. benzene) that cause mutations to occur.
5. Mutations can occur spontaneously and randomly, e.g. due to errors in replication, or they can be induced by mutagens such as radiation or certain chemicals.
6. For a mutation to be inherited it needs to be in a gonad cell, so that the sperm or ova will carry the mutation. Mutations in body cells are not inherited.
7. A point mutation changes the base sequences of a single gene and may form a new allele, while a chromosome mutation can involve either a change in the number of chromosomes or a rearrangement in the structure of a chromosome.
8. (a) Two types of point mutation are base pair substitutions and base pair insertions or deletions.
   (b) Base pair insertions and deletions can have a massive impact as they cause a frameshift and the triplet code is not read correctly, unless the shift is a multiple of three. To show the impact consider the sentence – *The dog ran up the road.* If you delete the 'e' from 'the', the sentence becomes – *thd ogr anu pth ero ad.* As the new sentence makes no sense, a deletion in a triplet code makes no sense and the polypeptide cannot be synthesised.
9. Darwin's theory of evolution states that there is variation within a population and those with favourable characteristics will survive to reproduce by natural selection. When Darwin proposed his theory he did not know how variations arose in populations – the discovery of mutations caused by radiation, certain chemicals etc. explains the source of variation.

10. (a) Variation can arise by random segregation during meiosis – each chromosome of the homologous pair sorts independently and randomly so that the gametes can have different combinations of chromosomes. Crossing over during meiosis can lead to a new combination of characteristics, increasing variation. Sexual reproduction involving the union of a sperm with an ovum means that the offspring have a new combination of genes and are not identical to the parents, thus increasing variation in the population.
    (b) Variation is mainly important when the environment is changing. If one variant has features more suited to the new environment, natural selection will lead to evolution and survival.
11. (a) Karyotype is the pairs of chromosomes obtained while the cell is divided and the chromosomes are sorted according to size from biggest to smallest.
    (b) No, twins are not identical as twin A is female and twin B is male.
    (c) One type of abnormality is having an incorrect number of chromosomes, e.g. some humans have 47 chromosomes instead of 46 chromosomes. These can be counted on the karyotype. For example, trisomy 21, or Down syndrome, is a genetic disease where there are three copies of chromosome 21; this causes specific symptoms in the phenotype of people with the disease.
12. B
13. D
14. C
15. D
16. C
17. A
18. B
19. B

## 3 Mutation and Polypeptide Synthesis

1. A mutation is a permanent change in the genetic information.
2. A somatic cell is any cell in a multicellular organism except gametes. A germ line cell is any cell that gives rise to the gametes.
3. Spontaneous mutations occur naturally and the causative agent cannot be identified, e.g. they occur in about one in every million to one in every billion divisions.
4. In an induced mutation the causative agent can be identified, e.g. exposure to radiation such as X-rays, gamma rays or certain chemical substances.
5. The activities of the cell can be disrupted if a particular protein cannot be produced, e.g. due to a mutation that causes a change in the DNA code which affects the production of a particular protein. This can even lead to disease. For example, if a mutation on the β-haemoglobin gene puts thymine instead of guanine at one particular location, the person will suffer from the disease thalassemia major, which results in abnormalities such as severe anaemia and growth retardation.
6. Both missense mutations and nonsense mutation are point mutations caused by a base pair substitution in a codon. In a missense mutation the codon still codes for an amino acid and a polypeptide is produced, though it is not the right amino acid in the polypeptide. In a nonsense mutation the substitution makes the stop codon which means the synthesis of the polypeptide stops at that point shortening the length of the polypeptide and usually making a non-functional protein.
7. The diagram shows polypeptide synthesis and the codon affected by the mutation on the mRNA is UCA. This is the codon for the amino acid serine which is shown in the diagram lining up to become part of the polypeptide. If the cytosine (C) is changed to adenine (A) in a substitution mutation the code becomes UAA. This is a stop codon which means the synthesis of the polypeptide will now stop at this point due to the nonsense mutation. This will result in a shortened polypeptide and probably form a non-functional protein. This could have a dramatic affect on cell metabolism and activities as it could interrupt and stop a chain of reactions, e.g. due to the lack of an enzyme to catalyse one of the steps in the metabolic pathway.
8. D

## 4 Point Mutations

1. A mutation is a permanent change in the genetic information and is a cause in genetic diversity.
2. A gene mutation is a permanent change in the genetic information in a gene. The mutation can involve one or more base pairs and can be anywhere in the gene.
3. A point mutation is a change in one base in a single nucleotide in a gene.
4. There are four basic types of point mutation – base pair substitution, insertion, deletion and inversion.
5. A silent mutation is a change that has no effect on the protein being produced.
6. (a) GUU (valine) to GUG (valine) is a silent mutation.
    (b) ACG (threonine – polar amino acid) to AAG (lysine – electrically charged amino acid) is not likely to be a silent mutation.
    (c) GGU (glycine a non-polar amino acid) to GAU (aspartic acid – electrically charged amino acid) is not likely to be a silent mutation.
    (d) CGG (arginine is an electrically charged amino acid) to UGG (tryptophan a non-polar amino acid) is not likely to be a silent mutation.
    (e) GCU (alanine) to GCC (alanine) is a silent mutation.
7. A missense mutation is a changed codon that codes for an amino acid but does not necessarily make the correct sense whereas a nonsense mutation is a point mutation that changes a codon for an amino acid into a stop codon. The stop codon causes translation to stop shortening the polypeptide chain that is being synthesised.
8. A frameshift mutation occurs when a deletion or insertion causes an incorrect reading of the nucleotide sequence. The sequence is read in multiples of three (the code for each amino acid) so the deletion or insertion of one nucleotide will change the reading sequence but it is not affected if the change was an insertion or deletion of three nucleotides.
9. If the point mutation affects the shape of the protein or is at the active site of an enzyme, the change can seriously affect the functioning of the protein. For example, disulfide bridges are formed when two cysteine amino acids are brought close together with their sulfhydryl groups (–SH) on their side chains in close proximity due to the folding of the protein and a change in the code from UGC (cysteine a polar amino) to UUC (phenylalanine a non-polar amino acid) can affect the structure and functioning of the protein. Disulfide bridges are important in maintaining the structure of immunoglobulins (antibodies) and the antigen receptor sites on lymphocytes. Inability to form functional antibodies will reduce the effectiveness of the immune system.
10. The mRNA stop codes are UAA, UAG and UGA.
11. 

12. A
13. D
14. B
15. C

## 5 Block Mutations

1. A block mutation is a permanent change to a segment of a chromosome that rearranges, deletes or disrupts many loci.
2. Transposons (transposable genetic elements) are DNA segments that can move from one position to another in the chromosome.
3. The term 'jumping genes' is not accurate as it does not explain how transposons move as they do not 'jump'. When the folding of the DNA molecule brings segments near each other transposons follow a 'cut and paste' mechanism to move to a new location or follow a 'copy and paste' mechanism replicating a section of DNA and adding it to another area.
4. Repetitive DNA is multiple copies of DNA sections which are found in the genome.
5. Barbara McClintock discovered transposons when studying corn in 1940s and she was awarded the Nobel Prize in Physiology or Medicine in 1983 for her discovery of 'mobile genetic elements'.
6. Block mutations can be duplications, inversions, deletions, insertions or translocations.

7. Hotspots are places that are more likely to undergo mutation than other places with an observable higher mutation frequency. Hotspots can be single nucleotides or short stretches of repeated nucleotides that have some basic instability or chemical tendency for nucleotide substitution.
8. The diagram shows a block deletion. In this type of mutation a section of DNA is removed from the sequence in a chromosome and in this case several genes have been removed near the centromere.
9.

| Change | What happens | Example |
|---|---|---|
| Deletion | A section of DNA is removed altering chromosome structure. | Cri du chat syndrome involves a deletion of the end of the short arm of chromosome 5. |
| Duplication | A section of DNA is repeated and added to the chromosome. | Charcot-Marie-Tooth disease type 1 with duplication of 17p12 – a large section on the short arm of chromosome 17. |
| Inversion | A section of DNA breaks and is reattached in the reverse orientation and order. | In factor VIII gene on X chromosome causing haemophilia A. |
| Translocation | A section of one chromosome moves to a non-homologous chromosome. In a reciprocal translocation the non-homologous chromosomes exchange segments. | Myeloproliferative syndrome caused by translocation of genetic material from chromosome 8 to other chromosomes, e.g. t(8;13)(p11;q12). |

10. Tandem repeats are copies of a repeat sequence that lie adjacent to each other
11. Satellite DNA or simple sequence DNA is a section of tandem, non-coding DNA that can be thousands of base pairs long and was discovered when DNA was centrifuged and the repetitive units appeared as a distinct band in the tube.
12. Microsatellite DNA is a short region of repeats that are used as genetic markers in DNA fingerprinting. In humans some microsatellites have 20 or more alleles which provides the variation to assist in identifying particular individuals by their DNA.
13. Trinucleotide disorders occur when there are too many trinucleotide repeats in a gene, e.g. Huntington's disease occurs when there are more than 35 CAG repeats on the gene coding for the protein HTT.
14. The diagram shows reciprocal translocation as the exchange is between non-homologous chromosomes (they are different lengths and have different genes) with blocks of DNA being swapped and two new different chromosomes are formed with different sets of genes.
15. (a) Multigene families arise after duplication of one or more genes to make a gene pair and both copies stay in the genome and are inherited by future generations. This means that the genes in the multigene family started with genes that coded for particular proteins which started the multigene family.
    (b) Sine the original genes that were duplicated and started the multigene family were involved in a particular body function, the future forms of the genes in the multigene family are most likely to be involved in the same process.
    (c) Comparative genomics studies compares genomic features of different organisms, e.g. looking for differences and similarities in DNA sequence, genes, gene order and regulatory sequences. Researchers are interested in multigene families such as the globin genes for haemoglobin as differences and similarities in the DNA sequences found in the genomes of different species can provide information about evolution and the interrelatedness of the species.
16. C
17. B
18. B
19. A

## 6 Transposable Genetic Elements

1. Transposable genetic elements are DNA segments that can move from one position to another in the chromosome.
2. Transposition is the movement of a transposon.
3. A prokaryote does not have membrane bound organelles, e.g. nucleus, while a eukaryote does have membrane bound organelles, e.g. has a nucleus.
4. Transposons have been found in prokaryotes and eukaryotes, being first found in plants.
5. The work of Barbara McClintock was not recognised until 1983 as the discovery of 'jumping genes' in the 1940s was quite ahead of its time. Watson and Crick did not propose the double helix structure of DNA until 1953 and thus the idea of genes moving location and changing the Mendelian laws of inheritance would have been unexplainable at that stage of genetic understanding. With the advent of genetic engineering her work could be fully recognised.
6. In humans, jumping genes have been linked with antibody production by the immune system and some of the rearrangements caused by jumping genes are believed to be the cause of some forms of cancer.
7. Retrotransposons use a 'cut and paste' method to move genetic information from one location on a chromosome to another but they copy RNA rather than DNA.
8. Transposons have been called 'junk' DNA as there is no clear indication how they benefit the host; and they have been called 'selfish' DNA as they often make multiple copies of themselves.
9. Since transposons have been found in all life forms, transposons may have been present in the last universal common ancestor. Or transposons may have evolved independently several times.
10.

| Feature | Transposable genetic element (TE) | Germ line mutation |
|---|---|---|
| Location | If the change in position occurs in a germ line cell then the gametes will have the change. If the change occurs in somatic cells then the gametes will not have the change. The gap in the DNA is not usually repaired. | Occurs in any cell that gives rise to the gametes and the gametes will have the mutation. |
| Inheritance | If the change is in a germ line cell then it will be transmitted to offspring and potentially future generations. If it is in a somatic cell then it will not be transmitted to offspring and future generations. If the move leads to chromosome elongation this can disrupt cell division, e.g. pairing in mitosis and meiosis. | Mutation is transmitted to offspring and potentially future generations. |
| Phenotype | When TE change their relative position on the genome or are replicated within the genome they change the coding of a functional gene and can disable the gene so that it cannot produce the correct polypeptide and often no polypeptide can be synthesised. This can change the functional protein and hence phenotype. | If the change is part of the coding portion of the genome then phenotype will be affected, e.g. base pair substitution to change an amino acid that changes the structure of a polypeptide. This can change the functional protein and hence phenotype. |
| Effect on organism | If the change is in a somatic cell and in coding that is not expressed then there will be minimal effect on the organism. If the change is in the gamete and stops the effective expression of a gene then the organism and the functioning of the organism could be highly affected. | If a functional protein can no longer be produced the metabolism of the organism would be dramatically affected with a disabled biochemical pathway and the proper functioning of the organism would be affected. |

11. A

## 7 Chromosome Abnormalities

1. Polyploidy is the possession of more than two sets of chromosomes per nucleus e.g. hexaploid ($6n$), while monoploidy is the loss of an entire set of chromosomes (the haploid number $n$) and aneuploidy is the addition of all or part of a chromosome.
2. Monosomy is the lack of one chromosome from the normal number of chromosomes and trisomy means there are three copies instead of the normal two copies of a particular chromosome.

3. Nondisjunction is an error during mitosis or meiosis when both members of a pair of homologous chromosomes or both sister chromatids fail to separate properly.
4. Nondisjunction leads to an abnormal number of chromosomes in a gamete. If it occurs in anaphase II there will be two affected daughter cells – one with two copies of the chromosome and the other with no copies of that chromosome. If the homologous chromosomes do not separate during anaphase I then all four daughter cells are affected – two daughter cells will have an additional chromosome and two cells will be missing one chromosome.
5. If the error and nondisjunction occurs during early embryonic development then some, but not all cells will have trisomy, e.g. in mosaic Down syndrome the person has some cells with trisomy 21 and some cells with the normal 46 chromosomes in a cell.
6. A changed number of sex chromosomes does not seem to upset the genetic balance as much as an aneuploidy autosome condition. The Y chromosome has few genes and in somatic cells the extra copies of the X chromosome are inactivated as Barr bodies.
7.

| Genetic disorder | Description |
|---|---|
| Klinefelter syndrome | Males have two or more X chromosomes, e.g. a male has XXY. It is a congenital disorder detected by karyotyping. Males have abnormally small testes and are sterile. |
| Turner syndrome | Females are missing part or a whole X chromosome, e.g. a female has XO. The female will have 45 chromosomes instead of 46 chromosomes. The person may show mosaicism with some cells normal and some cells lacking the second X chromosome. The person is sterile with immature sex organs. |
| XYY syndrome | Males have an extra Y chromosome in each cell. Individuals seem to have an increased risk for learning difficulties, often have heavy acne and have increased height. It is estimated to occur in about 1 in 1000 live births. The males have normal fertility. |
| Triple X syndrome | Has an X chromosome in each cell in a human female with a mosaic form being present in only some cells. Most cases are not diagnosed and since only one X is only activated at a time due to the formation of Barr bodies, there are few obvious signs of the condition and most are usually fertile. |

8. B
9. A
10. D

## 8 Gene Expression Affects Phenotype

1. Gene expression refers to a gene being 'turned on' to produce a polypeptide, i.e. it changes the DNA information into the functions and structures of a cell.
2. A promoter is a regulatory area at the start of a gene that acts as the binding site for RNA polymerase.
3. If there is a problem with the promoter there will be a problem in starting transcription and thus the polypeptide will not be synthesised and the gene will not be expressed.
4. A study of gene expression of children who have acute asthma has revealed that during an attack the children had a different gene expression profile to children in a stable (controlled) asthma state. They found one gene cluster that was active during an acute asthma attack but was not active in stable asthma and other gene clusters that were active in stable asthma and not during acute asthma attacks. This research focused on genes concerned with epithelial cells, mucosa and the underlying cells. This knowledge can be used to instigate treatment and control measures to reduce the incidence of acute asthma attacks for these children.
5. Cancer involves rapid cell division without cell differentiation so that a mass of cells accumulates in one place forming a tumour.
6. The *MYC* gene controls the production of the protein *MYC*. This protein controls cell division and thus the *MYC* gene controls cell division.
7. A study of the *MYC* gene in transgenic mice which had the *MYC* gene permanently 'switched on' caused the liver cancer cells in the mice to become harmless cells. The researchers injected the transgenic mice with the antibiotic doxycycline which 'switched off' the gene and the cancer cells became normal cells.

## 9 Allele Frequency

1. An allele is an alternative for a particular trait, e.g. there are two alleles for height in pea plants – tall (T) or short (t). Alleles are the alternative forms of a gene and occupy a particular locus on a chromosome.
2. The frequency of an allele in a population refers to the proportion of the population that have that allele.
3. Population genetics studies how populations change genetically over time and the Hardy-Weinberg equilibrium proposes that frequencies of alleles and genotypes remain constant from generation to generation provided that only Mendelian segregation and recombination of alleles occur. Thus the Hardy-Weinberg equilibrium is only useful in population genetics if there is no evolution occurring in the population.
4. A gene pool is the total aggregate of genes in a population at any one time.
5. A population is a group of organisms which can freely interbreed to produce fertile offspring living in a particular area at a particular time.
6. A genome is the complete complement of an organism's genes and genomics studies whole sets of genes and their interactions.
7. Genes are coded DNA which is transcribed into RNA that is used in protein formation. Controlling proteins means that when differentiation occurs the final cell structure and functioning is controlled by which genes are switched on or off in gene expression, e.g. the switching on of genes that control specific proteins such as keratin and collagen and the switching on of particular enzymes that regulate specific reactions will determine the shape, composition and functioning ability of the cell.
8. Phenotype is the observable physical and physiological traits of an organism and the outward appearance of an organism. Allele frequency of particular traits will then show the frequency of particular phenotypes in a population.
9. Phenotype is determine by the alleles present (genotype) and environmental influences.
10. C

## 10 Gene Flow

1. Gene pools are dynamic with changes in allele frequency caused by mutations, differing selection pressure, random genetic drift and changes in gene flow between adjoining groups.
2. Gene pools can be used to compare populations at different times or locations.
3. Gene flow occurs when alleles are added to or subtracted from a gene pool due to the movement of fertile individuals or gametes.
4. Migration will cause gene flow if fertile individuals with specific alleles move to another area. This means the allele is taken from one population and is now in another population.
5. Gene flow can reduce the number of differences between populations as alleles are transferred from one population to another making the two populations more genetically similar. The higher the gene flow between two populations the less likely the two populations will evolve into two species.
6. Gene flow can be affected by – 1. Mobility of fertile individuals or gametes. 2. Presence of barriers such as mountain ranges, oceans, deserts. 3. Cultural barriers.
7. Since the development of planes and faster transport there is greater international travel and gene flow between different human populations. Thus geographical isolation which is a primary cause of speciation for other species is less important for humans.

## 11 Genetic Drift

1. Genetic drift is a process that occurs when chance events cause unpredictable changes in allele frequencies from one generation to the next.
2. Genetic drift is due to random events while natural selection involves the survival and reproduction of those with favourable features for a particular environment. Thus while both genetic drift and natural selection lead to changes in allele frequencies in a population, natural selection leads to adaptations to a particular environment and genetic drift does not necessarily lead to adaptations to the environment.

3. Chance events that can lead to genetic drift include – 1. Natural disasters such as flash floods where it is chance which individuals survive. 2. Accidental death of some individuals that is not related to natural selection. 3. Chance events in fertilisation with the offspring having features that do not necessarily reflect the range of features of the parent population.
4. The high allele frequency for the recessive allele that causes a combination of dwarfism and polydactylism could have occurred as a result of one of the founding members of the Amish colony possessing this allele. If the colony has kept itself relatively isolated from the rest of the world then genetic drift and the founder effect have led to the high allele frequency with offspring likely to have two recessive alleles showing the trait.
5. If a population stays small and isolated there is a possibility that an allele can be 'lost' from the gene pool if, by chance, the individuals with this allele do not mate and pass the allele on to the next generation. If this occurs the population could become fixed for the other allele with its frequency becoming 1.0 and the frequency of the lost allele becoming zero.
6. The group of Aborigines that first colonised Australia were a small founder group. This group may have, by chance, not had the $I^B$ allele or if it was present the individuals with the $I^B$ allele did not have surviving offspring. Since the founding group was isolated on Australia the descendants did not have the $I^B$ allele and did not have blood groups B or AB. If Aborigines since European settlement have had partners from other groups that have the $I^B$ allele then some Aborigines today may have the $I^B$ allele.
7. If a population is small genetic drift can decrease genetic variation if only some individuals mate and produce surviving offspring. The allele frequency of features in the successive generations can change in a specific direction due to chance. These changes in the population mean that there will be increased differences between this population and other local populations. If the other populations are also small and changing due to genetic drift then the two populations will become dissimilar within a few generations.

## 12 Microevolution

1. Microevolution refers to evolutionary change below the species level with changes in the genetic composition of a population from one generation to the next generation.
2. Main processes of microevolution are – 1. Mutation. 2. Natural selection. 3. Genetic drift. 4. Gene flow.
3. There are several new strains of bacteria that are resistant to multiple antibiotics (often called superbugs), e.g. methicillin resistant *Staphylococcus aureus* (MRSA) and vancomycin resistant *Enterococcus* (VRE). If a person has a bacterial infection they are given antibiotics as part of their treatment. If some of the bacteria have a natural resistance to that dosage of antibiotic they will survive, reproduce and their offspring will inherit the resistance. Since bacteria can reproduce in a short time, e.g. 30 minutes, many generations can occur in a relatively short period and the frequency of the resistance allele can increase in the population. Thus the use of antibiotics has acted as a selective agent in natural selection for the evolution of bacteria resistant to many antibiotics.
4. The land bridges formed during the ice ages would have influenced gene flow as migration could occur between the land masses. Placental mammals such as rodents and bats could migrate to Australia and the land bridges also provided a pathway for the first humans to colonise the continent with a founder event.

## 13 Tay-Sachs Disease

1. The disease is named after Warren Tay (1843-1927) and Bernard Sachs (1858-1944) who both described the disorder.
2. Gene flow is the movement of alleles into or out of a population, e.g. due to immigration or emigration.
3. The founder effect occurs when a few individuals become isolated from a larger population and form a new population whose gene pool composition is not reflective of the original source population.
4. The founder effect with Tay-Sachs disease is shown by the Ashkenazi Jews who lived for many centuries in Eastern and Central Europe, some French-Canadian groups in Quebec, in the Old Order Amish community in Pennsylvania and in the Cajun population in Louisiana.
5. Tay-Sachs disease is an autosomal recessive disease.
6. Tay-Sachs disease is caused by a recessive allele on chromosome 15 for the *HEXA* gene.
7. Lysosomes are single-membrane cellular organelles that break down and digest complex molecules for recycling. Tay-Sachs disease affects the production of the enzyme hex-A which is found in lysosomes.
8. The allele for the disease is recessive, therefore a person with the disease must be homozygous for the phenotype to show.
9. The enzyme hex-A is involved in lipid metabolism. Without the enzyme lipids (GM2) accumulate in cells, especially nerve cells of the brain. Lipid metabolism and lipid storage are disrupted leading to a progressive deterioration of neurons in the nervous system.
10. The three types are: 1. Infantile (classic). 2. Juvenile onset (develops between ages 2 to 10). 3. Late onset (begins between adolescence and mid 30s).
11. D

## 14 Sickle Cell Anaemia

1. Sickle cell disease is a group of disorders that have misshapen red blood cells due to an atypical haemoglobin called haemoglobin S that makes the red blood cells more rigid and likely to stick together to block small blood vessels reducing life expectancy.
2. (a) A person with sickle cell is $Hb^SHb^S$.
   (b) A carrier for sickle cell is $Hb^NHb^S$.
3. Sickle cell anaemia is caused by a point mutation on chromosome 11 to the gene for beta haemoglobin. The gene codes for 287 amino acids and the affected allele has the code GTG (valine) rather than GAG (glutamic acid).
4. The terms recessive and dominant refer to the phenotype and the appearance caused by the genotype.
5. (a) Sickle cell anaemia requires two recessive alleles $Hb^SHb^S$ for the person to have the disease.
   (b) The shape of red blood cells is codominant with $Hb^AHb^S$ showing both sickle cells and normal red blood cell shape.
   (c) Malaria resistance is dominant with only one copy of the allele needed to provide resistance.
6. Red blood cells are distorted to the sickle cell shape and although sickle cells can easily pick up oxygen like normal red blood cells there is a loss in elasticity of the red blood cells.
7. A high proportion of abnormal haemoglobin molecules in a red blood cell will make the sickling permanent.
8. When the haemoglobin releases oxygen the molecules 'stick' together and form long polymer chains and the sickle cells clump together blocking capillaries causing vessel occlusion.
9. Most treatment involves relieving symptoms and preventing infection, e.g. children should be fully immunised including annual influenza vaccinations and immunisation for meningococcus and pneumococcus. There are medications for chronic pain and in episodes of anaemia blood transfusions are needed.
10. Hydroxyurea is an oral medication that inhibits DNA synthesis causing immature red blood cells to produce foetal haemoglobin instead of adult haemoglobin giving some protection against haemoglobin S and reducing some SCD complications such as inflammation.
11. The disease is common among people with ancestors from Africa and Mediterranean countries, e.g. Greece, Turkey, Italy, Middle East, India and Spanish-speaking areas of South America and Central America.

12. (a) Since the sickle cell disease is usually fatal or significantly reduces life expectancy, it would be expected that the frequency for the gene is low.
    (b) The countries with a high allele frequency also have a high incidence of malaria and abundance of the *Anopheles* mosquito which transmits the malarial parasite. Since the heterozygotes (carriers of the sickle cell allele) have some protection against malaria, the lethal allele stays in the population. In these countries, people homozygous for the normal haemoglobin can die of malaria. Thus the frequency of the sickle allele varies worldwide depending on the selective pressure of malaria.
13. (a) Sickle cell trait is the heterozygous condition ($Hb^AHb^S$). Heterozygotes are usually healthy and do not show the severe symptoms of sickle cell disease. Some may suffer some sickle cell symptoms during extended periods of reduced blood oxygen, e.g. strenuous exercise or at high altitudes.
    (b) The protozoan that causes malaria spends part of its life cycle in red blood cells and the presence of heterozygous amounts of sickle cell haemoglobin lowers the density of the parasite and reduces the symptoms of malaria.
    (c) You would expect to find a high frequency of the sickle cell allele in areas prone to malaria. In these areas there is a balanced natural selection and balanced polymorphism between the high selection against the homozygous sickle cell condition and the high selection against the normal red blood cell condition which can lead to death from malaria. The preference for the heterozygous condition shows heterozygous advantage.
14. In the USA, the frequency of the sickle cell allele among African Americans is lower than in Africa as there is no endemic malaria in the USA and the sickle cell trait does not have the same selective advantage.
15. C

## 15 Thalassemia

1. Thalassemia is an autosomal recessive disease.
2. Thalassemia involves a mutation either on chromosome 16 and/or chromosome 11.
3. Alpha thalassemia occurs when there is a mutation in the gene for the alpha globulin (two genes on each chromosome 16) and beta thalassemia occurs when there is a change in the beta globulin gene (one gene on each chromosome 11).
4. Thalassemia is most common in people from the Middle East, Mediterranean countries, e.g. Greece and Turkey, Africa and Asia. Alpha thalassemia is common in people of African, Middle Eastern, Chinese or South-East Asian descent, while beta thalassemia is common in people from Mediterranean areas, e.g. Africa, Greece, Malta and Italy.
5. The abnormal genes for haemoglobin cause the body to reduce globin synthesis and form less haemoglobin. There is increased friability of the cell membrane of the red blood cells leading to an excessive destruction of red blood cells and early haemolysis. This leads to mild or severe anaemia fatigue, jaundice (yellow skin), bone deformities, slow growth rates and cardiovascular problems. Each type of thalassemia has distinctive symptoms. Alpha thalassemia causes bone problems with cheeks, forehead and jaw overgrowing, malnourishment, extremely enlarged spleen and jaundice. Severe alpha thalassemia can cause stillbirth or early death due to heart failure. Beta thalassemia usually appears before 2 years of age causing paleness, frequent infections, poor appetite, jaundice, enlarged organs and requiring regular blood transfusions
6. Treatment depends on the type and severity of thalassemia with minimal treatment needed for people who are carriers or who have mild forms to blood transfusion programs with folate supplements and additional chelation therapy for people with moderate to severe forms of thalassemia.
7. Iron chelation therapy is needed to reduce the iron overload caused by the blood transfusions.

## 16 Natural Selection, Speciation and Evolution

1. Natural selection occurs when factors in the environment confer a selective advantage on specific phenotypes to enhance survival and reproduction.
2. Speciation is the evolution of a new species.
3. Allopatric speciation involves the geographical separation while sympatric speciation occurs in the new species forms in the same geographical regions as the parent species.
4. If the barrier in allopatric speciation is removed, there are three possible outcomes – 1. Speciation had led to sufficient changes that the two groups can no longer interbreed to produce fertile offspring and there are two distinct species. 2. Speciation had not occurred and the two groups can interbreed to produce fertile offspring making them one species. 3. The two groups have distinct differences and there is limited gene flow which can lead to sympatric speciation or a cline can develop when the differences change with location.
5. Evolution is the change in the gene pool from one generation to the next due to processes such as natural selection, mutation and genetic drift.
6. Mechanisms are – 1. Inherited variation – the ultimate source of variation within a population is mutation, e.g. due to mutagens. 2. Struggle for existence – in many populations more are born than can survive to reproduce causing 'survival of the fittest' with those with traits that best cope with pressures in a particular environment will survive and produce offspring. 3. Isolation and differential selection – isolation is very important in the processes of evolution and speciation. Isolating mechanisms reduce gene flow between populations and over time the cumulative changes in the gene pools of different groups can lead to speciation where interbreeding to produce fertile offspring is no longer possible. 4. Changes in gene pools – an allele becomes 'fixed' in a gene pool when all the individuals in the population are homozygous for that allele. If a new allele arises on one population, gene flow allows this allele to spread to other populations. This means the gene pool of the species can change over time due to mutation, changing environmental selective pressures and/or genetic drift.

## 17 Allele Frequencies and Natural Selection

1. The gene pool is the sum total of all the alleles of all the genes of all the individuals in a population.
2. Allele frequency is the proportion or percentage of a particular allele at a given locus in a population gene pool.
3. Evolution occurs as a result of accumulated changes in the gene pool over time with the frequency of some alleles increasing and the frequency of other alleles decreasing.
4. A closed population has effective boundaries that do not allow genetic interchange with other populations. An open population has boundaries that overlap with other popuations and allow genetic interchange.
5. (a) Determining frequency of genotypes.

| Genotype | AA | Aa | aa |
|---|---|---|---|
| Number of individuals | 195 | 351 | 104 |
| Genotype frequency (%) | 30% | 54% | 16% |
| Frequency | 0.30 | 0.54 | 0.16 |

   (b) Frequency of allele A = $0.30 + (\frac{1}{2} \times 0.54) + 0 = 0.57$

   Frequency of allele a = $0 + (\frac{1}{2} \times 0.54) + 0.16 = 0.43$

6.

| Genotype | BB | Bb | bb |
|---|---|---|---|
| Number of individuals | 53 | 158 | 39 |
| Genotype frequency | 0.212 | 0.632 | 0.156 |

   (b) Frequency of allele B = $0.212 + (\frac{1}{2} \times 0.632) + 0 = 0.528$

   Frequency of allele b = $0 + (\frac{1}{2} \times 0.632) + 0.156 = 0.472$

7. (a) If 0.8 A + 0.2 a male gametes randomly fertilise 0.8 A + 0.2 a female gametes then the genotype frequencies in the first generation will be
AA 0.64 Aa 0.32 aa 0.04
(b) The allele frequencies for the next mating will be:
Frequency allele A = $0.64 + (\frac{1}{2} \times 0.32) + 0 = 0.8$
Frequency allele a = $0 + (\frac{1}{2} \times 0.32) + 0.04 = 0.2$
(c) From the results in part (b), it can be concluded that gene frequencies will stay constant if it is a large population that sexually reproduces and has random matings.
8. Conditions that can lead to changes in gene frequencies include mutations, migrations, natural selection and random genetic drift.
9. A large proportion of recessive alleles will stay in a population as they are in heterozygotes. These alleles provide genetic variability which can be an evolutionary advantage if conditions change. The new conditions may favour some of these recessive alleles and the frequency of these alleles will then increase, survival is assisted and evolution occurs.
10. A

## 18 Activity – Model for Natural Selection

1. Selecting agent is the roll of the dice and the choice of colours on the side of the dice. This represents survival ability in the environment.
2. The hominin population began with equal numbers of green, red, yellow and orange hominins. Over time it changed, with yellow hominin gradually increasing and red hominin, followed by the yellow, decreasing and finally disappearing.
3. (a) To keep the population total constant, there were 20 hominins 'born' each year and 20 hominin 'non-survivor' by environmental selection pressure each year.
(b) It was important to keep the population total constant as this shows that there was no threat to the survival of the hominins – only the composition of the different populations changed over time.
4. Favourable characteristic is yellow colour which related to larger cranial capacity and use of stone cutters and choppers.
5. In a model for survival with large jaw muscles to eat tough fibrous bark and leaves the students should change the colours on the dice so that red is not represented and three sides of the dice are yellow. The other three sides of the dice can be split between green and orange, depending on which colour the students want to be the next favourable colour.
6. Changing the colour of the sides to 1 : 1 : 1 : 1 means that evolution would not occur. There may be changes from year to year in the numbers of each colour of hominin but that would be due to random probability. For natural selection to occur, you need a selection pressure which favours one particular characteristic and in a 1 : 1 : 1 : 1 situation none are favoured.
7. (a) To model natural selection you could use several different pieces of equipment, e.g. spoon, chopsticks, forks, forceps to collect a 'food', e.g. paper clip in a specified period of time. This can show how different tool making abilities can determine the amount of 'food' that can be gathered which will affect survival ability.
(b) In human evolution tool making was an important factor in the survival of different hominins. The ability to increase food production with hunter gatherer tools favoured the members of the genus *Homo* and then the agricultural revolution with the settlement of farms, agriculture and animal husbandry allowed a dramatic increase in the human population.
8. Isolation is usually needed for evolution to occur. If there is no contact with other populations of that species, e.g. due to geographical barriers such as desert, mountain range or ocean, natural selection will occur, the isolated population will change and eventually the changes will be so numerous they will not be able to interbreed with the other groups – that is, they form a new species.
9. Scientists use models as a way of representing scientific principles to make them easier to understand, give a visual representation and to be able to show and predict what will next happen.
10. (a) *Australopithecus africanus* = green, *Paranthropus robustus* = red, *Paranthropus boisei* = orange and *Homo habilis* = yellow.
(b) Since all four hominins were alive 2 million years ago the example shows evolution as *A. africanus* disappeared from the fossil record around 2 million years ago and the two *Paranthropus* species were a side branch of the human evolutionary tree existing longer than *H. habilis* in the fossil record and living at the same time as *H. erectus*. *H. habilis* is believed by some people to be a direct ancestor of *H. sapiens*. The example shows evolution with groups with more favourable features surviving and changes in the gene pool leading to evolution.
11. A
12. B
13. A
14. B

## 19 Endonucleases

1. An endonuclease is an enzyme that breaks the phosphodiester bond in the sugar phosphate backbone within a polynucleotide chain.
2. Deoxyribonuclease cuts DNA non-specifically at any nucleotide sequence.
3. Restriction endonucleases are degradative enzymes that recognise and cut up DNA, including DNA of certain phages that is foreign to a bacterium.
4. The substrates for the enzymes are the recognition sequences.
5. Found in bacteria and archaea the restriction endonucleases provide a form of protection for the prokaryotes against viral infection. When a virus injects DNA into the bacterium the restriction endonucleases cut up the viral DNA so that the virus cannot take over the bacterium and use the host for its own reproduction.
6. Restriction endonucleases are extremely useful in DNA manipulation as the ability to cut DNA at specific sites allows researchers to determine genetic diversity and to remove DNA from specific sources. Useful restriction endonucleases cut DNA at very specific nucleotide sequences.
7. The restriction site is where the restriction endonuclease cuts the sugar phosphate backbone by hydrolysing the phosphodiester bond between two nucleotides.
8.

| Type of restriction endonuclease | Description |
|---|---|
| Type I | Cuts DNA at a random distance from the recognition sequence. |
| Type II | Cuts DNA at specific positions close to their recognition sequence. |
| Type III | Cuts DNA away from their recognition sequence and need two of the recognition sequences in opposite orientations within the same DNA molecule. |
| Type IV | Recognise modified, usually methylated DNA. |

9. Type II restriction enzymes are the main type used in laboratories for DNA analysis and gene cloning as they produce specific discrete restriction fragments that will form distinct gel banding patterns.
10. B

## 20 DNA Ligases

1. DNA ligases are linking enzymes that are needed in DNA replication. They catalyse the bonding of the 3′ end of a new DNA fragment to the 5′ end of a growing chain.
2. In DNA replication DNA ligases join Okazaki fragments by forming a bond between their free ends to make a long continuous strand on the lagging strand.
3. DNA ligation is the process when linear fragments of DNA are joined together with covalent bonds.
4. DNA ligase will cause the formation of a phosphodiester bond between the phosphate and sugar to form the backbone of a DNA molecule. It is a covalent bond.
5. In DNA manipulation DNA is highly useful as it is an enzyme involved in DNA replication and DNA repair and thus can be used in genetic engineering to form recombinant DNA.

6. (a) Bacteriophage T4 is a virus that infects *Escherichia coli* bacteria and only undergoes the lytic life cycle which destroys the host cell.
   (b) T4 DNA ligase is frequently used in biotechnology as it is effective on *E. coli* plasmids which are used as vectors and used to produce 'designer' proteins which can then be mass produced in industrial fermentation, e.g. producing human insulin.
7. Ligases from thermophilic bacteria are enzymes that are stable at high temperatures and are often used in ligase amplification reactions as they do not denature at high temperatures.
8. (a) DNA ligation needs two or more DNA fragments as these are the pieces that will be joined together and they need either blunt or sticky ends so they will fit together.
   (b) The buffer will provide the correct pH and stabilise the pH for the reaction and the ATP is a source of energy.
   (c) The ligase that will join the DNA fragments is from the bacteriophage T4 which is effective on bacterial DNA, e.g. plasmids of *E. coli* that are used in DNA manipulation.
9. C

## 21 DNA Polymerases

1. DNA polymerases are enzymes that catalyse the elongation of new DNA at a replication fork by the addition of nucleotides to the existing chain.
2. The rate of elongation depends on the species, e.g. in human cells the rate is around 50 nucleotides per second while in bacteria it can be 500 nucleotides per second.
3. Being highly conserved means that the subunits of DNA polymerase and the functions they perform are similar from one species to another.
4. DNA polymerase 'reads' the existing DNA as a template code and adds free nucleotides to the 3′ end of a newly formed strand thus elongating the strand in a 5′ → 3′ direction. After one nucleotide is added the DNA polymerase 'proofreads" the code looking for mistakes so that misplaced pairs can be corrected.
5. Since DNA polymerase can only elongate in a 5′ → 3′ direction one side of the strand can be read continuously and elongated (leading strand) while the other strand (lagging strand) forms discontinuously in Okazaki fragments which then need to be linked together with DNA ligase.
6. DNA polymerase cannot initiate replication and thus a primer is needed to start the process.
7. The problem is that DNA in daughter cells gets shorter each time the DNA replicates and the cell divides because DNA polymerase only adds to the 3′ end of a pre-existing polynucleotide and it means it cannot complete the 5′ ends of daughter strands where the primer was attached.

## 22 The Polymerase Chain Reaction

1. The polymerase chain reaction (PCR) is a technique for amplifying DNA in vitro by incubating nucleotides with special primers and DNA polymerase.
2. PCR is important as it allows any chosen segment of DNA to be copied many times in a test tube in a relatively short time span. This means that tiny amounts of DNA from various sources can be amplified for analysis, e.g. studying genome sequence in archaeological remains.
3. The three main steps are – 1. Denaturing – when DNA sample is heated to separate the two strands. 2. Annealing – DNA primers attach to opposite ends of target sequence. 3. Elongation – DNA polymerase copies the strands.
4. Taq DNA polymerase is used in polymerase chain reactions (PCR) as it can withstand high temperatures without denaturing, e.g. a wide range of areas, e.g. in recombinant DNA technologies that need to produce many copies of small segments of DNA. It is used in mapping genomes, diagnosis of genetic disorders, forensics and clinical testing for bacteria or viruses.
5. (a) X1 and X2 are primers. Y is DNA polymerase.
   (b) The diagram shows the polymerase chain reaction. The two strands of the target DNA separate. The primers attach to either end and then DNA polymerase causes elongation leading to the formation of two copies of the target DNA. The process then repeats to make more copies (amplify).
6. DNA samples from crime scenes are often in small amounts and can be partially degraded. PCR is invaluable in amplifying the sample so that analysis testing can be carried out.
7. Collecting DNA samples for prenatal testing can put the baby and mother at risk and only small samples can be taken. Therefore PCR is important in amplifying the sample. It also allows rapid diagnosis of genetic disorders.
8. D
9. A
10. C

## 23 Gel Electrophoresis

1. Gel electrophoresis is a technique that separates nucleic acids or proteins, on the basis of their size and electrical charge, by measuring their rate of movement through an electrical field in a gel.
2. Substances include nucleic acids (DNA and RNA), proteins and nanoparticles. It relies on the fact that negatively charged particles will migrate from the cathode to the anode when placed in the gel and the current is switched on.
3. The medium needs to have anticonvective properties so that there is no thermal convection when the current is switched on and needs to have suitable polymer fibres to affect the rates of movement of different fragments.
4. When restriction enzymes are used to cut DNA into fragments, gel electrophoresis can be used to separate the fragments from each other.
5. The 'sieving' part of the process is when the electric field is applied. It is given the name as the different fragments separate from each other at this stage – somewhat like a sieve.
6. A post electrophoresis stain is applied so that the band pattern can be seen. This is often called the visualisation stage.
7. A
8. B

## 24 Developmental Biology

1. Developmental biology is the study of the process by which organisms grow and develop. It includes the study of the processes by which organisms grow and develop from a zygote to a fully grown adult, e.g. genetic control of cell growth, differentiation and morphogenesis.
2. Morphogenesis is the development of body shape and organisation.
3. During development there are changes in shape, form, size and degree of complexity. It involves differentiation where cells become specialised in structure and function with biochemical as well as structural changes.
4. Embryology is the study of the development of an embryo from fertilisation to the foetus stage.
5. Comparative embryology compares the embryos of different species and shows the relationship between different species of animals.
6. (a) Comparative embryology.
   (b) A study of the embryos of vertebrates shows that at an early stage they all possess similar features, e.g. all possess gill pouches on the sides of their throat. This shows that the different groups evolved from a common ancestor.
7. Basic stages are – 1. Cleavage – where the zygote divides into a mass of daughter cells with the progressive subdivisions occurring in particular geometric patterns depending on the species. 2. Gastrulation – the mechanical evolution forming the germ layers where the cells are rearranged into distinct layers and the developmental stage is called a gastrula. 3. Organogeny – where differentiation leads to the formation of tissues and organs.
8. Molecular systematics uses genetics to study the evolution of relationships among individuals and species to determine the history and processes involved in the history of life on Earth. Studies look at the nucleic acid sequence and the proteome to work out the systematic positions of different nucleotide sequences and the changes that have occurred over time.

## 25 Comparative Genomics

1. Genomics is the study of whole sets of genes and their interactions, e.g. it includes how particular genes are expressed, how they are regulated, their interactions with other genes and how gene expression affects growth, development and evolution.
2. The complementary base pairing properties of DNA is the basis for DNA hybridisation.
3. Denaturation of DNA means that the two strands of the double helix separate. It occurs under high temperatures when the hydrogen bonds between the complementary base pairs are broken.
4. In DNA hybridisation DNA is heated so that the hydrogen bonds holding the two strands of the double helix together are broken in denaturation. The two strands separate and restriction enzymes are used to cut the strands into smaller sections. Comparable segments from two different species are mixed and when they cool the hydrogen bonds re-form to make a double helix with random collisions causing some hybrid double helix molecules to form. Similar regions will form strong bonds. The DNA is reheated and the temperature needed to cause separation indicates the degree of similarity between the two species. The higher the temperature the greater the number of hydrogen bonds and the more genetically similar the species.
5. A probe is a labelled single stranded nucleic acid molecule that tags a particular nucleotide sequence in the nucleic acid sample.
6. The probe can be labelled with a radioactive isotope or a fluorescent tag so that its location can be detected and tracked.
7. Dish A shows DNA double helix for two different species – species 1 and species 2. These are heated so that denaturation occurs splitting the double helix into separate strands with broken hydrogen bonds. Dish C shows the mixed separated single strands. The DNA is cooled and the hydrogen bonds re-form showing some of the original strands and some hybrid double helix in dish B.
8. DNA hybridisation has been used to work out the relationships between many species. Differences in mitochondrial DNA have been used to work out the evolutionary tree of humans and other primates and when different groups shared a common ancestor.
9. Genetic screening involves testing specific DNA sequences for a particular condition in individuals, groups or populations without family history of the condition. Genetic testing also tests specific DNA sequences for a particular condition where an individual is suspected of being at increased risk due to their family history or the result of a genetic screening test.
10. B

## 26 Molecular Homology

1. Homology is the study of similarities in characteristics due to shared ancestry.
2. Molecular homology is the study of similarities in molecules such as DNA and proteins due to shared ancestry.
3. The genetic code is made from different combinations of three bases (triplet code) from a choice of four nitrogenous bases. There are 64 possible codes ($4^3$) with about 20 amino acids to code which means some codes are redundant. The genetic code is universal as it is shared by nearly all species from simple viruses and bacteria to complex animals.
4. Cytochrome c is a protein found on the inner membrane of mitochondria and is part of the electron transfer chain in aerobic respiration. It is present in organisms that use aerobic respiration and is used to compare different species and determine a molecular clock to work out the history of evolutionary relationships.
5. Immunological evidence uses the production of antibodies against the introduction of specific blood proteins to work out the relationship between species. For example, if human serum is introduced into a rabbit the rabbit will produce antibodies against the human (anti-human antibodies). If the anti-human antibodies are tested against other species the degree of precipitation shows the degree of closeness to humans, e.g. the anti-human antibodies have 100% precipitation with human serum and 8% precipitation with pig serum.
6. The immunological evidence is useful as it gives a broad indication of evolutionary relationships. However, it is limited as it shows that both gorilla and chimpanzee have 98% precipitation and thus it cannot clearly and accurately show closer relationships.
7. D

## 27 Ancestral Gene Homologues

1. Homologue genes or homeotic genes are DNA sequences that are similar in different organisms.
2. Molecular phylogenetics is a field of study that uses DNA sequencing technology to study evolutionary relationships.
3. The presence of similar DNA sequences suggests a common ancestor – the greater the similarity in the molecules, the more recent in time the common ancestor. This data allows relationships between species to be shown as an evolutionary tree with the length of each branch indicating a specific time frame.
4. The 'ssu rDNA' gene is a homologue and the similarities and differences in this gene in different species was used to construct an evolutionary tree of the eukaryotes showing the divergence of protists, fungi, plants and animals.
5. Cytochrome c is an essential protein that performs a key step in cellular respiration.
6. Cytochrome c is found in all aerobic eukaryotes, i.e. protists, plants and animals.
7. Homologue genes are DNA sequences that are similar in different organisms and since every aerobic eukaryote that has been sequenced has cytochrome c with at least 37 amino acids at equivalent positions, the genes for cytochrome c are homologues.
8. The DNA sequences for cytochrome c are useful for comparing the lower animal classes as the primitive common ancestor which had the precursor for cytochrome c existed over two billion years ago. Thus over a very long period of time there have been changes in the DNA sequences and these differences are quantifiable and can be used to construct evolutionary trees. More recent speciation is not detected using cytochrome c, e.g. humans and chimpanzees have identical cytochrome c.
9. Homologue genes are similar genes found in several different species. Since mutations occur at a certain rate in genes and random events spread the changes through the population, the gene pool will slowly change over time. If two populations are separated the gene changes will accumulate over time. A comparison of the homologue genes will show these differences and a time frame can be applied. Thus changes in molecules give a time frame – a 'molecular clock'.
10. B

## 28 Molecular Clocks

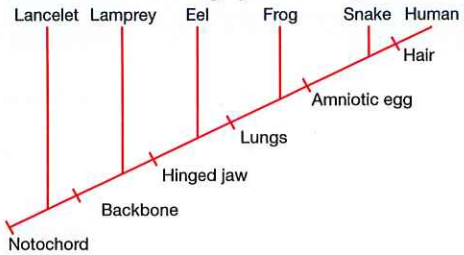
1. A molecular clock (or evolutionary clock) is an evolutionary timing method based on the observation that the code for some genes and particular regions of the genome seem to change at constant rates.
2. Molecular clocks are used to construct evolutionary trees with times of divergence from a common ancestor for different groups of organisms. The scientific principle behind this use is that if the rate of nucleotide change is constant then the number of changes is proportional to the time that has elapsed since the species branched from a common ancestor. Thus the number of differences in the nucleotide sequence gives a timeline since divergence.
3. Molecular clocks have been constructed using mtDNA, haemoglobin, cytochrome c, calmodulin, myoglobin and immunoglobulins.
4. Molecular clocks need to be calibrated against other independent evidence, e.g. using the fossil record and then a timing frame can be given to the molecular clock.
5. Molecular clocks using immunoglobulins, e.g. IgM and IgA have been used to show that monotremes and eutherians separated around 170 million years ago (million years ago) and marsupials and eutherians separated around 130 million years ago. Australian and American marsupials separated around 65 million years ago and the two monotremes (platypus and echidna) separated 22 million years ago.

## 29 Evolution of Genes

1. The biotechnology revolution caused by the advance of genetic engineering techniques has meant many scientists are now investigating gene similarities between different species to show evolutionary relationships.
2. Previously, evolutionary trees had been constructed using anatomy, physiology and embryology, with time lines based on fossil evidence to determine relationships between species.

3. If there is conflict between fossil evidence and molecular analysis, further research is needed as each technique has both advantages and disadvantages and more data is needed before final decisions can be made about the evolutionary tree.
4. An extant species has members still living.
5. Several evolutionary trees can be drawn from the same data as the data can be interpreted in different ways. Usually the different trees agree on certain aspects, but the finer details are disputable.
6. The 'serial endosymbiotic' theory proposes that mitochondria and chloroplasts were once free-living bacteria but were captured by primitive cells and turned into organelles for aerobic respiration and photosynthesis respectively.
7. Techniques used to compare the total genome of two species include DNA-DNA hybridisation and chromosome painting.
8. The collection of data about the DNA sequences of different species has greatly impacted on the development of technologies. Not only has there been an improvement in the DNA sequencing technologies to make them faster, more efficient and less costly, but the amount of data that needs to be stored and analysed has seen an improvement in computing technology increasing the capacity of computers to handle large volumes of data but also the creation of programs to manipulate the data.
9. The fossil record shows a sudden and explosive increase in the abundance and diversity of life forms in the Cambrian period. This was how geologists determined the beginning of the Palaeozoic era. All major groups in the animal kingdom had appeared before or during the Cambrian. Thus the base of the evolutionary tree is very clustered. This has created problems in the construction of a phylogenetic tree and more evidence is needed to help clarify relationships. The clustering at the base of the evolutionary tree in the Cambrian causes another problem as it means that many of the branches on the tree are very long and different branches have different mutation rates. Thus branches based on quickly evolving molecules could appear superficially long and cause problems in constructing a phylogenetic tree.
10. A back mutation occurs when there has been a change in an amino acid over time, but a later second mutation returns the code to its original state. Thus no change is recorded, but it should have counted as two changes.
11. Knowledge of the total genome and an analysis of the genes 'switched on' in particular types of cells provides data so that particular genes can be linked with particular activities.
12. When working out the evolution of genes, it is assumed that a form of the gene was present in an ancestor and over time it changed until it has its current components. 'Horizontal transfer' from one species to another means that the gene may not necessarily have been present in an ancestor. This means that there are considerable problems constructing an evolutionary tree based on this gene.
13. Convergent evolution is the independent evolution of similar features in different unrelated groups. For example, on a macroscopic level a shark and a dolphin look superficially similar with streamlined body and fins, however they evolved in different groups – fish with gills, ectotherm (shark) and mammal with lungs, endotherm (dolphin). On a microscopic level the production of the enzyme lysozyme found in lysosomes found in cows and langur monkeys show that both synthesise a lysozyme that share the same activity but have evolved from different ancestral molecules. Thus on a superficial level different groups could appear similar and the evolution of their genes could be closely linked. However, a more detailed study will show that it is convergent evolution and the groups are not as close as first suspected.
14. To gather secondary evidence to process, analyse and draw conclusions about evolutionary relationships from the analysis of genes you can use a search engine on the internet, and a library catalogue to locate information about DNA and genes. Secondary sources from the library provide information about the structure of DNA, genes and historical data, e.g. on *Drosophila* and mice which show the experiments which have identified common genes in different groups. This data allows the construction of evolutionary trees. Internet sources provide information on more recent experiments and current research projects showing which areas of the evolutionary tree are under further investigation.
15. When collecting secondary data, you can ensure the reliability of the source by checking the references cited in the source document, the qualifications of the author and by comparing the conclusions drawn by different sources. If the references are reputable, e.g. research institution, reputable university, government body, the person has appropriate qualifications in that area and if the conclusions from different sources are similar, then you can judge the data to be reliable.
16. D

## 30 Cladistics

1. Cladistics is a type of classification that places species into clades based on descent from a single common ancestor.
2. A clade is a group of species that belong to a single branch of the evolutionary tree. It includes the single common ancestor and all descendants. It is usually constructed by considering a single homologous characteristic, e.g. base sequences of a gene or the corresponding amino acid sequences of a protein.
3. A cladogram is a diagram used in cladistics to show ancestral relationships between species. It is an hypothesis of branching sequences and shows the most probable sequence of divergence in clades.
4. A node on a cladogram is a branch point indicating the evolution of shared derived characteristics.
5. A cladogram has branching points showing divergence along a single sequence whereas a phylogenetic tree has many branches with additional groupings.
6. Derived characters are the traits that tie the members of a clade together, e.g. the possession of feathers can be used to form a clade of birds whereas a shared ancestral character is a feature found in an ancestor of the taxon. For example, all primates have a backbone (a shared ancestral characteristic that was present in the ancestral vertebrate) and all primates have an opposable toe/thumb (not a feature found in their ancestors).
7. A morphological clade is based on structural features whereas a molecular clade is based on genetic information.
8. A clade is a monophyletic group as it gives the ancestor and all of its descendants.
9. In a monophyletic group all descendants are from a common ancestor whereas in a paraphyletic group the common ancestor is given and only some of the descendants given. While in a polyphyletic group all of the descendants are given but there is an unknown ancestor(s).
10. To construct a cladogram you need to follow the rules – 1. The original lineage should split into two when new species arise. 2. All species in the clade can be traced back to the ancestral species, which no longer exists. 3. All species derived from the common ancestor must be included in the taxon (must be monophyletic).
11. A scaled cladogram has a time axis so that each node indicates the approximate date of divergence of the two groups.
12. Molecular cladograms using the base sequence of a gene are very useful when constructing cladograms as the data is objectively quantifiable and the data can be used to compare very diverse organisms. For example, the protein cytochrome c has been used to construct clades of highly diverse organisms.
13. 

14. D
15. B

## 31 Homologous and Analogous Traits

1. Analogous structures perform the same or similar function often by a similar mechanism but have evolved separately whereas homologous structures are similar structural or biochemical features due to a common ancestor.

2.

| Feature | Homologous structure | Analogous structure |
|---|---|---|
| Structure | Similar fundamental structure | Different fundamental structure |
| Function | Similar in position and development but not necessarily in function | Similar function |
| Ancestry | Similarity due to common ancestry | Similarity is superficial resemblance |

3. Type 1 shows the C shaped spine of an ape while type 2 shows the S shaped spine of a human. They are homologous structures as they have similar structural features due to a common ancestor.
4. All mammals have a common ancestor that had seven cervical vertebrae. Divergent evolution and adaptive radiation led to the vertebrae in the giraffe reaching a combined length of over two metres while the house mouse vertebrae only span one centimetre.
5. C
6. A

## 32 Activity – Vertebrate Forelimbs

1. (a) Comparative anatomy is the study of the structures, e.g. organs or bones found in different groups.
   (b) The pentadactyl limb is a structure studied in comparative anatomy. It is the 5-digit limb found in some amphibians and reptiles, birds and mammals. It shows that these groups had a common ancestor and since that time adaptive radiation has led to different modifications of the limb.
2. (a) An homologous structure is a similar structure found in different organisms due to common ancestry but used for different purposes.
   (b) Whale – bones are short and thick to form a paddle which is strong and flat to overcome viscosity of water. Bat – phalanges are long and thin to form a wing with large surface area for flight. Pig – toes form a hoof to distribute weight when walking. Human – arm has highly opposable thumb, giving manual dexterity to make tools.
   (c) Homologous structures show how different groups have evolved from a common ancestor and since that time there has been adaptive radiation where the limb is modified for different purposes in each group.
3. C
4. C
5. A

## 33 Fossils

1. A fossil is anything that indicates the existence of past life. It can be part of or a whole organism or a trace left behind, such as a footprint or cast.
2. Palaeontology is the study of fossils.
3. The fossil record is incomplete as remains need to become fossilised, the fossil needs to be preserved over a long time and not destroyed in the rock cycle and the fossil needs to be found. This sequence of events does not always occur so that the entire biodiversity of a time or a location may or may not be complete.
4. If the body of the organism leaves an impression, the cavity it made may form a mould and if the cavity is filled with material, a cast is formed.
5. Traces of organisms include footprints, burrows, trails and animal dung.
6. Index fossils are fossils of organisms that lived for a short time and were found over a wide area.
7. Trilobites in the Palaeozoic era are used as index fossils to date rocks from different locations. Scientists have used the appearance and disappearance of different fossil species to divide geological time into eras, periods and epochs. The trilobites have been used to distinguish the different periods in the Palaeozoic era.
8. The sudden appearance of many organisms with shells around 540 million years ago was used by scientists to mark the beginning of the Palaeozoic (ancient life) era. The mass extinction of many invertebrates marks the end of the Palaeozoic era and the beginning of the Mesozoic era, just as the mass extinction of the dinosaurs marks the end of the Mesozoic era and the beginning of the Cainozoic era.
9. The eras were named as Palaeozoic ('ancient life'), Mesozoic ('middle life') and Cainozoic ('recent life'), i.e. according to the type of life form present. However, the periods were named by the early geologists who investigated the rocks from these different layers and they named the periods after the place or the type of rock found. For example, the first rocks of the Mesozoic era that were studied were limestones from the Jura Mountains and hence the age was given as 'Jurassic', and the chalk layers, called in French Terrain Crétacé, were called the 'Cretaceous'. Since the Mesozoic seemed to have three divisions, the earliest layer was then called the 'Triassic'.
10. Palaeontological evidence is the study of fossils. By comparing fossils with each other and with present-day species, any changes and any similarities provide important evidence to show how organisms have changed throughout the history of the Earth. We can then reconstruct ancient climates, environments and geography using this information. This shows how life has changed from simple to complex forms.
11. The oldest fossils that provide palaeontological evidence for the origin of life on Earth are bacteria fossils formed in stromatolites. Stromatolites are banded domes of sediment consisting of bacteria and cyanobacteria colonies. Geological evidence and the dating of these rocks gives an absolute date of around 3500 million years ago. Since these are colonial organisms, the actual first organisms would have existed before this date.
12. The study of fossil evidence shows several trends in the origin of life on Earth: 1. A trend from simple to more complex structures. 2. A trend from aquatic to terrestrial life forms. 3. A trend from asexual to sexual reproduction. 4. An increase in the degree of protection or care of the embryo as it develops.
13. (a) Oldest stratum is found at locality Y because you can use the leaf to find that Z is older than X and then use the shell below the fish to show that Y is older than Z.
    (b) Index fossils are organisms which only lived over a wide area, for a relatively short time span in geological history. Their presence therefore can be used to date any layer in which they are found.
14. A
15. D

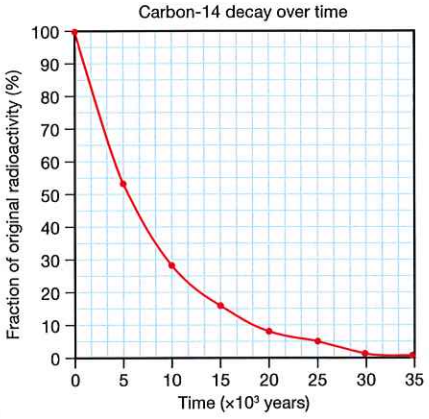
## 34 Absolute Age and Relative Age

1. The law of superposition states that the oldest layers are found at the bottom and the youngest layers are found at the top, unless there has been dynamic movement of the strata, e.g. folding and faulting.
2. (a) Oldest F, H, E, D, C, G, B, A youngest.
   (b) This is relative dating of rock layers.
3. An index fossil is widely distributed but only lived for a short time which means it clearly indicates the age of the rock layer.
4. The Precambrian is a supereon division for the geological time before the Cambrian period of the Palaeozoic era. The beginning of the Cambrian period (540 million years ago) is marked by a massive increase in the number and diversity of fossils and so many people use the term Precambrian for all the divisions before this time.
5. The Cainozoic era divided into the Palaeogene, Neogene and Quaternary periods.
6. Fluorine absorption analysis can only be used on bone and is a method of relative dating. When bone is in the ground, fluoride ions from the soil water replace some of the ions in the bone. The longer the bone is in the ground, the more fluorine present.
7. Natural variables affect fluorine absorption, e.g. rainfall, soil permeability and temperature so that bones being compared need to have been exposed to the same changing conditions over time. Different bones absorb fluorine at different rates which means that bones of similar density need to be compared. Differences can be due to age, e.g. adult bone/juvenile bone, part of the body, e.g. femur/jawbone or when comparing different animals.

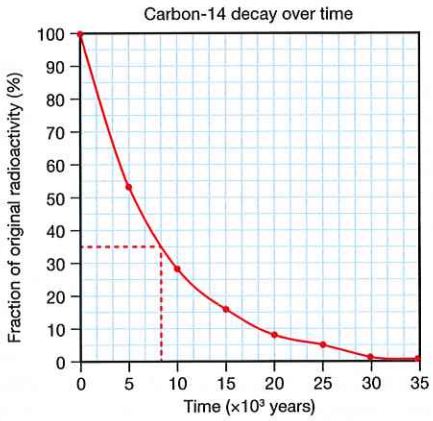
8. Absolute dating gives the age in years.
9. Radiometric dating uses radioisotopes to determine the age of rocks or fossils giving the actual age in years plus or minus the experimental error involved in the particular method used.
10. The half-life is the time it takes for half the atoms to decay.
11. The atom that undergoes radioactive decay is known as the parent nuclide and the atom which is formed is known as the daughter nuclide.
12. The rate of radioactive decay is not affected by changes in temperature, rainfall, soil permeability, pressure or the chemical environment while these factors can affect fluorine absorption dating.
13. Geological absolute dating requires long half-lives while medical applications requires short half-lives causing different radioisotopes to be useful for different time frames and for different purposes.
14. Useful geological radioisotopes include carbon-14, potassium-40, rubidium-87, samarium-147, uranium-238 ($^{238}U$ to $^{206}Pb$) and uranium-235 ($^{235}U$ to $^{207}Pb$).
15. Uranium is very useful for dating rocks as almost all igneous and metamorphic rocks contain enough uranium and lead for accurate dating and dating methods can be used on powdered whole rocks, mineral concentrates or single grains.
16. The oldest rocks of terrestrial origin dated so far come from Jack Hills, Western Australia dated at 4.4 billion years old.
17. As carbon-14 has a half-life of 5730 years the oldest date reliably measured by radiocarbon is around 50 000 years. This means that carbon-14 has limited use in dating fossils and is restricted for organisms that lived within the last 50 000 years.
18. Fission tracking occurs when uranium-238 decays (in fission process) and the released high speed atoms cause damage to nearby rocks and minerals (fission tracks). In a rock containing uranium-238 the number of fission tracks will increase over time at a rate that depends on the uranium content. The age of the rock can be calculated by measuring the uranium content and the density of the fission tracks.
19. D

## 35 Radiometric Dating

1. Radiometric dating uses radioisotopes to determine the absolute age of fossils or rocks plus or minus the experimental error involved in the particular method used.
2. Radiocarbon is used to date organic material such as bone, shell, charcoal, linen, wood, paper, and cave paintings which have pigments from honey, milk, or oil seed.
3. The half-life of carbon-14 is 5730 ±30 years.
4. The use of particle accelerators as mass spectrometers means it is possible to measure the number of carbon-14 atoms directly rather than waiting for their decay allowing dating up to 70 000 years ago.
5. When using radiocarbon dating for artefacts made by people the date of the use needs to be distinguished from the date the material being used was harvested or gathered. For example, with writing on paper such as the Dead Sea Scrolls, the radiocarbon dates give the time the flax was cut and the leaves died which is not necessarily the exact age of when the scrolls were written.
6. It has been discovered that the ratio in the atmosphere of one atom of carbon-14 to every million million ($10^{12}$) carbon-12 atoms is not a totally fixed ratio. The study of growth rings on trees and gas analysis of ice cores from glaciers and Antarctica has required corrections to be made to allow for fluctuations in the carbon-14 content of the atmosphere.
7. Potassium-40 has a half-life of $1.26 \times 10^9$ years.
8. Potasium-40 has a long half-life which means it is used in dating volcanic rocks and minerals older than 200 000 to 300 000 years and is limited to a small range of rocks past a certain age.
9. Many of the hominid fossils in East Africa have been found in layers associated with volcanism and sedimentation and potassium-argon dating has been used to date the volcanic ash deposits and thus the fossils in these deposits. For example, the footsteps at Laetoli were made by three hominin individuals walking in powdery volcanic ash from a nearby eruption and the tuff was dated using potassium-argon dating and stratigraphy.
10. (a) Zircon is a mineral ($ZrSiO_4$) that has no tracks when it forms in a volcanic eruption. As nearby uranium undergoes fission, the high speed atoms that are released leave fission tracks in the zircon. The number of tracks depends on the amount of uranium and the time of exposure. Zircon fission tracks can thus be used to date rocks.
    (b) Zircon grains from Jack Hills, WA are dated 4.4 million years ago and the oldest known rocks in the world of terrestrial origin.
11. Rb-Sr dating uses rubidium-87 which has a half-life of $48.8 \times 10^9$ years. With a long half-life it is usually only used to date rocks older than about 100 million years old, e.g. Moon rocks and is not suitable for a recent ice age 12 000 years ago.
12. (a)




(b) 8300 years

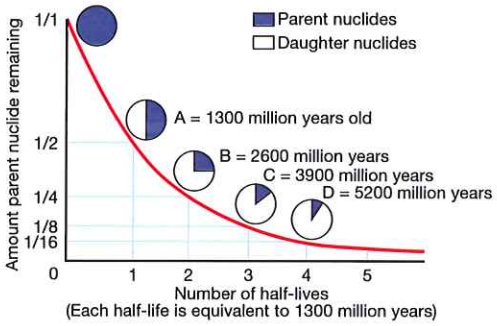



13. Fission tracking can be used on minerals and glass in a range from 100 years to 4500 million years before the present.
14. D
15. D

## 36 Changes in Technology

1. Absolute age gives a date for a rock or a fossil, e.g. 50 million years ago, while relative age compares the age with another rock or fossil, i.e. older than or younger than the other rock/fossil.
2. Stratigraphic correlation looks at the index fossils found in each rock layer. Since the index fossils are known to have lived at a certain time, you can correlate two layers from different areas. If they have the same index fossil, then the two layers were laid down at the same time, even though they were in different areas.
3. For many years scientists used stratigraphic correlation and the law of superposition to work out the relative ages of different rocks and fossils to indicate time frames. The discovery of radioactive isotopes and the invention of instruments to measure the amount of radioisotope present in a sample has enabled scientists to work out the absolute age of rocks and fossils. This gives a more precise date and age for the sample.

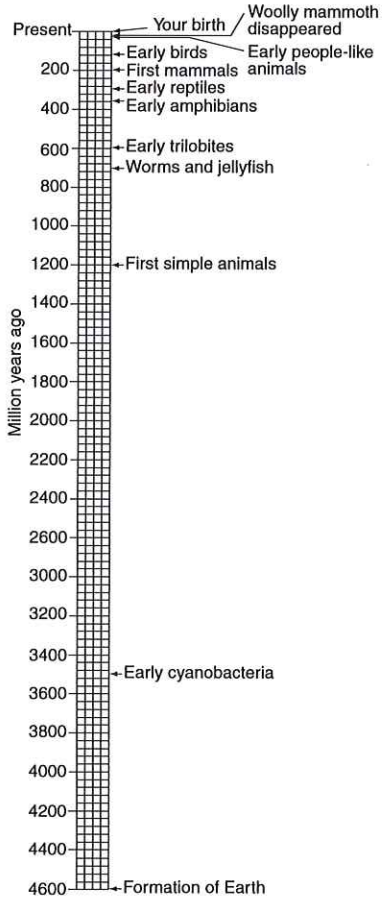
4. (a) Parent nuclide is potassium-40.
(b)




5. The use of the electron microscope is especially important when looking for the earliest signs of life on Earth as the earliest life form was unicellular and thus cannot be seen by the naked eye.
6. Genomics is the study of whole sets of genes and their interactions. Comparative genomics looks at the similarities and differences between the genomes of different species.
7. The invention of computer databases means that we can now store, handle and manipulate incredible amounts of information such as the code for DNA for humans. Using the computer, we can compare the human code with the code for other living livings such as other mammals, plants and bacteria. These comparisons show similarities and differences in DNA and provide information about the origin of life and the evolution of living things.
8. New techniques in biochemical analysis, e.g. radioactive tracers, have increased our understanding of the processes that occur in cells. Radioactive isotopes can be used to show particular metabolic pathways. A comparison of these pathways in different species, looking at similarities and differences, provides information about how changes have occurred as life evolved from simple forms to more complex forms. This increases our understanding of the origin of life on Earth.
9. (a) The homeobox DNA sequence would have evolved very early in the history of life on Earth, e.g. was probably present 600 million years ago.
   (b) If the homeobox DNA sequence has changed little for millions of years, it can be inferred that it is highly valuable and needed for survival.
10. Carbon-14 is formed from nitrogen in the atmosphere when bombarded by cosmic rays from the Sun. It combines with oxygen to form radioactive carbon dioxide. This is taken into autotrophs, e.g. plants in photosynthesis, and passed to heterotrophs through the food chain. While alive the ratio of carbon-12 to carbon-14 in the organism stays constant. When the organism dies, the amount of carbon-14 decreases as it decays without being replaced. Knowing the half-life of carbon-14 means that the amount of carbon-14 present can be used to date the fossil.
11. Comparing the amino acid sequences in DNA in different species shows similarities and differences between these species. From this, scientists gain an understanding of which genes are common and how changes have occurred over time. From this they can increase our understanding of the origin of life and the process of evolution through the history of life on Earth.
12. (a) PCR is polymerase chain reaction.
   (b) PCR is a technique for amplifying DNA by heating it with specific primers, a heat-resistant DNA polymerase and nucleotides.
   (c) PCR is used to make multiple copies (e.g. thousands to millions) of a particular DNA sequence for use in many situations, e.g. DNA fingerprinting, detection of bacteria and viruses, diagnosis of genetic disorders, preparing samples for DNA testing.
13. RNA not only stores genetic information but can catalyse certain chemical reactions and this led to the hypothesis that first life used RNA for basic cellular processes.
14. Bioinformatics is the use of computers, software and mathematical models to process and analyse large data sets and biological information.
15. C
16. B
17. A
18. B
19. A
20. C
21. B

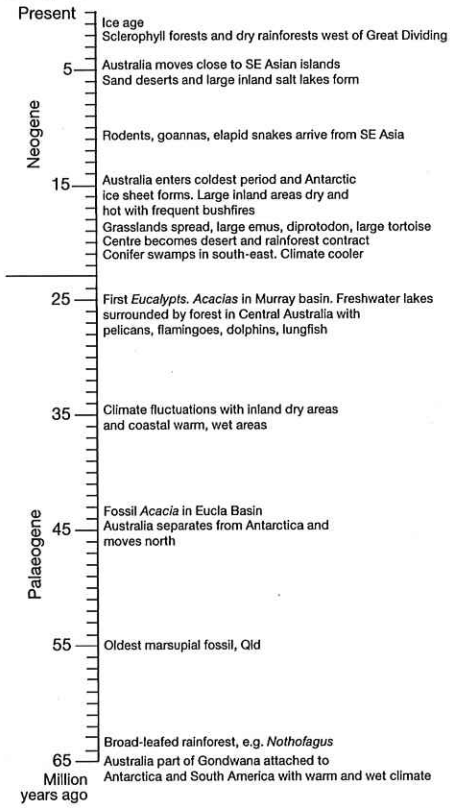
## 37 Activity – Fossils Give a Timeline for Life on Earth

1. For most of the history of the Earth (87%) it was the Precambrian era when organisms either did not exist or were soft bodied and left few fossils. Only 15% of the history of the Earth has fossils and hence we have little record for most of the history of the Earth.
2. Timeline.



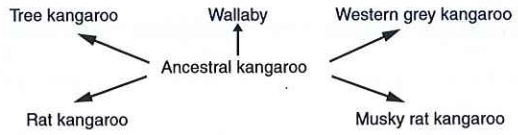

3. His classmates disagreed with his suggestion that the three fossils would have been in the same sedimentary bed as these fossils belong in different time frames in geological history. The trilobites (fossil X) were abundant during the Palaeozoic era and then died out in the Permian period. Humans (fossil Y) did not exist at this time and the dinosaur (fossil Z) lived during the Mesozoic era and hence they could not have been laid down at the same time.

4. Sample timeline.




## 38 Historical Development of Evolution Theories

1. Within the bacteria population there is variation so that some are naturally immune to the antibiotics. Natural selection occurs so that those that are not immune die due to antibiotics and those with the favourable characteristic survive, reproduce and their offspring inherit the favourable characteristic. Over time (and many generations of bacteria can occur within hours or days) the frequency of the characteristic increases in the population so that the population becomes immune to the antibiotic.
2. Charles Darwin queried why there would be special creation of two such species and suggested that the modern armadillo was related to or descended from the extinct glyptodon.
3. Convergent and divergent evolution both involve evolution – the change in a population over time – due to the mechanism of natural selection, whereby changes in the environment lead to changes in the population. They differ in that divergent evolution occurs when different groups arise from a common ancestor due to adaptive radiation so that groups look quite different. For example, adaptive radiation from the ancestral kangaroo has led to the musky rat kangaroo, the rat kangaroos, the red kangaroo, the tree kangaroo and the wallabies.




Convergent evolution leads to superficial similarities due to organisms living in the same habitat or having the same lifestyle although they are not closely related. For example, convergent evolution has led to a dolphin, which is a mammal, and a shark, which is a cartilaginous fish, looking superficially similar, although reproduction, body temperature control etc are quite different.

4. The Darwin/Wallace theory of evolution would explain the change in the kangaroo and the current 62 different species as an example of divergent evolution. From a common ancestor that lived 15 million years ago natural selection occurred and those with favourable features survived and reproduced. Their offspring inherited the favourable features and over time and with some groups becoming isolated the populations changed and speciation occurred.
5. C
6. C
7. C
8. A
9. D
10. A

## 39 Punctuated Equilibrium

1. Darwin proposed that there is a slight change in the population so that there is a gradual change over time, e.g. diagram (a) shows the butterfly fossils gradually changing in pattern on the wings and degree of colour shading in successive layers. Diagram (b) shows punctuated equilibrium where the butterfly species remains unchanged in a state of stasis for a long time and then the species is replaced by a new variation.
2. Since there are fossils for both mechanisms, both forms of evolution must exist. So far the fossil record has more examples of punctuated equilibrium where there are few changes for long periods and then times of rapid change. The argument then becomes a discussion of which method is more 'common'.
3. (a) The order would be: c, a, d, b. Originally they were one group living on one side of the mountain (c). They reproduced, spread out and inhabited both sides of the mountain range (a). A river formed and on the southern side a new variant appeared, probably by natural mutation (d). This variant was more suited to this environment and over time became more frequent in the population. The river provided a geographical barrier (b).
   (b) The diagrams show how biogeography occurs. The population is found in both areas, then a geographical barrier separates the group into two and then, due to isolation and natural selection, the two groups diverge so that eventually they cannot interbreed and have become two species.
4. B
5. B
6. A
7. A
8. B
9. C

## 40 Human Evolution

1. Humans are classified as a primate as they have – 1. An opposable thumb. 2. Shoulder joint with high rotating ability. 3. Forward facing eyes with stereoscopic vision. 4. Reduced snout and olfactory centre of the brain. 5. Enlarged skull with large cerebrum. 6. Five digits on limbs. 7. Nails (not claws). 8. Bicuspid teeth.
2. The hominoids includes humans and the great apes – the gibbons, gorillas, chimpanzees. While the hominins refers to species on the human branch of the evolutionary tree which all belong to the family Hominidae. Hominins are bipedal and have larger brains.
3. Fossils of hominins show that there was physical evolution with differences in hominins including different brain sizes, jaw structures, teeth, arm length in proportion to body length and degree of sexual dimorphism.
4. Human culture sets humans apart from other species as humans manufacture and use complex tools. These tools led to the agricultural revolution, Industrial Revolution, modern medicine and mass transportation.
5. Many people believe that the physical evolution of humans stopped as environmental pressures ceased to determine the 'fittest' in the population. They believe that the use of tools to collect and distribute food, build houses, construct cities and the use of modern drugs and medical procedures to sustain life has meant the human population has rapidly increased with no physical evolution, though there is constant and fast cultural evolution. However, recent research has suggested that there is a link with a woman's fertility and particular features, e.g. height, amount of body fat and cholesterol and blood pressure. This data predicts that there will be a slow gradual change in the physical appearance of women as allele frequencies change over generations due to some women having more children than other women.
6. Fixed genes have only one type of allele and thus that trait is 'fixed' in the gene pool. These traits are not polymorphic.

7. The 1000 Genomes Project is an international research project that started in 2008 to catalogue human genetic variations.
8. Ancestry information markers (AIMs) can be used to put people into broad, geographically based groups.

## 41 The Primates

1. The criteria used to classify the order Primates have changed over the last 300 years. Linnaeus in 1758 used the presence of four incisors (cutting front teeth), two clavicles (collarbones), two mammary glands on the chest and at least two prehensile extremities to identify primates. Robert Martin in *Primates: A Definition* (1986) included a wider range of features where subdivision depends on the degree or grade of a feature, e.g. compared with most other mammals primates have an increased brain size relative to their body size. Martin considered means of location, relative brain size, reproductive biology, dental patterns, geographic distribution, influence of major sense organs on the shape of the skull and habitat.
2. Physical features used to classify primates include – an opposable thumb and/or toe, shoulder joint with high rotating ability, forward facing eyes, reduced snout and olfactory centre of the brain, enlarged skull with large cerebrum, five digits on limbs, nails (not claws) and bicuspid teeth.
3. (a) Primates were traditionally subdivided into the prosimii and anthropoidea.
   (b) Primates are now subdivided into three groups – with the prosimians separated into the lemurs of Madagascar and the lorises and pottos of tropical Africa and southern Asia (wet noses) and the tarsiers (dry noses). The third group is the anthropoids.
4. The earliest primates were most likely similar to the lemurs, lorises and pottos.
5. More primitive features would involves smaller cerebral cortex, reduced parental care, gestation and maturation, less upright gait and greater reliance on scent.
6. Catarrhine means narrow nosed with nostrils that face downwards, e.g. Old World monkeys, apes and humans and platyrrhine means flat nosed with nostrils that face sideways, e.g. New World monkeys.
7.

| Feature | New World monkey | Old World monkey | Ape |
|---|---|---|---|
| Examples | Spider monkeys, squirrel monkeys, howler monkeys and capuchins | Baboons, leaf monkeys, Rhesus monkeys and vervets | Gibbons, gorillas, chimpanzees, orangutan and bonobo |
| Arboreal and/or terrestrial | Arboreal | Arboreal and/or terrestrial | Arboreal and/or terrestrial |
| Platyrrhine or catarrhine | Platyrrhine | Catarrhine | Catarrhine |
| Tail | Many have long prehensile tail | None have prehensile tail | No external tail |

8. Structural features that would identify a fossil as a primate include – opposable thumb that allowed objects to be grasped and manipulated and forward facing eyes that enabled stereoscopic binocular vision.
9. Captive breeding programs are extremely important for orangutans as it not only ensures survival of the species as humans can breed individuals but reduces the need for poaching and smuggling and hence there will be less interference with free orangutans living in native forests.
10. Diagram (a) shows the platyrrhine – flat nose nose with widely spaced nostrils of New World monkeys while diagram (b) shows the catarrhine – downward facing nose and close nostrils of Old World monkeys, apes and humans.
11. (a) Spider monkeys are New World monkeys and are found southern Mexico and South America, e.g. Brazil.
    (b) Baboons are Old World monkeys found in Africa.
    (c) Orangutans are apes and are found in western Borneo and the top of Sumatra.
    (d) Gorillas are apes and are found in Africa, e.g. west central Arica and east central Africa.
12. Social groups with group bonding provides protection for the mother and child, e.g. against predators and increases the chances of finding food and increases the chances of infant survival to adulthood. This aids evolution by securing the next generation to live in that area.
13. An opposable thumb means the thumb can be moved around to touch the other fingers. This allows the hand to grasp objects. Similarly an opposable toe can touch other toes. These features aid an arboreal lifestyle allowing branches to be grasped. Being able to grasp and manipulate objects has been an important feature in the development and evolution of the primates.
14. A prehensile tail acts as a fifth limb and aids movement through the trees as arms and legs may be free to grab food or other branches while the monkey 'hangs' by its tail. The reduction in size of the thumb of the spider monkey allows for greater brachiation as a thumb grasping a branch in the opposite direction to the fingers could be broken and would get in the way during the swinging motion of brachiation.

## 42 Primate Hands and Feet

1. An opposable thumb has the ability to be brought in contact with any of the fingers of the hand.
2. An opposable and mobile thumb gives the ability to grasp objects which is beneficial for an arboreal lifestyle and also increases the ability to manipulate objects.
3. All non-human primates have a divergent or partially opposable big toe.
4. A power grip is the ability to hold an object between the fingers and the palm. The thumb reinforces the fingers.
5. Different power grips include the squeeze grip, the disc grip, the spherical grip and the hook grip.
6. The precision grip relates to the ability of the index finger to touch the thumb. Unlike other primates the arrangement and length of the human thumb allows humans to form the precision grip, e.g. humans have the unique capacity to thread a needle.
7. Different precision grips include the five-jaw chuck, the two-jaw chuck pad to pad and the two-jaw chuck pad to side.
8. The hand of the orangutan has a long palm with long fingers and a short thumb. This is suited to an arboreal lifestyle using brachiation to swing through the trees grasping branches without a thumb holding the branch in the opposite direction to the other fingers. The hand of the gorilla has a long palm and shorter fingers. The gorilla uses its hand for walking as well as for handling objects. The human hand has a shortened palm and a long thumb which increases opposability and allows the formation of the precision grip with increased manipulation ability.
9. This evidence suggests that the human hand is more 'primitive' than the chimpanzee hand. Since the last common ancestor the chimpanzee hand has evolved, e.g. with longer palm and shorter thumb which aids swimming movement by brachiation in trees while the human hand has not dramatically changed and evolved in that time span.

## 43 Primate Head and Skull

1. Binocular vision means using two eyes and usually infers that both eyes can focus on the same target object. Stereoscopic vision involves the processing of images that overlap to give a 3D image with depth perception.
2. Forward facing eyes means there is less field of view. To overcome this problem primates have a mobile head and neck which can move quickly to get side images.
3. Photoreceptors called cones found in the retina at the back of the eye detect colour and provide colour vision.
4. Brain size will affect the shape of the skull. The human skull is higher and more rounded than the apes holding the larger braincase.
5. The particularly bright blue and red on the face and rump of the dominant alpha male mandrill provides colour communication establishing the hierarchy of the group. This reduces the need for in-fighting to maintain leadership and provides greater social cohesion.

6. The long snout of many mammals is often related to the sense of smell. Primates rely more on vision than their sense of smell and thus there are fewer olfactory receptors and nervous tissue in the primate face.

## 44 Primate Skeletons and Locomotion

1. The vertebral column is the backbone, which consists of a series of vertebrae (bony or cartilaginous segments) going from the head to the tail, enclosing the spinal cord.
2. (a) The clavicle is the collarbone.
   (b) The clavicle in primates gives rotation of the foramen which allows extensive arm movement and the ability to reach behind the body. This is important in moving along tree branches and is especially important in brachiation when swinging from branch to branch.
3. The earliest primates were probably arboreal quadrupeds.
4. An arboreal lifestyle requires an opposable thumb/big toe for gripping, mobile shoulders, hip and elbow joints and stereoscopic vision.
5. In knuckle walking the gorilla nd chimpanzees extend their arms and support their weight on their knuckles holding their body in a semi-upright position with their shoulders higher than their hips.
6. Brachiation is a kind of swinging movement used to get from tree to tree.
7. Species that run on the ground have longer limbs and some loss of gripping feet.
8. C

## 45 Primate Reproductive Features

1. Gestation means the length of the pregnancy.
2. The data shows that the relationship of female weight and gestation for other mammals, e.g. sheep, rabbit and cow is larger weight has longer gestation. However, this trend does not occur with the primates, e.g. gorilla has larger female weight but shorter gestation than humans.
3. Increased lengths of time for parental care means there is a greater dependency on learnt behaviour rather than innate or instinctive behaviour. Learnt behaviour can involve improving motor skills, social skills and cognitive skills, e.g. by observing the actions of others in their group, imitating these behaviours, practice and trial and error.
4.

| Social organisation | Description |
|---|---|
| Single female and offspring | The range of the adult male and female overlap and the adults come together for reproduction. The range of the male may be large overlapping the ranges of several females. |
| Monogamous social group | Consists of an adult female-male pair and their immature offspring, e.g. gibbons, tarsiers. The sizes of both sexes are relatively the same. |
| Polyandrous social group | Consists of one mature female and her offspring with several sexually active males, e.g. in tamarins, though this type of grouping is a relatively rare organisation. |
| One male social group | Consists of several females and their young and a single adult male, e.g. baboons, howler monkeys and gorillas. There are several different forms of this grouping with some being dominated by a stable core of females that choose the male who may join the group for a limited time and aid defence of the territory. Or it can be dominated by a silverback male who drives off males when they reach maturity. |
| Multimale-multifemale group | Consists of several adult males and many females and their offspring forming a large group, e.g. macaques. There is usually a dominance hierarchy among the males and females. |
| Fission-fusion group | Change size and composition throughout the year. Individuals leave or join groups and some separate to form a subgroup, e.g. while foraging for food or they all join together if a rich food source is found or to occupy a particular sleeping site. |

5. The size of a social group depends on the amount of food and shelter provided by a particular habitat. In a forest primates can disperse among the trees to find food and escape predators and can thus live in smaller groups while in grasslands larger groups provide protection against predators.

## 46 The Hominidae

1. The hominidae is a family in the primate order.
2. The ending '-idae' is given to the 'family' taxon in biological classification.
3. The definition of 'hominid' has changed over time with the term now having a broader meaning than it did in the past. Hominids now include the great apes and their ancestors which means it includes the modern humans, chimpanzees, gorillas and orangutans.
4. Changes in classification are due to improved technologies and increased knowledge and understanding about evolutionary relationships. Classification is a human construct which is used to show how we believe organisms should be grouped.
5. Technological advances include the use of DNA-DNA hybridisation to compare the DNA of different species, DNA sequencing to find differences in either nuclear or mitochondrial DNA (mtDNA) between two species to determine their likely time of divergence from a common ancestor and karyotype analysis to compare the number, shape and banding of chromosomes of different species to determine their degree of similarity.
6. The number of incisors, canines and molars are the same but the New World monkeys have three premolars while ho9minids only have two premolars.
7. Hominids are the largest primates and have – 1. No tail and no ischial callosities. 2. Upright or semi-upright stance. 3. Same dental formula $\frac{2,1,2,3}{2,1,2,3}$ with broad incisors. 4. Prominent face and prognathous jaw, 5. Arms shorter than legs. 5. Only the orangutan has an arboreal lifestyle. 6. Sexual dimorphism with males larger than females. 7. Large brain relative to body size.
8. Nails are a feature of primates and would have aided grasping ability for an arboreal lifestyle. Flattened nails improve grasping ability. Broader fingertips with tactile pads not only gives unique fingerprints in humans but also aids holding and manipulating objects for tool making to help food gathering and protection against predators.

## 47 The Hominins

1. The Hominins refers to the human branch of the evolutionary tree. It includes bipedal humans and extinct ancestors and close relatives, e.g. other species in the genus *Homo* and the australopithecines.
2. The earliest hominins were small in size, had brains similar in size to modern apes and had a set of characteristics that allowed both upright bipedal walking and climbing.
3. Fossils of *Sahelanthropus tchadensis*,, e.g. skull, jawbones and teeth show several hominin features, e.g. flatter face than apes and position of the foramen magnum, the opening for the spinal cord is oval, downward pointing and near the front of the skull suggesting an erect spine.
4. (a) Derived characters are features that evolved in the lineage leading up to a clade that sets the members of that clade apart from other individuals.
   (b) Shared derived characters of hominins include – reduced canine teeth, relatively flat face, bipedalism and more upright stance.
5. (a) Bipedalism means usually walking on two legs.
   (b) Bipedalism is advantageous in woodlands and grasslands as it raises the line of sight for seeing predators, it is an efficient way of travelling long distances, it minimises exposure to the Sun aiding temperature regulation and it frees the hands for carrying food and manipulating tools and objects.

6.

| Suggested theory | Adaptive advantage |
|---|---|
| Carrying theory | Those that could carry more food had better survival chances. Weapons could also be carried for defence and food gathering and tools held and manipulated. |
| Hunting theory | Those that could make weapons, e.g. spears and throw spears and catch prey had better survival chances. Upright walking is slower but long distances can be travelled to find food. |
| Better vision theory | Upright posture gave better vision over tall grasses and they could see approaching prey or predators giving better survival chances. |
| Eating grass seed theory | Freeing the hands to gather grass seeds means more food can be collected and carried ensuring a more permanent food supply and increased survival chances. |
| Feeding from bushes theory | Freeing the hands combined with taller reach means more food sources can be used and the different foods can be carried ensuring a more permanent food supply. |

7.

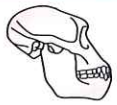
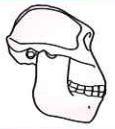
| Feature | Why needed for bipedalism |
|---|---|
| Position of foramen magnum | The entrance of the spine to the cranium, needs to be well forward for upright posture and balancing the head so that the spine begins vertically beneath the top of the skull. |
| Shape of pelvis | A short and bowl shaped pelvis is more stable for an upright stance and allows muscle attachment for bipedal locomotion. |
| S shaped spine | Allows the centre of gravity to fall over the hips and is able to withstand stresses and function as a spring. |
| Carrying angle of femur | The thighbone, is angled out from the knee so that weight distribution remains closer to the central axis giving more stability when walking upright. This changes the arrangement of the knee joint. |
| Foot | Big toe is not opposable and is rotated to be in line with the rest of the foot. The big toe is the last point of contact with the ground when walking and gives additional leverage. |
| Shape of rib cage | More barrel shaped chest so the torso can flex and the arms are free to swing and manipulate objects. |

8. Primate A is the chimpanzee, primate B is the australopithecine and C is the human. The top of the femur shows the carrying angle and how weight is carried from the pelvis to the feet. The thick bone with large head of the femur shows bipedalism and thus primate C is the human. Primate A has the smallest shaft and head of femur and is thus the chimpanzee. Primate B is the australopithecine being bipedal but not as upright as humans.
9. A

## 48 The Australopithecines

1. The australopithecines were hominins that lived 4 to 2 million years ago. They were bipedal with human-like hands and feet but had smaller brains about one third of the modern human brain.
2. Raymond Dart, an Australian scientist who was professor of anatomy at the Medical School of the University of Witwatersrand, South Africa discovered a fossil skull near Taung in 1924 which he named *Australopithecus africanus* (southern ape of Africa). He thus named the genus.
3. The Piltdown Man was a forgery and believed as the accepted theory of the time proposed that the evolution of humans involved the increase in brain size and then a shift to bipedal locomotion as we began using our hands to carry things, using our brain to think about it. The forgery used the cranium of a human with the jaw of an orangutan and hence supported this theory. Raymond Dart had found a real skull – *A. africanus* – but his findings were ignored as people preferred the Piltdown Man evidence, thus setting back the study of early humans by at least 20 years until the forgery was proven to be a fake in 1953 using fluorine dating.

4.

| Species | Front view of skull | Side view of skull |
|---|---|---|
| *Australopithecus afarensis* – rounded skull, small brain | |  |
| *Australopithecus africanus* – domed cranium, larger jaw bone | | |
| *Paranthropus robustus* – dish shaped face, sagittal crest on braincase | | |
| *Paranthropus boisei* – broad flaring cheekbones, robust upper jaw, skull short front to back | |  |

5. (a) Percentage difference can be calculated by the formula

$$\frac{\text{Mass of male} - \text{mass of female}}{\text{Male of male}} \times 100\%$$

| Australopithecine | Body mass (kg) | | Deviation (%) |
|---|---|---|---|
| | Female | Male | |
| *A. afarensis* | 29 | 45 | 36% |
| *A. africanus* | 30 | 41 | 27% |
| *P. robustus* | 32 | 40 | 20% |
| *P. boisei* | 34 | 49 | 31% |

From the calculations *A. afarensis* shows the greatest sexual dimorphism.

(b) Sexual dimorphism can be used to infer social organisation with difference in body size of males and females related to competition between members of the same sex vying for mates, e.g. males fighting for females to form a one-male social group with the dominant male defending and controlling the group. When there is less sexual dimorphism social organisation tends to be monogamous one male-one female and offspring groups. Thus the data infers that *A. afarensis* with the highest difference between males and females was most likely to live in a one-male group. The social organisation of the australopithecines is speculative and factors other than body size, e.g. canine size, diet, habitat need to also be considered.

6. (a) The large molars and large jaw muscles are used for grinding and suggests that *P. boisei* ate tough plants and other hard vegetation, e.g. seeds, nuts and roots. The small incisors and canines support this diet. (*P. boisei* was nicknamed the 'nutcracker man' due its enormous cheek teeth.)

(b) The relatively unspecialised teeth of *A. afarensis* infers a generalised diet of plant foods, e.g. fruit, seeds, nuts. It has usually been believed that the gracile forms ate 'softer' plant food than the robust forms. Recent microwear analysis of fossil teeth show *A. afarensis* ate tough plants even though their enamel is less thick than the enamel of teeth of the robust forms.

7. The *A. afarensis* dental arch is a more parabolic shape like humans than the U shape of apes and shows that the size of the canine teeth and cheek teeth cusps are in between the sizes of apes and humans. The size of the teeth in *A. afarensis* is more similar to ape than human.
8. The robust forms lived at the same time as early *Homo* species and there would need to be too many changes in too short a time for this branch to be the human ancestral branch.

9. The male and female skeletons found at Malapa, South Africa show some features of other australopithecines, e.g. body size, brain size upper limbs, finger bones but also show human characteristics, e.g. thighbone and locomotion and degree of sexual dimorphism. The number of differences means that these fossils can be classified into a separate species of australopithecine – *A. sediba*.
10. At the time, researchers and the public believed that human evolution dated back about half a million years. When the K/Ar dating came back 1.75 million years ago for the volcanic ash in which the fossils were found, everyone was extremely surprised and timelines for human evolution had to be significantly readjusted.
11. The footsteps at Laetoli show bipedal locomotion of *A. afarensis* at 3.6 million years ago. This is important evidence, besides body structural features, e.g. position of foramen magnum, shape of spine, carrying angle of thighbone that shows upright posture and human-like walking. There is debate as the footsteps show a slightly divergent big toe than found in humans but there is a heel-strike deep print with toe-push-off impression showing a human way of walking. The absence of any adult limb bones at the site also increases the debate about the source of the footsteps and any adaptations related to the making of the footsteps.

## 49 Genus – *Homo*

1. The genus *Homo* includes modern humans and many close ancestors, e.g. *Homo habilis*, *H. erectus, H. ergaster, H. heidelbergensis, H. georgicus, H. floresiensis, H. neanderthalensis* and *H. sapiens*.
2. The genus evolved in Africa with only one living (extant) species – *Homo sapiens* – modern humans who are found worldwide.
3. There is great variation in phenotypic features of humans in adult height, skin colour, body shape, hair and eye colour, nose and cheekbone structure, proportional finger length and fingerprint.
4. It is not easy to identify the features that mark the genus *Homo* as many features are behavioural, which leave little fossil evidence. Species in the genus *Homo* have – 1. Larger brains – though this must be measured relative to body size. 2. Prominent nose – with raised bone surfaces around the nasal opening and a bony spine at the centre of the base of the nasal opening. 3. Smaller teeth – especially the cheek teeth and tooth eruption is slow. 4. Flatter faces – with thin bone of lower jaw and there may be a chin. 5. Short base of skull – which can be folded and related to speech. 6. Late sexual maturity – with increased length of parental care.
5. Polymorphism means there are two or more forms of a particular trait that causes different phenotypes, e.g. different blood groups, different eye colour.
6. Sexual dimorphism occurs when there are differences between the sexes beside the sexual organs, e.g. secondary sexual features, size and height.
7. Species in the genus *Homo* have – 1. Larger brains – though this must be measured relative to body size, e.g. brain larger than 700 cc with a large cranial vault above the face to hold the brain with increased cranial vault thickness. 2. Prominent nose – with raised bone surfaces around the nasal opening and a bony spine at the centre of the base of the nasal opening. 3. Smaller teeth – especially the cheek teeth and tooth eruption is slow. 4. Flatter faces – with thin bone of lower jaw and there may be a chin. 5. Short base of skull – can be folded and related to speech. 6. Late sexual maturity – with increased parental care.
8. It is believed the genus *Homo* appeared around 2.8 million years ago.
9. The earliest member of the genus *Homo* in the current fossil record is *Homo habilis* – the 'handy man'.
10. It is believed that *H. sapiens* appeared 200 000 years ago.
11. A

## 50 *Homo habilis*

1. *Homo habilis* ('handy man') was named by Louis Leakey in 1964 from fossils found at Olduvai Gorge dating 1.75 million years ago.
2. Leakey named the fossil 'handy man' as stone tools were found near the remains.
3. Compared to the australopithecines *H. habilis* has a larger brain, about 700 cc, smaller teeth, stood more erect and was taller. *H. habilis* was around 120 cm tall, with a light build and weighed around 40 kg.
4. Some people feel *H. habilis* remains are too ape-like and show an advanced australopithecine while others feel it was an early *Homo erectus*. Skull 1470, found in 1972 was dated at 1.9 million years ago and had a large brain, 750 cc, with a more human braincase shape but has the australopithecine broad cheekbones. While Skull 1813, found in 1973, was dated 1.8 million years ago had a small brain, 510cc, but looks more 'human' than Skull 1470. Skull 1805 has teeth like 1813, a face like 1470 and a braincase of 580 cc with a primitive crest along the midline like robust australopithecines.
5. In 1991, Bernard Wood suggested that the finds from Olduvai should remain *H. habilis* while the fossils from Koobi Fora should be split into two different species, *H. habilis* and *H. rudolfensis*.
6. *H. habilis* has a smaller face while *H. rudolfensis* has a broader and flatter face. *H. habilis* has a strong curved brow ridge while *H. rudolfensis* has a slight brow ridge.
7. *Homo habilis* was placed in the genus *Homo* as stone tools were found near the remains. The construction of stone tools was an important development in human history. This means that the discovery of the fossil bones at Dikika puts the use of stone tools back to 3.4 million years ago and the australopithecines. Current theories do not place stone tool culture in this period. This means ideas and timelines may need to be revised and further finds are needed to resolve the issue and confirm new revisions of the evolutionary tree.
8. D

## 51 *Homo ergaster*

1. *Homo ergaster* was named the 'working man' as many stone tools have been found with the remains.
2. *Homo ergaster* had a human-like body shape with similar limb proportions and height which would have been quite different to the ape-like ancestors. The face projects forward but the lower jaw was receding with no external chin.
3. *H. ergaster* lived in East Africa and South Africa between 1.9 and 1.5 million years ago.
4. (a) Turkana Boy is well known as it was the first fossil to be described as *H. ergaster* and is a complete skeleton.
   (b) Turkana Boy had a slender body frame, tall height, shorter arms and longer legs and was fully bipedal but had a modest brain size. Analysis of the bones of the skull which had not fully fused together and teeth place the boy between 7 and 15 years old when he died.
5. Debate about the classification of *H. ergaster* is due to belief by some, e.g. Colin Groves that there is insufficient differentiation between *H. ergaster* and *H. erectus* while Bernard Wood suggested that *H. ergaster* is the sister group of *H. sapiens* and is sufficiently different *to H. erectus*. He believes that the differences between the East African *H. erectus* and the Asian *H. erectus* should be made into separate species with the African forms being called *H. ergaster* so that the name *H. erectus* would be used solely for the Asian hominins.
6. Stone knapping as used in Acheulian culture would have involved bifacial knapping in which one side is hit so that a flake chips off and then the stone would be turned over to flake off another piece. This forms a hand axe tool with a sharp and fairly straight edge.
7. The investigation shows that the premotor cortex would have been used to make the stone hand axes in the Acheulian culture. Since this brain area is also involved in understanding language the results could infer that tool making and language developed at a similar time in human evolution. Further data needs to be collected to confirm they have a shared evolutionary history.
8. Oldowan tool culture of *H. habilis* included choppers, flaked cobbles and choppers while the Acheulian culture included hand axes. Stone choppers were probably used to break up carcasses and break bones to get the marrow. A chopper would have been made when a flake was broken from a stone leaving a sharp edge. The Acheulian culture made hand axes by stone knapping and were used for butchering carcasses.

## 52 *Homo erectus*

1. *Homo erectus* was named the 'upright man'.
2. Other names given to fossils now classified as *H. erectus* include *Pithecanthropus* (Java remains) and *Sinanthropus* (Northern China remains).
3. *H. erectus* existed between 1.8 million years ago and 250 000 years ago, though some place *H. erectus* as late as 30 000 years ago.
4. *H. erectus* had prominent brow ridges; protruding jaw although a less projecting face and smaller teeth than *H. habilis*; an increase in body size; sexually dimorphic (East African specimens); receding forehead; no chin and long, low, thick skull. Cranial capacity ranged from 750 to 1250 cc.
5. *H. erectus* is important as it was successful and existed for a long time and was the first hominin to be identified outside of Africa and Europe.
6. The Acheulian culture is associated with *H. erectus* with the development of hand axes and cleavers with a biface (worked both sides) with sharper and straighter edges that were used for butchering and some believe this means *H. erectus* was a hunter while others believe he was a scavenger.
7. (a) The order is (youngest) A, C, B, D (oldest).
   (b) Features used to work out the order could include the flatness of the face (younger) compared to a protruding jawline (older), eyebrow ridges are reduced in more recent times, size of brain is larger in more recent.
   (c) *H. erectus* is likely to be skull C as it has large brain, prominent eyebrow ridges, no chin.

## 53 *Homo heidelbergensis* and *Homo floresiensis*

1. *Homo heidelbergenesis* has the receding chin and robust jaw of *H. erectus* but a larger brain (1100 to 1300 cc) with a taller braincase, more prominent face and nose, and smaller molars.
2. It is believed *H. heidelbergensis* is the last common ancestor of Neanderthals in Europe and modern humans in Africa.
3. It is believed *H. heidelbergensis* lived from 600 000 to 200 000 years ago with fossils found in East Africa, North Africa, South Africa and several parts of Europe.
4. *H. heidelbergensis* manufactured stone tools using the prepared-core technique in a manner similar to Neanderthals and early modern humans. Tools were made for specific functions, e.g. knapped hand axes and wooden spears for hunting and evidence suggests that tools were kept and reused rather than being thrown away after use.
5. The diagram suggests that *H. floresiensis* evolved from a *H. erectus* that migrated to South-East Asia. This means that the last common ancestor with *H. sapiens* would have been an African *H. erectus* that lived about 1 million years ago.

## 54 *Homo neanderthalensis*

1. The first recognised Neanderthal fossils were found in 1856 in a cave in the Neander Valley, Germany.
2. People for many years thought of Neanderthals as brutish and bent over as Marcellin Boule found a fossil in 1908 with these features and it became the popular image.
3. Neanderthals lived in Europe in Siberia and south-west Asia and they lived from 350 000 to 28 000 years ago.
4. Neanderthals had short stocky bodies (less than 170 cm tall); big faces; heavy brow ridges; large skulls with a distinctive 'bun' shape at the back; definite chin; cheekbones were swept back; and they had midfacial prognathism.
5. Neanderthals are associated with the Mousterian culture.
6. Neanderthals used a type of prepared core stone-tool knapping trimming a piece of flint into a disc shaped core then chipping flakes from the edge thus collecting more flakes per core. The flakes were used as scrapers, points, knives and some were hafted to wooden shafts. The technique is known as the Levallois technique.
7. Neanderthals lived in open sites, rock shelters and caves and were successful hunters. Mousterian sites show a range of habitats and animal remains show the Neanderthals were seasonal hunters, e.g. killing reindeer in winter and red deer in summer.
8. Head and neck injuries resemble rodeo performer injuries suggesting close contact with hoofed, clawed or taloned animals as well as care by more able members of the group for the wounded.
9. Deliberate burial shows complex thought and social organisation. It could suggest religious beliefs and rituals associated with the burial of the dead.
10. The Neanderthals have been classified as a subspecies *Homo sapiens neanderthalensis*; others have classified them as their own species *H. neanderthalensis*; and others have classified them as a form of archaic *H. sapiens*.
11. In 1997, Svante Paabo used genetic material extracted from the bones of the original Neanderthal specimen found in 1856 and tested the mtDNA. He reported that the Neanderthals could not be the ancestors of modern humans, they did not interbreed and the Neanderthal lineage is four times older than the human one meaning they split off much earlier than humans. Paabo found that there are 27 differences on one strand of Neanderthal DNA while the maximum genetic variation for the entire human race along the same strand has 8 mutations.
12. Neanderthal nuclear DNA suggests they had red hair, though a different colour red to modern human red hair and pale skin.
13. There was an overlap in time of around 6000 years of *H. sapiens* and *H. neanderthalensis*.
14. The current debate revolves around genome comparisons which suggest divergence occurred 440 000 years ago but there is between 1 and 4 per cent DNA of modern Europeans and Asians, but not Africans, that comes from Neanderthals. This infers there was some assimilation with interbreeding between archaic populations.
15. The first fossil find of the Denisovans gave them their name and they lived in a Denisova cave in Siberia. Their DNA shows archaic human features and that they are closely related to Neanderthals. Other fossil finds have shown they lived in Western Europe, e.g. Spain.
16. It is hard to classify the Denisovans as they were neither Neanderthal nor modern humans and have some genetic similarities with current Melanesians. Further study is needed before any absolute conclusions can be drawn about the relationship between the Neanderthals, Denisovans and *H. sapiens*.
17. Modern technologies such as DNA sequencing techniques that can determine the genome of mtDNA have been able to estimate the similarities and differences between different hominin species, e.g. between the Denisovans, Neanderthals and modern humans. The movement of the different species of *Homo* around the world can be traced through genetic changes and this increased knowledge has led to modifications of existing theories about the human evolutionary tree.
18. The graph shows that skin pigmentation decreases with distance from the equator with paler skin in the Northern Hemisphere compared to the Southern Hemisphere for similar distances. Current knowledge has Neanderthals with pale skin which agrees with this graph as the Neanderthals are found in Europe.
19. (a) Skull L is *H. neanderthalensis*, skull M is chimpanzee, skull N is *H. sapiens*, skull O is the australopithecine and skull P is *H. erectus*.
    (b) *H. neanderthalensis* has a large brain which means it is not skulls M, O or P and compared to modern humans has a 'bun' at the back of the skull which makes it skull L and modern humans skull N.

## 55 *Homo sapiens*

1. *Homo sapiens* is the only extant species.
2. Cro-Magnons were named after the site in France where fossils were found.
3. *H. sapiens* have sharply rising forehead; less prominent brow ridges; prominent chin; face is not forward thrusting; more rounded skulls shorter from front to back; shorter lower jaws and a taller less robust skeleton. The rib cage is barrel shaped and the hands are dextrous.
4. Oldest African fossils of modern humans are 200 000 years old.
5. *H. sapiens* has a larger braincase than other *Homo* species and this allowed greater motor dexterity and more complex thought processes. These features aided tool making and increased the ability to adapt to new environments and to change the environment.
6. A

## 56 Dating Hominin Fossils

1. Relative dating uses the law of superposition to determine if something is 'older than' or 'younger than' whereas absolute dating gives a date in years with the experimental error of the method used in determining the date.
2. An index fossil is widely distributed but only lived for a short time which means it clearly indicates the age of the rock layer.
3. The geological time scale shows major evolutionary events from 650 million years ago to the present day.
4.

| Property | Description for fluorine absorption dating |
|---|---|
| Definition | Finds the relative age of bone fossils and how long it has been in the soil. |
| Type of dating | Relative dating method. |
| How absolute age is determined | Absolute age can only be determined if the numerical age of another item from the same area is known. Otherwise the age is either 'younger than' or 'older than' the other object. |
| Processes involved in bone accumulating fluorine | Ground water contains fluorine and when a bone is in the ground, fluoride ions from the soil replace some of the ions in the bone. The longer the bone is in the ground, the more fluorine present. |
| Limitations of fluorine absorption dating | Bones from same area can only be compared due to fluctuations in the amount of fluoride in ground water. There are many natural variables that may not have remained constant, e.g. rainfall, temperature, soil permeability and the bones being compared need to have been exposed to the same changing conditions. Not all bones absorb fluorine at the same rate so that bones of similar density need to be compared, e.g. difficulties comparing a femur with a vertebrae, or adult bone with juvenile bone, or from different size mammals. |

5. Fission tracking was used to date the rocks in the Olduvai Gorge in the Great Rift Valley in East Africa at sites where remains of australopithecines and *Homo erectus* have been found.
6. The Hominins refers to the human branch of the evolutionary tree. It includes bipedal humans and extinct ancestors and close relatives, e.g. other species in the genus *Homo* and the australopithecines. According to the table hominins have existed since the Pliocene epoch.
7. The half-life is the time it takes for half the atoms to decay and if the half-life is plotted onto a graph, the decay forms an exponential curve with certain radioisotopes being useful for a certain time frame.
8. Useful radioisotopes include radiocarbon, potassium-argon dating, rubidium-strontium dating and uranium-235 dating.
9. Since all living things contain carbon radiocarbon dating is important in dating organic material, e.g. bone, shell, charcoal.
10. Using particle accelerators as mass spectrometers allows radiocarbon dating to date materials up to 70 000 years old.
11. It was discovered that the ratio in the atmosphere of one atom of carbon-14 to every million million ($10^{12}$) carbon-12 atoms is not a totally fixed ratio. An analysis of growth rings on trees was used to make corrections for fluctuations in the carbon-14 content of the atmosphere.
12. Potassium-argon dating is suitable for volcanic rocks and minerals older than 200 000 to 300 000 years, e.g. the hominid fossils in East Africa are in layers associated with volcanism and sedimentation and this method is used to date the volcanic ash deposits.
13. Fission tracking dating is based on the spontaneous fission of uranium-238 and can be used on minerals and glass in a range from 100 years to 4500 million years before present.
14. The ratios of stable carbon and nitrogen isotopes in human tissues show a relationship with diet. For example, bone collagen mainly comes from dietary protein and an analysis of samples from Neanderthals and early humans shows palaeodiet (food eaten in ancient times), e.g. it is estimated that Neanderthals ate around 80% meat and 20% vegetables.
15. Strontium and oxygen in food and drinking water become part of bones and teeth and thus the mineral component of bones and teeth can indicate if the hominin has migrated from its place of birth to its place of death.
16. Dates of climate changes provide dates for different hominin activities, e.g. during ice ages sea level is low and land bridges allow greater migration, e.g. Indonesian islands were occupied by *H. erectus* in the early Pleistocene (1.5 million years ago) and Aboriginal Australians reached Australia at least 50 000 years ago.

## 57 Models for Human Evolutionary Trees

1. Any model of human evolution is based on limited data and is open to interpretation. There are many different theories of evolution as different scientists place emphasis on different features and different aspects of the available evidence. As some areas of evidence are not particularly abundant, it is possible to construct alternative theories to explain the same evidence. Until we have a more complete fossil history and genome data, different lines of thought will propose different theories.
2. As new discoveries increase our knowledge, e.g. finding the remains of *H. floresiensis* on the island of Flores in Indonesia current theories need to be refined or replaced.
3.

| Technique | Description |
|---|---|
| Digital calipers | Used to measure dimensions of bones to compare features of different fossils to help determine the range of measurements for each species. |
| Scanning electron microscope | Studies surface features of bones, e.g. microwear patterns on teeth to help determine diets of different species and cut marks left by tools. Sections of bones and teeth can show the age at death and identify ancient diseases. |
| CT scans | Make internal images of fossils and can transform image to make solid replica, e.g. to make whole skull reconstructions. |

4.

| Type of DNA | Use |
|---|---|
| Nuclear DNA | Contains genome for individual, e.g. studies of CD4 locus on chromosome 12 which suggests non-African populations descended from ancestors who left Africa about 90 000 years ago and studies of the beta-globin gene on chromosome 11 which suggests Asians have been evolving distinctive patterns for 200 000 years. |
| Y Chromosome | Looks at amino acid sequences on Y chromosome which determines male sex in humans to show inheritance along the male line without inheritance from two parents. |
| mtDNA | Found in mitochondria in the cytoplasm and is inherited from the egg and through the female line. Studies of mtDNA of human populations around the world suggest that there was a common ancestor about 150 000 years ago in Africa (African Eve) and studies are investigating human migration patterns to each continent. |

5. B

## 58 Models for Evolution of *Homo sapiens*

1. The evidence for ancient migrations comes from fossil bones, artefacts, genetic material and the rock layers in which they are found.
2.

| Feature | Regional continuity | Out of Africa |
|---|---|---|
| Theory | *H. erectus* had initial migration out of Africa with continuous gene flow with successive migrations from Africa. There was simultaneous worldwide evolution from *H. erectus* to *H. sapiens*. | *H. erectus* had initial migration out of Africa followed by more recent migration of *H. sapiens* from Africa. |
| Approximate time of migration | 1 000 000 years ago | 100 000 years ago |
| Supporting evidence | Fossils found outside Africa and dated before the proposed second wave, e.g. the Jinniushan skeleton.<br>Mungo Man (dated around 60 000 years old) is not descended from mitochondrial Eve but is anatomically identical to *H. sapiens*. | Oldest fossil of *H. sapiens* is in Africa.<br>mtDNA molecular clock evidence suggests maternal inheritance from African Eve.<br>Greatest variety in mtDNA is in Africa. |

3. There are several intermediate theories which are a combination of the out of Africa theory and the regional continuity theory, e.g. one theory suggests that Africa was predominant and modern features spread gradually from Africa with interbreeding between adjacent populations. This accounts for gene flow with the Neanderthals.

## 59 Development of Culture

1. (a) The Palaeolithic (2.5 million to 12 000 years ago) is the 'Stone Age'.
   (b) It is divided into three stages – the Lower Palaeolithic (2.5 million to 300 000 years ago) with pebble tools and hand axes, the Middle Palaeolithic (300 000 to 40 000 years ago) with the Levallois technique and the Upper Palaeolithic (40 000 to 10 000 years ago) with microlithic industry (very small tools).
2.

| Hominin | Inferred culture |
|---|---|
| *Australopithecus afarensis* | Probably lived in small family groups. Did not have a 'tool kit' but probably used tools like modern chimpanzee. |
| *Paranthropus boisei* | No culture known. Tools found could belong to *Homo* found at the same site. |
| *Homo habilis* | Oldowan culture and earliest tools, e.g. pebble tools such as stone flakes used as cutters and choppers. Hunter gatherer with probable home base. |
| *Homo erectus* | Acheulian culture – with use of fire, hunting, systematic tool making, e.g. hand axes, home bases, first hominin confirmed out of Africa |
| *Homo heidelbergensis* | Tool culture with sharper edges, hunter-gather, home bases |
| *Homo neanderthalensis* | Mousterian culture using stone knapping to make hand axes and other flake tools, e.g. to be used as scrapers, points, knives and some were hafted to wooden shafts. They lived in open sites, rock shelters and caves and were successful hunters, with close contact with hoofed, clawed or taloned animals as well as care by more able members of the group for the wounded. |
| *Homo sapiens* | Used bone, antler, made blade tools, cave art, small figurines, then writing, domestication of animals, agriculture, industry, technology. |

3. (a) In 1948 Hallam Movius proposed the Movius Line dividing the Acheulian tool users of Europe and Africa from the chopping tool cultures of East Asia showing the most easterly extent of hand axes.
   (b) The Movius Line has recently been contested due to the discovery of stone tools in China in 2014 and thus shows that models and interpretations of the human fossil record are often contested and need to be refined or replaced.
4. The Neolithic Revolution is the agricultural revolution which occurred around 15 000 to 10 000 years ago and saw the development of farming with the domestication of plants and animals dramatically changing human lifestyle from nomadic hunter-gatherer to settlement.
5. The Neolithic Revolution was followed by the Bronze Age and then the Iron Age. The discovery of metal technology ended the New Stone Age (Neolithic).
6. The Levallois technique is most associated with *H. neanderthalensis* and the Mousterian culture. It was used to make flakes and points for hand axes and scrapers.
7. Cultural development has significantly impacted human evolution. The development of stone tools aided the butchering of carcasses and the digging and cutting of plant material such as bulbs and roots. This increased the range and amount of food available and placed a selective pressure on those who could make finer and better tools and work together in a cooperative community. The Neolithic Revolution changing from hunter gatherer to farming with crops and domesticated animals increased food, led to the development of more complex social structures, changed the diet, and allowed job specialisation. With assured food supplies cites grew and tools became more sophisticated with increased ability to travel long distances, effective communication through the use of verbal and written language and pottery for food storage and handling. These cultural developments affected human evolution leading to further developments such as the Industrial Revolution and technology revolution. Medicine and human inventions mean that human evolution is now being changed by our own technology, e.g. poor eyesight is rectified by glasses and/or lens surgery, vaccinations give immunity to certain diseases, replacement surgery removes diseased organs, artificial limbs give people mobility after accidents and cochlear implants aid the deaf. Thus the impact of cultural development is quite significant on human evolution.
8. The graph shows a rise in human population around 6000 years ago. This corresponds to the beginning of metal technology which improved weapons and tools with copper instruments. The very recent increase relates to the Industrial Revolution with machines increasing food production and increased employment opportunities in factories in cities. The reduction in infant mortality and increased life span led to larger families and increased population.
9. (a) Cro-Magnons (early *H. sapiens*) are associated with Palaeolithic art.
   (b) Early art is found in painting on cave walls with most showing diagrams of animals such as wild cattle, mammoths, deer and horses.
   (c) Small figurines in ivory or moulded clay, e.g. of plump women (called venuses) are found in early art.

## Topic Test

**Section A – Multiple Choice (35 marks)**

| | | | | |
|---|---|---|---|---|
| 1. C | 2. B | 3. D | 4. D | 5. C |
| 6. B | 7. A | 8. B | 9. B | 10. B |
| 11. A | 12. A | 13. D | 14. C | 15. D |
| 16. A | 17. A | 18. C | 19. A | 20. B |
| 21. B | 22. D | 23. A | 24. C | 25. A |
| 26. C | 27. D | 28. A | 29. D | 30. A |
| 31. C | 32. B | 33. C | 34. B | 35. A |

**Section B – Written Response (65 marks)**

36. (a) Marking Guidelines

| Criteria | Marks |
|---|---|
| Correctly identifies term | 1 |

*Sample answer:*
It is a point mutation or base substitution.

(b) Marking Guidelines

| Criteria | Marks |
|---|---|
| Relates DNA code to protein synthesis,<br>AND<br>how a change in proteins affects human health | 3 |
| Relates DNA code to protein synthesis,<br>OR<br>how a change in proteins affects human health | 1 to 2 |

*Sample answer:*
The diagram shows the original DNA code. This code relates to the production of particular proteins. Each protein is made of a sequence of amino acids. If the code changes, it changes an amino acid, e.g. in this case isoleucine is changed to phenylalanine. This in turn changes the protein produced. The proteins, e.g. enzymes control cell metabolism and structure. Thus a change in the code could change the processes occurring in the cell. Several diseases are caused by such changes, e.g. thalassemia major is caused by a base substitution.

37. (a) Marking Guidelines

| Criteria | Marks |
|---|---|
| Correctly defines chromosome mutation and gives at least two types | 3 |
| Correctly defines chromosome mutation and/or gives one or two types | 1 to 2 |

*Sample answer:*
Chromosome mutation is a change in the DNA that changes the chromosome themselves and can involve deletion, insertion, translocation or inversion of parts or all of a chromosome.

(b) Marking Guidelines

| Criteria | Marks |
|---|---|
| Gives at least three causes of chromosome mutation | 3 |
| Give one or two causes of chromosome mutation | 1 to 2 |

*Sample answer:*
Chromosome mutations can be caused by unequal crossing over, misalignment during replication, mutagens or incorrect repair. Sometimes incorrect separation during meiosis can cause a change in the chromosome number.

38. (a) Marking Guidelines

| Criteria | Marks |
|---|---|
| Identifies the composition of a nucleotide with the complementary pairing<br>AND<br>links importance in DNA replication | 2 |
| Identifies some aspect of nucleotides | 1 |

*Sample answer:*
A nucleotide consists of a sugar, a phosphate and a nitrogenous base. There are four nucleotides with the bases adenine (A), thymine (T), guanine (G) and cytosine (C). They form complementary pairs: A-T and C-G. This means that when the two strands of DNA separate in DNA replication each strand will form a template to produce a new strand. The correct coding will be maintained which means each gene will have the correct coding.

(b) Marking Guidelines

| Criteria | Marks |
|---|---|
| Identifies the type of bond that is weak allowing them to break during DNA replication<br>AND<br>links importance of bond in DNA replication | 2 |
| Identifies some aspect of the weak bond | 1 |

*Sample answer:*
The two nitrogenous bases that form the 'rung' of the DNA 'ladder' are held together by weak hydrogen bonds. These bonds break during DNA replication allowing the two strands to separate so that replication can occur. These bonds are thus important in DNA replication.

(c) Marking Guidelines

| Criteria | Marks |
|---|---|
| Identifies the function of DNA polymerase and relates this to DNA replication | 2 |
| Identifies some aspect of DNA polymerase | 1 |

*Sample answer:*
DNA polymerases are enzymes that catalyse the elongation of new DNA at a replication fork in DNA replication by the addition of nucleotides to the existing chain. The DNA polymerase adds nucleotides one by one 'reading' the existing DNA as a template code and adding free nucleotides to the 3′ end of a newly formed strand.

39. (a) Marking Guidelines

| Criteria | Marks |
|---|---|
| Defines a mutagen and explains its effect on genes | 2 |
| Defines a mutagen or explains its effect on genes | 1 |

*Sample answer:*
Mutagens are chemical or physical agents that interact with DNA and cause mutations. Genes and the DNA molecule can be damaged by chemical mutagens, e.g. benzene, formalin, Agent Orange altering the code so that dysfunctional proteins are produced or no protein is produced.

(b) Marking Guidelines

| Criteria | Marks |
|---|---|
| Constructs a table that clearly identifies the effect of the mutation on the chromosome number | 3 |
| Provides a table with some relevant information | 1 to 2 |

*Sample answer:*

| Mutation | Effect on chromosome number |
|---|---|
| Base substitution | No change in chromosome number. |
| Trisomy | Organism has three copies of one type of chromosome. |
| Polyploidy | Organism has three or more complete sets of chromosomes. |

40. Marking Guidelines

| Criteria | Marks |
|---|---|
| Constructs a closed table with suitable heading and columns to compare the pelvis, spine and big toe of human and gorilla | 6 |
| Some aspect of table given | 1 to 5 |

*Sample answer:*

| Feature | Human | Gorilla |
|---|---|---|
| Spine | Greater curvature with S shape | C shaped spine |
| Pelvis | Shorter, broader and more rounded | Long and narrow |
| Big toe | In line with other toes | Not in line with other toes |

41. (a) Marking Guidelines

| Criteria | Marks |
|---|---|
| Describes a founder population, relates to isolation and gene frequency | 3 |
| Recognises some aspects of why a disease has high prevalence | 1 to 2 |

*Sample answer:*
The settlement of Iceland involved a relatively small founder population. Geographical isolation kept the population relatively genetically isolated. This meant that the presence of a disease allele in the founder population will lead to higher incidence of that disease compared to the whole human population.

(b) Marking Guidelines

| Criteria | Marks |
|---|---|
| Defines a bottleneck population and explains its impact | 2 |
| Defines a bottleneck population | 1 |

*Sample answer:*
A population bottleneck occurs when the abundance and distribution of an existing population dramatically reduces. This means that many genetic lineages within the population become extinct and only a few remain. This increases the rate of genetic drift and decreases genetic diversity.

42. Marking Guidelines

| Criteria | Marks |
|---|---|
| Relates how fossils are formed and discovered to why the fossil record is incomplete | 4 |
| Gives some reason why the fossil record is incomplete | 1 to 3 |

*Sample answer:*
There are several reasons why the fossil record is incomplete – 1. Few organisms become fossils, e.g. most fossils are hard parts such as bone, shells and teeth as soft parts decompose. Suitable conditions for fossilisation include aquatic environments, e.g. shallow lakes, marshes, swamps or marine habitats where the organism is quickly covered by sediment, buried and decay stops. Other suitable conditions include dry caves or deserts or the organism becoming trapped in ice, tar or amber. Sometimes traces of organisms are found, e.g. footprints, tracks, burrows, nests, or hardened animal dung. 2. If the fossil forms, the rock needs to be in an area that is not actively involved in the rock cycle where the fossil is destroyed by plate subduction or igneous intrusions. 3. The fossil needs to be found. Most fossils are discovered when they are exposed by weathering and erosion. There are many remote and inaccessible areas that have yet to be searched for fossils.

43. Marking Guidelines

| Criteria | Marks |
|---|---|
| Refers to the diagram to give examples why there is difficulty in determining the exact pathway of the evolution of humans giving clear, logical and concise reasoning | 6 |
| Gives some explanation of pathway of evolution of humans with reference to the diagram | 1 to 5 |

*Sample answer:*
The construction of timelines shows that at various stages in hominin evolution several species coexisted, e.g. *H. habilis* coexisted with *A. africanus*. This overlap is important in the process of evolution as natural selection involves the survival of those most suited to the environment. If several groups inhabit the same region at the same time, natural selection leads to specialisation with different groups occupying different ecological niches. Over time, some groups become so specialised they cannot survive changing conditions or they may find themselves at a disadvantage compared to another group which has progressed in some way. The timeline shows the difficulty in determining the exact pathway of humans solely on timelines, e.g. *H. neanderthalensis* and *H. sapiens* coexisted in Europe for about half a million years with the numbers of *H. neanderthalensis* decreasing around 45 000 years ago. There is debate about the relationship between the two groups and whether the Neanderthals are a side branch of the evolutionary tree that became extinct or whether there was interbreeding and gene flow into the *H. sapiens* lineage. It is clear that the tool making ability of *H. sapiens* contemporary with *H. neaderthalensis* was superior and would have given *H. sapiens* a selective advantage. The exact pathway of the evolution of humans is still debated.

44. (a) Marking Guidelines

| Criteria | Marks |
|---|---|
| Outlines some of the evidence with examples of different hominins that shows there has been a change in diet during the evolution of humans | 4 |
| Gives limited evidence to show change in diet | 1 to 3 |

*Sample answer:*
Teeth provide evidence for changing diet from australopithecines to *Homo* and cultural artefacts, e.g. tools and bones found in middens show dietary developments since *Homo erectus*. Teeth fossilise well and a study of different teeth types, e.g. the relative prominence of the canines and the shape of the cheek teeth cusps, can be used to identify different hominins and determine their diet. australopithecines have typical ape-like first premolars but large flat cheek teeth similar to later hominins. The thickness of the enamel caps on the molars of australopithecines is used to infer diet and shows there was a change from being primarily herbivorous to omnivore (chimpanzees have thin enamel caps). Tools are often made for food collection, e.g. early stone tools were used for animal butchering and cutting soft vegetation. The bones of animals found with the stone artefacts have tell-tale marks to show meat had been removed using a stone tool. Thus the evidence shows a change in diet from herbivorous to omnivore with the killing and butchering of animals for food.

(b) Marking Guidelines

| Criteria | Marks |
|---|---|
| Relates the change in diet during the evolution of humans to increase in brain size | 3 |
| Shows some relationship between diet and brain size | 1 to 2 |

*Sample answer:*
Relative brain size (the ratio of brain to body size) distinguishes humans from other primates. Large brains require a high energy input and are a metabolically expensive organ, especially in humans who possess an enlarged temporal lobe and considerable sections of the brain are involved with complex voluntary behaviour and motor control. The shift to a diet higher in protein (meat) would have provided more energy, which can be digested more easily than vegetation. The long gestation period for humans, combined with a long period of parental care are also energy costing and the shift in diet would have complemented the critical growth times for the foetus and developing infant. Fossil evidence shows that there was a change in diet while brain size increased at the same time. Thus there is a positive relationship between change in diet and increase in brain size.

45. (a) Marking Guidelines

| Criteria | Marks |
|---|---|
| Defines culture | 1 |

*Sample answer:*
Culture refers to the behaviour or lifestyle passed by word of mouth, or written means from one generation to the next. It is learned behaviour acquired by individuals and reflects human social life.

(b) Marking Guidelines

| Criteria | Marks |
|---|---|
| Refers to both genetic and cultural evolution with examples and their relative importance during the evolution of humans in a logical and coherent manner | 6 |
| Discusses some aspects of genetic and/or cultural evolution of humans | 1 to 5 |

*Sample answer:*
The genetic evolution of humans was very important in the early evolution of humans, e.g. becoming upright, freeing the hands to carry food and larger brain size with particular development of the cerebrum. The increased brain size led to the development of tools, e.g. stone tools for chopping or scraping and then to flint and bone tools and more sophisticated tools with specialised functions. Since the emergence of *H. sapiens* cultural evolution has been more important than genetic evolution. Cultural development has been a significant feature of human development. The main stages in cultural evolution are tool making, hunter gatherer lifestyle with home bases, the use of fire, burial of the dead, the development of art, the agricultural revolution, the development of writing, the Industrial Revolution and the technological revolution.
Current tool making abilities enable humans to modify their environment, e.g. to survive under water in submarines, to survive on space station satellites, to live in deserts, high altitudes or subzero conditions. Being able to control disease, food supplies and the populations of other animals, e.g. predators means that many natural selection pressures are not as influential as they have been in the past. Humans, through their cultural evolution, have the potential of influencing their own evolution, e.g. applying genetic techniques, birth control, medicine and public health.

46. Marking Guidelines

| Criteria | Marks |
|---|---|
| Identifies technological advance<br>OR<br>Shows how there has been a change in thinking about evolutionary relationships | 2 |
| Identifies technological advance<br>OR<br>Shows how there has been a change in thinking about evolutionary relationships | 1 |

*Sample answer:*
DNA-DNA hybridisation is a new technological advance that shows the degree of similarity of DNA between two species. This procedure has changed our thinking about relationships in the primate order. It had been thought that gorillas were our closest relative but DNA-DNA hybridisation has shown that chimpanzees are closer to us.

47. (a) Marking Guidelines

| Criteria | Marks |
|---|---|
| Correctly distinguishes between binocular vision and stereoscopic vision | 2 |
| Only defines one term | 1 |

*Sample answer:*
Binocular vision means using two eyes and usually infers that both eyes can focus on the same target object. Stereoscopic vision involves the processing of images that overlap to give a 3-D image with depth perception.

(b) Marking Guidelines

| Criteria | Marks |
|---|---|
| Correctly links very large field of view to ability to adaptive advantage using an example from the diagram | 2 |
| Correctly links very large field of view to ability to adaptive advantage | 1 |

*Sample answer:*
Prey animals, e.g. the rabbit in the diagram have laterally placed eyes that give them an extremely wide field of view that can look forward and backward without moving the head. This means they can more easily see the approach of predators and escape in a direction away from the predators.

(c) Marking Guidelines

| Criteria | Marks |
|---|---|
| Correctly links forward facing eyes with overlapping field of view<br>AND<br>links to lifestyle of a named primate | 2 |
| Correctly links forward facing eyes with overlapping field of view<br>OR<br>links to lifestyle of a named primate | 1 |

*Sample answer:*
Forward facing eyes means that the field of view for each eye overlaps giving binocular vision. This is very important for arboreal lifestyles, e.g. gibbons, spider monkeys, orangutans as the brain interprets the two images to give a 3-D (stereoscopic) view so they can jump from branch to branch.

(d) Marking Guidelines

| Criteria | Marks |
|---|---|
| Correctly identifies another feature of vision<br>AND<br>suggests why this feature gives an adaptive evolutionary advantage | 2 |
| Correctly identifies another feature of vision | 1 |

*Sample answer:*
Most primates have colour vision with both rods and cones in the retina (nocturnal primates usually do not see in colour). Colour vision is an advantage for fruit eaters to tell when the fruit is ripe and colour is used for social communication which helps survival of a group as they act together as a unit.

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